

Codigo Fuente

NomiCore - Sistema de Gestion de Nomina

Proyecto: ProyFinal_LPI_Eq01_NomiCore

Tecnologia: C# 7.3, Windows Forms .NET Framework 4.8 y SQLite

Tema: Manejo de campos en las clases y PrintDocument

Contenido del documento

96 archivos C# incluidos

15,597 lineas de codigo y herramientas

Se excluyen: bin, obj, .git, .vs, packages y salidas de exportacion.

Data/AbsenceRequestRepository.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Data.SQLite;
4  using SistemaGestionNomina.Models;
5
6  namespace SistemaGestionNomina.Data
7  {
8      public sealed class AbsenceRequestRepository
9      {
10         private const string SelectSql = @"SELECT s.*, e.Codigo AS CodigoEmpleado,
11             e.Nombre || ' ' || e.Apellido AS EmpleadoNombre,
12             d.Nombre AS DepartamentoNombre
13         FROM SolicitudesAusencia s
14         INNER JOIN Empleados e ON e.IdEmpleado = s.IdEmpleado
15         INNER JOIN Departamentos d ON d.IdDepartamento = e.IdDepartamento ";
16
17         public List<SolicitudAusencia> Search(AbsenceQuery query, int? forcedEmployeeId,
18             int? scopeDepartmentId)
19         {
20             query = query ?? new AbsenceQuery();
21             List<SolicitudAusencia> items = new List<SolicitudAusencia>();
22             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
23             using (SQLiteCommand command = new SQLiteCommand(SelectSql + @"
24                 WHERE (@forcedEmployee = 0 OR s.IdEmpleado = @forcedEmployee)
25                 AND (@empleado = 0 OR s.IdEmpleado = @empleado)
26                 AND (@departamento = 0 OR e.IdDepartamento = @departamento)
27                 AND (@scopeDep = 0 OR e.IdDepartamento = @scopeDep)
28                 AND (@tipo = '' OR s.Tipo = @tipo)
29                 AND (@estado = '' OR s.Estado = @estado)
30                 AND (@desde = '' OR s.FechaFin >= @desde)
31                 AND (@hasta = '' OR s.FechaInicio <= @hasta)
32                 AND (@search = '' OR e.Codigo LIKE @like OR e.Nombre LIKE @like
33                     OR e.Apellido LIKE @like OR s.Motivo LIKE @like)
34                 ORDER BY s.FechaCreacion DESC, s.IdSolicitud DESC;", connection))
35             {
36                 command.Parameters.AddWithValue("@forcedEmployee", forcedEmployeeId ?? 0);
37                 command.Parameters.AddWithValue("@empleado", query.EmployeeId ?? 0);
38                 command.Parameters.AddWithValue("@departamento", query.DepartmentId ?? 0);
39                 command.Parameters.AddWithValue("@scopeDep", scopeDepartmentId ?? 0);
40                 command.Parameters.AddWithValue("@tipo", query.Type ?? string.Empty);
41                 command.Parameters.AddWithValue("@estado", query.Status ?? string.Empty);
42                 command.Parameters.AddWithValue("@desde", query.From.HasValue ? query.From.Value.ToString("yyyy-MM-dd") : string.Empty);
43                 command.Parameters.AddWithValue("@hasta", query.To.HasValue ? query.To.Value.ToString("yyyy-MM-dd") : string.Empty);
44                 command.Parameters.AddWithValue("@search", query.Search ?? string.Empty);
45                 command.Parameters.AddWithValue("@like", "%" + (query.Search ?? string.Empty) + "%");
46                 using (SQLiteDataReader reader = command.ExecuteReader())
47                 {
48                     while (reader.Read()) items.Add(Map(reader));
49                 }
50             }
51             return items;
52         }
53
54         public SolicitudAusencia GetById(int requestId)
55         {
56             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
57             {
58                 return GetById(connection, null, requestId);
59             }
60         }

```

Data/AbsenceRequestRepository.cs (continuacion)

```
61
62     public SolicitudAusencia GetById(SQLiteConnection connection, SQLiteTransaction transaction,
63         int requestId)
64     {
65         using (SQLiteCommand command = new SQLiteCommand(SelectSql +
66             " WHERE s.IdSolicitud = @id LIMIT 1;", connection, transaction))
67         {
68             command.Parameters.AddWithValue("@id", requestId);
69             using (SQLiteDataReader reader = command.ExecuteReader())
70             {
71                 return reader.Read() ? Map(reader) : null;
72             }
73         }
74     }
75
76     public int Add(SQLiteConnection connection, SQLiteTransaction transaction, SolicitudAusencia item)
77     {
78         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO SolicitudesAusencia
79             (IdEmpleado, Tipo, FechaInicio, FechaFin, Estado, Motivo,
80             UsuarioSolicitante, FechaCreacion)
81             VALUES (@empleado, @tipo, @inicio, @fin, @estado, @motivo, @usuario, @creacion);
82             SELECT last_insert_rowid();", connection, transaction))
83         {
84             command.Parameters.AddWithValue("@empleado", item.IdEmpleado);
85             command.Parameters.AddWithValue("@tipo", item.Tipo);
86             command.Parameters.AddWithValue("@inicio", item.FechaInicio.ToString("yyyy-MM-dd"));
87             command.Parameters.AddWithValue("@fin", item.FechaFin.ToString("yyyy-MM-dd"));
88             command.Parameters.AddWithValue("@estado", AbsenceStates.Pending);
89             command.Parameters.AddWithValue("@motivo", item.Motivo ?? string.Empty);
90             command.Parameters.AddWithValue("@usuario", item.UsuarioSolicitante ?? string.Empty);
91             command.Parameters.AddWithValue("@creacion", item.FechaCreacion.ToString("yyyy-MM-dd HH:mm:ss"));
92             return Convert.ToInt32(command.ExecuteScalar());
93         }
94     }
95
96     public void UpdateState(SQLiteConnection connection, SQLiteTransaction transaction, int requestId,
97         string expectedState, string newState, string resolvedBy, DateTime resolvedAt, string observation)
98     {
99         using (SQLiteCommand command = new SQLiteCommand(@"UPDATE SolicitudesAusencia
100             SET Estado = @nuevo,
101             AprobadoPor = CASE WHEN @resolver = '' THEN AprobadoPor ELSE @resolver END,
102             FechaAprobacion = CASE WHEN @resolver = '' THEN FechaAprobacion ELSE @fecha END,
103             ObservacionResolucion = @observacion
104             WHERE IdSolicitud = @id AND Estado = @anterior;", connection, transaction))
105         {
106             command.Parameters.AddWithValue("@nuevo", newState);
107             command.Parameters.AddWithValue("@resolver", resolvedBy ?? string.Empty);
108             command.Parameters.AddWithValue("@fecha", resolvedAt.ToString("yyyy-MM-dd HH:mm:ss"));
109             command.Parameters.AddWithValue("@observacion", observation ?? string.Empty);
110             command.Parameters.AddWithValue("@id", requestId);
111             command.Parameters.AddWithValue("@anterior", expectedState);
112             if (command.ExecuteNonQuery() != 1)
113                 throw new InvalidOperationException("La solicitud cambió de estado antes de completar la operación.");
114         }
115     }
116
117     public bool HasApprovedOverlap(SQLiteConnection connection, SQLiteTransaction transaction,
118         int employeeId, DateTime start, DateTime end, int ignoreRequestId)
119     {
120         using (SQLiteCommand command = new SQLiteCommand(@"SELECT COUNT(1) FROM SolicitudesAusencia
```

Data/AbsenceRequestRepository.cs (continuacion)

```

121         WHERE IdEmpleado = @empleado AND Estado = 'Aprobada' AND IdSolicitud <> @ignorar
122             AND FechaInicio <= @fin AND FechaFin >= @inicio;", connection, transaction))
123     {
124         command.Parameters.AddWithValue("@empleado", employeeId);
125         command.Parameters.AddWithValue("@inicio", start.ToString("yyyy-MM-dd"));
126         command.Parameters.AddWithValue("@fin", end.ToString("yyyy-MM-dd"));
127         command.Parameters.AddWithValue("@ignorar", ignoreRequestId);
128         return Convert.ToInt32(command.ExecuteScalar()) > 0;
129     }
130 }
131
132 public bool AttendanceExists(SQLiteConnection connection, SQLiteTransaction transaction,
133     int employeeId, DateTime date)
134 {
135     using (SQLiteCommand command = new SQLiteCommand(@"SELECT COUNT(1) FROM Asistencias
136         WHERE IdEmpleado = @empleado AND Fecha = @fecha;", connection, transaction))
137     {
138         command.Parameters.AddWithValue("@empleado", employeeId);
139         command.Parameters.AddWithValue("@fecha", date.ToString("yyyy-MM-dd"));
140         return Convert.ToInt32(command.ExecuteScalar()) > 0;
141     }
142 }
143
144 public void AddGeneratedAttendance(SQLiteConnection connection, SQLiteTransaction transaction,
145     int requestId, int employeeId, DateTime date, string status)
146 {
147     using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Asistencias
148         (IdEmpleado, Fecha, HoraEntrada, HoraSalida, HorasTrabajadas, Estado, IdSolicitudAusencia)
149         VALUES (@empleado, @fecha, NULL, NULL, 0, @estado, @solicitud);", connection, transaction))
150     {
151         command.Parameters.AddWithValue("@empleado", employeeId);
152         command.Parameters.AddWithValue("@fecha", date.ToString("yyyy-MM-dd"));
153         command.Parameters.AddWithValue("@estado", status);
154         command.Parameters.AddWithValue("@solicitud", requestId);
155         command.ExecuteNonQuery();
156     }
157 }
158
159 public int DeleteGeneratedAttendance(SQLiteConnection connection, SQLiteTransaction transaction,
160     int requestId)
161 {
162     using (SQLiteCommand command = new SQLiteCommand(
163         "DELETE FROM Asistencias WHERE IdSolicitudAusencia = @solicitud;", connection, transaction))
164     {
165         command.Parameters.AddWithValue("@solicitud", requestId);
166         return command.ExecuteNonQuery();
167     }
168 }
169
170 private static SolicitudAusencia Map(SQLiteDataReader reader)
171 {
172     return new SolicitudAusencia
173     {
174         IdSolicitud = Convert.ToInt32(reader["IdSolicitud"]),
175         IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
176        CodigoEmpleado = Convert.ToString(reader["CodigoEmpleado"]),
177         EmpleadoNombre = Convert.ToString(reader["EmpleadoNombre"]),
178         DepartamentoNombre = Convert.ToString(reader["DepartamentoNombre"]),
179         Tipo = Convert.ToString(reader["Tipo"]),
180         FechaInicio = DateTime.Parse(Convert.ToString(reader["FechaInicio"])),

```

Data/AbsenceRequestRepository.cs (continuacion)

```
181         FechaFin = DateTime.Parse(Convert.ToString(reader["FechaFin"])),
182         Estado = Convert.ToString(reader["Estado"]),
183         Motivo = Convert.ToString(reader["Motivo"]),
184         AprobadoPor = Convert.ToString(reader["AprobadoPor"]),
185         FechaAprobacion = reader["FechaAprobacion"] == DBNull.Value ||
186             string.IsNullOrEmpty(Convert.ToString(reader["FechaAprobacion"]))
187             ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaAprobacion"])),
188         ObservacionResolucion = Convert.ToString(reader["ObservacionResolucion"]),
189         UsuarioSolicitante = Convert.ToString(reader["UsuarioSolicitante"]),
190         FechaCreacion = DateTime.Parse(Convert.ToString(reader["FechaCreacion"]));
191     };
192 }
193 }
194 }
```

Data/AuditRepository.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using SistemaGestionNomina.Models;
5
6 namespace SistemaGestionNomina.Data
7 {
8     public sealed class AuditRepository
9     {
10         public void Add(AuditRecord record)
11         {
12             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
13             {
14                 Add(connection, null, record);
15             }
16         }
17
18         public void Add(SQLiteConnection connection, SQLiteTransaction transaction, AuditRecord record)
19         {
20             if (connection == null) throw new ArgumentNullException("connection");
21             if (record == null) throw new ArgumentNullException("record");
22
23             using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Auditoria
24                 (Usuario, Modulo, Accion, Detalle, Fecha)
25                 VALUES (@usuario, @modulo, @accion, @detalle, @fecha);", connection, transaction))
26             {
27                 command.Parameters.AddWithValue("@usuario", record.Usuario);
28                 command.Parameters.AddWithValue("@modulo", record.Modulo);
29                 command.Parameters.AddWithValue("@accion", record.Accion);
30                 command.Parameters.AddWithValue("@detalle", record.Detalle ?? string.Empty);
31                 command.Parameters.AddWithValue("@fecha", record.Fecha.ToString("yyyy-MM-dd HH:mm:ss"));
32                 command.ExecuteNonQuery();
33             }
34         }
35
36         public List<AuditRecord> Search(AuditQuery query)
37         {
38             List<AuditRecord> records = new List<AuditRecord>();
39             int limit = query == null || query.Limit <= 0 ? 100 : Math.Min(query.Limit, 500);
40             int offset = query == null ? 0 : Math.Max(0, query.Offset);
41             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
42             using (SQLiteCommand command = new SQLiteCommand(@"SELECT * FROM Auditoria
43                 WHERE (@usuario = '' OR Usuario LIKE @usuarioLike)
44                     AND (@modulo = '' OR Modulo = @modulo)
45                     AND (@accion = '' OR Accion = @accion)
46                     AND (@detalle = '' OR Detalle LIKE @detalleLike)
47                     AND (@desde = '' OR Fecha >= @desde)
48                     AND (@hasta = '' OR Fecha < @hasta)
49                 ORDER BY Fecha DESC, IdAuditoria DESC LIMIT @limit OFFSET @offset;", connection))
50             {
51                 string user = query == null ? string.Empty : query.Usuario ?? string.Empty;
52                 string module = query == null ? string.Empty : query.Modulo ?? string.Empty;
53                 string action = query == null ? string.Empty : query.Accion ?? string.Empty;
54                 string detail = query == null ? string.Empty : query.Detalle ?? string.Empty;
55                 command.Parameters.AddWithValue("@usuario", user);
56                 command.Parameters.AddWithValue("@usuarioLike", "%" + user + "%");
57                 command.Parameters.AddWithValue("@modulo", module);
58                 command.Parameters.AddWithValue("@accion", action);
59                 command.Parameters.AddWithValue("@detalle", detail);
60                 command.Parameters.AddWithValue("@detalleLike", "%" + detail + "%");
```

Data/AuditRepository.cs (continuación)

```
61         command.Parameters.AddWithValue("@desde", query != null && query.FechaDesde.HasValue
62             ? query.FechaDesde.Value.Date.ToString("yyyy-MM-dd HH:mm:ss") : string.Empty);
63         command.Parameters.AddWithValue("@hasta", query != null && query.FechaHasta.HasValue
64             ? query.FechaHasta.Value.Date.AddDays(1).ToString("yyyy-MM-dd HH:mm:ss") : string.Empty);
65         command.Parameters.AddWithValue("@limit", limit);
66         command.Parameters.AddWithValue("@offset", offset);
67         using (SQLiteDataReader reader = command.ExecuteReader())
68         {
69             while (reader.Read()) records.Add(Map(reader));
70         }
71     }
72
73     return records;
74 }
75
76 public List<string> GetDistinct(string column)
77 {
78     if (column != "Modulo" && column != "Accion") throw new ArgumentException("Columna de auditoría no válida.");
79     List<string> values = new List<string>();
80     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
81     using (SQLiteCommand command = new SQLiteCommand(
82         "SELECT DISTINCT " + column + " FROM Auditoria ORDER BY " + column + ";", connection))
83     using (SQLiteDataReader reader = command.ExecuteReader())
84     {
85         while (reader.Read()) values.Add(Convert.ToString(reader[0]));
86     }
87     return values;
88 }
89
90 private static AuditRecord Map(SQLiteDataReader reader)
91 {
92     return new AuditRecord
93     {
94         IdAuditoria = Convert.ToInt32(reader["IdAuditoria"]),
95         Usuario = Convert.ToString(reader["Usuario"]),
96         Modulo = Convert.ToString(reader["Modulo"]),
97         Accion = Convert.ToString(reader["Accion"]),
98         Detalle = Convert.ToString(reader["Detalle"]),
99         Fecha = DateTime.Parse(Convert.ToString(reader["Fecha"]));
100     };
101 }
102 }
103 }
```

Data/AuthenticationRepository.cs

```
1 using System;
2 using System.Data.SQLite;
3 using SistemaGestionNomina.Models;
4
5 namespace SistemaGestionNomina.Data
6 {
7     public sealed class AuthenticationRepository
8     {
9         public Usuario GetByUsername(string username)
10         {
11             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
12             using (SQLiteCommand command = new SQLiteCommand(
13                 "SELECT * FROM Usuarios WHERE NombreUsuario = @username LIMIT 1;", connection))
14             {
15                 command.Parameters.AddWithValue("@username", username);
16                 using (SQLiteDataReader reader = command.ExecuteReader())
17                 {
18                     return reader.Read() ? Map(reader) : null;
19                 }
20             }
21         }
22
23         public bool UpdatePassword(int userId, string passwordHash)
24         {
25             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
26             using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Usuarios SET PasswordHash = @hash
27                 WHERE IdUsuario = @id AND Estado = 'Activo';", connection))
28             {
29                 command.Parameters.AddWithValue("@hash", passwordHash);
30                 command.Parameters.AddWithValue("@id", userId);
31                 return command.ExecuteNonQuery() == 1;
32             }
33         }
34
35         public void RegisterFailedAttempt(int userId, int attempts, bool blocked, DateTime? blockedAt)
36         {
37             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
38             using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Usuarios SET
39                 IntentosFallidos = @attempts, Bloqueado = @blocked, FechaBloqueo = @blockedAt
40                 WHERE IdUsuario = @id;", connection))
41             {
42                 command.Parameters.AddWithValue("@attempts", attempts);
43                 command.Parameters.AddWithValue("@blocked", blocked ? 1 : 0);
44                 command.Parameters.AddWithValue("@blockedAt", blockedAt.HasValue
45                     ? (object)blockedAt.Value.ToString("yyyy-MM-dd HH:mm:ss") : DBNull.Value);
46                 command.Parameters.AddWithValue("@id", userId);
47                 command.ExecuteNonQuery();
48             }
49         }
50
51         public void CompleteLogin(int userId, DateTime lastAccess, string upgradedHash)
52         {
53             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
54             using (SQLiteTransaction transaction = connection.BeginTransaction())
55             using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Usuarios SET
56                 UltimoAcceso = @access, IntentosFallidos = 0, Bloqueado = 0, FechaBloqueo = NULL,
57                 PasswordHash = CASE WHEN @hash = '' THEN PasswordHash ELSE @hash END
58                 WHERE IdUsuario = @id;", connection, transaction))
59             {
60                 command.Parameters.AddWithValue("@access", lastAccess.ToString("yyyy-MM-dd HH:mm:ss"));
```


Data/AuthenticationRepository.cs (continuacion)

```
61         command.Parameters.AddWithValue("@hash", upgradedHash ?? string.Empty);
62         command.Parameters.AddWithValue("@id", userId);
63         command.ExecuteNonQuery();
64         transaction.Commit();
65     }
66 }
67
68 private static Usuario Map(SQLiteDataReader reader)
69 {
70     return new Usuario
71     {
72         IdUsuario = Convert.ToInt32(reader["IdUsuario"]),
73         NombreUsuario = Convert.ToString(reader["NombreUsuario"]),
74         PasswordHash = Convert.ToString(reader["PasswordHash"]),
75         Rol = Convert.ToString(reader["Rol"]),
76         Estado = Convert.ToString(reader["Estado"]),
77         IdEmpleado = reader["IdEmpleado"] == DBNull.Value ? (int?)null : Convert.ToInt32(reader["IdEmpleado"]),
78         UltimoAcceso = ParseNullableDate(reader["UltimoAcceso"]),
79         IntentosFallidos = Convert.ToInt32(reader["IntentosFallidos"]),
80         Bloqueado = Convert.ToInt32(reader["Bloqueado"]) == 1,
81         FechaBloqueo = ParseNullableDate(reader["FechaBloqueo"])
82     };
83 }
84
85 private static DateTime? ParseNullableDate(object value)
86 {
87     if (value == null || value == DBNull.Value || string.IsNullOrEmpty(Convert.ToString(value))) return null;
88     DateTime parsed;
89     return DateTime.TryParse(Convert.ToString(value), out parsed) ? parsed : (DateTime?)null;
90 }
91 }
92 }
```

Data/DatabaseInitializer.cs

```
1 using System;
2 using System.Data.SQLite;
3 using System.IO;
4 using SistemaGestionNomina.Helpers;
5
6 namespace SistemaGestionNomina.Data
7 {
8     public static class DatabaseInitializer
9     {
10         public static void Initialize()
11         {
12             string folder = Path.GetDirectoryName(SQLiteConnectionFactory.DatabasePath);
13             if (!Directory.Exists(folder))
14             {
15                 Directory.CreateDirectory(folder);
16             }
17
18             bool existed = File.Exists(SQLiteConnectionFactory.DatabasePath);
19             if (!existed)
20             {
21                 SQLiteConnection.CreateFile(SQLiteConnectionFactory.DatabasePath);
22             }
23
24             if (existed)
25             {
26                 using (SQLiteConnection versionConnection = SQLiteConnectionFactory.CreateConnection())
27                 {
28                     if (DatabaseMigrationRunner.GetCurrentVersion(versionConnection) < DatabaseMigrationRunner.LatestVersion)
29                     {
30                         versionConnection.Close();
31                         DatabaseMigrationRunner.BackupBeforeMigration(SQLiteConnectionFactory.DatabasePath);
32                     }
33                 }
34             }
35
36             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
37             {
38                 Execute(connection, "PRAGMA foreign_keys = ON;");
39                 CreateTable(connection);
40                 DatabaseMigrationRunner.RunPending(connection);
41                 SeedData(connection);
42             }
43         }
44
45         private static void CreateTable(SQLiteConnection connection)
46         {
47             Execute(connection, @"CREATE TABLE IF NOT EXISTS Usuarios (
48                 IdUsuario INTEGER PRIMARY KEY AUTOINCREMENT,
49                 NombreUsuario TEXT NOT NULL UNIQUE,
50                 PasswordHash TEXT NOT NULL,
51                 Rol TEXT NOT NULL,
52                 Estado TEXT NOT NULL
53             );");
54
55             Execute(connection, @"CREATE TABLE IF NOT EXISTS Departamentos (
56                 IdDepartamento INTEGER PRIMARY KEY AUTOINCREMENT,
57                 Nombre TEXT NOT NULL UNIQUE
58             );");
59
60             Execute(connection, @"CREATE TABLE IF NOT EXISTS Empleados (
```

Data/DatabaseInitializer.cs (continuacion)

```
61         IdEmpleado INTEGER PRIMARY KEY AUTOINCREMENT,
62         Codigo TEXT NOT NULL UNIQUE,
63         Nombre TEXT NOT NULL,
64         Apellido TEXT NOT NULL,
65         Cedula TEXT NOT NULL UNIQUE,
66         Cargo TEXT NOT NULL,
67         IdDepartamento INTEGER NOT NULL,
68         SalarioBase REAL NOT NULL,
69         Estado TEXT NOT NULL,
70         FechaIngreso TEXT NOT NULL,
71         FOREIGN KEY(IdDepartamento) REFERENCES Departamentos(IdDepartamento)
72     );";
73
74     Execute(connection, @"CREATE TABLE IF NOT EXISTS Asistencias (
75         IdAsistencia INTEGER PRIMARY KEY AUTOINCREMENT,
76         IdEmpleado INTEGER NOT NULL,
77         Fecha TEXT NOT NULL,
78         HoraEntrada TEXT,
79         HoraSalida TEXT,
80         HorasTrabajadas REAL,
81         Estado TEXT NOT NULL,
82         FOREIGN KEY(IdEmpleado) REFERENCES Empleados(IdEmpleado)
83     );");
84
85     Execute(connection, @"CREATE TABLE IF NOT EXISTS PeriodosNomina (
86         IdPeriodo INTEGER PRIMARY KEY AUTOINCREMENT,
87         Nombre TEXT NOT NULL,
88         FechaInicio TEXT NOT NULL,
89         FechaFin TEXT NOT NULL,
90         Estado TEXT NOT NULL
91     );");
92
93     Execute(connection, @"CREATE TABLE IF NOT EXISTS Nominas (
94         IdNomina INTEGER PRIMARY KEY AUTOINCREMENT,
95         IdPeriodo INTEGER NOT NULL,
96         FechaCalculo TEXT NOT NULL,
97         TotalIngresos REAL NOT NULL,
98         TotalDeducciones REAL NOT NULL,
99         TotalNeto REAL NOT NULL,
100        Estado TEXT NOT NULL,
101        FOREIGN KEY(IdPeriodo) REFERENCES PeriodosNomina(IdPeriodo)
102    );");
103
104     Execute(connection, @"CREATE TABLE IF NOT EXISTS NominaDetalle (
105         IdDetalle INTEGER PRIMARY KEY AUTOINCREMENT,
106         IdNomina INTEGER NOT NULL,
107         IdEmpleado INTEGER NOT NULL,
108         SueldoBase REAL NOT NULL,
109         Bonos REAL NOT NULL,
110         HorasExtra REAL NOT NULL,
111         MontoHorasExtra REAL NOT NULL,
112         TotalIngresos REAL NOT NULL,
113         TotalDeducciones REAL NOT NULL,
114         NetoPagar REAL NOT NULL,
115         FOREIGN KEY(IdNomina) REFERENCES Nominas(IdNomina),
116         FOREIGN KEY(IdEmpleado) REFERENCES Empleados(IdEmpleado)
117     );");
118
119     Execute(connection, @"CREATE TABLE IF NOT EXISTS Comprobantes (
120         IdComprobante INTEGER PRIMARY KEY AUTOINCREMENT,
```

Data/DatabaseInitializer.cs (continuacion)

```

121         IdNomina INTEGER NOT NULL,
122         IdEmpleado INTEGER NOT NULL,
123         NumeroComprobante TEXT NOT NULL UNIQUE,
124         FechaGeneracion TEXT NOT NULL,
125         RutaPdf TEXT,
126         FOREIGN KEY(IdNomina) REFERENCES Nominas(IdNomina),
127         FOREIGN KEY(IdEmpleado) REFERENCES Empleados(IdEmpleado)
128     );
129
130     Execute(connection, @"CREATE TABLE IF NOT EXISTS ConfiguracionNomina (
131         IdConfiguracion INTEGER PRIMARY KEY AUTOINCREMENT,
132         NombreParametro TEXT NOT NULL UNIQUE,
133         Valor REAL NOT NULL,
134         Descripcion TEXT
135     );");
136
137     Execute(connection, @"CREATE TABLE IF NOT EXISTS ReportesGenerados (
138         IdReporte INTEGER PRIMARY KEY AUTOINCREMENT,
139         NombreReporte TEXT NOT NULL,
140         Tipo TEXT NOT NULL,
141         GeneradoPor TEXT NOT NULL,
142         FechaGeneracion TEXT NOT NULL,
143         RutaArchivo TEXT
144     );");
145
146     Execute(connection, @"CREATE TABLE IF NOT EXISTS Auditoria (
147         IdAuditoria INTEGER PRIMARY KEY AUTOINCREMENT,
148         Usuario TEXT NOT NULL,
149         Modulo TEXT NOT NULL,
150         Accion TEXT NOT NULL,
151         Detalle TEXT,
152         Fecha TEXT NOT NULL
153     );");
154
155     Execute(connection, @"CREATE TABLE IF NOT EXISTS MigracionesLog (
156         IdMigracion INTEGER PRIMARY KEY AUTOINCREMENT,
157         Version INTEGER NOT NULL,
158         Nombre TEXT NOT NULL,
159         FechaAplicacion TEXT NOT NULL,
160         Detalle TEXT
161     );");
162 }
163
164 private static void SeedData(SQLiteConnection connection)
165 {
166     object adminExists = ExecuteScalar(connection,
167         "SELECT COUNT(1) FROM Usuarios WHERE NombreUsuario = @usuario;",
168         new SQLiteParameter("@usuario", "admin"));
169     if (Convert.ToInt32(adminExists) == 0)
170     {
171         ExecuteWithParameters(connection,
172             "INSERT INTO Usuarios (NombreUsuario, PasswordHash, Rol, Estado) VALUES (@u, @p, @r, @e);",
173             new SQLiteParameter("@u", "admin"),
174             new SQLiteParameter("@p", PasswordHelper.HashPassword("admin123")),
175             new SQLiteParameter("@r", "Admin"),
176             new SQLiteParameter("@e", "Activo"));
177     }
178
179     string[] departments = { "Tecnologia", "Ventas", "Recursos Humanos", "Finanzas", "Operaciones" };
180     for (int i = 0; i < departments.Length; i++)

```

Data/DatabaseInitializer.cs (continuacion)

```

181         {
182             ExecuteWithParameters(connection,
183                 "INSERT OR IGNORE INTO Departamentos (Nombre) VALUES (@nombre);",
184                 new SQLiteParameter("@nombre", departments[i]));
185         }
186
187         SeedEmployee(connection, "EMP-1001", "Carlos", "Mendoza", "8-765-123", "Desarrollador Senior", "Tecnología", 2450);
188         SeedEmployee(connection, "EMP-1002", "Ana", "López", "8-432-987", "Ejecutiva de Ventas", "Ventas", 1850);
189         SeedEmployee(connection, "EMP-1003", "Sofía", "Gómez", "9-111-222", "Analista de RRHH", "Recursos Humanos", 1700);
190         SeedEmployee(connection, "EMP-1004", "Javier", "Ramírez", "4-555-321", "Supervisor de Operaciones", "Operaciones", 2100);
191
192         SeedDemoUser(connection, "rrhh", "Rrhh2026*", "RRHH", null);
193         SeedDemoUser(connection, "contador", "Conta2026*", "Contabilidad", null);
194         SeedDemoUser(connection, "supervisor", "Super2026*", "Supervisor", "EMP-1004");
195         SeedDemoUser(connection, "trabajador", "Trabaja2026*", "Trabajador", "EMP-1001");
196
197         SeedConfig(connection, "SeguroSocial", 9.75m, "Porcentaje académico de seguro social.");
198         SeedConfig(connection, "ISR", 10, "Porcentaje académico de impuesto sobre la renta.");
199         SeedConfig(connection, "SeguroEducativo", 1.25m, "Porcentaje académico de seguro educativo.");
200         SeedConfig(connection, "RecargoHoraExtra", 1.25m, "Multiplicador académico de horas extra.");
201         SeedConfig(connection, "HorasMensualesBase", 160, "Horas mensuales base para cálculo por hora.");
202     }
203
204     private static void SeedEmployee(SQLiteConnection connection, string code, string name, string lastName, string cedula, string role, string \
department, decimal salary)
205     {
206         object departmentId = ExecuteScalar(connection, "SELECT IdDepartamento FROM Departamentos WHERE Nombre = @nombre;",
207             new SQLiteParameter("@nombre", department));
208
209         ExecuteWithParameters(connection, @"INSERT OR IGNORE INTO Empleados
210             (Codigo, Nombre, Apellido, Cedula, Cargo, IdDepartamento, SalarioBase, Estado, FechaIngreso)
211             VALUES (@codigo, @nombre, @apellido, @cedula, @cargo, @departamento, @salario, 'Activo', @fecha);",
212             new SQLiteParameter("@codigo", code),
213             new SQLiteParameter("@nombre", name),
214             new SQLiteParameter("@apellido", lastName),
215             new SQLiteParameter("@cedula", cedula),
216             new SQLiteParameter("@cargo", role),
217             new SQLiteParameter("@departamento", Convert.ToInt32(departmentId)),
218             new SQLiteParameter("@salario", salary),
219             new SQLiteParameter("@fecha", DateTime.Today.AddMonths(-8).ToString("yyyy-MM-dd")));
220     }
221
222     private static void SeedConfig(SQLiteConnection connection, string name, decimal value, string description)
223     {
224         ExecuteWithParameters(connection,
225             "INSERT OR IGNORE INTO ConfiguracionNomina (NombreParametro, Valor, Descripcion) VALUES (@n, @v, @d);",
226             new SQLiteParameter("@n", name),
227             new SQLiteParameter("@v", value),
228             new SQLiteParameter("@d", description));
229     }
230
231     private static void SeedDemoUser(SQLiteConnection connection, string username, string password,
232         string role, string employeeCode)
233     {
234         object existing = ExecuteScalar(connection,
235             "SELECT COUNT(1) FROM Usuarios WHERE NombreUsuario = @usuario;",
236             new SQLiteParameter("@usuario", username));
237         if (Convert.ToInt32(existing) > 0) return;
238
239         object employeeId = DBNull.Value;

```

Data/DatabaseInitializer.cs (continuacion)

```
240         if (!string.IsNullOrEmpty(employeeCode))
241         {
242             employeeId = ExecuteScalar(connection,
243                 "SELECT IdEmpleado FROM Empleados WHERE Codigo = @codigo AND Estado = 'Activo' LIMIT 1;",
244                 new SQLiteParameter("@codigo", employeeCode));
245             if (employeeId == null) employeeId = DBNull.Value;
246             if (employeeId != DBNull.Value)
247             {
248                 object associated = ExecuteScalar(connection,
249                     "SELECT COUNT(1) FROM Usuarios WHERE IdEmpleado = @empleado;",
250                     new SQLiteParameter("@empleado", employeeId));
251                 if (Convert.ToInt32(associated) > 0) return;
252             }
253         }
254
255         ExecuteWithParameters(connection, @"INSERT OR IGNORE INTO Usuarios
256             (NombreUsuario, PasswordHash, Rol, Estado, IdEmpleado, IntentosFallidos, Bloqueado)
257             VALUES (@usuario, @password, @rol, 'Activo', @empleado, 0, 0);",
258             new SQLiteParameter("@usuario", username),
259             new SQLiteParameter("@password", PasswordHelper.HashPassword(password)),
260             new SQLiteParameter("@rol", role),
261             new SQLiteParameter("@empleado", employeeId));
262     }
263
264     private static void Execute(SQLiteConnection connection, string sql)
265     {
266         using (SQLiteCommand command = new SQLiteCommand(sql, connection))
267         {
268             command.ExecuteNonQuery();
269         }
270     }
271
272     private static void ExecuteWithParameters(SQLiteConnection connection, string sql, params SQLiteParameter[] parameters)
273     {
274         using (SQLiteCommand command = new SQLiteCommand(sql, connection))
275         {
276             command.Parameters.AddRange(parameters);
277             command.ExecuteNonQuery();
278         }
279     }
280
281     private static object ExecuteScalar(SQLiteConnection connection, string sql, params SQLiteParameter[] parameters)
282     {
283         using (SQLiteCommand command = new SQLiteCommand(sql, connection))
284         {
285             command.Parameters.AddRange(parameters);
286             return command.ExecuteScalar();
287         }
288     }
289 }
290 }
```


Data/DatabaseMigrationRunner.cs (continuacion)

```

61         new SQLiteParameter("@fecha", DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss")),
62         new SQLiteParameter("@detalle", detail.ToString()));
63     transaction.Commit();
64 }
65 }
66
67 private static void MigrateUsers(SQLiteConnection connection, SQLiteTransaction transaction, StringBuilder detail)
68 {
69     AddColumnIfMissing(connection, transaction, "Usuarios", "IdEmpleado",
70         "INTEGER NULL REFERENCES Empleados(IdEmpleado)");
71     AddColumnIfMissing(connection, transaction, "Usuarios", "UltimoAcceso", "TEXT NULL");
72     AddColumnIfMissing(connection, transaction, "Usuarios", "IntentosFallidos", "INTEGER NOT NULL DEFAULT 0");
73     AddColumnIfMissing(connection, transaction, "Usuarios", "Bloqueado", "INTEGER NOT NULL DEFAULT 0");
74     AddColumnIfMissing(connection, transaction, "Usuarios", "FechaBloqueo", "TEXT NULL");
75     Execute(connection, transaction,
76         "CREATE INDEX IF NOT EXISTS IX_Usuarios_IdEmpleado ON Usuarios(IdEmpleado);");
77     detail.Append("Columnas de autenticación e índice de empleado agregados.");
78 }
79
80 private static void MigrateIndexesAndAttendance(SQLiteConnection connection, SQLiteTransaction transaction, StringBuilder detail)
81 {
82     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Empleados_Codigo ON Empleados(Codigo);");
83     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Empleados_Cedula ON Empleados(Cedula);");
84     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Asistencias_EmpleadoFecha ON Asistencias(IdEmpleado, Fecha);");
85     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Nominas_IdPeriodo ON Nominas(IdPeriodo);");
86     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Comprobantes_IdEmpleado ON Comprobantes(IdEmpleado);");
87     Execute(connection, transaction, "CREATE INDEX IF NOT EXISTS IX_Auditoria_Fecha ON Auditoria(Fecha DESC);");
88
89     int duplicateGroups = Convert.ToInt32(Scalar(connection, transaction, @"SELECT COUNT(1) FROM (
90         SELECT IdEmpleado, Fecha FROM Asistencias
91         GROUP BY IdEmpleado, Fecha HAVING COUNT(1) > 1);"));
92
93     if (duplicateGroups == 0)
94     {
95         Execute(connection, transaction,
96             "CREATE UNIQUE INDEX IF NOT EXISTS UX_Asistencias_EmpleadoFecha ON Asistencias(IdEmpleado, Fecha);");
97         detail.Append("No se detectaron asistencias duplicadas; se creó el índice único. ");
98     }
99     else
100     {
101         detail.Append("Se detectaron " + duplicateGroups +
102             " grupos históricos duplicados; se conservaron sin cambios. ");
103     }
104
105     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Asistencias_NoDuplicar_Insert
106     BEFORE INSERT ON Asistencias
107     WHEN EXISTS (SELECT 1 FROM Asistencias
108         WHERE IdEmpleado = NEW.IdEmpleado AND Fecha = NEW.Fecha)
109     BEGIN
110         SELECT RAISE(ABORT, 'Ya existe una asistencia para el empleado en esa fecha. ');
111     END;");
112     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Asistencias_NoDuplicar_Update
113     BEFORE UPDATE OF IdEmpleado, Fecha ON Asistencias
114     WHEN EXISTS (SELECT 1 FROM Asistencias
115         WHERE IdEmpleado = NEW.IdEmpleado AND Fecha = NEW.Fecha
116         AND IdAsistencia <> OLD.IdAsistencia)
117     BEGIN
118         SELECT RAISE(ABORT, 'Ya existe una asistencia para el empleado en esa fecha. ');
119     END;");
120     detail.Append("Se instalaron protecciones para impedir nuevas duplicaciones.");

```


Data/DatabaseMigrationRunner.cs (continuacion)

```
121     }
122
123     private static void MigrateIntegrityRules(SQLiteConnection connection, SQLiteTransaction transaction, StringBuilder detail)
124     {
125         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Usuarios_Validar_Insert
126         BEFORE INSERT ON Usuarios
127         WHEN NEW.Rol NOT IN ('Admin','RRHH','Contabilidad','Supervisor','Trabajador')
128         OR (NEW.Rol IN ('Supervisor','Trabajador') AND NEW.IdEmpleado IS NULL)
129         BEGIN
130             SELECT RAISE(ABORT, 'El rol o la asociación del usuario no es válida.');"");
131         END;");
132         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Usuarios_Validar_Update
133         BEFORE UPDATE OF Rol, IdEmpleado ON Usuarios
134         WHEN NEW.Rol NOT IN ('Admin','RRHH','Contabilidad','Supervisor','Trabajador')
135         OR (NEW.Rol IN ('Supervisor','Trabajador') AND NEW.IdEmpleado IS NULL)
136         BEGIN
137             SELECT RAISE(ABORT, 'El rol o la asociación del usuario no es válida.');"");
138         END;");
139         int duplicateUserLinks = Convert.ToInt32(Scalar(connection, transaction, @"SELECT COUNT(1) FROM (
140         SELECT IdEmpleado FROM Usuarios WHERE IdEmpleado IS NOT NULL
141         GROUP BY IdEmpleado HAVING COUNT(1) > 1);"));
142         if (duplicateUserLinks == 0)
143         {
144             Execute(connection, transaction, @"CREATE UNIQUE INDEX IF NOT EXISTS UX_Usuarios_IdEmpleado
145             ON Usuarios(IdEmpleado) WHERE IdEmpleado IS NOT NULL;");
146         }
147         else
148         {
149             detail.Append("Se conservaron " + duplicateUserLinks + " asociaciones históricas duplicadas. ");
150         }
151         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Usuarios_NoDuplicarEmpleado_Insert
152         BEFORE INSERT ON Usuarios
153         WHEN NEW.IdEmpleado IS NOT NULL AND EXISTS
154         (SELECT 1 FROM Usuarios WHERE IdEmpleado = NEW.IdEmpleado)
155         BEGIN
156             SELECT RAISE(ABORT, 'El empleado ya tiene un usuario asociado.');"");
157         END;");
158         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Usuarios_NoDuplicarEmpleado_Update
159         BEFORE UPDATE OF IdEmpleado ON Usuarios
160         WHEN NEW.IdEmpleado IS NOT NULL AND EXISTS
161         (SELECT 1 FROM Usuarios WHERE IdEmpleado = NEW.IdEmpleado AND IdUsuario <> OLD.IdUsuario)
162         BEGIN
163             SELECT RAISE(ABORT, 'El empleado ya tiene un usuario asociado.');"");
164         END;");
165
166         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Empleados_Validar_Insert
167         BEFORE INSERT ON Empleados
168         WHEN NEW.SalarioBase <= 0 OR NEW.IdDepartamento <= 0
169         BEGIN
170             SELECT RAISE(ABORT, 'El salario y el departamento del empleado deben ser válidos.');"");
171         END;");
172         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Empleados_Validar_Update
173         BEFORE UPDATE OF SalarioBase, IdDepartamento ON Empleados
174         WHEN NEW.SalarioBase <= 0 OR NEW.IdDepartamento <= 0
175         BEGIN
176             SELECT RAISE(ABORT, 'El salario y el departamento del empleado deben ser válidos.');"");
177         END;");
178
179         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Asistencias_Validar_Insert
180         BEFORE INSERT ON Asistencias
```

Data/DatabaseMigrationRunner.cs (continuacion)

```

181         WHEN NEW.HorasTrabajadas < 0
182         OR (NEW.HoraEntrada IS NOT NULL AND NEW.HoraSalida IS NOT NULL AND NEW.HoraSalida <= NEW.HoraEntrada)
183         BEGIN
184             SELECT RAISE(ABORT, 'Las horas de asistencia no son válidas.');
```

185 END;");

186 Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Asistencias_Validar_Update

187 BEFORE UPDATE OF HoraEntrada, HoraSalida, HorasTrabajadas ON Asistencias

188 WHEN NEW.HorasTrabajadas < 0

189 OR (NEW.HoraEntrada IS NOT NULL AND NEW.HoraSalida IS NOT NULL AND NEW.HoraSalida <= NEW.HoraEntrada)

190 BEGIN

191 SELECT RAISE(ABORT, 'Las horas de asistencia no son válidas.');

192 END;");

193 detail.Append("Roles, asociaciones, salarios, departamentos y horas protegidos mediante índices y triggers.");

194 }

195

196 private static void MigrateNominaRobusta(SQLiteConnection connection, SQLiteTransaction transaction, StringBuilder detail)

197 {

198 AddColumnIfMissing(connection, transaction, "PeriodosNomina", "Cerrado",

199 "INTEGER NOT NULL DEFAULT 0");

200 AddColumnIfMissing(connection, transaction, "PeriodosNomina", "FechaCierre", "TEXT NULL");

201 AddColumnIfMissing(connection, transaction, "PeriodosNomina", "CerradoPor", "TEXT NULL");

202

203 AddColumnIfMissing(connection, transaction, "Nominas", "FechaConfirmacion", "TEXT NULL");

204 AddColumnIfMissing(connection, transaction, "Nominas", "ConfirmadaPor", "TEXT NULL");

205 AddColumnIfMissing(connection, transaction, "Nominas", "FechaPago", "TEXT NULL");

206 AddColumnIfMissing(connection, transaction, "Nominas", "PagadaPor", "TEXT NULL");

207 AddColumnIfMissing(connection, transaction, "Nominas", "FechaAnulacion", "TEXT NULL");

208 AddColumnIfMissing(connection, transaction, "Nominas", "AnuladaPor", "TEXT NULL");

209 AddColumnIfMissing(connection, transaction, "Nominas", "MotivoAnulacion", "TEXT NULL");

210

211 Execute(connection, transaction, @"CREATE TABLE IF NOT EXISTS NominaVersiones (

212 IdVersion INTEGER PRIMARY KEY AUTOINCREMENT,

213 IdNominaOriginal INTEGER NOT NULL,

214 IdNominaNueva INTEGER NULL,

215 MotivoCambio TEXT NOT NULL,

216 UsuarioResponsable TEXT NOT NULL,

217 FechaCambio TEXT NOT NULL,

218 FOREIGN KEY(IdNominaOriginal) REFERENCES Nominas(IdNomina),

219 FOREIGN KEY(IdNominaNueva) REFERENCES Nominas(IdNomina)

220);");

221

222 Execute(connection, transaction,

223 "CREATE INDEX IF NOT EXISTS IX_Nominas_IdPeriodo_Estado ON Nominas(IdPeriodo, Estado);");

224 Execute(connection, transaction,

225 "CREATE INDEX IF NOT EXISTS IX_NominaVersiones_IdNominaOriginal ON NominaVersiones(IdNominaOriginal);");

226

227 Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Asistencias_PeriodoCerrado_Insert

228 BEFORE INSERT ON Asistencias

229 WHEN EXISTS (SELECT 1 FROM PeriodosNomina

230 WHERE Cerrado = 1

231 AND FechaInicio <= NEW.Fecha

232 AND FechaFin >= NEW.Fecha)

233 BEGIN

234 SELECT RAISE(ABORT, 'No se puede registrar asistencia en un período cerrado.');

235 END;");

236

237 detail.Append("PeriodosNomina (Cerrado, FechaCierre, CerradoPor), Nominas (fechas/usuarios de cada estado), ");

238 detail.Append("NominaVersiones, índices y trigger de bloqueo de asistencia creados.");

239 }

240

Data/DatabaseMigrationRunner.cs (continuacion)

```

241     private static void MigrateEmployeeHistoryAndPayrollProtection(SQLiteConnection connection,
242         SQLiteTransaction transaction, StringBuilder detail)
243     {
244         AddColumnIfMissing(connection, transaction, "Empleados", "FechaEfectivaLaboral", "TEXT NULL");
245         Execute(connection, transaction, @"UPDATE Empleados
246             SET FechaEfectivaLaboral = FechaIngreso
247             WHERE FechaEfectivaLaboral IS NULL OR FechaEfectivaLaboral = '';" );
248
249         AddColumnIfMissing(connection, transaction, "NominaDetalle", "CodigoEmpleadoSnapshot", "TEXT NULL");
250         AddColumnIfMissing(connection, transaction, "NominaDetalle", "NombreEmpleadoSnapshot", "TEXT NULL");
251         AddColumnIfMissing(connection, transaction, "NominaDetalle", "CargoEmpleadoSnapshot", "TEXT NULL");
252         AddColumnIfMissing(connection, transaction, "NominaDetalle", "DepartamentoSnapshot", "TEXT NULL");
253         Execute(connection, transaction, @"UPDATE NominaDetalle
254             SET CodigoEmpleadoSnapshot = COALESCE(CodigoEmpleadoSnapshot,
255                 (SELECT Codigo FROM Empleados WHERE IdEmpleado = NominaDetalle.IdEmpleado)),
256                 NombreEmpleadoSnapshot = COALESCE(NombreEmpleadoSnapshot,
257                 (SELECT Nombre || ' ' || Apellido FROM Empleados WHERE IdEmpleado = NominaDetalle.IdEmpleado)),
258                 CargoEmpleadoSnapshot = COALESCE(CargoEmpleadoSnapshot,
259                 (SELECT Cargo FROM Empleados WHERE IdEmpleado = NominaDetalle.IdEmpleado)),
260                 DepartamentoSnapshot = COALESCE(DepartamentoSnapshot,
261                 (SELECT d.Nombre FROM Empleados e INNER JOIN Departamentos d
262                     ON d.IdDepartamento = e.IdDepartamento WHERE e.IdEmpleado = NominaDetalle.IdEmpleado));");
263
264         Execute(connection, transaction, @"CREATE TABLE IF NOT EXISTS HistorialEmpleado (
265             IdHistorial INTEGER PRIMARY KEY AUTOINCREMENT,
266             IdEmpleado INTEGER NOT NULL,
267             CampoModificado TEXT NOT NULL,
268             ValorAnterior TEXT NULL,
269             ValorNuevo TEXT NULL,
270             ValorAnteriorTecnico TEXT NULL,
271             ValorNuevoTecnico TEXT NULL,
272             FechaCambio TEXT NOT NULL,
273             FechaEfectiva TEXT NOT NULL,
274             UsuarioResponsable TEXT NOT NULL,
275             Motivo TEXT NULL,
276             Aplicado INTEGER NOT NULL DEFAULT 1,
277             FechaAplicacion TEXT NULL,
278             FOREIGN KEY(IdEmpleado) REFERENCES Empleados(IdEmpleado)
279         );");
280         Execute(connection, transaction,
281             "CREATE INDEX IF NOT EXISTS IX_HistorialEmpleado_EmpleadoFecha ON HistorialEmpleado(IdEmpleado, FechaEfectiva DESC);");
282         Execute(connection, transaction,
283             "CREATE INDEX IF NOT EXISTS IX_HistorialEmpleado_Pendientes ON HistorialEmpleado(Aplicado, FechaEfectiva);");
284
285         // Los tres triggers cubren insercion, actualizacion y eliminacion de asistencia en periodos cerrados.
286         Execute(connection, transaction, "DROP TRIGGER IF EXISTS TR_Asstencias_PeriodoCerrado_Insert;");
287         Execute(connection, transaction, @"CREATE TRIGGER TR_Asstencias_PeriodoCerrado_Insert
288             BEFORE INSERT ON Asistencias
289             WHEN EXISTS (SELECT 1 FROM PeriodosNomina
290                 WHERE Cerrado = 1 AND FechaInicio <= NEW.Fecha AND FechaFin >= NEW.Fecha)
291             BEGIN
292                 SELECT RAISE(ABORT, 'No se puede registrar asistencia en un periodo cerrado.');
```

Data/DatabaseMigrationRunner.cs (continuacion)

```

301         BEGIN
302             SELECT RAISE(ABORT, 'No se puede modificar asistencia relacionada con un periodo cerrado.');
```

303 END;");

```

304     Execute(connection, transaction, "DROP TRIGGER IF EXISTS TR_Asigstencias_PeriodoCerrado_Delete;");
305     Execute(connection, transaction, @"CREATE TRIGGER TR_Asigstencias_PeriodoCerrado_Delete
306         BEFORE DELETE ON Asistencias
307         WHEN EXISTS (SELECT 1 FROM PeriodosNomina
308             WHERE Cerrado = 1 AND FechaInicio <= OLD.Fecha AND FechaFin >= OLD.Fecha)
309     BEGIN
310         SELECT RAISE(ABORT, 'No se puede eliminar asistencia relacionada con un periodo cerrado.');
```

311 END;");

312

```

313     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_PeriodosNomina_Rango_Insert
314         BEFORE INSERT ON PeriodosNomina WHEN NEW.FechaInicio > NEW.FechaFin
315     BEGIN SELECT RAISE(ABORT, 'La fecha inicial del periodo no puede superar la fecha final.');
```

316 END;");

```

317     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_PeriodosNomina_Rango_Update
318         BEFORE UPDATE OF FechaInicio, FechaFin ON PeriodosNomina WHEN NEW.FechaInicio > NEW.FechaFin
319     BEGIN SELECT RAISE(ABORT, 'La fecha inicial del periodo no puede superar la fecha final.');
```

320 END;");

// Dos rangos se solapan cuando el inicio de uno es anterior al fin del otro y viceversa.

```

321     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_PeriodosNomina_Solapado_Insert
322         BEFORE INSERT ON PeriodosNomina
323         WHEN (NEW.Cerrado = 1 OR NEW.Estado IN ('Confirmado', 'Pagado'))
324             AND EXISTS (SELECT 1 FROM PeriodosNomina
325                 WHERE (Cerrado = 1 OR Estado IN ('Confirmado', 'Pagado'))
326                     AND FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)
327     BEGIN SELECT RAISE(ABORT, 'El periodo se solapa con otro periodo cerrado o confirmado.');
```

328 END;");

```

329     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_PeriodosNomina_Solapado_Update
330         BEFORE UPDATE OF FechaInicio, FechaFin, Estado, Cerrado ON PeriodosNomina
331         WHEN (NEW.Cerrado = 1 OR NEW.Estado IN ('Confirmado', 'Pagado'))
332             AND EXISTS (SELECT 1 FROM PeriodosNomina
333                 WHERE IdPeriodo <> OLD.IdPeriodo
334                     AND (Cerrado = 1 OR Estado IN ('Confirmado', 'Pagado'))
335                     AND FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)
336     BEGIN SELECT RAISE(ABORT, 'El periodo se solapa con otro periodo cerrado o confirmado.');
```

337 END;");

338

```

339     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_HistorialEmpleado_PeriodoCerrado_Insert
340         BEFORE INSERT ON HistorialEmpleado
341         WHEN EXISTS (SELECT 1 FROM PeriodosNomina
342             WHERE Cerrado = 1 AND FechaFin >= NEW.FechaEfectiva)
343     BEGIN SELECT RAISE(ABORT, 'El cambio laboral afectaria un periodo de nomina cerrado.');
```

344 END;");

```

345     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Empleados_CambioLaboral_PeriodoCerrado
346         BEFORE UPDATE OF SalarioBase, Cargo, IdDepartamento, Estado ON Empleados
347         WHEN (OLD.SalarioBase <> NEW.SalarioBase OR OLD.Cargo <> NEW.Cargo
348             OR OLD.IdDepartamento <> NEW.IdDepartamento OR OLD.Estado <> NEW.Estado)
349             AND (NEW.FechaEfectivaLaboral IS NULL OR EXISTS (SELECT 1 FROM PeriodosNomina
350                 WHERE Cerrado = 1 AND FechaFin >= NEW.FechaEfectivaLaboral))
351     BEGIN SELECT RAISE(ABORT, 'El cambio laboral afectaria un periodo de nomina cerrado.');
```

352 END;");

353

```

354     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Nominas_Estado_Insert
355         BEFORE INSERT ON Nominas
356         WHEN NEW.Estado NOT IN ('Borrador', 'Calculada', 'Confirmada', 'Pagada', 'Anulada')
357     BEGIN SELECT RAISE(ABORT, 'El estado de nomina no es valido.');
```

358 END;");

```

359     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Nominas_Estado_Update
360         BEFORE UPDATE OF Estado ON Nominas
361         WHEN OLD.Estado <> NEW.Estado AND NOT (
362             (OLD.Estado = 'Borrador' AND NEW.Estado = 'Calculada') OR
363             (OLD.Estado = 'Calculada' AND NEW.Estado = 'Confirmada') OR
364             (OLD.Estado = 'Confirmada' AND NEW.Estado IN ('Pagada', 'Anulada')) OR
365             (OLD.Estado = 'Pagada' AND NEW.Estado = 'Anulada'))
366     BEGIN SELECT RAISE(ABORT, 'La transicion de estado de nomina no es valida.');
```

367 END;");

Data/DatabaseMigrationRunner.cs (continuacion)

```

361         Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_Nominas_ProtegerDatos_Update
362         BEFORE UPDATE OF IdPeriodo, TotalIngresos, TotalDeducciones, TotalNeto ON Nominas
363         WHEN OLD.Estado IN ('Confirmada','Pagada','Anulada') AND
364             (OLD.IdPeriodo <> NEW.IdPeriodo OR OLD.TotalIngresos <> NEW.TotalIngresos
365             OR OLD.TotalDeducciones <> NEW.TotalDeducciones OR OLD.TotalNeto <> NEW.TotalNeto)
366         BEGIN SELECT RAISE(ABORT, 'Una nomina confirmada, pagada o anulada no puede editarse.');"");
367     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_NominaDetalle_Proteger_Update
368     BEFORE UPDATE ON NominaDetalle
369     WHEN EXISTS (SELECT 1 FROM Nominas WHERE IdNomina = OLD.IdNomina
370     AND Estado IN ('Confirmada','Pagada','Anulada'))
371     BEGIN SELECT RAISE(ABORT, 'El detalle de una nomina cerrada no puede editarse.');"");
372     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_NominaDetalle_Proteger_Delete
373     BEFORE DELETE ON NominaDetalle
374     WHEN EXISTS (SELECT 1 FROM Nominas WHERE IdNomina = OLD.IdNomina
375     AND Estado IN ('Confirmada','Pagada','Anulada'))
376     BEGIN SELECT RAISE(ABORT, 'El detalle de una nomina cerrada no puede eliminarse.');"");
377
378     detail.Append("Historial laboral, cambios efectivos, snapshots, bloqueos por rango y estados de nomina instalados.");
379 }
380
381 private static void MigrateAbsenceRequests(SQLiteConnection connection,
382     SQLiteTransaction transaction, StringBuilder detail)
383 {
384     Execute(connection, transaction, @"CREATE TABLE IF NOT EXISTS SolicitudesAusencia (
385         IdSolicitud INTEGER PRIMARY KEY AUTOINCREMENT,
386         IdEmpleado INTEGER NOT NULL,
387         Tipo TEXT NOT NULL CHECK (Tipo IN ('Vacaciones','Permiso remunerado','Permiso no remunerado',
388         'Incapacidad médica','Licencia','Ausencia justificada','Ausencia injustificada','Suspensión')),
389         FechaInicio TEXT NOT NULL,
390         FechaFin TEXT NOT NULL,
391         Estado TEXT NOT NULL CHECK (Estado IN ('Pendiente','Aprobada','Rechazada','Cancelada')),
392         Motivo TEXT NULL,
393         AprobadoPor TEXT NULL,
394         FechaAprobacion TEXT NULL,
395         ObservacionResolucion TEXT NULL,
396         UsuarioSolicitante TEXT NULL,
397         FechaCreacion TEXT NOT NULL,
398         FOREIGN KEY(IdEmpleado) REFERENCES Empleados(IdEmpleado)
399     );");
400     AddColumnIfMissing(connection, transaction, "Asistencias", "IdSolicitudAusencia",
401         "INTEGER NULL REFERENCES SolicitudesAusencia(IdSolicitud)");
402     Execute(connection, transaction,
403         "CREATE INDEX IF NOT EXISTS IX_SolicitudesAusencia_EmpleadoFechas ON SolicitudesAusencia(IdEmpleado, FechaInicio, FechaFin);");
404     Execute(connection, transaction,
405         "CREATE INDEX IF NOT EXISTS IX_SolicitudesAusencia_Estado ON SolicitudesAusencia(Estado, FechaInicio);");
406     Execute(connection, transaction,
407         "CREATE INDEX IF NOT EXISTS IX_Asistencias_IdSolicitudAusencia ON Asistencias(IdSolicitudAusencia);");
408
409     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_Validar_Insert
410     BEFORE INSERT ON SolicitudesAusencia
411     WHEN NEW.FechaInicio > NEW.FechaFin OR NEW.Estado <> 'Pendiente'
412     BEGIN SELECT RAISE(ABORT, 'La solicitud debe tener fechas validas e iniciar pendiente.');"");
413     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_Transicion_Update
414     BEFORE UPDATE OF Estado ON SolicitudesAusencia
415     WHEN OLD.Estado <> NEW.Estado AND NOT (
416         (OLD.Estado = 'Pendiente' AND NEW.Estado IN ('Aprobada','Rechazada','Cancelada')) OR
417         (OLD.Estado = 'Aprobada' AND NEW.Estado = 'Cancelada'))
418     BEGIN SELECT RAISE(ABORT, 'La transicion de la solicitud de ausencia no es valida.');"");
419     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_PeriodoCerrado_Insert
420     BEFORE INSERT ON SolicitudesAusencia

```

Data/DatabaseMigrationRunner.cs (continuacion)

```

421         WHEN EXISTS (SELECT 1 FROM PeriodosNomina WHERE Cerrado = 1
422             AND FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)
423         BEGIN SELECT RAISE(ABORT, 'La ausencia afecta un periodo de nomina cerrado.');
```

END;");

```

424     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_PeriodoCerrado_Update
425     BEFORE UPDATE ON SolicitudesAusencia
426     WHEN EXISTS (SELECT 1 FROM PeriodosNomina WHERE Cerrado = 1
427         AND ((FechaInicio <= OLD.FechaFin AND FechaFin >= OLD.FechaInicio)
428             OR (FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)))
429     BEGIN SELECT RAISE(ABORT, 'La ausencia afecta un periodo de nomina cerrado.');
```

END;");

```

430     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_AprobadaSolapada_Insert
431     BEFORE INSERT ON SolicitudesAusencia
432     WHEN NEW.Estado = 'Aprobada' AND EXISTS (SELECT 1 FROM SolicitudesAusencia
433         WHERE IdEmpleado = NEW.IdEmpleado AND Estado = 'Aprobada'
434         AND FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)
435     BEGIN SELECT RAISE(ABORT, 'Ya existe una ausencia aprobada que se solapa con este rango.');
```

END;");

```

436     Execute(connection, transaction, @"CREATE TRIGGER IF NOT EXISTS TR_SolicitudesAusencia_AprobadaSolapada_Update
437     BEFORE UPDATE OF Estado, FechaInicio, FechaFin, IdEmpleado ON SolicitudesAusencia
438     WHEN NEW.Estado = 'Aprobada' AND EXISTS (SELECT 1 FROM SolicitudesAusencia
439         WHERE IdSolicitud <> OLD.IdSolicitud AND IdEmpleado = NEW.IdEmpleado AND Estado = 'Aprobada'
440         AND FechaInicio <= NEW.FechaFin AND FechaFin >= NEW.FechaInicio)
441     BEGIN SELECT RAISE(ABORT, 'Ya existe una ausencia aprobada que se solapa con este rango.');
```

END;");

```

442
443     detail.Append("Solicitudes de ausencia, alcance de asistencia, indices y reglas de integridad instalados.");
444 }
445
446 private static void AddColumnIfMissing(SQLiteConnection connection, SQLiteTransaction transaction,
447     string table, string column, string definition)
448 {
449     if (ColumnExists(connection, transaction, table, column)) return;
450     Execute(connection, transaction, "ALTER TABLE " + table + " ADD COLUMN " + column + " " + definition + ";");
451 }
452
453 private static bool ColumnExists(SQLiteConnection connection, SQLiteTransaction transaction, string table, string column)
454 {
455     using (SQLiteCommand command = new SQLiteCommand("PRAGMA table_info(" + table + ");", connection, transaction))
456     using (SQLiteDataReader reader = command.ExecuteReader())
457     {
458         while (reader.Read())
459         {
460             if (string.Equals(Convert.ToString(reader["name"]), column, StringComparison.OrdinalIgnoreCase))
461             {
462                 return true;
463             }
464         }
465     }
466     return false;
467 }
468
469 private static object Scalar(SQLiteConnection connection, SQLiteTransaction transaction, string sql)
470 {
471     using (SQLiteCommand command = new SQLiteCommand(sql, connection, transaction))
472     {
473         return command.ExecuteScalar();
474     }
475 }
476
477 private static void Execute(SQLiteConnection connection, SQLiteTransaction transaction, string sql,
478     params SQLiteParameter[] parameters)
479 {
480

```

Data/DatabaseMigrationRunner.cs (continuacion)

```
481         using (SQLiteCommand command = new SQLiteCommand(sql, connection, transaction))
482         {
483             if (parameters != null && parameters.Length > 0) command.Parameters.AddRange(parameters);
484             command.ExecuteNonQuery();
485         }
486     }
487
488     private static string ComputeSha256(string path)
489     {
490         using (SHA256 sha = SHA256.Create())
491         using (FileStream stream = File.OpenRead(path))
492         {
493             byte[] hash = sha.ComputeHash(stream);
494             StringBuilder builder = new StringBuilder(hash.Length * 2);
495             for (int i = 0; i < hash.Length; i++) builder.Append(hash[i].ToString("x2", CultureInfo.InvariantCulture));
496             return builder.ToString();
497         }
498     }
499 }
500 }
```

Data/EmployeeHistoryRepository.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Data.SQLite;
4  using SistemaGestionNomina.Models;
5
6  namespace SistemaGestionNomina.Data
7  {
8      public sealed class EmployeeHistoryRepository
9      {
10         private const string SelectSql = @"SELECT h.*, e.Codigo AS CodigoEmpleado,
11             e.Nombre || ' ' || e.Apellido AS EmpleadoNombre
12             FROM HistorialEmpleado h
13             INNER JOIN Empleados e ON e.IdEmpleado = h.IdEmpleado ";
14
15         public int Add(SQLiteConnection connection, SQLiteTransaction transaction, HistorialEmpleado item)
16         {
17             using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO HistorialEmpleado
18                 (IdEmpleado, CampoModificado, ValorAnterior, ValorNuevo,
19                 ValorAnteriorTecnico, ValorNuevoTecnico, FechaCambio, FechaEfectiva,
20                 UsuarioResponsable, Motivo, Aplicado, FechaAplicacion)
21                 VALUES (@empleado, @campo, @anterior, @nuevo, @anteriorTecnico, @nuevoTecnico,
22                 @cambio, @efectiva, @usuario, @motivo, @aplicado, @aplicacion);
23                 SELECT last_insert_rowid();", connection, transaction))
24             {
25                 command.Parameters.AddWithValue("@empleado", item.IdEmpleado);
26                 command.Parameters.AddWithValue("@campo", item.CampoModificado);
27                 command.Parameters.AddWithValue("@anterior", item.ValorAnterior ?? string.Empty);
28                 command.Parameters.AddWithValue("@nuevo", item.ValorNuevo ?? string.Empty);
29                 command.Parameters.AddWithValue("@anteriorTecnico", item.ValorAnteriorTecnico ?? item.ValorAnterior ?? string.Empty);
30                 command.Parameters.AddWithValue("@nuevoTecnico", item.ValorNuevoTecnico ?? item.ValorNuevo ?? string.Empty);
31                 command.Parameters.AddWithValue("@cambio", item.FechaCambio.ToString("yyyy-MM-dd HH:mm:ss"));
32                 command.Parameters.AddWithValue("@efectiva", item.FechaEfectiva.ToString("yyyy-MM-dd"));
33                 command.Parameters.AddWithValue("@usuario", item.UsuarioResponsable);
34                 command.Parameters.AddWithValue("@motivo", item.Motivo ?? string.Empty);
35                 command.Parameters.AddWithValue("@aplicado", item.Aplicado ? 1 : 0);
36                 command.Parameters.AddWithValue("@aplicacion", item.FechaAplicacion.HasValue
37                     ? (object)item.FechaAplicacion.Value.ToString("yyyy-MM-dd HH:mm:ss") : DBNull.Value);
38                 return Convert.ToInt32(command.ExecuteScalar());
39             }
40         }
41
42         public List<HistorialEmpleado> Search(EmployeeHistoryQuery query, int? scopeDepartmentId)
43         {
44             query = query ?? new EmployeeHistoryQuery();
45             List<HistorialEmpleado> items = new List<HistorialEmpleado>();
46             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
47             using (SQLiteCommand command = new SQLiteCommand(SelectSql + @"
48                 WHERE (@empleado = 0 OR h.IdEmpleado = @empleado)
49                 AND (@scopeDep = 0 OR e.IdDepartamento = @scopeDep)
50                 AND (@campo = '' OR h.CampoModificado = @campo)
51                 AND (@usuario = '' OR h.UsuarioResponsable LIKE @usuarioLike)
52                 AND (@desde = '' OR h.FechaEfectiva >= @desde)
53                 AND (@hasta = '' OR h.FechaEfectiva <= @hasta)
54                 ORDER BY h.FechaEfectiva DESC, h.FechaCambio DESC, h.IdHistorial DESC;", connection))
55             {
56                 command.Parameters.AddWithValue("@empleado", query.EmployeeId ?? 0);
57                 command.Parameters.AddWithValue("@scopeDep", scopeDepartmentId ?? 0);
58                 command.Parameters.AddWithValue("@campo", query.Field ?? string.Empty);
59                 command.Parameters.AddWithValue("@usuario", query.User ?? string.Empty);
60                 command.Parameters.AddWithValue("@usuarioLike", "%" + (query.User ?? string.Empty) + "%");

```


Data/EmployeeHistoryRepository.cs (continuacion)

```
61         command.Parameters.AddWithValue("@desde", query.From.HasValue ? query.From.Value.ToString("yyyy-MM-dd") : string.Empty);
62         command.Parameters.AddWithValue("@hasta", query.To.HasValue ? query.To.Value.ToString("yyyy-MM-dd") : string.Empty);
63         using (SQLiteDataReader reader = command.ExecuteReader())
64         {
65             while (reader.Read()) items.Add(Map(reader));
66         }
67     }
68     return items;
69 }
70
71 public List<HistorialEmpleado> GetByEmployee(int employeeId)
72 {
73     List<HistorialEmpleado> items = new List<HistorialEmpleado>();
74     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
75     using (SQLiteCommand command = new SQLiteCommand(SelectSql + @"
76         WHERE h.IdEmpleado = @empleado
77         ORDER BY h.FechaEfectiva ASC, h.FechaCambio ASC, h.IdHistorial ASC;", connection))
78     {
79         command.Parameters.AddWithValue("@empleado", employeeId);
80         using (SQLiteDataReader reader = command.ExecuteReader())
81         {
82             while (reader.Read()) items.Add(Map(reader));
83         }
84     }
85     return items;
86 }
87
88 public List<HistorialEmpleado> GetPendingDue(SQLiteConnection connection,
89     SQLiteTransaction transaction, DateTime effectiveThrough)
90 {
91     List<HistorialEmpleado> items = new List<HistorialEmpleado>();
92     using (SQLiteCommand command = new SQLiteCommand(SelectSql + @"
93         WHERE h.Aplicado = 0 AND h.FechaEfectiva <= @fecha
94         ORDER BY h.FechaEfectiva, h.FechaCambio, h.IdHistorial;", connection, transaction))
95     {
96         command.Parameters.AddWithValue("@fecha", effectiveThrough.ToString("yyyy-MM-dd"));
97         using (SQLiteDataReader reader = command.ExecuteReader())
98         {
99             while (reader.Read()) items.Add(Map(reader));
100         }
101     }
102     return items;
103 }
104
105 public void ApplyChange(SQLiteConnection connection, SQLiteTransaction transaction,
106     HistorialEmpleado item, DateTime appliedAt)
107 {
108     string column;
109     if (item.CampoModificado == EmployeeHistoryFields.BaseSalary) column = "SalarioBase";
110     else if (item.CampoModificado == EmployeeHistoryFields.Position) column = "Cargo";
111     else if (item.CampoModificado == EmployeeHistoryFields.Department) column = "IdDepartamento";
112     else if (item.CampoModificado == EmployeeHistoryFields.Status) column = "Estado";
113     else throw new InvalidOperationException("El campo de historial laboral no es compatible.");
114
115     using (SQLiteCommand command = new SQLiteCommand("UPDATE Empleados SET " + column +
116         " = @valor, FechaEfectivaLaboral = @efectiva WHERE IdEmpleado = @empleado;", connection, transaction))
117     {
118         object value = item.ValorNuevoTecnico ?? item.ValorNuevo;
119         if (item.CampoModificado == EmployeeHistoryFields.BaseSalary)
120             value = decimal.Parse(Convert.ToString(value), System.Globalization.CultureInfo.InvariantCulture);
```

Data/EmployeeHistoryRepository.cs (continuacion)

```
121         else if (item.CampoModificado == EmployeeHistoryFields.Department)
122             value = int.Parse(Convert.ToString(value), System.Globalization.CultureInfo.InvariantCulture);
123         command.Parameters.AddWithValue("@valor", value);
124         command.Parameters.AddWithValue("@efectiva", item.FechaEfectiva.ToString("yyyy-MM-dd"));
125         command.Parameters.AddWithValue("@empleado", item.IdEmpleado);
126         if (command.ExecuteNonQuery() != 1) throw new InvalidOperationException("No se pudo aplicar el cambio laboral programado.");
127     }
128
129     using (SQLiteCommand command = new SQLiteCommand(@"UPDATE HistorialEmpleado
130         SET Aplicado = 1, FechaAplicacion = @fecha WHERE IdHistorial = @id AND Aplicado = 0;",
131         connection, transaction))
132     {
133         command.Parameters.AddWithValue("@fecha", appliedAt.ToString("yyyy-MM-dd HH:mm:ss"));
134         command.Parameters.AddWithValue("@id", item.IdHistorial);
135         if (command.ExecuteNonQuery() != 1) throw new InvalidOperationException("El cambio laboral ya fue procesado.");
136     }
137 }
138
139 private static HistorialEmpleado Map(SQLiteDataReader reader)
140 {
141     return new HistorialEmpleado
142     {
143         IdHistorial = Convert.ToInt32(reader["IdHistorial"]),
144         IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
145         CodigoEmpleado = Convert.ToString(reader["CodigoEmpleado"]),
146         EmpleadoNombre = Convert.ToString(reader["EmpleadoNombre"]),
147         CampoModificado = Convert.ToString(reader["CampoModificado"]),
148         ValorAnterior = Convert.ToString(reader["ValorAnterior"]),
149         ValorNuevo = Convert.ToString(reader["ValorNuevo"]),
150         ValorAnteriorTecnico = Convert.ToString(reader["ValorAnteriorTecnico"]),
151         ValorNuevoTecnico = Convert.ToString(reader["ValorNuevoTecnico"]),
152         FechaCambio = DateTime.Parse(Convert.ToString(reader["FechaCambio"])),
153         FechaEfectiva = DateTime.Parse(Convert.ToString(reader["FechaEfectiva"])),
154         UsuarioResponsable = Convert.ToString(reader["UsuarioResponsable"]),
155         Motivo = Convert.ToString(reader["Motivo"]),
156         Aplicado = Convert.ToInt32(reader["Aplicado"]) == 1,
157         FechaAplicacion = reader["FechaAplicacion"] == DBNull.Value ||
158             string.IsNullOrWhiteSpace(Convert.ToString(reader["FechaAplicacion"]))
159             ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaAplicacion"]));
160     };
161 }
162 }
163 }
```

Data/Repositories.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using SistemaGestionNomina.Models;
5
6 namespace SistemaGestionNomina.Data
7 {
8     public class UsuarioRepository
9     {
10         public Usuario GetByUsername(string username)
11         {
12             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
13             using (SQLiteCommand command = new SQLiteCommand("SELECT * FROM Usuarios WHERE NombreUsuario = @u AND Estado = 'Activo';", connection))
14             {
15                 command.Parameters.AddWithValue("@u", username);
16                 using (SQLiteDataReader reader = command.ExecuteReader())
17                 {
18                     if (reader.Read())
19                     {
20                         return new Usuario
21                         {
22                             IdUsuario = Convert.ToInt32(reader["IdUsuario"]),
23                             NombreUsuario = Convert.ToString(reader["NombreUsuario"]),
24                             PasswordHash = Convert.ToString(reader["PasswordHash"]),
25                             Rol = Convert.ToString(reader["Rol"]),
26                             Estado = Convert.ToString(reader["Estado"]),
27                             IdEmpleado = reader["IdEmpleado"] == DBNull.Value ? (int?)null : Convert.ToInt32(reader["IdEmpleado"]),
28                             UltimoAcceso = ParseNullableDate(reader["UltimoAcceso"]),
29                             IntentosFallidos = Convert.ToInt32(reader["IntentosFallidos"]),
30                             Bloqueado = Convert.ToInt32(reader["Bloqueado"]) == 1,
31                             FechaBloqueo = ParseNullableDate(reader["FechaBloqueo"])
32                         };
33                     }
34                 }
35             }
36             return null;
37         }
38     }
39
40     private static DateTime? ParseNullableDate(object value)
41     {
42         if (value == null || value == DBNull.Value || string.IsNullOrEmpty(Convert.ToString(value))) return null;
43         DateTime parsed;
44         return DateTime.TryParse(Convert.ToString(value), out parsed) ? parsed : (DateTime?)null;
45     }
46 }
47
48 public class DepartamentoRepository
49 {
50     public List<Departamento> GetAll()
51     {
52         List<Departamento> items = new List<Departamento>();
53         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
54         using (SQLiteCommand command = new SQLiteCommand("SELECT IdDepartamento, Nombre FROM Departamentos ORDER BY Nombre;", connection))
55         using (SQLiteDataReader reader = command.ExecuteReader())
56         {
57             while (reader.Read())
58             {
59                 items.Add(new Departamento
60                 {
```

Data/Repositories.cs (continuacion)

```
61             IdDepartamento = Convert.ToInt32(reader["IdDepartamento"]),
62             Nombre = Convert.ToString(reader["Nombre"])
63         });
64     }
65 }
66
67     return items;
68 }
69 }
70
71 public class EmpleadoRepository
72 {
73     public List<Empleado> GetAll(string search, int? departmentId, string status)
74     {
75         return GetAll(search, departmentId, status, null);
76     }
77
78     public List<Empleado> GetAll(string search, int? departmentId, string status, int? scopeDepartmentId)
79     {
80         List<Empleado> items = new List<Empleado>();
81         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
82         using (SQLiteCommand command = new SQLiteCommand(@"SELECT e.*, d.Nombre AS DepartamentoNombre
83             FROM Empleados e
84             INNER JOIN Departamentos d ON d.IdDepartamento = e.IdDepartamento
85             WHERE (@search = '' OR e.Codigo LIKE @like OR e.Nombre LIKE @like OR e.Apellido LIKE @like OR e.Cedula LIKE @like)
86                 AND (@dep = 0 OR e.IdDepartamento = @dep)
87                 AND (@scopeDep = 0 OR e.IdDepartamento = @scopeDep)
88                 AND (@estado = '' OR e.Estado = @estado)
89             ORDER BY e.Nombre, e.Apellido;", connection))
90         {
91             command.Parameters.AddWithValue("@search", search ?? string.Empty);
92             command.Parameters.AddWithValue("@like", "%" + (search ?? string.Empty) + "%");
93             command.Parameters.AddWithValue("@dep", departmentId.HasValue ? departmentId.Value : 0);
94             command.Parameters.AddWithValue("@scopeDep", scopeDepartmentId.HasValue ? scopeDepartmentId.Value : 0);
95             command.Parameters.AddWithValue("@estado", status ?? string.Empty);
96             using (SQLiteDataReader reader = command.ExecuteReader())
97             {
98                 while (reader.Read())
99                 {
100                     items.Add(MapEmpleado(reader));
101                 }
102             }
103         }
104
105         return items;
106     }
107
108     public List<Empleado> GetActiveByDepartment(int? departmentId)
109     {
110         return GetAll(string.Empty, departmentId, "Activo");
111     }
112
113     public List<Empleado> GetActiveByDepartment(int? departmentId, int? scopeDepartmentId)
114     {
115         return GetAll(string.Empty, departmentId, "Activo", scopeDepartmentId);
116     }
117
118     public List<Empleado> GetAllForEffectiveDate(int? departmentId)
119     {
120         return GetAll(string.Empty, departmentId, string.Empty, null);
121     }
122 }
```

Data/Repositories.cs (continuacion)

```
121     }
122
123     public Empleado GetById(int id)
124     {
125         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
126         {
127             return GetById(connection, null, id);
128         }
129     }
130
131     public Empleado GetById(SQLiteConnection connection, SQLiteTransaction transaction, int id)
132     {
133         using (SQLiteCommand command = new SQLiteCommand(@"SELECT e.*, d.Nombre AS DepartamentoNombre
134             FROM Empleados e INNER JOIN Departamentos d ON d.IdDepartamento = e.IdDepartamento
135             WHERE e.IdEmpleado = @id;", connection, transaction))
136         {
137             command.Parameters.AddWithValue("@id", id);
138             using (SQLiteDataReader reader = command.ExecuteReader())
139             {
140                 return reader.Read() ? MapEmpleado(reader) : null;
141             }
142         }
143     }
144
145     public int? GetDepartmentId(int employeeId)
146     {
147         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
148         using (SQLiteCommand command = new SQLiteCommand(
149             "SELECT IdDepartamento FROM Empleados WHERE IdEmpleado = @id LIMIT 1;", connection))
150         {
151             command.Parameters.AddWithValue("@id", employeeId);
152             object value = command.ExecuteScalar();
153             return value == null || value == DBNull.Value ? (int?)null : Convert.ToInt32(value);
154         }
155     }
156
157     public bool ExistsCode(string code, int ignoreId)
158     {
159         return Exists("Codigo", code, ignoreId);
160     }
161
162     public bool ExistsCedula(string cedula, int ignoreId)
163     {
164         return Exists("Cedula", cedula, ignoreId);
165     }
166
167     public int Add(Empleado empleado)
168     {
169         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
170         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Empleados
171             (Codigo, Nombre, Apellido, Cedula, Cargo, IdDepartamento, SalarioBase, Estado, FechaIngreso, FechaEfectivaLaboral)
172             VALUES (@codigo, @nombre, @apellido, @cedula, @cargo, @dep, @salario, @estado, @fecha, @efectiva);
173             SELECT last_insert_rowid();", connection))
174         {
175             FillEmpleadoParameters(command, empleado);
176             return Convert.ToInt32(command.ExecuteScalar());
177         }
178     }
179
180     public void Update(Empleado empleado)
```

Data/Repositories.cs (continuacion)

```
181     {
182         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
183         using (SQLiteTransaction transaction = connection.BeginTransaction())
184         {
185             Update(connection, transaction, empleado);
186             transaction.Commit();
187         }
188     }
189
190     public void Update(SQLiteConnection connection, SQLiteTransaction transaction, Empleado empleado)
191     {
192         using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Empleados SET
193             Codigo = @codigo, Nombre = @nombre, Apellido = @apellido, Cedula = @cedula,
194             Cargo = @cargo, IdDepartamento = @dep, SalarioBase = @salario,
195             Estado = @estado, FechaIngreso = @fecha, FechaEfectivaLaboral = @efectiva
196             WHERE IdEmpleado = @id;", connection, transaction))
197         {
198             FillEmpleadoParameters(command, empleado);
199             command.Parameters.AddWithValue("@id", empleado.IdEmpleado);
200             if (command.ExecuteNonQuery() != 1)
201                 throw new InvalidOperationException("No se pudo actualizar el empleado.");
202         }
203     }
204
205     public string GetDepartmentName(int departmentId)
206     {
207         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
208         using (SQLiteCommand command = new SQLiteCommand(
209             "SELECT Nombre FROM Departamentos WHERE IdDepartamento = @id LIMIT 1;", connection))
210         {
211             command.Parameters.AddWithValue("@id", departmentId);
212             object value = command.ExecuteScalar();
213             return value == null || value == DBNull.Value ? string.Empty : Convert.ToString(value);
214         }
215     }
216
217     public void Deactivate(int id)
218     {
219         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
220         using (SQLiteCommand command = new SQLiteCommand("UPDATE Empleados SET Estado = 'Inactivo' WHERE IdEmpleado = @id;", connection))
221         {
222             command.Parameters.AddWithValue("@id", id);
223             command.ExecuteNonQuery();
224         }
225     }
226
227     private bool Exists(string field, string value, int ignoreId)
228     {
229         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
230         using (SQLiteCommand command = new SQLiteCommand("SELECT COUNT(1) FROM Empleados WHERE " + field + " = @value AND IdEmpleado <> @id;", \
connection))
231         {
232             command.Parameters.AddWithValue("@value", value);
233             command.Parameters.AddWithValue("@id", ignoreId);
234             return Convert.ToInt32(command.ExecuteScalar()) > 0;
235         }
236     }
237
238     private static Empleado MapEmpleado(SQLiteDataReader reader)
239     {
```

Data/Repositories.cs (continuacion)

```

240         return new Empleado
241         {
242             IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
243            Codigo = Convert.ToString(reader["Codigo"]),
244             Nombre = Convert.ToString(reader["Nombre"]),
245             Apellido = Convert.ToString(reader["Apellido"]),
246             Cedula = Convert.ToString(reader["Cedula"]),
247             Cargo = Convert.ToString(reader["Cargo"]),
248             IdDepartamento = Convert.ToInt32(reader["IdDepartamento"]),
249             DepartamentoNombre = Convert.ToString(reader["DepartamentoNombre"]),
250             SalarioBase = Convert.ToDecimal(reader["SalarioBase"]),
251             Estado = Convert.ToString(reader["Estado"]),
252             FechaIngreso = DateTime.Parse(Convert.ToString(reader["FechaIngreso"])),
253             FechaEfectivaLaboral = reader["FechaEfectivaLaboral"] == DBNull.Value ||
254                 string.IsNullOrEmpty(Convert.ToString(reader["FechaEfectivaLaboral"]))
255                 ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaEfectivaLaboral"]));
256         };
257     }
258
259     private static void FillEmpleadoParameters(SQLiteCommand command, Empleado empleado)
260     {
261         command.Parameters.AddWithValue("@codigo", empleado.Codigo);
262         command.Parameters.AddWithValue("@nombre", empleado.Nombre);
263         command.Parameters.AddWithValue("@apellido", empleado.Apellido);
264         command.Parameters.AddWithValue("@cedula", empleado.Cedula);
265         command.Parameters.AddWithValue("@cargo", empleado.Cargo);
266         command.Parameters.AddWithValue("@dep", empleado.IdDepartamento);
267         command.Parameters.AddWithValue("@salario", empleado.SalarioBase);
268         command.Parameters.AddWithValue("@estado", empleado.Estado);
269         command.Parameters.AddWithValue("@fecha", empleado.FechaIngreso.ToString("yyyy-MM-dd"));
270         command.Parameters.AddWithValue("@efectiva", (empleado.FechaEfectivaLaboral ?? empleado.FechaIngreso).ToString("yyyy-MM-dd"));
271     }
272 }
273
274 public class AsistenciaRepository
275 {
276     public bool Exists(int employeeId, DateTime date)
277     {
278         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
279         using (SQLiteCommand command = new SQLiteCommand(@"SELECT COUNT(1) FROM Asistencias
280             WHERE IdEmpleado = @empleado AND Fecha = @fecha;", connection))
281         {
282             command.Parameters.AddWithValue("@empleado", employeeId);
283             command.Parameters.AddWithValue("@fecha", date.ToString("yyyy-MM-dd"));
284             return Convert.ToInt32(command.ExecuteScalar()) > 0;
285         }
286     }
287
288     public int Add(Asistencia asistencia)
289     {
290         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
291         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Asistencias
292             (IdEmpleado, Fecha, HoraEntrada, HoraSalida, HorasTrabajadas, Estado)
293             VALUES (@empleado, @fecha, @entrada, @salida, @horas, @estado);
294             SELECT last_insert_rowid();", connection))
295         {
296             command.Parameters.AddWithValue("@empleado", asistencia.IdEmpleado);
297             command.Parameters.AddWithValue("@fecha", asistencia.Fecha.ToString("yyyy-MM-dd"));
298             command.Parameters.AddWithValue("@entrada", asistencia.HoraEntrada.HasValue ? asistencia.HoraEntrada.Value.ToString(@"hh:mm") : null);
299             command.Parameters.AddWithValue("@salida", asistencia.HoraSalida.HasValue ? asistencia.HoraSalida.Value.ToString(@"hh:mm") : null);

```

Data/Repositories.cs (continuacion)

```
300         command.Parameters.AddWithValue("@horas", asistencia.HorasTrabajadas);
301         command.Parameters.AddWithValue("@estado", asistencia.Estado);
302         return Convert.ToInt32(command.ExecuteScalar());
303     }
304 }
305
306 public List<Asistencia> GetAll(DateTime? start, DateTime? end, int? employeeId, string status)
307 {
308     return GetAll(start, end, employeeId, status, null);
309 }
310
311 public List<Asistencia> GetAll(DateTime? start, DateTime? end, int? employeeId, string status,
312     int? scopeDepartmentId)
313 {
314     List<Asistencia> items = new List<Asistencia>();
315     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
316     using (SQLiteCommand command = new SQLiteCommand(@"SELECT a.*, e.Nombre || ' ' || e.Apellido AS EmpleadoNombre
317         FROM Asistencias a
318         INNER JOIN Empleados e ON e.IdEmpleado = a.IdEmpleado
319         WHERE (@inicio = '' OR a.Fecha >= @inicio)
320             AND (@fin = '' OR a.Fecha <= @fin)
321             AND (@empleado = 0 OR a.IdEmpleado = @empleado)
322             AND (@scopeDep = 0 OR e.IdDepartamento = @scopeDep)
323             AND (@estado = '' OR a.Estado = @estado)
324         ORDER BY a.Fecha DESC, EmpleadoNombre;", connection))
325     {
326         command.Parameters.AddWithValue("@inicio", start.HasValue ? start.Value.ToString("yyyy-MM-dd") : string.Empty);
327         command.Parameters.AddWithValue("@fin", end.HasValue ? end.Value.ToString("yyyy-MM-dd") : string.Empty);
328         command.Parameters.AddWithValue("@empleado", employeeId.HasValue ? employeeId.Value : 0);
329         command.Parameters.AddWithValue("@scopeDep", scopeDepartmentId.HasValue ? scopeDepartmentId.Value : 0);
330         command.Parameters.AddWithValue("@estado", status ?? string.Empty);
331         using (SQLiteDataReader reader = command.ExecuteReader())
332         {
333             while (reader.Read())
334             {
335                 items.Add(MapAsistencia(reader));
336             }
337         }
338     }
339
340     return items;
341 }
342
343 public List<Asistencia> GetByEmployee(int employeeId, DateTime? start, DateTime? end, string status)
344 {
345     List<Asistencia> items = new List<Asistencia>();
346     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
347     using (SQLiteCommand command = new SQLiteCommand(@"SELECT a.*, e.Nombre || ' ' || e.Apellido AS EmpleadoNombre
348         FROM Asistencias a
349         INNER JOIN Empleados e ON e.IdEmpleado = a.IdEmpleado
350         WHERE a.IdEmpleado = @empleado
351             AND (@inicio = '' OR a.Fecha >= @inicio)
352             AND (@fin = '' OR a.Fecha <= @fin)
353             AND (@estado = '' OR a.Estado = @estado)
354         ORDER BY a.Fecha DESC;", connection))
355     {
356         command.Parameters.AddWithValue("@empleado", employeeId);
357         command.Parameters.AddWithValue("@inicio", start.HasValue ? start.Value.ToString("yyyy-MM-dd") : string.Empty);
358         command.Parameters.AddWithValue("@fin", end.HasValue ? end.Value.ToString("yyyy-MM-dd") : string.Empty);
359         command.Parameters.AddWithValue("@estado", status ?? string.Empty);
```


Data/Repositories.cs (continuacion)

```
360         using (SQLiteDataReader reader = command.ExecuteReader())
361         {
362             while (reader.Read())
363             {
364                 items.Add(MapAsistencia(reader));
365             }
366         }
367     }
368
369     return items;
370 }
371
372 public decimal GetExtraHours(int employeeId, DateTime start, DateTime end)
373 {
374     decimal total = 0;
375     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
376     using (SQLiteCommand command = new SQLiteCommand(@"SELECT HorasTrabajadas FROM Asistencias
377         WHERE IdEmpleado = @id AND Fecha BETWEEN @inicio AND @fin AND Estado IN ('Puntual','Tardanza');", connection))
378     {
379         command.Parameters.AddWithValue("@id", employeeId);
380         command.Parameters.AddWithValue("@inicio", start.ToString("yyyy-MM-dd"));
381         command.Parameters.AddWithValue("@fin", end.ToString("yyyy-MM-dd"));
382         using (SQLiteDataReader reader = command.ExecuteReader())
383         {
384             while (reader.Read())
385             {
386                 decimal hours = Convert.ToDecimal(reader["HorasTrabajadas"]);
387                 if (hours > 8)
388                 {
389                     total += hours - 8;
390                 }
391             }
392         }
393     }
394
395     return total;
396 }
397
398 private static Asistencia MapAsistencia(SQLiteDataReader reader)
399 {
400     string entrada = Convert.ToString(reader["HoraEntrada"]);
401     string salida = Convert.ToString(reader["HoraSalida"]);
402     return new Asistencia
403     {
404         IdAsistencia = Convert.ToInt32(reader["IdAsistencia"]),
405         IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
406         EmpleadoNombre = Convert.ToString(reader["EmpleadoNombre"]),
407         Fecha = DateTime.Parse(Convert.ToString(reader["Fecha"])),
408         HoraEntrada = string.IsNullOrEmpty(entrada) ? (TimeSpan?)null : TimeSpan.Parse(entrada),
409         HoraSalida = string.IsNullOrEmpty(salida) ? (TimeSpan?)null : TimeSpan.Parse(salida),
410         HorasTrabajadas = Convert.ToDecimal(reader["HorasTrabajadas"]),
411         Estado = Convert.ToString(reader["Estado"]),
412         IdSolicitudAusencia = reader["IdSolicitudAusencia"] == DBNull.Value
413             ? (int?)null : Convert.ToInt32(reader["IdSolicitudAusencia"])
414     };
415 }
416 }
417
418 public class NominaRepository
419 {
```

Data/Repositories.cs (continuacion)

```
420     public int CreatePeriodo(PeriodoNomina periodo)
421     {
422         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
423         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO PeriodosNomina (Nombre, FechaInicio, FechaFin, Estado)
424             VALUES (@nombre, @inicio, @fin, @estado); SELECT last_insert_rowid();", connection))
425         {
426             command.Parameters.AddWithValue("@nombre", periodo.Nombre);
427             command.Parameters.AddWithValue("@inicio", periodo.FechaInicio.ToString("yyyy-MM-dd"));
428             command.Parameters.AddWithValue("@fin", periodo.FechaFin.ToString("yyyy-MM-dd"));
429             command.Parameters.AddWithValue("@estado", periodo.Estado);
430             return Convert.ToInt32(command.ExecuteScalar());
431         }
432     }
433
434     public int CreateNomina(Nomina nomina)
435     {
436         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
437         using (SQLiteTransaction transaction = connection.BeginTransaction())
438         {
439             using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Nominas
440                 (IdPeriodo, FechaCalculo, TotalIngresos, TotalDeducciones, TotalNeto, Estado)
441                 VALUES (@periodo, @fecha, @ingresos, @deducciones, @neto, @estado);
442                 SELECT last_insert_rowid();", connection, transaction))
443             {
444                 command.Parameters.AddWithValue("@periodo", nomina.IdPeriodo);
445                 command.Parameters.AddWithValue("@fecha", nomina.FechaCalculo.ToString("yyyy-MM-dd HH:mm:ss"));
446                 command.Parameters.AddWithValue("@ingresos", nomina.TotalIngresos);
447                 command.Parameters.AddWithValue("@deducciones", nomina.TotalDeducciones);
448                 command.Parameters.AddWithValue("@neto", nomina.TotalNeto);
449                 command.Parameters.AddWithValue("@estado", nomina.Estado);
450                 int idNomina = Convert.ToInt32(command.ExecuteScalar());
451
452                 foreach (NominaDetalle detalle in nomina.Detalles)
453                 {
454                     using (SQLiteCommand detailCommand = new SQLiteCommand(@"INSERT INTO NominaDetalle
455                         (IdNomina, IdEmpleado, SueldoBase, Bonos, HorasExtra, MontoHorasExtra, TotalIngresos, TotalDeducciones, NetoPagar)
456                         VALUES (@nomina, @empleado, @sueldo, @bonos, @horas, @montoHoras, @ingresos, @deducciones, @neto);", connection, transaction))
457                     {
458                         detailCommand.Parameters.AddWithValue("@nomina", idNomina);
459                         detailCommand.Parameters.AddWithValue("@empleado", detalle.IdEmpleado);
460                         detailCommand.Parameters.AddWithValue("@sueldo", detalle.SueldoBase);
461                         detailCommand.Parameters.AddWithValue("@bonos", detalle.Bonos);
462                         detailCommand.Parameters.AddWithValue("@horas", detalle.HorasExtra);
463                         detailCommand.Parameters.AddWithValue("@montoHoras", detalle.MontoHorasExtra);
464                         detailCommand.Parameters.AddWithValue("@ingresos", detalle.TotalIngresos);
465                         detailCommand.Parameters.AddWithValue("@deducciones", detalle.TotalDeducciones);
466                         detailCommand.Parameters.AddWithValue("@neto", detalle.NetoPagar);
467                         detailCommand.ExecuteNonQuery();
468                     }
469                 }
470
471                 transaction.Commit();
472                 return idNomina;
473             }
474         }
475     }
476
477     public List<Nomina> GetAll()
478     {
479         List<Nomina> items = new List<Nomina>();
```

Data/Repositories.cs (continuación)

```
480         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
481         using (SQLiteCommand command = new SQLiteCommand(@"SELECT n.*, p.Nombre AS PeriodoNombre
482             FROM Nominas n INNER JOIN PeriodosNomina p ON p.IdPeriodo = n.IdPeriodo
483             WHERE n.Estado <> 'Anulada'
484             ORDER BY n.FechaCalculo DESC;", connection))
485         using (SQLiteDataReader reader = command.ExecuteReader())
486         {
487             while (reader.Read())
488             {
489                 items.Add(MapNomina(reader));
490             }
491         }
492         return items;
493     }
494 }
495
496 public List<NominaDetalle> GetDetalles(int idNomina)
497 {
498     List<NominaDetalle> items = new List<NominaDetalle>();
499     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
500     using (SQLiteCommand command = new SQLiteCommand(@"SELECT nd.*,
501         COALESCE(NULLIF(nd.CodigoEmpleadoSnapshot, ''), e.Codigo) AS Codigo,
502         COALESCE(NULLIF(nd.NombreEmpleadoSnapshot, ''), e.Nombre || ' ' || e.Apellido) AS EmpleadoNombre,
503         COALESCE(NULLIF(nd.CargoEmpleadoSnapshot, ''), e.Cargo) AS CargoEmpleado,
504         COALESCE(NULLIF(nd.DepartamentoSnapshot, ''), d.Nombre) AS Departamento
505     FROM NominaDetalle nd
506     INNER JOIN Empleados e ON e.IdEmpleado = nd.IdEmpleado
507     INNER JOIN Departamentos d ON d.IdDepartamento = e.IdDepartamento
508     WHERE nd.IdNomina = @id
509     ORDER BY EmpleadoNombre;", connection))
510     {
511         command.Parameters.AddWithValue("@id", idNomina);
512         using (SQLiteDataReader reader = command.ExecuteReader())
513         {
514             while (reader.Read())
515             {
516                 items.Add(MapDetalle(reader));
517             }
518         }
519     }
520     return items;
521 }
522
523 public Nomina GetById(int idNomina)
524 {
525     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
526     using (SQLiteCommand command = new SQLiteCommand(@"SELECT n.*, p.Nombre AS PeriodoNombre
527         FROM Nominas n INNER JOIN PeriodosNomina p ON p.IdPeriodo = n.IdPeriodo
528         WHERE n.IdNomina = @id;", connection))
529     {
530         command.Parameters.AddWithValue("@id", idNomina);
531         using (SQLiteDataReader reader = command.ExecuteReader())
532         {
533             if (reader.Read()) return MapNomina(reader);
534         }
535     }
536     return null;
537 }
538
539 }
```

Data/Repositories.cs (continuacion)

```
540     public void UpdateEstadoNomina(int idNomina, string estado,
541         string campoFecha, DateTime? fecha,
542         string campoUsuario, string usuario,
543         string motivo = null)
544     {
545         string sql = "UPDATE Nominas SET Estado = @estado";
546         if (!string.IsNullOrEmpty(campoFecha) && fecha.HasValue)
547             sql += ", " + campoFecha + " = @fecha";
548         if (!string.IsNullOrEmpty(campoUsuario) && !string.IsNullOrEmpty(usuario))
549             sql += ", " + campoUsuario + " = @usuario";
550         if (motivo != null)
551             sql += ", MotivoAnulacion = @motivo";
552         sql += " WHERE IdNomina = @id;";
553
554         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
555         using (SQLiteCommand command = new SQLiteCommand(sql, connection))
556         {
557             command.Parameters.AddWithValue("@id", idNomina);
558             command.Parameters.AddWithValue("@estado", estado);
559             if (fecha.HasValue) command.Parameters.AddWithValue("@fecha", fecha.Value.ToString("yyyy-MM-dd HH:mm:ss"));
560             if (!string.IsNullOrEmpty(usuario)) command.Parameters.AddWithValue("@usuario", usuario);
561             if (motivo != null) command.Parameters.AddWithValue("@motivo", motivo);
562             command.ExecuteNonQuery();
563         }
564     }
565
566     public void CerrarPeriodo(int idPeriodo, string usuario)
567     {
568         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
569         using (SQLiteCommand command = new SQLiteCommand(@"UPDATE PeriodosNomina
570             SET Cerrado = 1, FechaCierre = @fecha, CerradoPor = @usuario
571             WHERE IdPeriodo = @id;", connection))
572         {
573             command.Parameters.AddWithValue("@id", idPeriodo);
574             command.Parameters.AddWithValue("@fecha", DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss"));
575             command.Parameters.AddWithValue("@usuario", usuario);
576             command.ExecuteNonQuery();
577         }
578     }
579
580     public void ReabrirPeriodo(int idPeriodo)
581     {
582         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
583         using (SQLiteCommand command = new SQLiteCommand(@"UPDATE PeriodosNomina
584             SET Cerrado = 0, FechaCierre = NULL, CerradoPor = NULL
585             WHERE IdPeriodo = @id;", connection))
586         {
587             command.Parameters.AddWithValue("@id", idPeriodo);
588             command.ExecuteNonQuery();
589         }
590     }
591
592     public PeriodoNomina GetPeriodoById(int idPeriodo)
593     {
594         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
595         using (SQLiteCommand command = new SQLiteCommand("SELECT * FROM PeriodosNomina WHERE IdPeriodo = @id;", connection))
596         {
597             command.Parameters.AddWithValue("@id", idPeriodo);
598             using (SQLiteDataReader reader = command.ExecuteReader())
599             {
```

Data/Repositories.cs (continuacion)

```
600         if (reader.Read())
601         {
602             return new PeriodoNomina
603             {
604                 IdPeriodo = Convert.ToInt32(reader["IdPeriodo"]),
605                 Nombre = Convert.ToString(reader["Nombre"]),
606                 FechaInicio = DateTime.Parse(Convert.ToString(reader["FechaInicio"])),
607                 FechaFin = DateTime.Parse(Convert.ToString(reader["FechaFin"])),
608                 Estado = Convert.ToString(reader["Estado"]),
609                 Cerrado = Convert.ToInt32(reader["Cerrado"]) == 1,
610                 FechaCierre = reader["FechaCierre"] == DBNull.Value ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaCierre"])),
611                 CerradoPor = reader["CerradoPor"] == DBNull.Value ? null : Convert.ToString(reader["CerradoPor"])
612             };
613         }
614     }
615 }
616 return null;
617 }
618
619 public PeriodoNomina GetPeriodoByFechas(DateTime fechaInicio, DateTime fechaFin)
620 {
621     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
622     using (SQLiteCommand command = new SQLiteCommand(@"SELECT * FROM PeriodosNomina
623         WHERE FechaInicio <= @fin AND FechaFin >= @inicio
624         ORDER BY Cerrado DESC, FechaInicio LIMIT 1;", connection))
625     {
626         command.Parameters.AddWithValue("@inicio", fechaInicio.ToString("yyyy-MM-dd"));
627         command.Parameters.AddWithValue("@fin", fechaFin.ToString("yyyy-MM-dd"));
628         using (SQLiteDataReader reader = command.ExecuteReader())
629         {
630             if (reader.Read())
631             {
632                 return new PeriodoNomina
633                 {
634                     IdPeriodo = Convert.ToInt32(reader["IdPeriodo"]),
635                     Nombre = Convert.ToString(reader["Nombre"]),
636                     FechaInicio = DateTime.Parse(Convert.ToString(reader["FechaInicio"])),
637                     FechaFin = DateTime.Parse(Convert.ToString(reader["FechaFin"])),
638                     Estado = Convert.ToString(reader["Estado"]),
639                     Cerrado = Convert.ToInt32(reader["Cerrado"]) == 1
640                 };
641             }
642         }
643     }
644     return null;
645 }
646
647 public PeriodoNomina GetOverlappingProtectedPeriod(DateTime fechaInicio, DateTime fechaFin)
648 {
649     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
650     using (SQLiteCommand command = new SQLiteCommand(@"SELECT * FROM PeriodosNomina
651         WHERE FechaInicio <= @fin AND FechaFin >= @inicio
652         AND (Cerrado = 1 OR Estado IN ('Confirmado', 'Pagado'))
653         ORDER BY FechaInicio LIMIT 1;", connection))
654     {
655         command.Parameters.AddWithValue("@inicio", fechaInicio.ToString("yyyy-MM-dd"));
656         command.Parameters.AddWithValue("@fin", fechaFin.ToString("yyyy-MM-dd"));
657         using (SQLiteDataReader reader = command.ExecuteReader())
658         {
659             if (reader.Read()) return MapPeriodo(reader);
```

Data/Repositories.cs (continuacion)

```

660         }
661     }
662     return null;
663 }
664
665 public bool HasClosedPeriodAffectedByEmployeeChange(DateTime effectiveDate)
666 {
667     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
668     {
669         return HasClosedPeriodAffectedByEmployeeChange(connection, null, effectiveDate);
670     }
671 }
672
673 public bool HasClosedPeriodAffectedByEmployeeChange(SQLiteConnection connection,
674     SQLiteTransaction transaction, DateTime effectiveDate)
675 {
676     using (SQLiteCommand command = new SQLiteCommand(@"SELECT COUNT(1) FROM PeriodosNomina
677         WHERE Cerrado = 1 AND FechaFin >= @efectiva;", connection, transaction))
678     {
679         command.Parameters.AddWithValue("@efectiva", effectiveDate.ToString("yyyy-MM-dd"));
680         return Convert.ToInt32(command.ExecuteScalar()) > 0;
681     }
682 }
683
684 private static PeriodoNomina MapPeriodo(SQLiteDataReader reader)
685 {
686     return new PeriodoNomina
687     {
688         IdPeriodo = Convert.ToInt32(reader["IdPeriodo"]),
689         Nombre = Convert.ToString(reader["Nombre"]),
690         FechaInicio = DateTime.Parse(Convert.ToString(reader["FechaInicio"])),
691         FechaFin = DateTime.Parse(Convert.ToString(reader["FechaFin"])),
692         Estado = Convert.ToString(reader["Estado"]),
693         Cerrado = Convert.ToInt32(reader["Cerrado"]) == 1,
694         FechaCierre = reader["FechaCierre"] == DBNull.Value ? (DateTime?)null :
695             DateTime.Parse(Convert.ToString(reader["FechaCierre"])),
696         CerradoPor = reader["CerradoPor"] == DBNull.Value ? null : Convert.ToString(reader["CerradoPor"])
697     };
698 }
699
700 public int CreateNominaEnTransaccion(SQLiteConnection connection, SQLiteTransaction transaction, Nomina nomina)
701 {
702     using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Nominas
703         (IdPeriodo, FechaCalculo, TotalIngresos, TotalDeducciones, TotalNeto, Estado,
704         FechaConfirmacion, ConfirmadaPor)
705         VALUES (@periodo, @fecha, @ingresos, @deducciones, @neto, @estado, @fecConf, @usrConf);
706         SELECT last_insert_rowid();", connection, transaction))
707     {
708         command.Parameters.AddWithValue("@periodo", nomina.IdPeriodo);
709         command.Parameters.AddWithValue("@fecha", nomina.FechaCalculo.ToString("yyyy-MM-dd HH:mm:ss"));
710         command.Parameters.AddWithValue("@ingresos", nomina.TotalIngresos);
711         command.Parameters.AddWithValue("@deducciones", nomina.TotalDeducciones);
712         command.Parameters.AddWithValue("@neto", nomina.TotalNeto);
713         command.Parameters.AddWithValue("@estado", nomina.Estado);
714         command.Parameters.AddWithValue("@fecConf", nomina.FechaConfirmacion?.ToString("yyyy-MM-dd HH:mm:ss") ?? \
715             DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss"));
716         command.Parameters.AddWithValue("@usrConf", nomina.ConfirmadaPor ?? string.Empty);
717         return Convert.ToInt32(command.ExecuteScalar());
718     }
719 }

```

Data/Repositories.cs (continuacion)

```

719
720     private static Nomina MapNomina(SQLiteDataReader reader)
721     {
722         return new Nomina
723         {
724             IdNomina = Convert.ToInt32(reader["IdNomina"]),
725             IdPeriodo = Convert.ToInt32(reader["IdPeriodo"]),
726             PeriodoNombre = Convert.ToString(reader["PeriodoNombre"]),
727             FechaCalculo = DateTime.Parse(Convert.ToString(reader["FechaCalculo"])),
728             TotalIngresos = Convert.ToDecimal(reader["TotalIngresos"]),
729             TotalDeducciones = Convert.ToDecimal(reader["TotalDeducciones"]),
730             TotalNeto = Convert.ToDecimal(reader["TotalNeto"]),
731             Estado = Convert.ToString(reader["Estado"]),
732             FechaConfirmacion = reader["FechaConfirmacion"] == DBNull.Value ? (DateTime?)null : \
DateTime.Parse(Convert.ToString(reader["FechaConfirmacion"])),
733             ConfirmadaPor = reader["ConfirmadaPor"] == DBNull.Value ? null : Convert.ToString(reader["ConfirmadaPor"]),
734             FechaPago = reader["FechaPago"] == DBNull.Value ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaPago"])),
735             PagadaPor = reader["PagadaPor"] == DBNull.Value ? null : Convert.ToString(reader["PagadaPor"]),
736             FechaAnulacion = reader["FechaAnulacion"] == DBNull.Value ? (DateTime?)null : DateTime.Parse(Convert.ToString(reader["FechaAnulacion"])),
737             AnuladaPor = reader["AnuladaPor"] == DBNull.Value ? null : Convert.ToString(reader["AnuladaPor"]),
738             MotivoAnulacion = reader["MotivoAnulacion"] == DBNull.Value ? null : Convert.ToString(reader["MotivoAnulacion"])
739         };
740     }
741
742     private static NominaDetalle MapDetalle(SQLiteDataReader reader)
743     {
744         return new NominaDetalle
745         {
746             IdDetalle = Convert.ToInt32(reader["IdDetalle"]),
747             IdNomina = Convert.ToInt32(reader["IdNomina"]),
748             IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
749             CodigoEmpleado = Convert.ToString(reader["Codigo"]),
750             EmpleadoNombre = Convert.ToString(reader["EmpleadoNombre"]),
751             CargoEmpleado = Convert.ToString(reader["CargoEmpleado"]),
752             Departamento = Convert.ToString(reader["Departamento"]),
753             SueldoBase = Convert.ToDecimal(reader["SueldoBase"]),
754             Bonos = Convert.ToDecimal(reader["Bonos"]),
755             HorasExtra = Convert.ToDecimal(reader["HorasExtra"]),
756             MontoHorasExtra = Convert.ToDecimal(reader["MontoHorasExtra"]),
757             TotalIngresos = Convert.ToDecimal(reader["TotalIngresos"]),
758             TotalDeducciones = Convert.ToDecimal(reader["TotalDeducciones"]),
759             NetoPagar = Convert.ToDecimal(reader["NetoPagar"])
760         };
761     }
762 }
763
764 public class NominaVersionRepository
765 {
766     public void Add(NominaVersion version)
767     {
768         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
769         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO NominaVersiones
770             (IdNominaOriginal, IdNominaNueva, MotivoCambio, UsuarioResponsable, FechaCambio)
771             VALUES (@original, @nueva, @motivo, @usuario, @fecha);", connection))
772         {
773             command.Parameters.AddWithValue("@original", version.IdNominaOriginal);
774             command.Parameters.AddWithValue("@nueva", version.IdNominaNueva.HasValue ? (object)version.IdNominaNueva.Value : DBNull.Value);
775             command.Parameters.AddWithValue("@motivo", version.MotivoCambio);
776             command.Parameters.AddWithValue("@usuario", version.UsuarioResponsable);
777             command.Parameters.AddWithValue("@fecha", version.FechaCambio.ToString("yyyy-MM-dd HH:mm:ss"));

```

Data/Repositories.cs (continuacion)

```
778         command.ExecuteNonQuery();
779     }
780 }
781
782 public List<NominaVersion> GetByIdNominaOriginal(int idNominaOriginal)
783 {
784     List<NominaVersion> items = new List<NominaVersion>();
785     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
786     using (SQLiteCommand command = new SQLiteCommand(@"SELECT * FROM NominaVersiones
787         WHERE IdNominaOriginal = @id ORDER BY FechaCambio DESC;", connection))
788     {
789         command.Parameters.AddWithValue("@id", idNominaOriginal);
790         using (SQLiteDataReader reader = command.ExecuteReader())
791         {
792             while (reader.Read())
793             {
794                 items.Add(new NominaVersion
795                 {
796                     IdVersion = Convert.ToInt32(reader["IdVersion"]),
797                     IdNominaOriginal = Convert.ToInt32(reader["IdNominaOriginal"]),
798                     IdNominaNueva = reader["IdNominaNueva"] == DBNull.Value ? (int?)null : Convert.ToInt32(reader["IdNominaNueva"]),
799                     MotivoCambio = Convert.ToString(reader["MotivoCambio"]),
800                     UsuarioResponsable = Convert.ToString(reader["UsuarioResponsable"]),
801                     FechaCambio = DateTime.Parse(Convert.ToString(reader["FechaCambio"]))
802                 });
803             }
804         }
805     }
806     return items;
807 }
808
809 public void UpdateIdNominaNueva(int idVersion, int idNominaNueva)
810 {
811     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
812     using (SQLiteCommand command = new SQLiteCommand(@"UPDATE NominaVersiones
813         SET IdNominaNueva = @nueva WHERE IdVersion = @id;", connection))
814     {
815         command.Parameters.AddWithValue("@id", idVersion);
816         command.Parameters.AddWithValue("@nueva", idNominaNueva);
817         command.ExecuteNonQuery();
818     }
819 }
820
821 public List<NominaVersion> GetHistorialChain(int idNomina)
822 {
823     List<NominaVersion> result = new List<NominaVersion>();
824     int? currentId = idNomina;
825     while (currentId.HasValue)
826     {
827         NominaVersion version = GetByNominaNueva(currentId.Value);
828         if (version == null)
829             break;
830         result.Add(version);
831         currentId = version.IdNominaOriginal;
832     }
833     return result;
834 }
835
836 private NominaVersion GetByNominaNueva(int idNominaNueva)
837 {
```


Data/Repositories.cs (continuacion)

```

838         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
839         using (SQLiteCommand command = new SQLiteCommand(@"SELECT * FROM NominaVersiones
840             WHERE IdNominaNueva = @id LIMIT 1;", connection))
841         {
842             command.Parameters.AddWithValue("@id", idNominaNueva);
843             using (SQLiteDataReader reader = command.ExecuteReader())
844             {
845                 if (reader.Read())
846                 {
847                     return new NominaVersion
848                     {
849                         IdVersion = Convert.ToInt32(reader["IdVersion"]),
850                         IdNominaOriginal = Convert.ToInt32(reader["IdNominaOriginal"]),
851                         IdNominaNueva = reader["IdNominaNueva"] == DBNull.Value ? (int?)null : Convert.ToInt32(reader["IdNominaNueva"]),
852                         MotivoCambio = Convert.ToString(reader["MotivoCambio"]),
853                         UsuarioResponsable = Convert.ToString(reader["UsuarioResponsable"]),
854                         FechaCambio = DateTime.Parse(Convert.ToString(reader["FechaCambio"]))
855                     };
856                 }
857             }
858         }
859         return null;
860     }
861 }
862
863 public class ComprobanteRepository
864 {
865     private const string ComprobanteSelect = @"SELECT c.*,
866         COALESCE(NULLIF(nd.CodigoEmpleadoSnapshot, ''), e.Codigo) AS Codigo,
867         COALESCE(NULLIF(nd.NombreEmpleadoSnapshot, ''), e.Nombre || ' ' || e.Apellido) AS EmpleadoNombre,
868         p.Nombre AS PeriodoNombre, nd.TotalIngresos, nd.TotalDeducciones, nd.NetoPagar
869     FROM Comprobantes c
870     INNER JOIN Empleados e ON e.IdEmpleado = c.IdEmpleado
871     INNER JOIN Nominas n ON n.IdNomina = c.IdNomina
872     INNER JOIN PeriodosNomina p ON p.IdPeriodo = n.IdPeriodo
873     INNER JOIN NominaDetalle nd ON nd.IdNomina = c.IdNomina AND nd.IdEmpleado = c.IdEmpleado ";
874
875     public void Add(Comprobante comprobante)
876     {
877         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
878         using (SQLiteCommand command = new SQLiteCommand(@"INSERT OR IGNORE INTO Comprobantes
879             (IdNomina, IdEmpleado, NumeroComprobante, FechaGeneracion, RutaPdf)
880             VALUES (@nomina, @empleado, @numero, @fecha, @ruta);", connection))
881         {
882             command.Parameters.AddWithValue("@nomina", comprobante.IdNomina);
883             command.Parameters.AddWithValue("@empleado", comprobante.IdEmpleado);
884             command.Parameters.AddWithValue("@numero", comprobante.NumeroComprobante);
885             command.Parameters.AddWithValue("@fecha", comprobante.FechaGeneracion.ToString("yyyy-MM-dd HH:mm:ss"));
886             command.Parameters.AddWithValue("@ruta", comprobante.RutaPdf ?? string.Empty);
887             command.ExecuteNonQuery();
888         }
889     }
890
891     public void UpdateRutaPdf(int idComprobante, string ruta)
892     {
893         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
894         using (SQLiteCommand command = new SQLiteCommand("UPDATE Comprobantes SET RutaPdf = @ruta WHERE IdComprobante = @id;", connection))
895         {
896             command.Parameters.AddWithValue("@ruta", ruta ?? string.Empty);
897             command.Parameters.AddWithValue("@id", idComprobante);

```

Data/Repositories.cs (continuacion)

```
898         command.ExecuteNonQuery();
899     }
900 }
901
902 public bool UpdateRutaPdfForEmployee(int idComprobante, int employeeId, string ruta)
903 {
904     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
905     using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Comprobantes SET RutaPdf = @ruta
906         WHERE IdComprobante = @id AND IdEmpleado = @empleado;", connection))
907     {
908         command.Parameters.AddWithValue("@ruta", ruta ?? string.Empty);
909         command.Parameters.AddWithValue("@id", idComprobante);
910         command.Parameters.AddWithValue("@empleado", employeeId);
911         return command.ExecuteNonQuery() == 1;
912     }
913 }
914
915 public List<Comprobante> GetAll(string search)
916 {
917     List<Comprobante> items = new List<Comprobante>();
918     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
919     using (SQLiteCommand command = new SQLiteCommand(ComprobanteSelect + @"
920         WHERE (@search = '' OR c.NumeroComprobante LIKE @like OR e.Nombre LIKE @like OR p.Nombre LIKE @like)
921         ORDER BY c.FechaGeneracion DESC;", connection))
922     {
923         command.Parameters.AddWithValue("@search", search ?? string.Empty);
924         command.Parameters.AddWithValue("@like", "%" + (search ?? string.Empty) + "%");
925         using (SQLiteDataReader reader = command.ExecuteReader())
926         {
927             while (reader.Read())
928             {
929                 items.Add(MapComprobante(reader));
930             }
931         }
932     }
933
934     return items;
935 }
936
937 public Comprobante GetById(int id)
938 {
939     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
940     using (SQLiteCommand command = new SQLiteCommand(ComprobanteSelect +
941         " WHERE c.IdComprobante = @id LIMIT 1;", connection))
942     {
943         command.Parameters.AddWithValue("@id", id);
944         using (SQLiteDataReader reader = command.ExecuteReader())
945         {
946             return reader.Read() ? MapComprobante(reader) : null;
947         }
948     }
949 }
950
951 public List<Comprobante> GetByEmployee(int employeeId, string search)
952 {
953     List<Comprobante> items = new List<Comprobante>();
954     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
955     using (SQLiteCommand command = new SQLiteCommand(ComprobanteSelect + @"
956         WHERE c.IdEmpleado = @empleado
957         AND (@search = '' OR c.NumeroComprobante LIKE @like OR p.Nombre LIKE @like)
```

Data/Repositories.cs (continuacion)

```
958         ORDER BY c.FechaGeneracion DESC;", connection))
959     {
960         command.Parameters.AddWithValue("@empleado", employeeId);
961         command.Parameters.AddWithValue("@search", search ?? string.Empty);
962         command.Parameters.AddWithValue("@like", "%" + (search ?? string.Empty) + "%");
963         using (SQLiteDataReader reader = command.ExecuteReader())
964         {
965             while (reader.Read())
966             {
967                 items.Add(MapComprobante(reader));
968             }
969         }
970     }
971
972     return items;
973 }
974
975 public Comprobante GetByIdAndEmployee(int payslipId, int employeeId)
976 {
977     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
978     using (SQLiteCommand command = new SQLiteCommand(ComprobanteSelect + @"
979         WHERE c.IdComprobante = @id AND c.IdEmpleado = @empleado LIMIT 1;", connection))
980     {
981         command.Parameters.AddWithValue("@id", payslipId);
982         command.Parameters.AddWithValue("@empleado", employeeId);
983         using (SQLiteDataReader reader = command.ExecuteReader())
984         {
985             return reader.Read() ? MapComprobante(reader) : null;
986         }
987     }
988 }
989
990 private static Comprobante MapComprobante(SQLiteDataReader reader)
991 {
992     return new Comprobante
993     {
994         IdComprobante = Convert.ToInt32(reader["IdComprobante"]),
995         IdNomina = Convert.ToInt32(reader["IdNomina"]),
996         IdEmpleado = Convert.ToInt32(reader["IdEmpleado"]),
997         NumeroComprobante = Convert.ToString(reader["NumeroComprobante"]),
998         FechaGeneracion = DateTime.Parse(Convert.ToString(reader["FechaGeneracion"])),
999         RutaPdf = Convert.ToString(reader["RutaPdf"]),
1000         CodigoEmpleado = Convert.ToString(reader["Codigo"]),
1001         EmpleadoNombre = Convert.ToString(reader["EmpleadoNombre"]),
1002         PeriodoNombre = Convert.ToString(reader["PeriodoNombre"]),
1003         TotalIngresos = Convert.ToDecimal(reader["TotalIngresos"]),
1004         TotalDeducciones = Convert.ToDecimal(reader["TotalDeducciones"]),
1005         NetoPagar = Convert.ToDecimal(reader["NetoPagar"])
1006     };
1007 }
1008 }
1009
1010 public class ConfiguracionRepository
1011 {
1012     public List<ConfiguracionNomina> GetAll()
1013     {
1014         List<ConfiguracionNomina> items = new List<ConfiguracionNomina>();
1015         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
1016         using (SQLiteCommand command = new SQLiteCommand("SELECT * FROM ConfiguracionNomina ORDER BY NombreParametro;", connection))
1017         using (SQLiteDataReader reader = command.ExecuteReader())
```

Data/Repositories.cs (continuacion)

```

1018         {
1019             while (reader.Read())
1020             {
1021                 items.Add(new ConfiguracionNomina
1022                 {
1023                     IdConfiguracion = Convert.ToInt32(reader["IdConfiguracion"]),
1024                     NombreParametro = Convert.ToString(reader["NombreParametro"]),
1025                     Valor = Convert.ToDecimal(reader["Valor"]),
1026                     Descripcion = Convert.ToString(reader["Descripcion"])
1027                 });
1028             }
1029         }
1030
1031         return items;
1032     }
1033
1034     public decimal GetValue(string name, decimal fallback)
1035     {
1036         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
1037         using (SQLiteCommand command = new SQLiteCommand("SELECT Valor FROM ConfiguracionNomina WHERE NombreParametro = @n;", connection))
1038         {
1039             command.Parameters.AddWithValue("@n", name);
1040             object result = command.ExecuteScalar();
1041             return result == null || result == DBNull.Value ? fallback : Convert.ToDecimal(result);
1042         }
1043     }
1044
1045     public void Save(string name, decimal value, string description)
1046     {
1047         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
1048         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO ConfiguracionNomina (NombreParametro, Valor, Descripcion)
1049             VALUES (@n, @v, @d)
1050             ON CONFLICT(NombreParametro) DO UPDATE SET Valor = excluded.Valor, Descripcion = excluded.Descripcion;", connection))
1051         {
1052             command.Parameters.AddWithValue("@n", name);
1053             command.Parameters.AddWithValue("@v", value);
1054             command.Parameters.AddWithValue("@d", description);
1055             command.ExecuteNonQuery();
1056         }
1057     }
1058 }
1059
1060 public class ReporteRepository
1061 {
1062     public int Add(ReporteGenerado reporte)
1063     {
1064         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
1065         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO ReportesGenerados
1066             (NombreReporte, Tipo, GeneradoPor, FechaGeneracion, RutaArchivo)
1067             VALUES (@nombre, @tipo, @por, @fecha, @ruta); SELECT last_insert_rowid();", connection))
1068         {
1069             command.Parameters.AddWithValue("@nombre", reporte.NombreReporte);
1070             command.Parameters.AddWithValue("@tipo", reporte.Tipo);
1071             command.Parameters.AddWithValue("@por", reporte.GeneradoPor);
1072             command.Parameters.AddWithValue("@fecha", reporte.FechaGeneracion.ToString("yyyy-MM-dd HH:mm:ss"));
1073             command.Parameters.AddWithValue("@ruta", reporte.RutaArchivo ?? string.Empty);
1074             return Convert.ToInt32(command.ExecuteScalar());
1075         }
1076     }
1077 }

```

Data/Repositories.cs (continuacion)

```
1078     public List<ReporteGenerado> GetAll()
1079     {
1080         List<ReporteGenerado> items = new List<ReporteGenerado>();
1081         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
1082         using (SQLiteCommand command = new SQLiteCommand("SELECT * FROM ReportesGenerados ORDER BY FechaGeneracion DESC;", connection))
1083         using (SQLiteDataReader reader = command.ExecuteReader())
1084         {
1085             while (reader.Read())
1086             {
1087                 items.Add(new ReporteGenerado
1088                 {
1089                     IdReporte = Convert.ToInt32(reader["IdReporte"]),
1090                     NombreReporte = Convert.ToString(reader["NombreReporte"]),
1091                     Tipo = Convert.ToString(reader["Tipo"]),
1092                     GeneradoPor = Convert.ToString(reader["GeneradoPor"]),
1093                     FechaGeneracion = DateTime.Parse(Convert.ToString(reader["FechaGeneracion"])),
1094                     RutaArchivo = Convert.ToString(reader["RutaArchivo"])
1095                 });
1096             }
1097         }
1098         return items;
1099     }
1100 }
1101 }
1102 }
```

Data/SQLiteConnectionFactory.cs

```
1 using System;
2 using System.Data.SQLite;
3 using System.IO;
4
5 namespace SistemaGestionNomina.Data
6 {
7     public static class SQLiteConnectionFactory
8     {
9         public static string DatabasePath
10         {
11             get
12             {
13                 string overridePath = Environment.GetEnvironmentVariable("NOMINA_DB_PATH");
14                 return string.IsNullOrEmpty(overridePath)
15                     ? Path.Combine(GetApplicationFolder(), "nomina.db")
16                     : Path.GetFullPath(overridePath);
17             }
18         }
19
20         public static SQLiteConnection CreateConnection()
21         {
22             SQLiteConnection connection = new SQLiteConnection("Data Source=" + DatabasePath + ";Version=3;Foreign Keys=True;");
23             connection.Open();
24             return connection;
25         }
26
27         private static string GetApplicationFolder()
28         {
29             string location = typeof(SQLiteConnectionFactory).Assembly.Location;
30             if (!string.IsNullOrEmpty(location))
31             {
32                 return Path.GetDirectoryName(location);
33             }
34
35             return AppDomain.CurrentDomain.BaseDirectory;
36         }
37     }
38 }
```

Helpers/ComprobantePrintRender.cs

```
1 using System.Drawing;
2 using SistemaGestionNomina.Models;
3
4 namespace SistemaGestionNomina.Helpers
5 {
6     public static class ComprobantePrintRender
7     {
8         public static void Draw(Graphics graphics, Rectangle bounds, Comprobante comprobante)
9         {
10             // El mismo diseno se reutiliza en vista previa, impresion fisica y comprobante PDF.
11             int y = bounds.Top;
12             using (Font titleFont = new Font("Segoe UI", 18F, FontStyle.Bold))
13             using (Font headerFont = new Font("Segoe UI", 11F, FontStyle.Bold))
14             using (Font bodyFont = new Font("Segoe UI", 10F))
15             using (Font totalFont = new Font("Segoe UI", 13F, FontStyle.Bold))
16             using (Pen linePen = new Pen(Color.Black, 1F))
17             {
18                 graphics.DrawString("ProyFinal_LPI_Eq01_NomiCore", titleFont, Brushes.Black, bounds.Left, y);
19                 y += 42;
20                 graphics.DrawString("Comprobante de Pago", headerFont, Brushes.Black, bounds.Left, y);
21                 y += 34;
22                 graphics.DrawLine(linePen, bounds.Left, y, bounds.Right, y);
23                 y += 22;
24                 DrawRow(graphics, bodyFont, "Documento:", comprobante.NumeroComprobante, bounds.Left, ref y);
25                 DrawRow(graphics, bodyFont, "Fecha emision:", comprobante.FechaGeneracion.ToString("dd/MM/yyyy HH:mm"), bounds.Left, ref y);
26                 DrawRow(graphics, bodyFont, "Periodo:", comprobante.PeriodoNombre, bounds.Left, ref y);
27                 y += 8;
28                 graphics.DrawLine(linePen, bounds.Left, y, bounds.Right, y);
29                 y += 22;
30                 DrawRow(graphics, bodyFont, "Empleado:", comprobante.EmpleadoNombre, bounds.Left, ref y);
31                 DrawRow(graphics, bodyFont, "Codigo:", comprobante.CodigoEmpleado, bounds.Left, ref y);
32                 y += 8;
33                 graphics.DrawLine(linePen, bounds.Left, y, bounds.Right, y);
34                 y += 22;
35                 DrawRow(graphics, bodyFont, "Ingresos:", "B/. " + comprobante.TotalIngresos.ToString("0.00"), bounds.Left, ref y);
36                 DrawRow(graphics, bodyFont, "Deducciones:", "B/. " + comprobante.TotalDeducciones.ToString("0.00"), bounds.Left, ref y);
37                 y += 12;
38                 graphics.DrawString("NETO A PAGAR: B/. " + comprobante.Netopagar.ToString("0.00"), totalFont, Brushes.Black, bounds.Left, y);
39                 y += 54;
40                 graphics.DrawLine(linePen, bounds.Left, y, bounds.Left + 300, y);
41                 graphics.DrawString("Firma autorizada", bodyFont, Brushes.Black, bounds.Left, y + 8);
42             }
43         }
44
45         private static void DrawRow(Graphics graphics, Font font, string label, string value, int x, ref int y)
46         {
47             graphics.DrawString(label, font, Brushes.Black, x, y);
48             graphics.DrawString(value ?? string.Empty, font, Brushes.Black, x + 140, y);
49             y += 26;
50         }
51     }
52 }
```

Helpers/ControlStyleHelper.cs

```
1 using System;
2 using System.Drawing;
3 using System.Drawing.Drawing2D;
4 using System.Windows.Forms;
5 using FontAwesome.Sharp;
6 using MaterialSkin;
7
8 namespace SistemaGestionNomina.Helpers
9 {
10     public static class ControlStyleHelper
11     {
12         private const int PanelRadius = 12;
13         private const int ButtonRadius = 8;
14         private static bool materialThemeApplied;
15
16         public static void ApplyModernForm(Form form)
17         {
18             TryApplyMaterialTheme();
19             form.BackColor = ThemeHelper.Background;
20             form.ForeColor = ThemeHelper.Text;
21             form.Font = ThemeHelper.BodyFont(9F, FontStyle.Regular);
22             StyleContainer(form);
23         }
24
25         public static void StyleContainer(Control parent)
26         {
27             foreach (Control control in parent.Controls)
28             {
29                 if (control is DataGridView)
30                 {
31                     StyleGrid((DataGridView)control);
32                 }
33                 else if (control is IconButton)
34                 {
35                     StyleIconButton((IconButton)control, false);
36                 }
37                 else if (control is Button)
38                 {
39                     StyleButton((Button)control, IsPrimaryButton((Button)control));
40                 }
41                 else if (control is TextBox)
42                 {
43                     StyleTextBox((TextBox)control);
44                 }
45                 else if (control is ComboBox)
46                 {
47                     StyleComboBox((ComboBox)control);
48                 }
49                 else if (control is NumericUpDown)
50                 {
51                     StyleNumeric((NumericUpDown)control);
52                 }
53                 else if (control is DateTimePicker)
54                 {
55                     StyleDatePicker((DateTimePicker)control);
56                 }
57                 else if (control is CheckBox)
58                 {
59                     StyleCheckBox((CheckBox)control);
60                 }
61             }
62         }
63     }
64 }
```


Helpers/ControlStyleHelper.cs (continuacion)

```
61         else if (control is Panel)
62         {
63             StylePanel((Panel)control);
64         }
65         else if (control is Label)
66         {
67             StyleLabel((Label)control);
68         }
69
70         if (control.HasChildren)
71         {
72             StyleContainer(control);
73         }
74     }
75 }
76
77 public static void StyleGrid(DataGridView grid)
78 {
79     grid.BackgroundColor = ThemeHelper.CardDeep;
80     grid.BorderStyle = BorderStyle.None;
81     grid.EnableHeadersVisualStyles = false;
82     grid.RowHeadersVisible = false;
83     grid.AllowUserToAddRows = false;
84     grid.AllowUserToDeleteRows = false;
85     grid.ReadOnly = true;
86     grid.SelectionMode = DataGridViewSelectionMode.FullRowSelect;
87     grid.MultiSelect = false;
88     grid.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
89     grid.GridColor = ThemeHelper.BorderSoft;
90     grid.CellBorderStyle = DataGridViewCellBorderStyle.SingleHorizontal;
91     grid.ColumnHeadersBorderStyle = DataGridViewHeaderBorderStyle.None;
92     grid.ColumnHeadersDefaultCellStyle.BackColor = ThemeHelper.CardAlt;
93     grid.ColumnHeadersDefaultCellStyle.ForeColor = ThemeHelper.Text;
94     grid.ColumnHeadersDefaultCellStyle.SelectionBackColor = ThemeHelper.Hover;
95     grid.ColumnHeadersDefaultCellStyle.Font = ThemeHelper.BodyFont(9F, FontStyle.Bold);
96     grid.ColumnHeadersHeight = 40;
97     grid.DefaultCellStyle.BackColor = ThemeHelper.CardDeep;
98     grid.DefaultCellStyle.ForeColor = ThemeHelper.Text;
99     grid.DefaultCellStyle.SelectionBackColor = ThemeHelper.Hover;
100    grid.DefaultCellStyle.SelectionForeColor = Color.White;
101    grid.DefaultCellStyle.Font = ThemeHelper.BodyFont(9.25F, FontStyle.Regular);
102    grid.AlternatingRowsDefaultCellStyle.BackColor = Color.FromArgb(20, 23, 34);
103    grid.AlternatingRowsDefaultCellStyle.ForeColor = ThemeHelper.Text;
104    grid.RowTemplate.Height = 36;
105 }
106
107 public static void StyleSidebarButton(IconButton button, bool active)
108 {
109     StyleIconButton(button, active);
110     button.TextAlign = ContentAlignment.MiddleLeft;
111     button.ImageAlign = ContentAlignment.MiddleLeft;
112     button.TextImageRelation = TextImageRelation.ImageBeforeText;
113     button.Padding = new Padding(16, 0, 0, 0);
114     button.IconSize = 20;
115 }
116
117 public static void SetActiveSidebarButton(Control sidebar, IconButton activeButton)
118 {
119     StyleSidebarButtons(sidebar, activeButton);
120 }
```

Helpers/ControlStyleHelper.cs (continuacion)

```
121
122     private static void StyleSidebarButtons(Control parent, IconButton activeButton)
123     {
124         foreach (Control control in parent.Controls)
125         {
126             IconButton button = control as IconButton;
127             if (button != null)
128             {
129                 StyleSidebarButton(button, button == activeButton);
130             }
131
132             if (control.HasChildren)
133             {
134                 StyleSidebarButtons(control, activeButton);
135             }
136         }
137     }
138
139     private static void StyleButton(Button button, bool primary)
140     {
141         button.Cursor = Cursors.Hand;
142         button.FlatStyle = FlatStyle.Flat;
143         button.FlatAppearance.BorderSize = 1;
144         button.FlatAppearance.BorderColor = primary ? ThemeHelper.Accent : ThemeHelper.Border;
145         button.BackColor = primary ? ThemeHelper.AccentAlt : ThemeHelper.CardDeep;
146         button.ForeColor = ThemeHelper.Text;
147         button.Font = ThemeHelper.BodyFont(9F, FontStyle.Bold);
148         button.UseVisualStyleBackColor = false;
149         button.MouseEnter -= Button_MouseEnter;
150         button.MouseLeave -= Button_MouseLeave;
151         button.MouseEnter += Button_MouseEnter;
152         button.MouseLeave += Button_MouseLeave;
153         button.Resize -= Button_Resize;
154         button.Resize += Button_Resize;
155         button.Tag = primary ? "primary" : "secondary";
156         ApplyRoundedRegion(button, ButtonRadius);
157     }
158
159     private static void StyleIconButton(IconButton button, bool active)
160     {
161         button.Cursor = Cursors.Hand;
162         button.FlatStyle = FlatStyle.Flat;
163         button.FlatAppearance.BorderSize = 0;
164         button.BackColor = active ? ThemeHelper.AccentSoft : ThemeHelper.Sidebar;
165         button.ForeColor = active ? ThemeHelper.Text : ThemeHelper.TextMuted;
166         button.IconColor = active ? ThemeHelper.Accent : ThemeHelper.TextMuted;
167         button.Font = ThemeHelper.BodyFont(10F, FontStyle.Bold);
168         button.UseVisualStyleBackColor = false;
169         button.MouseEnter -= IconButton_MouseEnter;
170         button.MouseLeave -= IconButton_MouseLeave;
171         button.MouseEnter += IconButton_MouseEnter;
172         button.MouseLeave += IconButton_MouseLeave;
173         button.Tag = active ? "active" : "inactive";
174         ApplyRoundedRegion(button, ButtonRadius);
175     }
176
177     private static void StyleTextBox(TextBox textBox)
178     {
179         textBox.BackColor = ThemeHelper.Input;
180         textBox.ForeColor = ThemeHelper.Text;
```

Helpers/ControlStyleHelper.cs (continuacion)

```
181         textBox.BorderStyle = BorderStyle.FixedSingle;
182         textBox.Font = ThemeHelper.BodyFont(9.5F, FontStyle.Regular);
183     }
184
185     private static void StyleComboBox(ComboBox comboBox)
186     {
187         comboBox.BackColor = ThemeHelper.Input;
188         comboBox.ForeColor = ThemeHelper.Text;
189         comboBox.FlatStyle = FlatStyle.Flat;
190         comboBox.Font = ThemeHelper.BodyFont(9.5F, FontStyle.Regular);
191     }
192
193     private static void StyleNumeric(NumericUpDown numeric)
194     {
195         numeric.BackColor = ThemeHelper.Input;
196         numeric.ForeColor = ThemeHelper.Text;
197         numeric.BorderStyle = BorderStyle.FixedSingle;
198         numeric.Font = ThemeHelper.BodyFont(9.5F, FontStyle.Regular);
199     }
200
201     private static void StyleDatePicker(DateTimePicker picker)
202     {
203         picker.CalendarForeColor = ThemeHelper.Text;
204         picker.CalendarMonthBackground = ThemeHelper.Input;
205         picker.CalendarTitleBackColor = ThemeHelper.CardAlt;
206         picker.CalendarTitleForeColor = ThemeHelper.Text;
207         picker.CalendarTrailingForeColor = ThemeHelper.TextMuted;
208         picker.Font = ThemeHelper.BodyFont(9.5F, FontStyle.Regular);
209     }
210
211     private static void StyleCheckBox(CheckBox checkBox)
212     {
213         checkBox.BackColor = Color.Transparent;
214         checkBox.ForeColor = ThemeHelper.TextMuted;
215         checkBox.Font = ThemeHelper.BodyFont(9F, FontStyle.Regular);
216         checkBox.Cursor = Cursors.Hand;
217         checkBox.UseVisualStyleBackColor = false;
218     }
219
220     private static void StylePanel(Panel panel)
221     {
222         if (IsLayoutPanel(panel))
223         {
224             panel.ForeColor = ThemeHelper.Text;
225             return;
226         }
227
228         if (panel.BackColor == Color.Black || panel.BackColor == ThemeHelper.Card || panel.BackColor == ThemeHelper.CardDeep)
229         {
230             panel.BackColor = ThemeHelper.Card;
231         }
232         else if (panel.BackColor == ThemeHelper.CardAlt)
233         {
234             panel.BackColor = ThemeHelper.CardAlt;
235         }
236
237         panel.ForeColor = ThemeHelper.Text;
238         panel.Paint -= Panel_Paint;
239         panel.Resize -= Panel_Resize;
240         panel.Paint += Panel_Paint;
```

Helpers/ControlStyleHelper.cs (continuacion)

```
241         panel.Resize += Panel_Resize;
242         ApplyRoundedRegion(panel, PanelRadius);
243     }
244
245     private static void StyleLabel(Label label)
246     {
247         string name = label.Name.ToLowerInvariant();
248         if (label.ForeColor == SystemColors.ControlText)
249         {
250             label.ForeColor = ThemeHelper.Text;
251         }
252
253         if (name.Contains("ingreso"))
254         {
255             label.ForeColor = ThemeHelper.Success;
256         }
257         else if (name.Contains("deduccion") || name.Contains("deducciones") || name.Contains("alerta"))
258         {
259             label.ForeColor = ThemeHelper.Error;
260         }
261         else if (name.Contains("totalneto") || name.Contains("neto"))
262         {
263             label.ForeColor = ThemeHelper.Accent;
264         }
265
266         label.BackColor = Color.Transparent;
267     }
268
269     private static bool IsPrimaryButton(Button button)
270     {
271         string name = button.Name.ToLowerInvariant();
272         return name.Contains("guardar") ||
273             name.Contains("nuevo") ||
274             name.Contains("editar") ||
275             name.Contains("calcular") ||
276             name.Contains("confirmar") ||
277             name.Contains("ingresar") ||
278             name.Contains("registrar") ||
279             name.Contains("procesar") ||
280             name.Contains("imprimir") ||
281             name.Contains("email");
282     }
283
284     private static void Button_MouseEnter(object sender, EventArgs e)
285     {
286         Button button = (Button)sender;
287         button.BackColor = ThemeHelper.Hover;
288         button.ForeColor = Color.White;
289     }
290
291     private static void Button_MouseLeave(object sender, EventArgs e)
292     {
293         Button button = (Button)sender;
294         bool primary = Convert.ToString(button.Tag) == "primary";
295         button.BackColor = primary ? ThemeHelper.AccentAlt : ThemeHelper.CardDeep;
296         button.ForeColor = ThemeHelper.Text;
297     }
298
299     private static void IconButton_MouseEnter(object sender, EventArgs e)
300     {
```

Helpers/ControlStyleHelper.cs (continuacion)

```
301         IconButton button = (IconButton)sender;
302         button.BackColor = ThemeHelper.AccentSoft;
303         button.ForeColor = ThemeHelper.Text;
304         button.IconColor = ThemeHelper.Accent;
305     }
306
307     private static void IconButton_MouseLeave(object sender, EventArgs e)
308     {
309         IconButton button = (IconButton)sender;
310         bool active = Convert.ToString(button.Tag) == "active";
311         button.BackColor = active ? ThemeHelper.AccentSoft : ThemeHelper.Sidebar;
312         button.ForeColor = active ? ThemeHelper.Text : ThemeHelper.TextMuted;
313         button.IconColor = active ? ThemeHelper.Accent : ThemeHelper.TextMuted;
314     }
315
316     private static void Panel_Paint(object sender, PaintEventArgs e)
317     {
318         Panel panel = (Panel)sender;
319         if (panel.Width < 2 || panel.Height < 2)
320         {
321             return;
322         }
323
324         e.Graphics.SmoothingMode = SmoothingMode.AntiAlias;
325         using (GraphicsPath path = CreateRoundPath(new Rectangle(0, 0, panel.Width - 1, panel.Height - 1), PanelRadius))
326         using (Pen pen = new Pen(ThemeHelper.Border, 1))
327         {
328             e.Graphics.DrawPath(pen, path);
329         }
330     }
331
332     private static void Panel_Resize(object sender, EventArgs e)
333     {
334         ApplyRoundedRegion((Control)sender, PanelRadius);
335     }
336
337     private static void Button_Resize(object sender, EventArgs e)
338     {
339         ApplyRoundedRegion((Control)sender, ButtonRadius);
340     }
341
342     private static void ApplyRoundedRegion(Control control, int radius)
343     {
344         if (control.Width <= 0 || control.Height <= 0)
345         {
346             return;
347         }
348
349         using (GraphicsPath path = CreateRoundPath(new Rectangle(0, 0, control.Width, control.Height), radius))
350         {
351             control.Region = new Region(path);
352         }
353     }
354
355     private static GraphicsPath CreateRoundPath(Rectangle bounds, int radius)
356     {
357         int diameter = radius * 2;
358         Rectangle rect = new Rectangle(bounds.X, bounds.Y, bounds.Width - 1, bounds.Height - 1);
359         GraphicsPath path = new GraphicsPath();
360         path.AddArc(rect.X, rect.Y, diameter, diameter, 180, 90);
```

Helpers/ControlStyleHelper.cs (continuacion)

```
361         path.AddArc(rect.Right - diameter, rect.Y, diameter, diameter, 270, 90);
362         path.AddArc(rect.Right - diameter, rect.Bottom - diameter, diameter, diameter, 0, 90);
363         path.AddArc(rect.X, rect.Bottom - diameter, diameter, diameter, 90, 90);
364         path.CloseFigure();
365         return path;
366     }
367
368     private static bool IsLayoutPanel(Panel panel)
369     {
370         string name = panel.Name.ToLowerInvariant();
371         return name == "panelsidebar" ||
372            name == "panelcontent" ||
373            name == "paneltopbar" ||
374            name == "panelbrand" ||
375            name == "flowmenu" ||
376            name == "flowaccount";
377     }
378
379     private static void TryApplyMaterialTheme()
380     {
381         if (materialThemeApplied)
382         {
383             return;
384         }
385
386         try
387         {
388             MaterialSkinManager manager = MaterialSkinManager.Instance;
389             manager.Theme = MaterialSkinManager.Themes.DARK;
390             manager.ColorScheme = new ColorScheme(
391                 Primary.DeepPurple400,
392                 Primary.DeepPurple700,
393                 Primary.DeepPurple200,
394                 Accent.DeepPurple200,
395                 TextShade.WHITE);
396             materialThemeApplied = true;
397         }
398         catch
399         {
400             // MaterialSkin is optional here; standard WinForms styling keeps the designer stable.
401         }
402     }
403 }
404 }
```

Helpers/PasswordHelper.cs

```
1 using System;
2 using System.Security.Cryptography;
3 using System.Text;
4
5 namespace SistemaGestionNomina.Helpers
6 {
7     public static class PasswordHelper
8     {
9         private const string Prefix = "pbkdf2-sha256$v1";
10        private const int Iterations = 120000;
11        private const int SaltSize = 16;
12        private const int HashSize = 32;
13
14        public static string HashPassword(string password)
15        {
16            if (string.IsNullOrEmpty(password))
17            {
18                throw new ArgumentException("La contraseña es obligatoria.", "password");
19            }
20
21            byte[] salt = new byte[SaltSize];
22            using (RandomNumberGenerator random = RandomNumberGenerator.Create())
23            {
24                random.GetBytes(salt);
25            }
26
27            byte[] hash = Derive(password, salt, Iterations, HashSize);
28            return Prefix + "$" + Iterations + "$" + Convert.ToBase64String(salt) + "$" + Convert.ToBase64String(hash);
29        }
30
31        public static bool IsLegacyHash(string storedHash)
32        {
33            if (string.IsNullOrEmpty(storedHash) || storedHash.Length != 64)
34            {
35                return false;
36            }
37
38            for (int i = 0; i < storedHash.Length; i++)
39            {
40                if (!Uri.IsHexDigit(storedHash[i]))
41                {
42                    return false;
43                }
44            }
45
46            return true;
47        }
48
49        public static bool Verify(string password, string hash)
50        {
51            if (string.IsNullOrEmpty(password) || string.IsNullOrEmpty(hash))
52            {
53                return false;
54            }
55
56            if (IsLegacyHash(hash))
57            {
58                return FixedTimeEquals(ComputeLegacyHash(password), hash.ToLowerInvariant());
59            }
60        }
61    }
62 }
```

Helpers/PasswordHelper.cs (continuacion)

```
61         string[] parts = hash.Split('$');
62         if (parts.Length != 5 || !string.Equals(parts[0] + "$" + parts[1], Prefix, StringComparison.Ordinal))
63         {
64             return false;
65         }
66
67         int iterations;
68         byte[] salt;
69         byte[] expected;
70         if (!int.TryParse(parts[2], out iterations) || iterations < 10000 || iterations > 1000000)
71         {
72             return false;
73         }
74
75         try
76         {
77             salt = Convert.FromBase64String(parts[3]);
78             expected = Convert.FromBase64String(parts[4]);
79         }
80         catch (FormatException)
81         {
82             return false;
83         }
84
85         if (salt.Length < SaltSize || expected.Length != HashSize)
86         {
87             return false;
88         }
89
90         byte[] actual = Derive(password, salt, iterations, expected.Length);
91         return FixedTimeEquals(actual, expected);
92     }
93
94     private static byte[] Derive(string password, byte[] salt, int iterations, int length)
95     {
96         using (Rfc2898DeriveBytes derive = new Rfc2898DeriveBytes(
97             password, salt, iterations, HashAlgorithmName.SHA256))
98         {
99             return derive.GetBytes(length);
100         }
101     }
102
103     private static string ComputeLegacyHash(string password)
104     {
105         using (SHA256 sha = SHA256.Create())
106         {
107             byte[] bytes = sha.ComputeHash(Encoding.UTF8.GetBytes(password));
108             StringBuilder builder = new StringBuilder(bytes.Length * 2);
109             for (int i = 0; i < bytes.Length; i++)
110             {
111                 builder.Append(bytes[i].ToString("x2"));
112             }
113
114             return builder.ToString();
115         }
116     }
117
118     private static bool FixedTimeEquals(string left, string right)
119     {
120         return FixedTimeEquals(Encoding.UTF8.GetBytes(left ?? string.Empty), Encoding.UTF8.GetBytes(right ?? string.Empty));
```


Helpers/PasswordHelper.cs (continuacion)

```
121     }
122
123     private static bool FixedTimeEquals(byte[] left, byte[] right)
124     {
125         if (left == null || right == null || left.Length != right.Length)
126         {
127             return false;
128         }
129
130         int difference = 0;
131         for (int i = 0; i < left.Length; i++)
132         {
133             difference |= left[i] ^ right[i];
134         }
135
136         return difference == 0;
137     }
138 }
139 }
```

Helpers/PathHelper.cs

```
1 using System;
2 using System.IO;
3 using System.Windows.Forms;
4
5 namespace SistemaGestionNomina.Helpers
6 {
7     public static class PathHelper
8     {
9         public static string Timestamp()
10         {
11             return DateTime.Now.ToString("yyyy_MM_dd_HH:mm");
12         }
13
14         public static string RequestExportPath(string prefix, string extension, string filter)
15         {
16             string cleanExtension = NormalizeExtension(extension);
17             using (SaveFileDialog dialog = new SaveFileDialog())
18             {
19                 dialog.Title = "Guardar exportación";
20                 dialog.FileName = BuildExportFileName(prefix, cleanExtension);
21                 dialog.Filter = filter + "|Todos los archivos (*.*)|*.*";
22                 dialog.DefaultExt = cleanExtension.TrimStart('.');
23                 dialog.AddExtension = true;
24                 dialog.OverwritePrompt = true;
25                 dialog.InitialDirectory = GetSafeInitialDirectory();
26
27                 return dialog.ShowDialog() == DialogResult.OK ? dialog.FileName : string.Empty;
28             }
29         }
30
31         public static string BuildUserDocumentPath(string prefix, string extension)
32         {
33             string folder = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments);
34             if (string.IsNullOrEmpty(folder))
35             {
36                 folder = Path.GetTempPath();
37             }
38
39             Directory.CreateDirectory(folder);
40             return Path.Combine(folder, BuildExportFileName(prefix, NormalizeExtension(extension)));
41         }
42
43         private static string BuildExportFileName(string prefix, string extension)
44         {
45             return SanitizeFileName(prefix) + "_" + Timestamp() + extension;
46         }
47
48         private static string NormalizeExtension(string extension)
49         {
50             if (string.IsNullOrEmpty(extension))
51             {
52                 return ".dat";
53             }
54
55             return extension.StartsWith(".") ? extension : "." + extension;
56         }
57
58         private static string GetSafeInitialDirectory()
59         {
60             string documents = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments);
```

Helpers/PathHelper.cs (continuacion)

```
61         return Directory.Exists(documents) ? documents : Environment.GetFolderPath(Environment.SpecialFolder.DesktopDirectory);
62     }
63
64     private static string SanitizeFileName(string value)
65     {
66         string safe = string.IsNullOrEmpty(value) ? "Exportacion" : value.Trim();
67         foreach (char c in Path.GetInvalidFileNameChars())
68         {
69             safe = safe.Replace(c, '_');
70         }
71         return safe.Replace(' ', '_');
72     }
73 }
74
75 }
```

Helpers/RememberedUsernameHelper.cs

```
1 using System;
2 using System.IO;
3
4 namespace SistemaGestionNomina.Helpers
5 {
6     public static class RememberedUsernameHelper
7     {
8         private static string FilePath
9         {
10             get
11             {
12                 return Path.Combine(Environment.GetFolderPath(Environment.SpecialFolder.LocalApplicationData),
13                     "ProyFinal_LPI_Eq01_NomiCore", "usuario.recordado");
14             }
15         }
16
17         public static string Load()
18         {
19             try
20             {
21                 return File.Exists(FilePath) ? File.ReadAllText(FilePath).Trim() : string.Empty;
22             }
23             catch
24             {
25                 return string.Empty;
26             }
27         }
28
29         public static void Save(string username)
30         {
31             string folder = Path.GetDirectoryName(FilePath);
32             Directory.CreateDirectory(folder);
33             File.WriteAllText(FilePath, (username ?? string.Empty).Trim());
34         }
35
36         public static void Clear()
37         {
38             if (File.Exists(FilePath)) File.Delete(FilePath);
39         }
40     }
41 }
```

Helpers/ThemeHelper.cs

```
1 using System.Drawing;
2 using System.Windows.Forms;
3
4 namespace SistemaGestionNomina.Helpers
5 {
6     public static class ThemeHelper
7     {
8         public static readonly Color Background = Color.FromArgb(13, 14, 20);
9         public static readonly Color BackgroundAlt = Color.FromArgb(17, 24, 39);
10        public static readonly Color Sidebar = Color.FromArgb(11, 15, 25);
11        public static readonly Color SidebarAlt = Color.FromArgb(8, 10, 17);
12        public static readonly Color Card = Color.FromArgb(24, 25, 34);
13        public static readonly Color CardAlt = Color.FromArgb(31, 41, 55);
14        public static readonly Color CardDeep = Color.FromArgb(17, 18, 26);
15        public static readonly Color Input = Color.FromArgb(20, 23, 35);
16        public static readonly Color Border = Color.FromArgb(45, 51, 64);
17        public static readonly Color BorderSoft = Color.FromArgb(38, 43, 55);
18        public static readonly Color Text = Color.FromArgb(243, 244, 246);
19        public static readonly Color TextMuted = Color.FromArgb(161, 161, 170);
20        public static readonly Color Accent = Color.FromArgb(139, 92, 246);
21        public static readonly Color AccentAlt = Color.FromArgb(124, 58, 237);
22        public static readonly Color AccentSoft = Color.FromArgb(46, 32, 84);
23        public static readonly Color Hover = Color.FromArgb(91, 62, 168);
24        public static readonly Color Success = Color.FromArgb(34, 197, 94);
25        public static readonly Color Warning = Color.FromArgb(251, 191, 36);
26        public static readonly Color Error = Color.FromArgb(239, 68, 68);
27        public static readonly Color Info = Color.FromArgb(59, 130, 246);
28
29        public static Font TitleFont(float size)
30        {
31            return new Font("Segoe UI", size, FontStyle.Bold, GraphicsUnit.Point);
32        }
33
34        public static Font BodyFont(float size, FontStyle style)
35        {
36            return new Font("Segoe UI", size, style, GraphicsUnit.Point);
37        }
38
39        public static void ApplyForm(Form form)
40        {
41            form.BackColor = Background;
42            form.ForeColor = Text;
43            form.Font = BodyFont(10F, FontStyle.Regular);
44        }
45
46        public static Label Label(string text, int x, int y, int width, int height, float size, FontStyle style, Color color)
47        {
48            return new Label
49            {
50                Text = text,
51                Location = new Point(x, y),
52                Size = new Size(width, height),
53                Font = BodyFont(size, style),
54                ForeColor = color,
55                BackColor = Color.Transparent
56            };
57        }
58
59        public static TextBox TextBox(string name, int x, int y, int width)
60        {
61            return new TextBox
62            {
63                Name = name,
64                Location = new Point(x, y),
65                Width = width,
66                Height = 20,
67                Font = BodyFont(10F, FontStyle.Regular),
68                ForeColor = Text,
69                BackColor = Background,
70                Border = true,
71                BorderColor = Border,
72                BorderStyle = BorderStyle.FixedSingle,
73                Text = "",
74                TextColor = Text,
75                TextAlign = TextAlign.Left,
76                Multiline = false,
77                PasswordChar = '\0'
78            };
79        }
80    }
81 }
```

Helpers/ThemeHelper.cs (continuacion)

```
61         TextBox textBox = new TextBox();
62         textBox.Name = name;
63         textBox.Location = new Point(x, y);
64         textBox.Size = new Size(width, 30);
65         textBox.BorderStyle = BorderStyle.FixedSingle;
66         textBox.BackColor = CardAlt;
67         textBox.ForeColor = Text;
68         textBox.Font = BodyFont(10F, FontStyle.Regular);
69         return textBox;
70     }
71
72     public static ComboBox ComboBox(string name, int x, int y, int width)
73     {
74         ComboBox combo = new ComboBox();
75         combo.Name = name;
76         combo.Location = new Point(x, y);
77         combo.Size = new Size(width, 30);
78         combo.DropDownStyle = ComboBoxStyle.DropDownList;
79         combo.BackColor = CardAlt;
80         combo.ForeColor = Text;
81         combo.FlatStyle = FlatStyle.Flat;
82         combo.Font = BodyFont(10F, FontStyle.Regular);
83         return combo;
84     }
85
86     public static void ApplyGrid(DataGridView grid)
87     {
88         grid.BackgroundColor = Color.Black;
89         grid.BorderStyle = BorderStyle.None;
90         grid.EnableHeadersVisualStyles = false;
91         grid.AllowUserToAddRows = false;
92         grid.AllowUserToDeleteRows = false;
93         grid.ReadOnly = true;
94         grid.SelectionMode = DataGridViewSelectionMode.FullRowSelect;
95         grid.MultiSelect = false;
96         grid.RowHeadersVisible = false;
97         grid.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
98         grid.GridColor = Border;
99         grid.ColumnHeadersDefaultCellStyle.BackColor = CardAlt;
100        grid.ColumnHeadersDefaultCellStyle.ForeColor = TextMuted;
101        grid.ColumnHeadersDefaultCellStyle.Font = BodyFont(9F, FontStyle.Bold);
102        grid.ColumnHeadersHeight = 38;
103        grid.DefaultCellStyle.BackColor = Color.Black;
104        grid.DefaultCellStyle.ForeColor = Text;
105        grid.DefaultCellStyle.SelectionBackColor = Color.FromArgb(36, 31, 52);
106        grid.DefaultCellStyle.SelectionForeColor = Text;
107        grid.DefaultCellStyle.Font = BodyFont(9.5F, FontStyle.Regular);
108        grid.RowTemplate.Height = 36;
109    }
110 }
111 }
```

Helpers/ValidationHelper.cs

```
1 using System;
2 using System.Windows.Forms;
3
4 namespace SistemaGestionNomina.Helpers
5 {
6     public static class ValidationHelper
7     {
8         public static bool RequireText(TextBox textBox, string fieldName)
9         {
10             if (textBox == null || !string.IsNullOrEmpty(textBox.Text))
11             {
12                 return true;
13             }
14
15             MessageBox.Show(fieldName + " es obligatorio.", "Validación", MessageBoxButtons.OK, MessageBoxIcon.Warning);
16             if (textBox != null)
17             {
18                 textBox.Focus();
19             }
20
21             return false;
22         }
23
24         public static bool TryPositiveDecimal(TextBox textBox, string fieldName, out decimal value)
25         {
26             value = 0;
27             if (textBox != null && decimal.TryParse(textBox.Text, out value) && value > 0)
28             {
29                 return true;
30             }
31
32             MessageBox.Show(fieldName + " debe ser mayor que cero.", "Validación", MessageBoxButtons.OK, MessageBoxIcon.Warning);
33             if (textBox != null)
34             {
35                 textBox.Focus();
36             }
37
38             return false;
39         }
40
41         public static bool ValidateDateRange(DateTime start, DateTime end)
42         {
43             if (start.Date <= end.Date)
44             {
45                 return true;
46             }
47
48             MessageBox.Show("La fecha de inicio debe ser menor o igual que la fecha fin.",
49                 "Validación", MessageBoxButtons.OK, MessageBoxIcon.Warning);
50             return false;
51         }
52
53         public static bool ValidateHourRange(TimeSpan entry, TimeSpan exit)
54         {
55             if (exit > entry)
56             {
57                 return true;
58             }
59
60             MessageBox.Show("La hora de salida debe ser mayor que la hora de entrada.",
```

Helpers/ValidationHelper.cs (continuación)

```
61         "Validación", MessageBoxButtons.OK, MessageBoxIcon.Warning);
62         return false;
63     }
64 }
65 }
```


Models/AbsenceModels.cs

```
1 using System;
2 using System.Collections.Generic;
3
4 namespace SistemaGestionNomina.Models
5 {
6     public static class AbsenceTypes
7     {
8         public const string Vacation = "Vacaciones";
9         public const string PaidLeave = "Permiso remunerado";
10        public const string UnpaidLeave = "Permiso no remunerado";
11        public const string MedicalLeave = "Incapacidad médica";
12        public const string License = "Licencia";
13        public const string JustifiedAbsence = "Ausencia justificada";
14        public const string UnjustifiedAbsence = "Ausencia injustificada";
15        public const string Suspension = "Suspensión";
16
17        public static readonly string[] All =
18        {
19            Vacation, PaidLeave, UnpaidLeave, MedicalLeave, License,
20            JustifiedAbsence, UnjustifiedAbsence, Suspension
21        };
22
23        public static bool IsValid(string value)
24        {
25            for (int i = 0; i < All.Length; i++)
26            {
27                if (string.Equals(All[i], value, StringComparison.OrdinalIgnoreCase)) return true;
28            }
29            return false;
30        }
31
32        public static string ToAttendanceStatus(string value)
33        {
34            if (string.Equals(value, UnpaidLeave, StringComparison.OrdinalIgnoreCase) ||
35                string.Equals(value, UnjustifiedAbsence, StringComparison.OrdinalIgnoreCase) ||
36                string.Equals(value, Suspension, StringComparison.OrdinalIgnoreCase))
37                return "Falta";
38            return "Permiso";
39        }
40    }
41
42    public static class AbsenceStates
43    {
44        public const string Pending = "Pendiente";
45        public const string Approved = "Aprobada";
46        public const string Rejected = "Rechazada";
47        public const string Cancelled = "Cancelada";
48
49        public static readonly string[] All = { Pending, Approved, Rejected, Cancelled };
50
51        public static bool IsValid(string value)
52        {
53            for (int i = 0; i < All.Length; i++)
54            {
55                if (string.Equals(All[i], value, StringComparison.OrdinalIgnoreCase)) return true;
56            }
57            return false;
58        }
59    }
60
```

Models/AbsenceModels.cs (continuacion)

```
61 public static class AbsenceStateMachine
62 {
63     public static bool CanTransition(string currentState, string targetState)
64     {
65         if (string.Equals(currentState, AbsenceStates.Pending, StringComparison.OrdinalIgnoreCase))
66         {
67             return string.Equals(targetState, AbsenceStates.Approved, StringComparison.OrdinalIgnoreCase) ||
68                 string.Equals(targetState, AbsenceStates.Rejected, StringComparison.OrdinalIgnoreCase) ||
69                 string.Equals(targetState, AbsenceStates.Cancelled, StringComparison.OrdinalIgnoreCase);
70         }
71         return string.Equals(currentState, AbsenceStates.Approved, StringComparison.OrdinalIgnoreCase) &&
72             string.Equals(targetState, AbsenceStates.Cancelled, StringComparison.OrdinalIgnoreCase);
73     }
74
75     public static void DemandTransition(string currentState, string targetState)
76     {
77         if (!CanTransition(currentState, targetState))
78             throw new InvalidOperationException("La solicitud no puede cambiar de " + currentState + " a " + targetState + ".");
79     }
80 }
81
82 public sealed class SolicitudAusencia
83 {
84     public int IdSolicitud { get; set; }
85     public int IdEmpleado { get; set; }
86     public string CodigoEmpleado { get; set; }
87     public string EmpleadoNombre { get; set; }
88     public string DepartamentoNombre { get; set; }
89     public string Tipo { get; set; }
90     public DateTime FechaInicio { get; set; }
91     public DateTime FechaFin { get; set; }
92     public string Estado { get; set; }
93     public string Motivo { get; set; }
94     public string AprobadoPor { get; set; }
95     public DateTime? FechaAprobacion { get; set; }
96     public string ObservacionResolucion { get; set; }
97     public string UsuarioSolicitante { get; set; }
98     public DateTime FechaCreacion { get; set; }
99 }
100
101 public sealed class AbsenceQuery
102 {
103     public int? EmployeeId { get; set; }
104     public int? DepartmentId { get; set; }
105     public string Type { get; set; }
106     public string Status { get; set; }
107     public DateTime? From { get; set; }
108     public DateTime? To { get; set; }
109     public string Search { get; set; }
110 }
111
112 public sealed class OwnAbsenceRequest
113 {
114     public string Type { get; set; }
115     public DateTime StartDate { get; set; }
116     public DateTime EndDate { get; set; }
117     public string Reason { get; set; }
118 }
119 }
```

Models/AuditModels.cs

```
1 using System;
2
3 namespace SistemaGestionNomina.Models
4 {
5     public sealed class AuditRecord
6     {
7         public int IdAuditoria { get; set; }
8         public string Usuario { get; set; }
9         public string Modulo { get; set; }
10        public string Accion { get; set; }
11        public string Detalle { get; set; }
12        public DateTime Fecha { get; set; }
13    }
14
15    public sealed class AuditQuery
16    {
17        public string Usuario { get; set; }
18        public string Modulo { get; set; }
19        public string Accion { get; set; }
20        public string Detalle { get; set; }
21        public DateTime? FechaDesde { get; set; }
22        public DateTime? FechaHasta { get; set; }
23        public int Offset { get; set; }
24        public int Limit { get; set; }
25
26        public AuditQuery()
27        {
28            Limit = 100;
29        }
30    }
31 }
```

Models/AuthenticationModels.cs

```
1 using System;
2
3 namespace SistemaGestionNomina.Models
4 {
5     public enum AuthenticationStatus
6     {
7         Success,
8         InvalidCredentials,
9         TemporarilyBlocked
10    }
11
12    public sealed class AuthenticationResult
13    {
14        public AuthenticationStatus Status { get; private set; }
15        public Usuario User { get; private set; }
16        public string Message { get; private set; }
17        public DateTime? BlockedUntil { get; private set; }
18
19        public bool IsSuccess
20        {
21            get { return Status == AuthenticationStatus.Success && User != null; }
22        }
23
24        public static AuthenticationResult Successful(Usuario user)
25        {
26            return new AuthenticationResult
27            {
28                Status = AuthenticationStatus.Success,
29                User = user,
30                Message = string.Empty
31            };
32        }
33
34        public static AuthenticationResult Invalid()
35        {
36            return new AuthenticationResult
37            {
38                Status = AuthenticationStatus.InvalidCredentials,
39                Message = "Usuario o contraseña incorrectos."
40            };
41        }
42
43        public static AuthenticationResult Blocked(DateTime blockedUntil)
44        {
45            return new AuthenticationResult
46            {
47                Status = AuthenticationStatus.TemporarilyBlocked,
48                BlockedUntil = blockedUntil,
49                Message = "El acceso está bloqueado temporalmente. Intente nuevamente después de " +
50                    blockedUntil.ToString("HH:mm") + "."
51            };
52        }
53    }
54 }
```

Models/EmployeeHistoryModels.cs

```
1 using System;
2
3 namespace SistemaGestionNomina.Models
4 {
5     public static class EmployeeHistoryFields
6     {
7         public const string BaseSalary = "SalarioBase";
8         public const string Position = "Cargo";
9         public const string Department = "Departamento";
10        public const string Status = "Estado";
11    }
12
13    public sealed class HistorialEmpleado
14    {
15        public int IdHistorial { get; set; }
16        public int IdEmpleado { get; set; }
17        public string CodigoEmpleado { get; set; }
18        public string EmpleadoNombre { get; set; }
19        public string CampoModificado { get; set; }
20        public string ValorAnterior { get; set; }
21        public string ValorNuevo { get; set; }
22        public string ValorAnteriorTecnico { get; set; }
23        public string ValorNuevoTecnico { get; set; }
24        public DateTime FechaCambio { get; set; }
25        public DateTime FechaEfectiva { get; set; }
26        public string UsuarioResponsable { get; set; }
27        public string Motivo { get; set; }
28        public bool Aplicado { get; set; }
29        public DateTime? FechaAplicacion { get; set; }
30    }
31
32    public sealed class EmployeeChangeContext
33    {
34        public DateTime EffectiveDate { get; set; }
35        public string Reason { get; set; }
36    }
37
38    public sealed class EmployeeSaveResult
39    {
40        public int EmployeeId { get; set; }
41        public int HistoryRecords { get; set; }
42        public bool AppliedImmediately { get; set; }
43        public bool HasScheduledChanges { get; set; }
44    }
45
46    public sealed class EmployeeHistoryQuery
47    {
48        public int? EmployeeId { get; set; }
49        public string Field { get; set; }
50        public string User { get; set; }
51        public DateTime? From { get; set; }
52        public DateTime? To { get; set; }
53    }
54 }
```

Models/EmployeePortalModels.cs

```
1 using System;
2 using System.Collections.Generic;
3
4 namespace SistemaGestionNomina.Models
5 {
6     public sealed class EmployeeProfileViewModel
7     {
8         public int IdEmpleado { get; set; }
9         public string Codigo { get; set; }
10        public string NombreCompleto { get; set; }
11        public string Cedula { get; set; }
12        public string Cargo { get; set; }
13        public string Departamento { get; set; }
14        public decimal SalarioBase { get; set; }
15        public string Estado { get; set; }
16        public DateTime FechaIngreso { get; set; }
17    }
18
19    public sealed class EmployeeAttendanceSummaryViewModel
20    {
21        public int Puntuales { get; set; }
22        public int Tardanzas { get; set; }
23        public int Faltas { get; set; }
24        public int Permisos { get; set; }
25        public decimal HorasTrabajadas { get; set; }
26        public decimal HorasExtra { get; set; }
27    }
28
29    public sealed class EmployeePortalDashboardViewModel
30    {
31        public EmployeeProfileViewModel Profile { get; set; }
32        public Comprobante LatestPayslip { get; set; }
33        public List<Asistencia> RecentAttendance { get; set; }
34        public EmployeeAttendanceSummaryViewModel AttendanceSummary { get; set; }
35    }
36 }
```

Models/Entities.cs

```
1 using System;
2 using System.Collections.Generic;
3
4 namespace SistemaGestionNomina.Models
5 {
6     public class Usuario
7     {
8         public int IdUsuario { get; set; }
9         public string NombreUsuario { get; set; }
10        public string PasswordHash { get; set; }
11        public string Rol { get; set; }
12        public string Estado { get; set; }
13        public int? IdEmpleado { get; set; }
14        public DateTime? UltimoAcceso { get; set; }
15        public int IntentosFallidos { get; set; }
16        public bool Bloqueado { get; set; }
17        public DateTime? FechaBloqueo { get; set; }
18    }
19
20    public class Departamento
21    {
22        public int IdDepartamento { get; set; }
23        public string Nombre { get; set; }
24
25        public override string ToString()
26        {
27            return Nombre;
28        }
29    }
30
31    public class Empleado
32    {
33        private int idEmpleado;
34        private string codigo;
35        private string nombre;
36        private string apellido;
37        private string cedula;
38        private string cargo;
39        private int idDepartamento;
40        private string departamentoNombre;
41        private decimal salarioBase;
42        private string estado;
43        private DateTime fechaIngreso;
44        private DateTime? fechaEfectivaLaboral;
45
46        public int IdEmpleado
47        {
48            get { return idEmpleado; }
49            set { idEmpleado = value; }
50        }
51
52        public string Codigo
53        {
54            get { return codigo; }
55            set { codigo = EntityValidation.NormalizeRequired(value, "El codigo del empleado"); }
56        }
57
58        public string Nombre
59        {
60            get { return nombre; }
```

Models/Entities.cs (continuacion)

```
61         set { nombre = EntityValidation.NormalizeRequired(value, "El nombre del empleado"); }
62     }
63
64     public string Apellido
65     {
66         get { return apellido; }
67         set { apellido = EntityValidation.NormalizeOptional(value); }
68     }
69
70     public string Cedula
71     {
72         get { return cedula; }
73         set { cedula = EntityValidation.NormalizeRequired(value, "La cedula del empleado"); }
74     }
75
76     public string Cargo
77     {
78         get { return cargo; }
79         set { cargo = EntityValidation.NormalizeRequired(value, "El cargo del empleado"); }
80     }
81
82     public int IdDepartamento
83     {
84         get { return idDepartamento; }
85         set { idDepartamento = value; }
86     }
87
88     public string DepartamentoNombre
89     {
90         get { return departamentoNombre; }
91         set { departamentoNombre = EntityValidation.NormalizeOptional(value); }
92     }
93
94     public decimal SalarioBase
95     {
96         get { return salarioBase; }
97         set
98         {
99             EntityValidation.RequireNonNegative(value, "El salario base");
100             salarioBase = value;
101         }
102     }
103
104     public string Estado
105     {
106         get { return estado; }
107         set { estado = EntityValidation.NormalizeAllowedRequired(value, "El estado del empleado", "Activo", "Inactivo"); }
108     }
109
110     public DateTime FechaIngreso
111     {
112         get { return fechaIngreso; }
113         set
114         {
115             EntityValidation.RequireValidDate(value, "La fecha de ingreso");
116             fechaIngreso = value;
117         }
118     }
119
120     public DateTime? FechaEfectivaLaboral
```


Models/Entities.cs (continuacion)

```
121     {
122         get { return fechaEfectivaLaboral; }
123         set { fechaEfectivaLaboral = value; }
124     }
125
126     public string NombreCompleto
127     {
128         get { return (Nombre + " " + Apellido).Trim(); }
129     }
130
131     public override string ToString()
132     {
133         returnCodigo + " - " + NombreCompleto;
134     }
135 }
136
137 public class Asistencia
138 {
139     private int idAsistencia;
140     private int idEmpleado;
141     private string empleadoNombre;
142     private DateTime fecha;
143     private TimeSpan? horaEntrada;
144     private TimeSpan? horaSalida;
145     private decimal horasTrabajadas;
146     private string estado;
147     private int? idSolicitudAusencia;
148
149     public int IdAsistencia
150     {
151         get { return idAsistencia; }
152         set { idAsistencia = value; }
153     }
154
155     public int IdEmpleado
156     {
157         get { return idEmpleado; }
158         set
159         {
160             EntityValidation.RequirePositiveId(value, "El empleado");
161             idEmpleado = value;
162         }
163     }
164
165     public string EmpleadoNombre
166     {
167         get { return empleadoNombre; }
168         set { empleadoNombre = EntityValidation.NormalizeOptional(value); }
169     }
170
171     public DateTime Fecha
172     {
173         get { return fecha; }
174         set
175         {
176             EntityValidation.RequireValidDate(value, "La fecha de asistencia");
177             fecha = value;
178         }
179     }
180 }
```

Models/Entities.cs (continuacion)

```
181     public TimeSpan? HoraEntrada
182     {
183         get { return horaEntrada; }
184         set { horaEntrada = value; }
185     }
186
187     public TimeSpan? HoraSalida
188     {
189         get { return horaSalida; }
190         set { horaSalida = value; }
191     }
192
193     public decimal HorasTrabajadas
194     {
195         get { return horasTrabajadas; }
196         set
197         {
198             EntityValidation.RequireNonNegative(value, "Las horas trabajadas");
199             horasTrabajadas = value;
200         }
201     }
202
203     public string Estado
204     {
205         get { return estado; }
206         set { estado = EntityValidation.NormalizeAllowedRequired(value, "El estado de asistencia", "Puntual", "Tardanza", "Falta", "Permiso"); }
207     }
208
209     public int? IdSolicitudAusencia
210     {
211         get { return idSolicitudAusencia; }
212         set { idSolicitudAusencia = value; }
213     }
214 }
215
216 public class PeriodoNomina
217 {
218     public int IdPeriodo { get; set; }
219     public string Nombre { get; set; }
220     public DateTime FechaInicio { get; set; }
221     public DateTime FechaFin { get; set; }
222     public string Estado { get; set; }
223     public bool Cerrado { get; set; }
224     public DateTime? FechaCierre { get; set; }
225     public string CerradoPor { get; set; }
226 }
227
228 public class Nomina
229 {
230     private int idNomina;
231     private int idPeriodo;
232     private string periodoNombre;
233     private DateTime fechaCalculo;
234     private decimal totalIngresos;
235     private decimal totalDeducciones;
236     private decimal totalNeto;
237     private string estado;
238     private DateTime? fechaConfirmacion;
239     private string confirmadaPor;
240     private DateTime? fechaPago;
```

Models/Entities.cs (continuacion)

```
241     private string pagadaPor;
242     private DateTime? fechaAnulacion;
243     private string anuladaPor;
244     private string motivoAnulacion;
245     private List<NominaDetalle> detalles;
246
247     public Nomina()
248     {
249         Detalles = new List<NominaDetalle>();
250     }
251
252     public int IdNomina
253     {
254         get { return idNomina; }
255         set { idNomina = value; }
256     }
257
258     public int IdPeriodo
259     {
260         get { return idPeriodo; }
261         set { idPeriodo = value; }
262     }
263
264     public string PeriodoNombre
265     {
266         get { return periodoNombre; }
267         set { periodoNombre = EntityValidation.NormalizeOptional(value); }
268     }
269
270     public DateTime FechaCalculo
271     {
272         get { return fechaCalculo; }
273         set
274         {
275             EntityValidation.RequireValidDate(value, "La fecha de calculo");
276             fechaCalculo = value;
277         }
278     }
279
280     public decimal TotalIngresos
281     {
282         get { return totalIngresos; }
283         set
284         {
285             EntityValidation.RequireNonNegative(value, "El total de ingresos");
286             totalIngresos = value;
287         }
288     }
289
290     public decimal TotalDeducciones
291     {
292         get { return totalDeducciones; }
293         set
294         {
295             EntityValidation.RequireNonNegative(value, "El total de deducciones");
296             totalDeducciones = value;
297         }
298     }
299
300     public decimal TotalNeto
```

Models/Entities.cs (continuacion)

```
301     {
302         get { return totalNeto; }
303         set
304         {
305             EntityValidation.RequireNonNegative(value, "El total neto");
306             totalNeto = value;
307         }
308     }
309     public string Estado
310     {
311         get { return estado; }
312         set { estado = EntityValidation.NormalizeOptional(value); }
313     }
314
315     public DateTime? FechaConfirmacion
316     {
317         get { return fechaConfirmacion; }
318         set { fechaConfirmacion = value; }
319     }
320
321     public string ConfirmadaPor
322     {
323         get { return confirmadaPor; }
324         set { confirmadaPor = EntityValidation.NormalizeOptional(value); }
325     }
326
327     public DateTime? FechaPago
328     {
329         get { return fechaPago; }
330         set { fechaPago = value; }
331     }
332
333     public string PagadaPor
334     {
335         get { return pagadaPor; }
336         set { pagadaPor = EntityValidation.NormalizeOptional(value); }
337     }
338
339     public DateTime? FechaAnulacion
340     {
341         get { return fechaAnulacion; }
342         set { fechaAnulacion = value; }
343     }
344
345     public string AnuladaPor
346     {
347         get { return anuladaPor; }
348         set { anuladaPor = EntityValidation.NormalizeOptional(value); }
349     }
350
351     public string MotivoAnulacion
352     {
353         get { return motivoAnulacion; }
354         set { motivoAnulacion = EntityValidation.NormalizeOptional(value); }
355     }
356
357     public List<NominaDetalle> Detalles
358     {
359         get { return detalles; }
360     }
```

Models/Entities.cs (continuacion)

```
361         set { detalles = value ?? new List<NominaDetalle>(); }
362     }
363 }
364
365 public class NominaDetalle
366 {
367     private int idDetalle;
368     private int idNomina;
369     private int idEmpleado;
370     private string codigoEmpleado;
371     private string empleadoNombre;
372     private string departamento;
373     private decimal sueldoBase;
374     private decimal bonos;
375     private decimal horasExtra;
376     private decimal montoHorasExtra;
377     private decimal totalIngresos;
378     private decimal totalDeducciones;
379     private decimal netoPagar;
380     private string cargoEmpleado;
381
382     public int IdDetalle
383     {
384         get { return idDetalle; }
385         set { idDetalle = value; }
386     }
387
388     public int IdNomina
389     {
390         get { return idNomina; }
391         set { idNomina = value; }
392     }
393
394     public int IdEmpleado
395     {
396         get { return idEmpleado; }
397         set { idEmpleado = value; }
398     }
399
400     public string CodigoEmpleado
401     {
402         get { return codigoEmpleado; }
403         set { codigoEmpleado = EntityValidation.NormalizeOptional(value); }
404     }
405
406     public string EmpleadoNombre
407     {
408         get { return empleadoNombre; }
409         set { empleadoNombre = EntityValidation.NormalizeOptional(value); }
410     }
411
412     public string Departamento
413     {
414         get { return departamento; }
415         set { departamento = EntityValidation.NormalizeOptional(value); }
416     }
417
418     public string CargoEmpleado
419     {
420         get { return cargoEmpleado; }
```

Models/Entities.cs (continuacion)

```
421         set { cargoEmpleado = EntityValidation.NormalizeOptional(value); }
422     }
423
424     public decimal SueldoBase
425     {
426         get { return sueldoBase; }
427         set
428         {
429             EntityValidation.RequireNonNegative(value, "El sueldo base");
430             sueldoBase = value;
431         }
432     }
433
434     public decimal Bonos
435     {
436         get { return bonos; }
437         set
438         {
439             EntityValidation.RequireNonNegative(value, "Los bonos");
440             bonos = value;
441         }
442     }
443
444     public decimal HorasExtra
445     {
446         get { return horasExtra; }
447         set
448         {
449             EntityValidation.RequireNonNegative(value, "Las horas extra");
450             horasExtra = value;
451         }
452     }
453
454     public decimal MontoHorasExtra
455     {
456         get { return montoHorasExtra; }
457         set
458         {
459             EntityValidation.RequireNonNegative(value, "El monto de horas extra");
460             montoHorasExtra = value;
461         }
462     }
463
464     public decimal TotalIngresos
465     {
466         get { return totalIngresos; }
467         set
468         {
469             EntityValidation.RequireNonNegative(value, "El total de ingresos");
470             totalIngresos = value;
471         }
472     }
473
474     public decimal TotalDeducciones
475     {
476         get { return totalDeducciones; }
477         set
478         {
479             EntityValidation.RequireNonNegative(value, "El total de deducciones");
480             totalDeducciones = value;
481         }
482     }
483 }
```

Models/Entities.cs (continuacion)

```
481     }
482 }
483
484     public decimal NetoPagar
485     {
486         get { return netoPagar; }
487         set
488         {
489             EntityValidation.RequireNonNegative(value, "El neto a pagar");
490             netoPagar = value;
491         }
492     }
493 }
494
495 public class Comprobante
496 {
497     private int idComprobante;
498     private int idNomina;
499     private int idEmpleado;
500     private string numeroComprobante;
501     private DateTime fechaGeneracion;
502     private string rutaPdf;
503     private string empleadoNombre;
504     private string codigoEmpleado;
505     private string periodoNombre;
506     private decimal totalIngresos;
507     private decimal totalDeducciones;
508     private decimal netoPagar;
509
510     public int IdComprobante
511     {
512         get { return idComprobante; }
513         set { idComprobante = value; }
514     }
515
516     public int IdNomina
517     {
518         get { return idNomina; }
519         set { idNomina = value; }
520     }
521
522     public int IdEmpleado
523     {
524         get { return idEmpleado; }
525         set { idEmpleado = value; }
526     }
527
528     public string NumeroComprobante
529     {
530         get { return numeroComprobante; }
531         set { numeroComprobante = EntityValidation.NormalizeRequired(value, "El numero de comprobante"); }
532     }
533
534     public DateTime FechaGeneracion
535     {
536         get { return fechaGeneracion; }
537         set
538         {
539             EntityValidation.RequireValidDate(value, "La fecha de generacion");
540             fechaGeneracion = value;
541         }
542     }
543 }
```

Models/Entities.cs (continuacion)

```
541     }
542 }
543
544 public string RutaPdf
545 {
546     get { return rutaPdf; }
547     set { rutaPdf = EntityValidation.NormalizeOptional(value); }
548 }
549
550 public string EmpleadoNombre
551 {
552     get { return empleadoNombre; }
553     set { empleadoNombre = EntityValidation.NormalizeOptional(value); }
554 }
555
556 public string CodigoEmpleado
557 {
558     get { return codigoEmpleado; }
559     set { codigoEmpleado = EntityValidation.NormalizeOptional(value); }
560 }
561
562 public string PeriodoNombre
563 {
564     get { return periodoNombre; }
565     set { periodoNombre = EntityValidation.NormalizeOptional(value); }
566 }
567
568 public decimal TotalIngresos
569 {
570     get { return totalIngresos; }
571     set
572     {
573         EntityValidation.RequireNonNegative(value, "El total de ingresos");
574         totalIngresos = value;
575     }
576 }
577
578 public decimal TotalDeducciones
579 {
580     get { return totalDeducciones; }
581     set
582     {
583         EntityValidation.RequireNonNegative(value, "El total de deducciones");
584         totalDeducciones = value;
585     }
586 }
587
588 public decimal NetoPagar
589 {
590     get { return netoPagar; }
591     set
592     {
593         EntityValidation.RequireNonNegative(value, "El neto a pagar");
594         netoPagar = value;
595     }
596 }
597 }
598
599 public class ConfiguracionNomina
600 {
```


Models/Entities.cs (continuacion)

```
601     public int IdConfiguracion { get; set; }
602     public string NombreParametro { get; set; }
603     public decimal Valor { get; set; }
604     public string Descripcion { get; set; }
605 }
606
607 public class ReporteGenerado
608 {
609     public int IdReporte { get; set; }
610     public string NombreReporte { get; set; }
611     public string Tipo { get; set; }
612     public string GeneradoPor { get; set; }
613     public DateTime FechaGeneracion { get; set; }
614     public string RutaArchivo { get; set; }
615 }
616
617 public class NominaVersion
618 {
619     public int IdVersion { get; set; }
620     public int IdNominaOriginal { get; set; }
621     public int? IdNominaNueva { get; set; }
622     public string MotivoCambio { get; set; }
623     public string UsuarioResponsable { get; set; }
624     public DateTime FechaCambio { get; set; }
625 }
626
627 internal static class EntityValidation
628 {
629     public static string NormalizeRequired(string value, string fieldName)
630     {
631         if (string.IsNullOrEmpty(value))
632         {
633             throw new ArgumentException(fieldName + " es obligatorio.");
634         }
635
636         return value.Trim();
637     }
638
639     public static string NormalizeOptional(string value)
640     {
641         return value == null ? null : value.Trim();
642     }
643
644     public static string NormalizeAllowedRequired(string value, string fieldName, params string[] allowedValues)
645     {
646         string normalized = NormalizeRequired(value, fieldName);
647         for (int i = 0; i < allowedValues.Length; i++)
648         {
649             if (string.Equals(normalized, allowedValues[i], StringComparison.OrdinalIgnoreCase))
650             {
651                 return allowedValues[i];
652             }
653         }
654
655         throw new ArgumentException(fieldName + " debe ser: " + string.Join(", ", allowedValues) + ".");
656     }
657
658     public static void RequireNonNegative(decimal value, string fieldName)
659     {
660         if (value < 0)
```

Models/Entities.cs (continuacion)

```
661         {
662             throw new ArgumentException(fieldName + " no puede ser negativo.");
663         }
664     }
665
666     public static void RequirePositiveId(int value, string fieldName)
667     {
668         if (value <= 0)
669         {
670             throw new ArgumentException(fieldName + " debe ser mayor que cero.");
671         }
672     }
673
674     public static void RequireValidDate(DateTime value, string fieldName)
675     {
676         if (value == DateTime.MinValue)
677         {
678             throw new ArgumentException(fieldName + " debe ser una fecha valida.");
679         }
680     }
681 }
682 }
```

Models/PayrollDomain.cs

```
1 using System;
2 using System.Collections.Generic;
3
4 namespace SistemaGestionNomina.Models
5 {
6     public static class PayrollStates
7     {
8         public const string Draft = "Borrador";
9         public const string Calculated = "Calculada";
10        public const string Confirmed = "Confirmada";
11        public const string Paid = "Pagada";
12        public const string Annulled = "Anulada";
13
14        public static bool IsValid(string state)
15        {
16            return string.Equals(state, Draft, StringComparison.OrdinalIgnoreCase) ||
17                string.Equals(state, Calculated, StringComparison.OrdinalIgnoreCase) ||
18                string.Equals(state, Confirmed, StringComparison.OrdinalIgnoreCase) ||
19                string.Equals(state, Paid, StringComparison.OrdinalIgnoreCase) ||
20                string.Equals(state, Annulled, StringComparison.OrdinalIgnoreCase);
21        }
22    }
23
24    public static class PayrollPeriodStates
25    {
26        public const string Open = "Abierto";
27        public const string Confirmed = "Confirmado";
28        public const string Paid = "Pagado";
29        public const string Reopened = "Reabierto";
30    }
31
32    public static class PayrollStateMachine
33    {
34        private static readonly IDictionary<string, string[]> AllowedTransitions =
35            new Dictionary<string, string[]>(StringComparer.OrdinalIgnoreCase)
36            {
37                { PayrollStates.Draft, new[] { PayrollStates.Calculated } },
38                { PayrollStates.Calculated, new[] { PayrollStates.Confirmed } },
39                { PayrollStates.Confirmed, new[] { PayrollStates.Paid, PayrollStates.Annulled } },
40                { PayrollStates.Paid, new[] { PayrollStates.Annulled } },
41                { PayrollStates.Annulled, new string[0] }
42            };
43
44        public static bool CanTransition(string currentState, string targetState)
45        {
46            if (string.IsNullOrWhiteSpace(currentState) || string.IsNullOrWhiteSpace(targetState))
47                return false;
48
49            string[] targets;
50            if (!AllowedTransitions.TryGetValue(currentState.Trim(), out targets)) return false;
51            for (int i = 0; i < targets.Length; i++)
52            {
53                if (string.Equals(targets[i], targetState.Trim(), StringComparison.OrdinalIgnoreCase)) return true;
54            }
55            return false;
56        }
57
58        public static void DemandTransition(string currentState, string targetState)
59        {
60            if (!CanTransition(currentState, targetState))
```

Models/PayrollDomain.cs (continuación)

```
61         {
62             throw new InvalidOperationException("No se permite cambiar la nómina de " +
63                 (currentState ?? "sin estado") + " a " + (targetState ?? "sin estado") + ".");
64         }
65     }
66 }
67 }
```

Program.cs

```
1 using System;
2 using System.Windows.Forms;
3 using SistemaGestionNomina.Data;
4 using SistemaGestionNomina.UI;
5 using SistemaGestionNomina.Security;
6 using SistemaGestionNomina.Services;
7
8 namespace SistemaGestionNomina
9 {
10     internal static class Program
11     {
12         [STAThread]
13         private static void Main()
14         {
15             Application.EnableVisualStyles();
16             Application.SetCompatibleTextRenderingDefault(false);
17
18             try
19             {
20                 DatabaseInitializer.Initialize();
21                 new EmployeeHistoryService().ApplyDueChanges(DateTime.Today);
22             }
23             catch (Exception ex)
24             {
25                 MessageBox.Show("No se pudo inicializar la base de datos.\n\n" + ex.Message,
26                     "Error de inicio", MessageBoxButtons.OK, MessageBoxIcon.Error);
27                 return;
28             }
29
30             bool continueLogin = true;
31             while (continueLogin)
32             {
33                 using (FrmLogin login = new FrmLogin())
34                 {
35                     if (login.ShowDialog() != DialogResult.OK)
36                     {
37                         break;
38                     }
39
40                     using (FrmMain main = new FrmMain(login.AuthenticatedUser))
41                     {
42                         Application.Run(main);
43                         continueLogin = main.LogoutRequested;
44                         if (SessionContext.IsAuthenticated)
45                         {
46                             new AuditTrailService().RegistrarAccion("Seguridad", "Cerrar aplicación", string.Empty);
47                             SessionContext.Clear();
48                         }
49                     }
50                 }
51             }
52         }
53     }
54 }
```

Properties/Resources.Designer.cs

```

1  //-----
2  // <auto-generated>
3  //     This code was generated by a tool.
4  //     Runtime Version:4.0.30319.42000
5  //
6  //     Changes to this file may cause incorrect behavior and will be lost if
7  //     the code is regenerated.
8  // </auto-generated>
9  //-----
10
11 namespace SistemaGestionNomina.Properties {
12     using System;
13
14
15     /// <summary>
16     ///     A strongly-typed resource class, for looking up localized strings, etc.
17     /// </summary>
18     // This class was auto-generated by the StronglyTypedResourceBuilder
19     // class via a tool like ResGen or Visual Studio.
20     // To add or remove a member, edit your .ResX file then rerun ResGen
21     // with the /str option, or rebuild your VS project.
22     [global::System.CodeDom.Compiler.GeneratedCodeAttribute("System.Resources.Tools.StronglyTypedResourceBuilder", "18.0.0.0")]
23     [global::System.Diagnostics.DebuggerNonUserCodeAttribute()]
24     [global::System.Runtime.CompilerServices.CompilerGeneratedAttribute()]
25     internal class Resources {
26
27         private static global::System.Resources.ResourceManager resourceMan;
28
29         private static global::System.Globalization.CultureInfo resourceCulture;
30
31         [global::System.Diagnostics.CodeAnalysis.SuppressMessageAttribute("Microsoft.Performance", "CA1811:AvoidUncalledPrivateCode")]
32         internal Resources() {
33         }
34
35         /// <summary>
36         ///     Returns the cached ResourceManager instance used by this class.
37         /// </summary>
38         [global::System.ComponentModel.EditorBrowsableAttribute(global::System.ComponentModel.EditorBrowsableState.Advanced)]
39         internal static global::System.Resources.ResourceManager ResourceManager {
40             get {
41                 if (object.ReferenceEquals(resourceMan, null)) {
42                     global::System.Resources.ResourceManager temp = new global::System.Resources.ResourceManager("SistemaGestionNomina.Properties.Resource \
43 s", typeof(Resources).Assembly);
44                     resourceMan = temp;
45                 }
46                 return resourceMan;
47             }
48
49             /// <summary>
50             ///     Overrides the current thread's CurrentUICulture property for all
51             ///     resource lookups using this strongly typed resource class.
52             /// </summary>
53             [global::System.ComponentModel.EditorBrowsableAttribute(global::System.ComponentModel.EditorBrowsableState.Advanced)]
54             internal static global::System.Globalization.CultureInfo Culture {
55                 get {
56                     return resourceCulture;
57                 }
58                 set {
59                     resourceCulture = value;

```

Properties/Resources.Designer.cs (continuacion)

```
60     }
61 }
62
63 /// <summary>
64 /// Looks up a localized resource of type System.Drawing.Bitmap.
65 /// </summary>
66 internal static System.Drawing.Bitmap ChatGPT_Image_28_jun_2026__16_41_58 {
67     get {
68         object obj = ResourceManager.GetObject("ChatGPT Image 28 jun 2026, 16_41_58", resourceCulture);
69         return ((System.Drawing.Bitmap)(obj));
70     }
71 }
72
73 /// <summary>
74 /// Looks up a localized resource of type System.Drawing.Bitmap.
75 /// </summary>
76 internal static System.Drawing.Bitmap logo_aplicacion {
77     get {
78         object obj = ResourceManager.GetObject("logo aplicacion", resourceCulture);
79         return ((System.Drawing.Bitmap)(obj));
80     }
81 }
82
83 /// <summary>
84 /// Looks up a localized resource of type System.Drawing.Bitmap.
85 /// </summary>
86 internal static System.Drawing.Bitmap logo_transparent {
87     get {
88         object obj = ResourceManager.GetObject("logo transparent", resourceCulture);
89         return ((System.Drawing.Bitmap)(obj));
90     }
91 }
92 }
93 }
```

Security/AuthorizationService.cs

```
1 using System;
2 using SistemaGestionNomina.Services;
3
4 namespace SistemaGestionNomina.Security
5 {
6     public sealed class AuthorizationService
7     {
8         private readonly RolePermissionService rolePermissionService = new RolePermissionService();
9         private readonly AuditTrailService auditTrailService = new AuditTrailService();
10
11         public bool HasPermission(string permission)
12         {
13             return SessionContext.IsAuthenticated &&
14                 rolePermissionService.TienePermiso(SessionContext.Role, permission);
15         }
16
17         public void DemandPermission(string permission)
18         {
19             if (HasPermission(permission)) return;
20             auditTrailService.RegistrarAccion("Seguridad", "Acceso rechazado", permission ?? string.Empty);
21             throw new UnauthorizedAccessException("No tiene permisos para realizar esta operación.");
22         }
23
24         public void DemandAny(params string[] permissions)
25         {
26             if (permissions != null)
27             {
28                 for (int i = 0; i < permissions.Length; i++)
29                 {
30                     if (HasPermission(permissions[i])) return;
31                 }
32             }
33
34             auditTrailService.RegistrarAccion("Seguridad", "Acceso rechazado",
35                 permissions == null ? string.Empty : string.Join(", ", permissions));
36             throw new UnauthorizedAccessException("No tiene permisos para realizar esta operación.");
37         }
38     }
39 }
```


Security/EmployeeScopeService.cs

```
1 using System;
2 using SistemaGestionNomina.Data;
3 using SistemaGestionNomina.Models;
4
5 namespace SistemaGestionNomina.Security
6 {
7     public sealed class EmployeeScopeService
8     {
9         private readonly EmpleadoRepository employeeRepository = new EmpleadoRepository();
10        private readonly AuditRepository auditRepository = new AuditRepository();
11
12        public int RequireCurrentEmployeeId()
13        {
14            if (!SessionContext.IsAuthenticated ||
15                !string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase))
16            {
17                throw Reject("Esta operacion requiere una sesion de trabajador.", "Rol o sesion no validos");
18            }
19
20            if (!SessionContext.EmployeeId.HasValue || SessionContext.EmployeeId.Value <= 0)
21            {
22                throw Reject("El usuario no tiene un empleado asociado.", "Cuenta sin empleado asociado");
23            }
24
25            Empleado employee = employeeRepository.GetById(SessionContext.EmployeeId.Value);
26            if (employee == null)
27            {
28                throw Reject("No se encontro el empleado asociado a la cuenta.", "Empleado asociado inexistente");
29            }
30
31            if (!string.Equals(employee.Estado, "Activo", StringComparison.OrdinalIgnoreCase))
32            {
33                throw Reject("El empleado asociado no se encuentra activo.", "Empleado asociado inactivo");
34            }
35
36            return employee.IdEmpleado;
37        }
38
39        public void DemandCurrentEmployee(int employeeId)
40        {
41            int currentEmployeeId = RequireCurrentEmployeeId();
42            if (employeeId <= 0 || employeeId != currentEmployeeId)
43            {
44                throw Reject("Solo puede consultar su propia informacion laboral.", "Intento de acceso a otro empleado");
45            }
46        }
47
48        private UnauthorizedAccessException Reject(string message, string detail)
49        {
50            auditRepository.Add(new AuditRecord
51            {
52                Usuario = SessionContext.IsAuthenticated ? SessionContext.Username : "anonimo",
53                Modulo = "Portal trabajador",
54                Accion = "Acceso rechazado",
55                Detalle = detail,
56                Fecha = DateTime.Now
57            });
58            return new UnauthorizedAccessException(message);
59        }
60    }
```

Security/EmployeeScopeService.cs (continuación)

```
61     public int? GetDepartmentScope()
62     {
63         if (!string.Equals(SessionContext.Role, Roles.Supervisor, StringComparison.OrdinalIgnoreCase))
64         {
65             return null;
66         }
67
68         if (!SessionContext.EmployeeId.HasValue)
69         {
70             throw new UnauthorizedAccessException("El supervisor no tiene un empleado asociado.");
71         }
72
73         int? departmentId = employeeRepository.GetDepartmentId(SessionContext.EmployeeId.Value);
74         if (!departmentId.HasValue)
75         {
76             throw new UnauthorizedAccessException("No se encontró el departamento asociado al supervisor.");
77         }
78
79         return departmentId;
80     }
81
82     public void DemandEmployeeInScope(int employeeId)
83     {
84         int? scopeDepartmentId = GetDepartmentScope();
85         if (!scopeDepartmentId.HasValue) return;
86
87         int? employeeDepartmentId = employeeRepository.GetDepartmentId(employeeId);
88         if (!employeeDepartmentId.HasValue || employeeDepartmentId.Value != scopeDepartmentId.Value)
89         {
90             throw new UnauthorizedAccessException("No puede consultar o modificar empleados de otro departamento.");
91         }
92     }
93 }
94 }
```

```
1 namespace SistemaGestionNomina.Security
2 {
3     public static class Permissions
4     {
5         public const string DashboardView = "dashboard.ver";
6         public const string EmployeesView = "empleados.ver";
7         public const string EmployeesCreate = "empleados.crear";
8         public const string EmployeesEdit = "empleados.editar";
9         public const string EmployeesDeactivate = "empleados.desactivar";
10        public const string EmployeesExport = "empleados.exportar";
11        public const string EmployeeHistoryView = "empleados.historial.ver";
12        public const string EmployeeHistoryExport = "empleados.historial.exportar";
13        public const string EmployeeChangesSchedule = "empleados.cambios.programar";
14        public const string AttendanceView = "asistencia.ver";
15        public const string AttendanceRegister = "asistencia.registrar";
16        public const string AttendanceImport = "asistencia.importar";
17        public const string AttendanceExport = "asistencia.exportar";
18        public const string PayrollView = "nomina.ver";
19        public const string PayrollCalculate = "nomina.calcular";
20        public const string PayrollConfirm = "nomina.confirmar";
21        public const string PayrollExport = "nomina.exportar";
22        public const string PayslipsView = "comprobantes.ver";
23        public const string PayslipsExport = "comprobantes.exportar";
24        public const string PayslipsPrint = "comprobantes.imprimir";
25        public const string PayslipsEmail = "comprobantes.email";
26        public const string ReportsView = "reportes.ver";
27        public const string ReportsPersonal = "reportes.personal";
28        public const string ReportsFinancial = "reportes.financiero";
29        public const string ReportsExport = "reportes.exportar";
30        public const string ConfigurationView = "configuracion.ver";
31        public const string ConfigurationEdit = "configuracion.editar";
32        public const string BackupsManage = "backups.gestionar";
33        public const string AuditView = "auditoria.ver";
34        public const string PortalView = "portal.ver";
35        public const string OwnProfileView = "portal.perfil.ver";
36        public const string OwnAttendanceView = "portal.asistencia.ver";
37        public const string OwnPayslipsView = "portal.comprobantes.ver";
38        public const string OwnPayslipsDownload = "portal.comprobantes.descargar";
39        public const string OwnPasswordChange = "cuenta.password.cambiar";
40
41        public const string PayrollPay = "nomina.pagar";
42        public const string PayrollAnnul = "nomina.anular";
43        public const string PayrollRecalculate = "nomina.recalcular";
44        public const string PayrollHistoryView = "nomina.historial.ver";
45
46        public const string AbsencesView = "ausencias.ver";
47        public const string AbsencesCreate = "ausencias.crear";
48        public const string AbsencesApprove = "ausencias.aprobar";
49        public const string AbsencesReject = "ausencias.rechazar";
50        public const string AbsencesCancel = "ausencias.cancelar";
51        public const string AbsencesExport = "ausencias.exportar";
52        public const string OwnAbsencesView = "portal.ausencias.ver";
53        public const string OwnAbsencesCreate = "portal.ausencias.crear";
54        public const string OwnAbsencesCancel = "portal.ausencias.cancelar";
55    }
56 }
```

Security/Roles.cs

```
1 namespace SistemaGestionNomina.Security
2 {
3     public static class Roles
4     {
5         public const string Admin = "Admin";
6         public const string RRHH = "RRHH";
7         public const string Contabilidad = "Contabilidad";
8         public const string Supervisor = "Supervisor";
9         public const string Trabajador = "Trabajador";
10
11         public static bool IsValid(string role)
12         {
13             return role == Admin || role == RRHH || role == Contabilidad ||
14                 role == Supervisor || role == Trabajador;
15         }
16     }
17 }
```

Security/SessionContext.cs

```
1 using System;
2 using SistemaGestionNomina.Models;
3
4 namespace SistemaGestionNomina.Security
5 {
6     public static class SessionContext
7     {
8         private static readonly object Sync = new object();
9         private static Usuario currentUser;
10
11         public static Usuario CurrentUser
12         {
13             get { lock (Sync) { return currentUser; } }
14         }
15
16         public static bool IsAuthenticated
17         {
18             get { return CurrentUser != null; }
19         }
20
21         public static string Username
22         {
23             get { return CurrentUser == null ? string.Empty : CurrentUser.NombreUsuario; }
24         }
25
26         public static string Role
27         {
28             get { return CurrentUser == null ? string.Empty : CurrentUser.Rol; }
29         }
30
31         public static int? EmployeeId
32         {
33             get { return CurrentUser == null ? null : CurrentUser.IdEmpleado; }
34         }
35
36         public static void Begin(Usuario user)
37         {
38             if (user == null || user.IdUsuario <= 0 || string.IsNullOrEmpty(user.NombreUsuario))
39             {
40                 throw new ArgumentException("El usuario autenticado no es válido.", "user");
41             }
42
43             lock (Sync)
44             {
45                 currentUser = user;
46             }
47         }
48
49         public static void Clear()
50         {
51             lock (Sync)
52             {
53                 currentUser = null;
54             }
55         }
56     }
57 }
```

Services/AbsenceRequestService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using SistemaGestionNomina.Data;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7
8 namespace SistemaGestionNomina.Services
9 {
10     public sealed class AbsenceRequestService
11     {
12         private readonly AbsenceRequestRepository repository = new AbsenceRequestRepository();
13         private readonly EmpleadoRepository employeeRepository = new EmpleadoRepository();
14         private readonly EmployeeEffectiveDataService effectiveEmployeeService = new EmployeeEffectiveDataService();
15         private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         private readonly PayrollPeriodPolicyService periodPolicyService = new PayrollPeriodPolicyService();
18         private readonly AuditRepository auditRepository = new AuditRepository();
19         private readonly AuditTrailService auditTrailService = new AuditTrailService();
20
21         public List<SolicitudAusencia> Search(AbsenceQuery query)
22         {
23             query = query ?? new AbsenceQuery();
24             ValidateQuery(query);
25             int? forcedEmployeeId = null;
26             int? scopeDepartmentId = null;
27             if (IsWorker())
28             {
29                 authorizationService.DemandPermission(Permissions.OwnAbsencesView);
30                 forcedEmployeeId = employeeScopeService.RequireCurrentEmployeeId();
31                 if (query.EmployeeId.HasValue) employeeScopeService.DemandCurrentEmployee(query.EmployeeId.Value);
32             }
33             else
34             {
35                 authorizationService.DemandPermission(Permissions.AbsencesView);
36                 if (query.EmployeeId.HasValue) employeeScopeService.DemandEmployeeInScope(query.EmployeeId.Value);
37                 scopeDepartmentId = employeeScopeService.GetDepartmentScope();
38             }
39
40             List<SolicitudAusencia> items = repository.Search(query, forcedEmployeeId, scopeDepartmentId);
41             auditTrailService.RegistrarAccion("Ausencias", "Consultar", "Registros=" + items.Count);
42             return items;
43         }
44
45         public SolicitudAusencia GetById(int requestId)
46         {
47             if (requestId <= 0) throw new ArgumentException("Seleccione una solicitud válida.");
48             SolicitudAusencia item = repository.GetById(requestId);
49             if (item == null) throw new InvalidOperationException("La solicitud no existe.");
50             DemandRequestScope(item, Permissions.AbsencesView, Permissions.OwnAbsencesView);
51             return item;
52         }
53
54         public int CreateForEmployee(SolicitudAusencia item)
55         {
56             authorizationService.DemandPermission(Permissions.AbsencesCreate);
57             if (item == null) throw new ArgumentNullException("item");
58             employeeScopeService.DemandEmployeeInScope(item.IdEmpleado);
59             return Create(item.IdEmpleado, item.Tipo, item.FechaInicio, item.FechaFin, item.Motivo);
60         }
61     }
62 }
```

Services/AbsenceRequestService.cs (continuacion)

```
61
62     public int CreateOwn(OwnAbsenceRequest request)
63     {
64         authorizationService.DemandPermission(Permissions.OwnAbsencesCreate);
65         if (request == null) throw new ArgumentNullException("request");
66         int employeeId = employeeScopeService.RequireCurrentEmployeeId();
67         return Create(employeeId, request.Type, request.StartDate, request.EndDate, request.Reason);
68     }
69
70     public void Approve(int requestId, string observation)
71     {
72         authorizationService.DemandPermission(Permissions.AbsencesApprove);
73         SolicitudAusencia current = repository.GetById(requestId);
74         if (current == null) throw new InvalidOperationException("La solicitud no existe.");
75         employeeScopeService.DemandEmployeeInScope(current.IdEmpleado);
76         AbsenceStateMachine.DemandTransition(current.Estado, AbsenceStates.Approved);
77         periodPolicyService.VerificarFechasAbiertas(current.FechaInicio, current.FechaFin);
78
79         List<DateTime> workdays = GetWorkdays(current.FechaInicio, current.FechaFin);
80         if (workdays.Count == 0) throw new InvalidOperationException("La solicitud no contiene días laborables de lunes a viernes.");
81         string user = CurrentUsername();
82         DateTime now = DateTime.Now;
83
84         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
85         using (SQLiteTransaction transaction = connection.BeginTransaction())
86         {
87             try
88             {
89                 current = repository.GetById(connection, transaction, requestId);
90                 if (current == null) throw new InvalidOperationException("La solicitud no existe.");
91                 AbsenceStateMachine.DemandTransition(current.Estado, AbsenceStates.Approved);
92                 if (repository.HasApprovedOverlap(connection, transaction, current.IdEmpleado,
93                     current.FechaInicio, current.FechaFin, current.IdSolicitud))
94                     throw new InvalidOperationException("Ya existe una ausencia aprobada que se solapa con el rango.");
95
96                 // Se valida todo el rango antes de escribir para no reemplazar asistencias existentes.
97                 for (int i = 0; i < workdays.Count; i++)
98                 {
99                     if (repository.AttendanceExists(connection, transaction, current.IdEmpleado, workdays[i]))
100                         throw new InvalidOperationException("Ya existe asistencia para " +
101                             workdays[i].ToString("dd/MM/yyyy") + ". No se reemplazó ningún registro.");
102                 }
103
104                 repository.UpdateState(connection, transaction, requestId, AbsenceStates.Pending,
105                     AbsenceStates.Approved, user, now, observation);
106                 string attendanceStatus = AbsenceTypes.ToAttendanceStatus(current.Tipo);
107                 // La aprobacion refleja la ausencia como asistencia y mantiene el vinculo para poder revertirla.
108                 for (int i = 0; i < workdays.Count; i++)
109                     repository.AddGeneratedAttendance(connection, transaction, requestId,
110                         current.IdEmpleado, workdays[i], attendanceStatus);
111                 AddAudit(connection, transaction, user, "Aprobar",
112                     "IdSolicitud=" + requestId + ", Días=" + workdays.Count, now);
113                 transaction.Commit();
114             }
115             catch
116             {
117                 transaction.Rollback();
118                 throw;
119             }
120         }
121     }
```

Services/AbsenceRequestService.cs (continuacion)

```
121     }
122
123     public void Reject(int requestId, string observation)
124     {
125         authorizationService.DemandPermission(Permissions.AbsencesReject);
126         if (string.IsNullOrWhiteSpace(observation))
127             throw new ArgumentException("Debe indicar la razón del rechazo.");
128         SolicitudAusencia current = repository.GetById(requestId);
129         if (current == null) throw new InvalidOperationException("La solicitud no existe.");
130         employeeScopeService.DemandEmployeeInScope(current.IdEmpleado);
131         AbsenceStateMachine.DemandTransition(current.Estado, AbsenceStates.Rejected);
132         periodPolicyService.VerificarFechasAbiertas(current.FechaInicio, current.FechaFin);
133         ChangeState(current, AbsenceStates.Rejected, observation.Trim(), "Rechazar");
134     }
135
136     public void Cancel(int requestId, string reason)
137     {
138         if (string.IsNullOrWhiteSpace(reason))
139             throw new ArgumentException("Debe indicar el motivo de cancelación.");
140         SolicitudAusencia current = repository.GetById(requestId);
141         if (current == null) throw new InvalidOperationException("La solicitud no existe.");
142
143         if (IsWorker())
144         {
145             authorizationService.DemandPermission(Permissions.OwnAbsencesCancel);
146             employeeScopeService.DemandCurrentEmployee(current.IdEmpleado);
147             if (!string.Equals(current.Estado, AbsenceStates.Pending, StringComparison.OrdinalIgnoreCase))
148                 throw new InvalidOperationException("El trabajador solo puede cancelar solicitudes propias pendientes.");
149         }
150         else
151         {
152             authorizationService.DemandPermission(Permissions.AbsencesCancel);
153             employeeScopeService.DemandEmployeeInScope(current.IdEmpleado);
154         }
155
156         AbsenceStateMachine.DemandTransition(current.Estado, AbsenceStates.Cancelled);
157         periodPolicyService.VerificarFechasAbiertas(current.FechaInicio, current.FechaFin);
158         string user = CurrentUsername();
159         DateTime now = DateTime.Now;
160         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
161         using (SQLiteTransaction transaction = connection.BeginTransaction())
162         {
163             try
164             {
165                 if (string.Equals(current.Estado, AbsenceStates.Approved, StringComparison.OrdinalIgnoreCase))
166                     // Solo se eliminan las asistencias creadas automaticamente por esta solicitud.
167                     repository.DeleteGeneratedAttendance(connection, transaction, current.IdSolicitud);
168                 repository.UpdateState(connection, transaction, current.IdSolicitud, current.Estado,
169                     AbsenceStates.Cancelled, user, now, reason.Trim());
170                 AddAudit(connection, transaction, user, "Cancelar",
171                     "IdSolicitud=" + current.IdSolicitud + ", EstadoAnterior=" + current.Estado, now);
172                 transaction.Commit();
173             }
174             catch
175             {
176                 transaction.Rollback();
177                 throw;
178             }
179         }
180     }
```


Services/AbsenceRequestService.cs (continuacion)

```
181
182     private int Create(int employeeId, string type, DateTime start, DateTime end, string reason)
183     {
184         ValidateRequest(employeeId, type, start, end, reason);
185         Empleado employee = employeeRepository.GetById(employeeId);
186         Empleado effectiveEmployee = effectiveEmployeeService.GetEmployeeAsOf(employeeId, start.Date);
187         if (employee == null || !string.Equals(employee.Estado, "Activo", StringComparison.OrdinalIgnoreCase) ||
188             effectiveEmployee == null || !string.Equals(effectiveEmployee.Estado, "Activo", StringComparison.OrdinalIgnoreCase))
189             throw new InvalidOperationException("El empleado debe estar activo al crear la solicitud y en su fecha inicial.");
190
191         periodPolicyService.VerificarFechasAbiertas(start.Date, end.Date);
192         string user = CurrentUsername();
193         DateTime now = DateTime.Now;
194         SolicitudAusencia item = new SolicitudAusencia
195         {
196             IdEmpleado = employeeId,
197             Tipo = NormalizeType(type),
198             FechaInicio = start.Date,
199             FechaFin = end.Date,
200             Estado = AbsenceStates.Pending,
201             Motivo = reason.Trim(),
202             UsuarioSolicitante = user,
203             FechaCreacion = now
204         };
205
206         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
207         using (SQLiteTransaction transaction = connection.BeginTransaction())
208         {
209             try
210             {
211                 int id = repository.Add(connection, transaction, item);
212                 AddAudit(connection, transaction, user, "Crear",
213                     "IdSolicitud=" + id + ", IdEmpleado=" + employeeId + ", Tipo=" + item.Tipo, now);
214                 transaction.Commit();
215                 return id;
216             }
217             catch
218             {
219                 transaction.Rollback();
220                 throw;
221             }
222         }
223     }
224
225     private void ChangeState(SolicitudAusencia current, string newState, string observation, string action)
226     {
227         string user = CurrentUsername();
228         DateTime now = DateTime.Now;
229         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
230         using (SQLiteTransaction transaction = connection.BeginTransaction())
231         {
232             try
233             {
234                 repository.UpdateState(connection, transaction, current.IdSolicitud, current.Estado,
235                     newState, user, now, observation);
236                 AddAudit(connection, transaction, user, action, "IdSolicitud=" + current.IdSolicitud, now);
237                 transaction.Commit();
238             }
239             catch
240             {

```

Services/AbsenceRequestService.cs (continuacion)

```
241         transaction.Rollback();
242         throw;
243     }
244 }
245 }
246
247 private void DemandRequestScope(SolicitudAusencia item, string generalPermission, string ownPermission)
248 {
249     if (IsWorker())
250     {
251         authorizationService.DemandPermission(ownPermission);
252         employeeScopeService.DemandCurrentEmployee(item.IdEmpleado);
253     }
254     else
255     {
256         authorizationService.DemandPermission(generalPermission);
257         employeeScopeService.DemandEmployeeInScope(item.IdEmpleado);
258     }
259 }
260
261 private static void ValidateQuery(AbsenceQuery query)
262 {
263     if (query.From.HasValue && query.To.HasValue && query.From.Value.Date > query.To.Value.Date)
264         throw new ArgumentException("La fecha inicial no puede superar la fecha final.");
265     if (!string.IsNullOrEmpty(query.Type) && !AbsenceTypes.IsValid(query.Type))
266         throw new ArgumentException("El tipo de ausencia no es válido.");
267     if (!string.IsNullOrEmpty(query.Status) && !AbsenceStates.IsValid(query.Status))
268         throw new ArgumentException("El estado de ausencia no es válido.");
269 }
270
271 private static void ValidateRequest(int employeeId, string type, DateTime start, DateTime end, string reason)
272 {
273     if (employeeId <= 0) throw new ArgumentException("Debe seleccionar un empleado.");
274     if (!AbsenceTypes.IsValid(type)) throw new ArgumentException("Seleccione un tipo de ausencia válido.");
275     if (start.Date > end.Date) throw new ArgumentException("La fecha inicial no puede superar la fecha final.");
276     if (GetWorkdays(start, end).Count == 0)
277         throw new ArgumentException("La solicitud debe incluir al menos un día de lunes a viernes.");
278     if (string.IsNullOrEmpty(reason)) throw new ArgumentException("Debe indicar el motivo de la solicitud.");
279 }
280
281 private static string NormalizeType(string type)
282 {
283     for (int i = 0; i < AbsenceTypes.All.Length; i++)
284     {
285         if (string.Equals(AbsenceTypes.All[i], type, StringComparison.OrdinalIgnoreCase))
286             return AbsenceTypes.All[i];
287     }
288     return type;
289 }
290
291 private static List<DateTime> GetWorkdays(DateTime start, DateTime end)
292 {
293     List<DateTime> dates = new List<DateTime>();
294     for (DateTime date = start.Date; date <= end.Date; date = date.AddDays(1))
295     {
296         if (date.DayOfWeek != DayOfWeek.Saturday && date.DayOfWeek != DayOfWeek.Sunday)
297             dates.Add(date);
298     }
299     return dates;
300 }
```

Services/AbsenceRequestService.cs (continuacion)

```
301
302     private void AddAudit(SQLiteConnection connection, SQLiteTransaction transaction, string user,
303         string action, string detail, DateTime date)
304     {
305         auditRepository.Add(connection, transaction, new AuditRecord
306         {
307             Usuario = user,
308             Modulo = "Ausencias",
309             Accion = action,
310             Detalle = detail,
311             Fecha = date
312         });
313     }
314
315     private static bool IsWorker()
316     {
317         return string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase);
318     }
319
320     private static string CurrentUsername()
321     {
322         if (!SessionContext.IsAuthenticated || string.IsNullOrEmpty(SessionContext.Username))
323             throw new UnauthorizedAccessException("Debe iniciar sesión para gestionar solicitudes.");
324         return SessionContext.Username;
325     }
326 }
327 }
```

Services/AccountService.cs

```
1 using System;
2 using SistemaGestionNomina.Data;
3 using SistemaGestionNomina.Helpers;
4 using SistemaGestionNomina.Models;
5 using SistemaGestionNomina.Security;
6
7 namespace SistemaGestionNomina.Services
8 {
9     public sealed class AccountService
10    {
11        private readonly AuthenticationRepository authenticationRepository = new AuthenticationRepository();
12        private readonly AuthorizationService authorizationService = new AuthorizationService();
13        private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
14        private readonly AuditTrailService auditTrailService = new AuditTrailService();
15
16        public void ChangeOwnPassword(string currentPassword, string newPassword, string confirmation)
17        {
18            authorizationService.DemandPermission(Permissions.OwnPasswordChange);
19            employeeScopeService.RequireCurrentEmployeeId();
20
21            Usuario sessionUser = SessionContext.CurrentUser;
22            if (sessionUser == null)
23            {
24                throw new UnauthorizedAccessException("No existe una sesion autenticada.");
25            }
26
27            Usuario storedUser = authenticationRepository.GetByUsername(SessionContext.Username);
28            if (storedUser == null || storedUser.IdUsuario != sessionUser.IdUsuario)
29            {
30                throw new UnauthorizedAccessException("No se pudo validar la cuenta de la sesion.");
31            }
32
33            if (!PasswordHelper.Verify(currentPassword, storedUser.PasswordHash))
34            {
35                AuditRejected("Contraseña actual invalida");
36                throw new ArgumentException("La contraseña actual no es correcta.");
37            }
38
39            ValidateNewPassword(newPassword, confirmation, storedUser.PasswordHash);
40            string newHash = PasswordHelper.HashPassword(newPassword);
41            if (!authenticationRepository.UpdatePassword(storedUser.IdUsuario, newHash))
42            {
43                throw new InvalidOperationException("No se pudo actualizar la contraseña.");
44            }
45
46            sessionUser.PasswordHash = newHash;
47            auditTrailService.RegistrarAccion("Cuenta", "Cambiar contraseña", "Actualizacion completada");
48        }
49
50        private void ValidateNewPassword(string newPassword, string confirmation, string currentHash)
51        {
52            if (string.IsNullOrEmpty(newPassword))
53            {
54                AuditRejected("Nueva contraseña vacia");
55                throw new ArgumentException("La nueva contraseña es obligatoria.");
56            }
57
58            if (!string.Equals(newPassword, confirmation, StringComparison.Ordinal))
59            {
60                AuditRejected("Confirmacion diferente");
```

Services/AccountService.cs (continuacion)

```
61         throw new ArgumentException("La confirmacion no coincide con la nueva contrasena.");
62     }
63
64     if (PasswordHelper.Verify(newPassword, currentHash))
65     {
66         AuditRejected("Contrasena reutilizada");
67         throw new ArgumentException("La nueva contrasena debe ser diferente de la actual.");
68     }
69
70     bool hasUpper = false;
71     bool hasLower = false;
72     bool hasDigit = false;
73     for (int i = 0; i < newPassword.Length; i++)
74     {
75         hasUpper |= char.IsUpper(newPassword[i]);
76         hasLower |= char.IsLower(newPassword[i]);
77         hasDigit |= char.IsDigit(newPassword[i]);
78     }
79
80     if (newPassword.Length < 8 || !hasUpper || !hasLower || !hasDigit)
81     {
82         AuditRejected("Politica de contrasena incumplida");
83         throw new ArgumentException("Use al menos 8 caracteres, con mayuscula, minuscula y numero.");
84     }
85 }
86
87 private void AuditRejected(string reason)
88 {
89     auditTrailService.RegistrarAccion("Cuenta", "Cambio de contrasena rechazado", reason);
90 }
91 }
92 }
```

Services/AdvancedNominaRules.cs

```
1 using System;
2 using System.Data.SQLite;
3 using SistemaGestionNomina.Data;
4
5 namespace SistemaGestionNomina.Services
6 {
7     /// <summary>
8     /// Contiene reglas avanzadas opcionales de nómina.
9     /// </summary>
10    public class AdvancedNominaRules
11    {
12        /// <summary>
13        /// Calcula ajustes especiales no incluidos en la nómina académica básica.
14        /// </summary>
15        public decimal CalcularAjusteEspecial(int idEmpleado, decimal ingresoBase)
16        {
17            DateTime today = DateTime.Today;
18            DateTime start = new DateTime(today.Year, today.Month, 1);
19            DateTime end = start.AddMonths(1).AddDays(-1);
20            return CalcularAjusteEspecial(idEmpleado, ingresoBase, start, end);
21        }
22
23        /// <summary>
24        /// Calcula ajuste por antigüedad, asistencia perfecta y descuentos disciplinarios del periodo.
25        /// </summary>
26        public decimal CalcularAjusteEspecial(int idEmpleado, decimal ingresoBase, DateTime fechaInicio, DateTime fechaFin)
27        {
28            if (idEmpleado <= 0)
29            {
30                throw new ArgumentException("Debe indicar un empleado válido.");
31            }
32
33            if (ingresoBase < 0)
34            {
35                throw new ArgumentException("El ingreso base no puede ser negativo.");
36            }
37
38            if (fechaInicio.Date > fechaFin.Date)
39            {
40                throw new ArgumentException("La fecha de inicio no puede ser mayor que la fecha fin.");
41            }
42
43            using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
44            {
45                DateTime fechaIngreso = ObtenerFechaIngreso(connection, idEmpleado);
46                int years = Math.Max(0, (DateTime.Today - fechaIngreso.Date).Days / 365);
47                decimal porcentajeAntigüedad = Math.Min(years * 0.005m, 0.03m);
48
49                int faltas;
50                int tardanzas;
51                ObtenerIndicadoresAsistencia(connection, idEmpleado, fechaInicio, fechaFin, out faltas, out tardanzas);
52
53                decimal bonoAntigüedad = ingresoBase * porcentajeAntigüedad;
54                decimal bonoAsistencia = faltas == 0 && tardanzas == 0 ? ingresoBase * 0.01m : 0m;
55                decimal descuentoFaltas = ingresoBase * Math.Min(faltas * 0.02m, 0.10m);
56                decimal descuentoTardanzas = ingresoBase * Math.Min(tardanzas * 0.005m, 0.03m);
57
58                return Math.Round(bonoAntigüedad + bonoAsistencia - descuentoFaltas - descuentoTardanzas, 2);
59            }
60        }
61    }
62 }
```

Services/AdvancedNominaRules.cs (continuacion)

```
61
62     private static DateTime ObtenerFechaIngreso(SQLiteConnection connection, int idEmpleado)
63     {
64         using (SQLiteCommand command = new SQLiteCommand(
65             "SELECT FechaIngreso FROM Empleados WHERE IdEmpleado = @id AND Estado = 'Activo';", connection))
66         {
67             command.Parameters.AddWithValue("@id", idEmpleado);
68             object result = command.ExecuteScalar();
69             if (result == null || result == DBNull.Value)
70             {
71                 throw new ArgumentException("No se encontró un empleado activo para calcular el ajuste.");
72             }
73             return DateTime.Parse(Convert.ToString(result));
74         }
75     }
76 }
77
78 private static void ObtenerIndicadoresAsistencia(SQLiteConnection connection, int idEmpleado, DateTime fechaInicio, DateTime fechaFin, out int \
faltas, out int tardanzas)
79 {
80     faltas = 0;
81     tardanzas = 0;
82
83     using (SQLiteCommand command = new SQLiteCommand(@"SELECT Estado, COUNT(1) AS Total
84 FROM Asistencias
85 WHERE IdEmpleado = @id AND Fecha BETWEEN @inicio AND @fin
86 GROUP BY Estado;", connection))
87     {
88         command.Parameters.AddWithValue("@id", idEmpleado);
89         command.Parameters.AddWithValue("@inicio", fechaInicio.ToString("yyyy-MM-dd"));
90         command.Parameters.AddWithValue("@fin", fechaFin.ToString("yyyy-MM-dd"));
91
92         using (SQLiteDataReader reader = command.ExecuteReader())
93         {
94             while (reader.Read())
95             {
96                 string estado = Convert.ToString(reader["Estado"]);
97                 int total = Convert.ToInt32(reader["Total"]);
98                 if (estado == "Falta")
99                 {
100                     faltas = total;
101                 }
102                 else if (estado == "Tardanza")
103                 {
104                     tardanzas = total;
105                 }
106             }
107         }
108     }
109 }
110 }
111 }
```

Services/AdvancedReportBuilder.cs

```
1 using System;
2 using System.Data.SQLite;
3 using System.IO;
4 using System.Text;
5 using SistemaGestionNomina.Data;
6 using SistemaGestionNomina.Helpers;
7 using SistemaGestionNomina.Security;
8
9 namespace SistemaGestionNomina.Services
10 {
11     /// <summary>
12     /// Constructor de reportes avanzados personalizados.
13     /// </summary>
14     public class AdvancedReportBuilder
15     {
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         private readonly AuditTrailService auditTrailService = new AuditTrailService();
18
19         /// <summary>
20         /// Construye un reporte personalizado según filtros dinámicos.
21         /// </summary>
22         public string ConstruirReportePersonalizado(string nombreReporte)
23         {
24             authorizationService.DemandPermission(Permissions.ReportsView);
25             if (string.IsNullOrEmpty(nombreReporte))
26             {
27                 throw new ArgumentException("Debe indicar el nombre del reporte.");
28             }
29
30             StringBuilder builder = new StringBuilder();
31             builder.AppendLine("# " + nombreReporte.Trim());
32             builder.AppendLine();
33             builder.AppendLine("Generado: " + DateTime.Now.ToString("dd/MM/yyyy HH:mm"));
34             builder.AppendLine();
35
36             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
37             {
38                 AppendMetric(builder, connection, "Empleados activos", "SELECT COUNT(1) FROM Empleados WHERE Estado = 'Activo'");
39                 AppendMetric(builder, connection, "Departamentos", "SELECT COUNT(1) FROM Departamentos");
40                 AppendMetric(builder, connection, "Nominas vigentes", "SELECT COUNT(1) FROM Nominas WHERE Estado <> 'Anulada'");
41                 AppendMetric(builder, connection, "Comprobantes generados", "SELECT COUNT(1) FROM Comprobantes");
42                 AppendMetric(builder, connection, "Total neto pagado", "SELECT COALESCE(SUM(TotalNeto), 0) FROM Nominas WHERE Estado = 'Pagada'", true);
43
44                 builder.AppendLine();
45                 builder.AppendLine("## Nómina por departamento");
46                 builder.AppendLine();
47                 using (SQLiteCommand command = new SQLiteCommand(@"SELECT d.Nombre AS Departamento,
48                     COUNT(DISTINCT e.IdEmpleado) AS Empleados,
49                     COALESCE(SUM(nd.NetoPagar), 0) AS Neto
50                     FROM Departamentos d
51                     LEFT JOIN Empleados e ON e.IdDepartamento = d.IdDepartamento AND e.Estado = 'Activo'
52                     LEFT JOIN NominaDetalle nd ON nd.IdEmpleado = e.IdEmpleado
53                     AND EXISTS (SELECT 1 FROM Nominas n
54                         WHERE n.IdNomina = nd.IdNomina AND n.Estado <> 'Anulada')
55                     GROUP BY d.Nombre
56                     ORDER BY d.Nombre;", connection))
57                 using (SQLiteDataReader reader = command.ExecuteReader())
58                 {
59                     while (reader.Read())
60                     {
```


Services/AdvancedReportBuilder.cs (continuacion)

```
61         builder.AppendLine("- " + Convert.ToString(reader["Departamento"]) +
62             ": " + Convert.ToInt32(reader["Empleados"]) + " empleados, B/. " +
63             Convert.ToDecimal(reader["Neto"]).ToString("0.00"));
64     }
65 }
66
67 builder.AppendLine();
68 builder.AppendLine("## Asistencia reciente");
69 builder.AppendLine();
70 using (SQLiteCommand command = new SQLiteCommand(@"SELECT Estado, COUNT(1) AS Total
71     FROM Asistencias
72     WHERE Fecha >= date('now', '-30 day')
73     GROUP BY Estado
74     ORDER BY Estado;", connection))
75 using (SQLiteDataReader reader = command.ExecuteReader())
76 {
77     bool hasRows = false;
78     while (reader.Read())
79     {
80         hasRows = true;
81         builder.AppendLine("- " + Convert.ToString(reader["Estado"]) + ": " + Convert.ToInt32(reader["Total"]));
82     }
83
84     if (!hasRows)
85     {
86         builder.AppendLine("- Sin registros de asistencia en los últimos 30 días.");
87     }
88 }
89 }
90
91 return builder.ToString();
92 }
93
94 /// <summary>
95 /// Genera un reporte ejecutivo en archivo Markdown para revisión o conversión a PDF.
96 /// </summary>
97 public string ExportarReportePersonalizado(string nombreReporte)
98 {
99     authorizationService.DemandPermission(Permissions.ReportsExport);
100     string content = ConstruirReportePersonalizado(nombreReporte);
101     string path = PathHelper.RequestExportPath(SanitizeFileName(nombreReporte), ".md", "Archivo Markdown (*.md)|*.md");
102     if (string.IsNullOrEmpty(path)) return string.Empty;
103
104     File.WriteAllText(path, content, Encoding.UTF8);
105     auditTrailService.RegistrarAccion("Reportes", "Exportar personalizado", Path.GetFileName(path));
106     return path;
107 }
108
109 private static void AppendMetric(StringBuilder builder, SQLiteConnection connection, string label, string sql)
110 {
111     AppendMetric(builder, connection, label, sql, false);
112 }
113
114 private static void AppendMetric(StringBuilder builder, SQLiteConnection connection, string label, string sql, bool currency)
115 {
116     using (SQLiteCommand command = new SQLiteCommand(sql, connection))
117     {
118         object value = command.ExecuteScalar();
119         if (currency)
120         {
```

Services/AdvancedReportBuilder.cs (continuacion)

```
121         builder.AppendLine("- " + label + ": B/. " + Convert.ToDecimal(value).ToString("0.00"));
122     }
123     else
124     {
125         builder.AppendLine("- " + label + ": " + Convert.ToInt32(value));
126     }
127 }
128 }
129
130 private static string SanitizeFileName(string value)
131 {
132     string safe = string.IsNullOrEmpty(value) ? "ReporteAvanzado" : value.Trim();
133     foreach (char c in Path.GetInvalidFileNameChars())
134     {
135         safe = safe.Replace(c, '_');
136     }
137
138     return safe.Replace(' ', '_');
139 }
140 }
141 }
```

Services/AttendanceDeviceImportService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using System.Globalization;
5 using System.IO;
6 using SistemaGestionNomina.Data;
7 using SistemaGestionNomina.Security;
8
9 namespace SistemaGestionNomina.Services
10 {
11     /// <summary>
12     /// Servicio destinado a importar asistencia desde reloj marcador o archivos externos.
13     /// </summary>
14     public class AttendanceDeviceImportService
15     {
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         private readonly AuditTrailService auditTrailService = new AuditTrailService();
18
19         /// <summary>
20         /// Importa registros externos de asistencia desde CSV:Codigo, Fecha, HoraEntrada, HoraSalida, Estado.
21         /// </summary>
22         public int ImportarDesdeArchivo(string rutaArchivo)
23         {
24             authorizationService.DemandPermission(Permissions.AttendanceImport);
25             if (string.IsNullOrEmpty(rutaArchivo))
26             {
27                 throw new ArgumentException("Debe indicar el archivo de asistencia.");
28             }
29
30             if (!File.Exists(rutaArchivo))
31             {
32                 throw new FileNotFoundException("No se encontró el archivo de asistencia.", rutaArchivo);
33             }
34
35             string[] lines = File.ReadAllLines(rutaArchivo);
36             if (lines.Length == 0)
37             {
38                 return 0;
39             }
40
41             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
42             using (SQLiteTransaction transaction = connection.BeginTransaction())
43             {
44                 Dictionary<string, int> empleados = CargarEmpleados(connection, transaction);
45                 int imported = 0;
46
47                 for (int i = 0; i < lines.Length; i++)
48                 {
49                     if (string.IsNullOrEmpty(lines[i]))
50                     {
51                         continue;
52                     }
53
54                     string[] values = SplitCsvLine(lines[i]);
55                     if (i == 0 && values.Length > 0 && values[0].Trim().Equals("Codigo", StringComparison.OrdinalIgnoreCase))
56                     {
57                         continue;
58                     }
59
60                     if (values.Length < 5)
```

Services/AttendanceDeviceImportService.cs (continuación)

```
61         {
62             throw new InvalidDataException("Formato inválido en línea " + (i + 1) + ". Use Código, Fecha, HoraEntrada, HoraSalida, Estado.");
63         }
64
65         string codigo = values[0].Trim();
66         if (!empleados.ContainsKey(codigo))
67         {
68             throw new InvalidDataException("Empleado no encontrado en línea " + (i + 1) + ": " + codigo);
69         }
70
71         DateTime fecha = ParseDate(values[1], i + 1);
72         TimeSpan? entrada = ParseTime(values[2], i + 1, false);
73         TimeSpan? salida = ParseTime(values[3], i + 1, false);
74         string estado = NormalizeStatus(values[4], i + 1);
75         decimal horas = CalculateHours(estado, entrada, salida, i + 1);
76
77         UpsertAsistencia(connection, transaction, empleados[codigo], fecha, entrada, salida, horas, estado);
78         imported++;
79     }
80
81     transaction.Commit();
82     auditTrailService.RegistrarAccion("Asistencia", "Importar",
83         Path.GetFileName(rutaArchivo) + ", Registros=" + imported);
84     return imported;
85 }
86
87
88 private static Dictionary<string, int> CargarEmpleados(SQLiteConnection connection, SQLiteTransaction transaction)
89 {
90     Dictionary<string, int> empleados = new Dictionary<string, int>(StringComparer.OrdinalIgnoreCase);
91     using (SQLiteCommand command = new SQLiteCommand("SELECT IdEmpleado, Código FROM Empleados WHERE Estado = 'Activo';", connection, transaction))
92     using (SQLiteDataReader reader = command.ExecuteReader())
93     {
94         while (reader.Read())
95         {
96             empleados[Convert.ToString(reader["Código"])] = Convert.ToInt32(reader["IdEmpleado"]);
97         }
98     }
99
100     return empleados;
101 }
102
103 private static string[] SplitCsvLine(string line)
104 {
105     List<string> values = new List<string>();
106     bool insideQuotes = false;
107     string current = string.Empty;
108
109     for (int i = 0; i < line.Length; i++)
110     {
111         char c = line[i];
112         if (c == '"')
113         {
114             insideQuotes = !insideQuotes;
115         }
116         else if (c == ',' && !insideQuotes)
117         {
118             values.Add(current);
119             current = string.Empty;
120         }
121     }
122 }
```

Services/AttendanceDeviceImportService.cs (continuacion)

```
121         else
122         {
123             current += c;
124         }
125     }
126
127     values.Add(current);
128     return values.ToArray();
129 }
130
131 private static DateTime ParseDate(string value, int lineNumber)
132 {
133     DateTime date;
134     string text = (value ?? string.Empty).Trim();
135     string[] formats = { "yyyy-MM-dd", "dd/MM/yyyy", "d/M/yyyy", "MM/dd/yyyy" };
136     if (DateTime.TryParseExact(text, formats, CultureInfo.InvariantCulture, DateTimeStyles.None, out date) ||
137         DateTime.TryParse(text, out date))
138     {
139         return date.Date;
140     }
141
142     throw new InvalidDataException("Fecha inválida en línea " + lineNumber + ".");
143 }
144
145 private static TimeSpan? ParseTime(string value, int lineNumber, bool required)
146 {
147     string text = (value ?? string.Empty).Trim();
148     if (text.Length == 0)
149     {
150         if (required)
151         {
152             throw new InvalidDataException("Hora obligatoria en línea " + lineNumber + ".");
153         }
154
155         return null;
156     }
157
158     TimeSpan time;
159     if (TimeSpan.TryParse(text, out time))
160     {
161         return time;
162     }
163
164     DateTime dateTime;
165     if (DateTime.TryParse(text, out dateTime))
166     {
167         return dateTime.TimeOfDay;
168     }
169
170     throw new InvalidDataException("Hora inválida en línea " + lineNumber + ".");
171 }
172
173 private static string NormalizeStatus(string value, int lineNumber)
174 {
175     string status = (value ?? string.Empty).Trim();
176     string[] allowed = { "Puntual", "Tardanza", "Falta", "Permiso" };
177     for (int i = 0; i < allowed.Length; i++)
178     {
179         if (string.Equals(status, allowed[i], StringComparison.OrdinalIgnoreCase))
180         {
```

Services/AttendanceDeviceImportService.cs (continuación)

```
181         return allowed[i];
182     }
183 }
184
185     throw new InvalidDataException("Estado inválido en línea " + lineNumber + ". Use Puntual, Tardanza, Falta o Permiso.");
186 }
187
188 private static decimal CalculateHours(string estado, TimeSpan? entrada, TimeSpan? salida, int lineNumber)
189 {
190     if (estado == "Falta" || estado == "Permiso")
191     {
192         return 0m;
193     }
194
195     if (!entrada.HasValue || !salida.HasValue)
196     {
197         throw new InvalidDataException("Debe indicar entrada y salida en línea " + lineNumber + ".");
198     }
199
200     if (salida.Value <= entrada.Value)
201     {
202         throw new InvalidDataException("La salida debe ser mayor que la entrada en línea " + lineNumber + ".");
203     }
204
205     return Math.Round(Convert.ToDecimal((salida.Value - entrada.Value).TotalHours), 2);
206 }
207
208 private static void UpsertAsistencia(SQLiteConnection connection, SQLiteTransaction transaction, int idEmpleado, DateTime fecha, TimeSpan? \
entrada, TimeSpan? salida, decimal horas, string estado)
209 {
210     object existingId;
211     using (SQLiteCommand find = new SQLiteCommand(
212         "SELECT IdAsistencia FROM Asistencias WHERE IdEmpleado = @empleado AND Fecha = @fecha LIMIT 1;", connection, transaction))
213     {
214         find.Parameters.AddWithValue("@empleado", idEmpleado);
215         find.Parameters.AddWithValue("@fecha", fecha.ToString("yyyy-MM-dd"));
216         existingId = find.ExecuteScalar();
217     }
218
219     string entradaText = entrada.HasValue ? entrada.Value.ToString(@"hh\:mm") : null;
220     string salidaText = salida.HasValue ? salida.Value.ToString(@"hh\:mm") : null;
221
222     if (existingId == null || existingId == DBNull.Value)
223     {
224         using (SQLiteCommand insert = new SQLiteCommand(@"INSERT INTO Asistencias
225             (IdEmpleado, Fecha, HoraEntrada, HoraSalida, HorasTrabajadas, Estado)
226             VALUES (@empleado, @fecha, @entrada, @salida, @horas, @estado);", connection, transaction))
227         {
228             FillParameters(insert, idEmpleado, fecha, entradaText, salidaText, horas, estado);
229             insert.ExecuteNonQuery();
230         }
231     }
232     else
233     {
234         using (SQLiteCommand update = new SQLiteCommand(@"UPDATE Asistencias SET
235             HoraEntrada = @entrada, HoraSalida = @salida, HorasTrabajadas = @horas, Estado = @estado
236             WHERE IdAsistencia = @id;", connection, transaction))
237         {
238             FillParameters(update, idEmpleado, fecha, entradaText, salidaText, horas, estado);
239             update.Parameters.AddWithValue("@id", Convert.ToInt32(existingId));
```

Services/AttendanceDeviceImportService.cs (continuacion)

```
240         update.ExecuteNonQuery();
241     }
242 }
243 }
244
245     private static void FillParameters(SQLiteCommand command, int idEmpleado, DateTime fecha, string entrada, string salida, decimal horas, string \
estado)
246     {
247         command.Parameters.AddWithValue("@empleado", idEmpleado);
248         command.Parameters.AddWithValue("@fecha", fecha.ToString("yyyy-MM-dd"));
249         command.Parameters.AddWithValue("@entrada", (object)entrada ?? DBNull.Value);
250         command.Parameters.AddWithValue("@salida", (object)salida ?? DBNull.Value);
251         command.Parameters.AddWithValue("@horas", horas);
252         command.Parameters.AddWithValue("@estado", estado);
253     }
254 }
255 }
```

Services/AuditTrailService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using SistemaGestionNomina.Data;
4 using SistemaGestionNomina.Models;
5 using SistemaGestionNomina.Security;
6
7 namespace SistemaGestionNomina.Services
8 {
9     public class AuditTrailService
10    {
11        private readonly AuditRepository repository = new AuditRepository();
12
13        public void RegistrarAccion(string modulo, string accion, string detalle)
14        {
15            RegistrarCambio(SessionContext.IsAuthenticated ? SessionContext.Username : "sistema",
16                            modulo, accion, detalle);
17        }
18
19        public void RegistrarCambio(string usuario, string modulo, string accion)
20        {
21            RegistrarCambio(usuario, modulo, accion, string.Empty);
22        }
23
24        public void RegistrarCambio(string usuario, string modulo, string accion, string detalle)
25        {
26            repository.Add(new AuditRecord
27            {
28                Usuario = Normalize(usuario, "sistema"),
29                Modulo = Normalize(modulo, "General"),
30                Accion = Normalize(accion, "Acción"),
31                Detalle = Normalize(detalle, string.Empty),
32                Fecha = DateTime.Now
33            });
34        }
35
36        public List<AuditRecord> Buscar(AuditQuery query)
37        {
38            return repository.Search(query ?? new AuditQuery());
39        }
40
41        public List<string> ObtenerModulos()
42        {
43            return repository.GetDistinct("Modulo");
44        }
45
46        public List<string> ObtenerAcciones()
47        {
48            return repository.GetDistinct("Accion");
49        }
50
51        public List<string> ObtenerUltimos(int cantidad)
52        {
53            List<string> items = new List<string>();
54            List<AuditRecord> records = repository.Search(new AuditQuery
55            {
56                Limit = cantidad <= 0 ? 20 : Math.Min(cantidad, 200)
57            });
58            for (int i = 0; i < records.Count; i++)
59            {
60                AuditRecord record = records[i];
```


Services/AuditTrailService.cs (continuacion)

```
61         items.Add(record.Fecha.ToString("yyyy-MM-dd HH:mm:ss") + " | " + record.Usuario + " | " +
62             record.Modulo + " | " + record.Accion + " | " + record.Detalle);
63     }
64     return items;
65 }
66
67 private static string Normalize(string value, string fallback)
68 {
69     return string.IsNullOrEmpty(value) ? fallback : value.Trim();
70 }
71 }
72 }
```

Services/BackupService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.IO;
4 using System.Security.Cryptography;
5 using System.Text;
6 using SistemaGestionNomina.Data;
7 using SistemaGestionNomina.Security;
8
9 namespace SistemaGestionNomina.Services
10 {
11     /// <summary>
12     /// Servicio reservado para crear copias de seguridad de la base de datos.
13     /// </summary>
14     public class BackupService
15     {
16         private readonly AuditTrailService auditTrailService = new AuditTrailService();
17         private readonly AuthorizationService authorizationService = new AuthorizationService();
18
19         /// <summary>
20         /// Crea una copia de seguridad de nomina.db en la ruta indicada.
21         /// </summary>
22         public string CrearBackup(string carpetaDestino)
23         {
24             authorizationService.DemandPermission(Permissions.BackupsManage);
25             string source = SQLiteConnectionFactory.DatabasePath;
26             if (!File.Exists(source))
27             {
28                 throw new FileNotFoundException("No se encontró la base de datos para respaldar.", source);
29             }
30
31             string destinationFolder = string.IsNullOrEmpty(carpetaDestino)
32                 ? Path.Combine(AppDomain.CurrentDomain.BaseDirectory, "Backups")
33                 : carpetaDestino;
34
35             Directory.CreateDirectory(destinationFolder);
36             string destination = Path.Combine(destinationFolder, "nomina_" + DateTime.Now.ToString("yyyyMMdd_HH:mm:ss") + ".db");
37             File.Copy(source, destination, false);
38
39             string hash = ComputeSha256(destination);
40             File.WriteAllText(destination + ".sha256", hash, Encoding.ASCII);
41             auditTrailService.RegistrarAccion("Backup", "Crear", Path.GetFileName(destination));
42             return destination;
43         }
44
45         /// <summary>
46         /// Lista respaldos disponibles ordenados del más reciente al más antiguo.
47         /// </summary>
48         public List<string> ListarBackups(string carpeta)
49         {
50             authorizationService.DemandPermission(Permissions.BackupsManage);
51             string folder = string.IsNullOrEmpty(carpeta)
52                 ? Path.Combine(AppDomain.CurrentDomain.BaseDirectory, "Backups")
53                 : carpeta;
54
55             List<string> backups = new List<string>();
56             if (!Directory.Exists(folder))
57             {
58                 return backups;
59             }
60 }
```

Services/BackupService.cs (continuacion)

```
61         string[] files = Directory.GetFiles(folder, "nomina_*.db");
62         Array.Sort(files);
63         Array.Reverse(files);
64         backups.AddRange(files);
65         return backups;
66     }
67
68     /// <summary>
69     /// Verifica que el hash del backup coincida con su archivo .sha256.
70     /// </summary>
71     public bool VerificarBackup(string rutaBackup)
72     {
73         authorizationService.DemandPermission(Permissions.BackupsManage);
74         if (string.IsNullOrEmpty(rutaBackup) || !File.Exists(rutaBackup))
75         {
76             return false;
77         }
78
79         string hashFile = rutaBackup + ".sha256";
80         if (!File.Exists(hashFile))
81         {
82             return false;
83         }
84
85         string expected = File.ReadAllText(hashFile).Trim();
86         return string.Equals(expected, ComputeSha256(rutaBackup), StringComparison.OrdinalIgnoreCase);
87     }
88
89     private static string ComputeSha256(string path)
90     {
91         using (SHA256 sha = SHA256.Create())
92         using (FileStream stream = File.OpenRead(path))
93         {
94             byte[] hash = sha.ComputeHash(stream);
95             StringBuilder builder = new StringBuilder();
96             for (int i = 0; i < hash.Length; i++)
97             {
98                 builder.Append(hash[i].ToString("x2"));
99             }
100
101             return builder.ToString();
102         }
103     }
104 }
105 }
```

Services/BusinessServices.cs

```
1 using System;
2 using System.Collections.Generic;
3 using SistemaGestionNomina.Data;
4 using SistemaGestionNomina.Helpers;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7
8 namespace SistemaGestionNomina.Services
9 {
10     public class AuthService
11     {
12         private const int MaximumAttempts = 5;
13         private static readonly TimeSpan LockDuration = TimeSpan.FromMinutes(15);
14         private readonly AuthenticationRepository authenticationRepository = new AuthenticationRepository();
15         private readonly AuditTrailService auditTrailService = new AuditTrailService();
16
17         public AuthenticationResult Authenticate(string username, string password)
18         {
19             string normalizedUsername = (username ?? string.Empty).Trim();
20             if (normalizedUsername.Length == 0 || string.IsNullOrEmpty(password))
21             {
22                 return AuthenticationResult.Invalid();
23             }
24
25             Usuario usuario = authenticationRepository.GetByUsername(normalizedUsername);
26             if (usuario == null || !string.Equals(usuario.Estado, "Activo", StringComparison.OrdinalIgnoreCase) ||
27                 !Roles.IsValid(usuario.Rol))
28             {
29                 auditTrailService.RegistrarCambio(normalizedUsername, "Seguridad", "Inicio de sesión fallido",
30                     "Credenciales no válidas.");
31                 return AuthenticationResult.Invalid();
32             }
33
34             DateTime now = DateTime.Now;
35             if (usuario.Bloqueado && usuario.FechaBloqueo.HasValue)
36             {
37                 DateTime blockedUntil = usuario.FechaBloqueo.Value.Add(LockDuration);
38                 if (now < blockedUntil)
39                 {
40                     auditTrailService.RegistrarCambio(normalizedUsername, "Seguridad", "Acceso bloqueado",
41                         "Intento durante bloqueo temporal.");
42                     return AuthenticationResult.Blocked(blockedUntil);
43                 }
44
45                 authenticationRepository.RegisterFailedAttempt(usuario.IdUsuario, 0, false, null);
46                 usuario.IntentosFallidos = 0;
47                 usuario.Bloqueado = false;
48                 usuario.FechaBloqueo = null;
49             }
50
51             if (!PasswordHelper.Verify(password, usuario.PasswordHash))
52             {
53                 int attempts = usuario.IntentosFallidos + 1;
54                 bool blocked = attempts >= MaximumAttempts;
55                 DateTime? blockedAt = blocked ? now : (DateTime?)null;
56                 authenticationRepository.RegisterFailedAttempt(usuario.IdUsuario, attempts, blocked, blockedAt);
57                 auditTrailService.RegistrarCambio(normalizedUsername, "Seguridad",
58                     blocked ? "Usuario bloqueado" : "Inicio de sesión fallido",
59                     blocked ? "Se alcanzó el máximo de intentos." : "Intento " + attempts + " de " + MaximumAttempts + ".");
60                 return blocked ? AuthenticationResult.Blocked(now.Add(LockDuration)) : AuthenticationResult.Invalid();
61             }
62         }
63     }
64 }
```

Services/BusinessServices.cs (continuacion)

```
61         }
62
63         bool upgradeLegacyHash = PasswordHelper.IsLegacyHash(usuario.PasswordHash);
64         string upgradedHash = upgradeLegacyHash ? PasswordHelper.HashPassword(password) : string.Empty;
65         authenticationRepository.CompleteLogin(usuario.IdUsuario, now, upgradedHash);
66         if (upgradeLegacyHash) usuario.PasswordHash = upgradedHash;
67         usuario.UltimoAcceso = now;
68         usuario.IntentosFallidos = 0;
69         usuario.Bloqueado = false;
70         usuario.FechaBloqueo = null;
71         SessionContext.Begin(usuario);
72         auditTrailService.RegistrarCambio(usuario.NombreUsuario, "Seguridad", "Inicio de sesión correcto",
73             upgradeLegacyHash ? "Contraseña actualizada a PBKDF2." : string.Empty);
74         return AuthenticationResult.Successful(usuario);
75     }
76 }
77
78 public class EmpleadoService
79 {
80     private readonly EmpleadoRepository empleadoRepository = new EmpleadoRepository();
81     private readonly DepartamentoRepository departamentoRepository = new DepartamentoRepository();
82     private readonly AuditTrailService auditTrailService = new AuditTrailService();
83     private readonly AuthorizationService authorizationService = new AuthorizationService();
84     private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
85     private readonly EmployeeHistoryService employeeHistoryService = new EmployeeHistoryService();
86
87     public List<Empleado> GetAll(string search, int? departmentId, string status)
88     {
89         authorizationService.DemandPermission(Permissions.EmployeesView);
90         employeeHistoryService.ApplyDueChanges(DateTime.Today);
91         return empleadoRepository.GetAll(search, departmentId, status, employeeScopeService.GetDepartmentScope());
92     }
93
94     public List<Empleado> GetActive(int? departmentId)
95     {
96         authorizationService.DemandAny(Permissions.EmployeesView, Permissions.AttendanceView,
97             Permissions.PayrollView, Permissions.ReportsPersonal);
98         employeeHistoryService.ApplyDueChanges(DateTime.Today);
99         int? scope = employeeScopeService.GetDepartmentScope();
100         return empleadoRepository.GetActiveByDepartment(scope ?? departmentId, scope);
101     }
102
103     public Empleado GetById(int id)
104     {
105         authorizationService.DemandPermission(Permissions.EmployeesView);
106         employeeScopeService.DemandEmployeeInScope(id);
107         employeeHistoryService.ApplyDueChanges(DateTime.Today);
108         return empleadoRepository.GetById(id);
109     }
110
111     public List<Departamento> GetDepartments()
112     {
113         authorizationService.DemandAny(Permissions.EmployeesView, Permissions.AttendanceView,
114             Permissions.PayrollView);
115         return departamentoRepository.GetAll();
116     }
117
118     public int Save(Empleado empleado)
119     {
120         return Save(empleado, null).EmployeeId;
```

Services/BusinessServices.cs (continuacion)

```
121     }
122
123     public EmployeeSaveResult Save(Empleado empleado, EmployeeChangeContext changeContext)
124     {
125         if (empleado == null)
126         {
127             throw new ArgumentException("Debe indicar los datos del empleado.");
128         }
129
130         authorizationService.DemandPermission(empleado.IdEmpleado == 0
131             ? Permissions.EmployeesCreate : Permissions.EmployeesEdit);
132
133         if (empleado.IdDepartamento <= 0)
134         {
135             throw new ArgumentException("Debe seleccionar un departamento válido.");
136         }
137
138         if (empleado.SalarioBase <= 0)
139         {
140             throw new ArgumentException("El salario base debe ser mayor que cero.");
141         }
142
143         if (empleadoRepository.ExistsCode(empleado.Codigo, empleado.IdEmpleado))
144         {
145             throw new InvalidOperationException("Ya existe un empleado con ese código.");
146         }
147
148         if (empleadoRepository.ExistsCedula(empleado.Cedula, empleado.IdEmpleado))
149         {
150             throw new InvalidOperationException("Ya existe un empleado con esa cédula.");
151         }
152
153         if (empleado.IdEmpleado == 0)
154         {
155             empleado.FechaEfectivaLaboral = empleado.FechaIngreso;
156             int id = empleadoRepository.Add(empleado);
157             auditTrailService.RegistrarAccion("Empleados", "Crear", empleado.Codigo);
158             return new EmployeeSaveResult
159             {
160                 EmployeeId = id,
161                 AppliedImmediately = true
162             };
163         }
164
165         EmployeeSaveResult result = employeeHistoryService.SaveEmployee(empleado, changeContext);
166         if (result.HistoryRecords == 0)
167             auditTrailService.RegistrarAccion("Empleados", "Actualizar", empleado.Codigo);
168         return result;
169     }
170
171     public void Deactivate(int id)
172     {
173         throw new ArgumentException("Debe indicar la fecha efectiva y el motivo de desactivación.");
174     }
175
176     public EmployeeSaveResult Deactivate(int id, EmployeeChangeContext changeContext)
177     {
178         authorizationService.DemandPermission(Permissions.EmployeesDeactivate);
179         Empleado employee = empleadoRepository.GetById(id);
180         if (employee == null) throw new InvalidOperationException("El empleado no existe.");
```

Services/BusinessServices.cs (continuacion)

```
181         employee.Estado = "Inactivo";
182         return employeeHistoryService.SaveEmployee(employee, changeContext);
183     }
184 }
185
186 public class AsistenciaService
187 {
188     private readonly AsistenciaRepository asistenciaRepository = new AsistenciaRepository();
189     private readonly AuditTrailService auditTrailService = new AuditTrailService();
190     private readonly AuthorizationService authorizationService = new AuthorizationService();
191     private readonly PayrollPeriodPolicyService periodPolicyService = new PayrollPeriodPolicyService();
192     private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
193
194     public int Register(Asistencia asistencia)
195     {
196         if (asistencia == null)
197         {
198             throw new ArgumentException("Debe indicar los datos de asistencia.");
199         }
200
201         authorizationService.DemandPermission(Permissions.AttendanceRegister);
202         employeeScopeService.DemandEmployeeInScope(asistencia.IdEmpleado);
203         periodPolicyService.VerificarFechasAbiertas(asistencia.Fecha.Date, asistencia.Fecha.Date);
204
205         if (asistenciaRepository.Exists(asistencia.IdEmpleado, asistencia.Fecha.Date))
206         {
207             throw new InvalidOperationException("Ya existe una asistencia para el empleado en esa fecha.");
208         }
209
210         if (asistencia.Estado == "Falta" || asistencia.Estado == "Permiso")
211         {
212             asistencia.HoraEntrada = null;
213             asistencia.HoraSalida = null;
214             asistencia.HorasTrabajadas = 0;
215         }
216         else if (asistencia.HoraEntrada.HasValue && asistencia.HoraSalida.HasValue)
217         {
218             // Para asistencia valida, la salida debe cerrar una jornada real.
219             if (asistencia.HoraSalida.Value <= asistencia.HoraEntrada.Value)
220             {
221                 throw new InvalidOperationException("La hora de salida debe ser mayor que la hora de entrada.");
222             }
223
224             asistencia.HorasTrabajadas = Convert.ToDecimal((asistencia.HoraSalida.Value - asistencia.HoraEntrada.Value).TotalHours);
225         }
226         else
227         {
228             throw new InvalidOperationException("Debe registrar hora de entrada y salida.");
229         }
230
231         int id = asistenciaRepository.Add(asistencia);
232         auditTrailService.RegistrarAccion("Asistencia", "Registrar",
233             "IdEmpleado=" + asistencia.IdEmpleado + ", Fecha=" + asistencia.Fecha.ToString("yyyy-MM-dd"));
234         return id;
235     }
236
237     public List<Asistencia> GetAll(DateTime? start, DateTime? end, int? employeeId, string status)
238     {
239         authorizationService.DemandPermission(Permissions.AttendanceView);
240         if (start.HasValue && end.HasValue && start.Value.Date > end.Value.Date)
```

Services/BusinessServices.cs (continuacion)

```

241         {
242             throw new ArgumentException("La fecha inicial no puede ser mayor que la fecha final.");
243         }
244
245         if (employeeId.HasValue) employeeScopeService.DemandEmployeeInScope(employeeId.Value);
246         return asistenciaRepository.GetAll(start, end, employeeId, status,
247             employeeScopeService.GetDepartmentScope());
248     }
249 }
250
251 public class ConfiguracionService
252 {
253     private readonly ConfiguracionRepository configuracionRepository = new ConfiguracionRepository();
254     private readonly AuthorizationService authorizationService = new AuthorizationService();
255     private readonly AuditTrailService auditTrailService = new AuditTrailService();
256
257     public List<ConfiguracionNomina> GetAll()
258     {
259         authorizationService.DemandPermission(Permissions.ConfigurationView);
260         return configuracionRepository.GetAll();
261     }
262
263     public decimal GetValue(string name, decimal fallback)
264     {
265         authorizationService.DemandPermission(Permissions.ConfigurationView);
266         return configuracionRepository.GetValue(name, fallback);
267     }
268
269     public void SaveDefaults(Dictionary<string, decimal> values)
270     {
271         authorizationService.DemandPermission(Permissions.ConfigurationEdit);
272         if (values == null || !values.ContainsKey("SeguroSocial") || !values.ContainsKey("ISR") ||
273             !values.ContainsKey("SeguroEducativo") || !values.ContainsKey("RecargoHoraExtra") ||
274             !values.ContainsKey("HorasMensualesBase"))
275         {
276             throw new ArgumentException("La configuración de nómina está incompleta.");
277         }
278         configuracionRepository.Save("SeguroSocial", values["SeguroSocial"], "Porcentaje académico de seguro social.");
279         configuracionRepository.Save("ISR", values["ISR"], "Porcentaje académico de impuesto sobre la renta.");
280         configuracionRepository.Save("SeguroEducativo", values["SeguroEducativo"], "Porcentaje académico de seguro educativo.");
281         configuracionRepository.Save("RecargoHoraExtra", values["RecargoHoraExtra"], "Multiplicador académico de horas extra.");
282         configuracionRepository.Save("HorasMensualesBase", values["HorasMensualesBase"], "Horas mensuales base para cálculo por hora.");
283         auditTrailService.RegistrarAccion("Configuración", "Actualizar", "Parámetros de nómina.");
284     }
285 }
286
287 public class NominaService
288 {
289     private readonly EmpleadoRepository empleadoRepository = new EmpleadoRepository();
290     private readonly AsistenciaRepository asistenciaRepository = new AsistenciaRepository();
291     private readonly ConfiguracionRepository configuracionRepository = new ConfiguracionRepository();
292     private readonly NominaRepository nominaRepository = new NominaRepository();
293     private readonly ComprobanteRepository comprobanteRepository = new ComprobanteRepository();
294     private readonly AuditTrailService auditTrailService = new AuditTrailService();
295     private readonly AuthorizationService authorizationService = new AuthorizationService();
296     private readonly EmployeeEffectiveDataService effectiveEmployeeService = new EmployeeEffectiveDataService();
297     private readonly PayrollPeriodPolicyService periodPolicyService = new PayrollPeriodPolicyService();
298
299     public Nomina CalcularNomina(DateTime fechaInicio, DateTime fechaFin, int? departamentoId)
300     {

```


Services/BusinessServices.cs (continuacion)

```
301         authorizationService.DemandPermission(Permissions.PayrollCalculate);
302         if (fechaInicio.Date > fechaFin.Date)
303         {
304             throw new InvalidOperationException("La fecha de inicio debe ser menor o igual que la fecha fin.");
305         }
306
307         periodPolicyService.VerificarFechasAbiertas(fechaInicio.Date, fechaFin.Date);
308
309         List<Empleado> empleados = effectiveEmployeeService.GetActiveEmployeesAsOf(fechaFin.Date, departamentoId);
310         if (empleados.Count == 0)
311         {
312             throw new InvalidOperationException("No hay empleados activos para calcular la nómina.");
313         }
314
315         decimal seguroSocial = configuracionRepository.GetValue("SeguroSocial", 9.75m);
316         decimal isr = configuracionRepository.GetValue("ISR", 10m);
317         decimal seguroEducativo = configuracionRepository.GetValue("SeguroEducativo", 1.25m);
318         decimal recargoHoraExtra = configuracionRepository.GetValue("RecargoHoraExtra", 1.25m);
319         decimal horasMensualesBase = configuracionRepository.GetValue("HorasMensualesBase", 160m);
320
321         Nomina nomina = new Nomina();
322         nomina.FechaCalculo = DateTime.Now;
323         nomina.PeriodoNombre = "Nómina " + fechaInicio.ToString("dd/MM/yyyy") + " - " + fechaFin.ToString("dd/MM/yyyy");
324         nomina.Estado = PayrollStates.Calculated;
325
326         foreach (Empleado empleado in empleados)
327         {
328             // Formula academica: sueldo base + bono + horas extra - deducciones configurables.
329             decimal horasExtra = asistenciaRepository.GetExtraHours(empleado.IdEmpleado, fechaInicio, fechaFin);
330             decimal pagoHora = horasMensualesBase <= 0 ? 0 : empleado.SalarioBase / horasMensualesBase;
331             decimal montoHorasExtra = Math.Round(horasExtra * pagoHora * recargoHoraExtra, 2);
332             decimal bonos = Math.Round(empleado.SalarioBase * 0.03m, 2);
333             decimal totalIngresos = empleado.SalarioBase + bonos + montoHorasExtra;
334             decimal totalDeducciones = Math.Round(
335                 totalIngresos * (seguroSocial / 100m) +
336                 totalIngresos * (isr / 100m) +
337                 totalIngresos * (seguroEducativo / 100m), 2);
338             decimal neto = totalIngresos - totalDeducciones;
339
340             NominaDetalle detalle = new NominaDetalle();
341             detalle.IdEmpleado = empleado.IdEmpleado;
342             detalle.CodigoEmpleado = empleado.Codigo;
343             detalle.EmpleadoNombre = empleado.NombreCompleto;
344             detalle.CargoEmpleado = empleado.Cargo;
345             detalle.Departamento = empleado.DepartamentoNombre;
346             detalle.SueldoBase = empleado.SalarioBase;
347             detalle.Bonos = bonos;
348             detalle.HorasExtra = horasExtra;
349             detalle.MontoHorasExtra = montoHorasExtra;
350             detalle.TotalIngresos = totalIngresos;
351             detalle.TotalDeducciones = totalDeducciones;
352             detalle.Netopagar = neto;
353             nomina.Detalles.Add(detalle);
354
355             nomina.TotalIngresos += totalIngresos;
356             nomina.TotalDeducciones += totalDeducciones;
357             nomina.TotalNeto += neto;
358         }
359
360         auditTrailService.RegistrarAccion("Nómina", "Calcular",
```

Services/BusinessServices.cs (continuacion)

```
361         fechaInicio.ToString("yyyy-MM-dd") + " a " + fechaFin.ToString("yyyy-MM-dd") +
362         ", Empleados=" + nomina.Detalles.Count);
363
364     return nomina;
365 }
366
367 private readonly PayrollLifecycleService payrollLifecycleService = new PayrollLifecycleService();
368
369 public int ConfirmarPago(Nomina nomina, DateTime fechaInicio, DateTime fechaFin)
370 {
371     return payrollLifecycleService.Confirmar(nomina, fechaInicio, fechaFin);
372 }
373
374 public void PagarNomina(int idNomina)
375 {
376     payrollLifecycleService.Pagar(idNomina);
377 }
378
379 public void AnularNomina(int idNomina, string motivo)
380 {
381     payrollLifecycleService.Anular(idNomina, motivo);
382 }
383
384 public int RecalcularNomina(int idNominaAnulada, DateTime fechaInicio, DateTime fechaFin, Nomina nominaRecalculada)
385 {
386     return payrollLifecycleService.Recalcular(idNominaAnulada, fechaInicio, fechaFin, nominaRecalculada);
387 }
388
389 public List<NominaVersion> ObtenerHistorialNomina(int idNomina)
390 {
391     return payrollLifecycleService.ObtenerHistorial(idNomina);
392 }
393
394 public List<Nomina> GetNominas()
395 {
396     authorizationService.DemandPermission(Permissions.PayrollView);
397     return nominaRepository.GetAll();
398 }
399
400 public List<NominaDetalle> GetDetalles(int idNomina)
401 {
402     authorizationService.DemandPermission(Permissions.PayrollView);
403     return nominaRepository.GetDetalles(idNomina);
404 }
405 }
406
407 public class ComprobanteService
408 {
409     private readonly ComprobanteRepository comprobanteRepository = new ComprobanteRepository();
410     private readonly AuditTrailService auditTrailService = new AuditTrailService();
411     private readonly AuthorizationService authorizationService = new AuthorizationService();
412
413     public List<Comprobante> GetAll(string search)
414     {
415         authorizationService.DemandPermission(Permissions.PayslipsView);
416         return comprobanteRepository.GetAll(search);
417     }
418
419     public Comprobante GetById(int id)
420     {
```

Services/BusinessServices.cs (continuacion)

```
421         authorizationService.DemandPermission(Permissions.PayslipsView);
422         return comprobanteRepository.GetById(id);
423     }
424
425     public void SaveRutaPdf(int idComprobante, string ruta)
426     {
427         authorizationService.DemandPermission(Permissions.PayslipsExport);
428         comprobanteRepository.UpdateRutaPdf(idComprobante, ruta);
429         auditTrailService.RegistrarAccion("Comprobantes", "Guardar PDF",
430             "IdComprobante=" + idComprobante);
431     }
432 }
433
434 public class ReporteService
435 {
436     private readonly ReporteRepository reporteRepository = new ReporteRepository();
437     private readonly AuditTrailService auditTrailService = new AuditTrailService();
438     private readonly AuthorizationService authorizationService = new AuthorizationService();
439
440     public List<ReporteGenerado> GetAll()
441     {
442         authorizationService.DemandPermission(Permissions.ReportsView);
443         List<ReporteGenerado> reports = reporteRepository.GetAll();
444         if (string.Equals(SessionContext.Role, Roles.Admin, StringComparison.OrdinalIgnoreCase)) return reports;
445
446         List<ReporteGenerado> filtered = new List<ReporteGenerado>();
447         bool personalOnly = authorizationService.HasPermission(Permissions.ReportsPersonal) &&
448             !authorizationService.HasPermission(Permissions.ReportsFinancial);
449         for (int i = 0; i < reports.Count; i++)
450         {
451             bool isPersonal = (reports[i].Tipo ?? string.Empty).IndexOf("Personal",
452                 StringComparison.OrdinalIgnoreCase) >= 0;
453             if ((personalOnly && isPersonal) || (!personalOnly && !isPersonal)) filtered.Add(reports[i]);
454         }
455         return filtered;
456     }
457
458     public ReporteGenerado Register(string name, string type, string path)
459     {
460         authorizationService.DemandPermission(Permissions.ReportsExport);
461         ReporteGenerado reporte = new ReporteGenerado();
462         reporte.NombreReporte = name;
463         reporte.Tipo = type;
464         reporte.GeneradoPor = SessionContext.Username;
465         reporte.FechaGeneracion = DateTime.Now;
466         reporte.RutaArchivo = path;
467         reporte.IdReporte = reporteRepository.Add(reporte);
468         auditTrailService.RegistrarAccion("Reportes", "Generar", name + " - " + type);
469         return reporte;
470     }
471 }
472 }
```

Services/EmailService.cs

```
1 using System;
2 using System.IO;
3 using System.Text;
4 using SistemaGestionNomina.Helpers;
5 using SistemaGestionNomina.Security;
6
7 namespace SistemaGestionNomina.Services
8 {
9     /// <summary>
10    /// Servicio responsable de enviar comprobantes por correo electrónico.
11    /// </summary>
12    public class EmailService
13    {
14        private readonly AuthorizationService authorizationService = new AuthorizationService();
15        private readonly AuditTrailService auditTrailService = new AuditTrailService();
16
17        /// <summary>
18        /// Envía un comprobante generado a un destinatario.
19        /// </summary>
20        public void EnviarComprobante(string destinatario, string rutaPdf)
21        {
22            CrearBorradorComprobante(destinatario, rutaPdf, "Comprobante de pago");
23        }
24
25        /// <summary>
26        /// Crea un borrador .eml con el comprobante adjunto para abrirlo con el cliente de correo.
27        /// </summary>
28        public string CrearBorradorComprobante(string destinatario, string rutaPdf, string asunto)
29        {
30            authorizationService.DemandPermission(Permissions.PayslipsEmail);
31            if (string.IsNullOrEmpty(destinatario) || string.IsNullOrEmpty(rutaPdf))
32            {
33                throw new ArgumentException("Debe indicar destinatario y ruta del comprobante.");
34            }
35
36            if (!IsValidEmail(destinatario))
37            {
38                throw new ArgumentException("El correo del destinatario no tiene un formato válido.");
39            }
40
41            if (!File.Exists(rutaPdf))
42            {
43                throw new FileNotFoundException("No se encontró el PDF del comprobante.", rutaPdf);
44            }
45
46            string fileName = "EmailComprobante_" + DateTime.Now.ToString("yyyyMMdd_HHMMss") + ".eml";
47            string path = PathHelper.RequestExportPath(Path.GetFileNameWithoutExtension(fileName), ".eml", "Mensaje de correo (*.eml)|*.eml");
48            if (string.IsNullOrEmpty(path)) return string.Empty;
49
50            string boundary = "---=_Nomina_" + Guid.NewGuid().ToString("N");
51            string subject = string.IsNullOrEmpty(asunto) ? "Comprobante de pago" : asunto.Trim();
52
53            StringBuilder builder = new StringBuilder();
54            builder.AppendLine("To: " + destinatario.Trim());
55            builder.AppendLine("Subject: " + subject);
56            builder.AppendLine("X-Unsent: 1");
57            builder.AppendLine("MIME-Version: 1.0");
58            builder.AppendLine("Content-Type: multipart/mixed; boundary=\"" + boundary + "\"");
59            builder.AppendLine();
60            builder.AppendLine("--" + boundary);
```

Services/EmailService.cs (continuacion)

```
61         builder.AppendLine("Content-Type: text/plain; charset=utf-8");
62         builder.AppendLine("Content-Transfer-Encoding: 8bit");
63         builder.AppendLine();
64         builder.AppendLine("Adjunto encontrará su comprobante de pago generado por el Sistema de Gestión de Nómina.");
65         builder.AppendLine();
66         builder.AppendLine("--" + boundary);
67         builder.AppendLine("Content-Type: application/pdf; name=\"\" + Path.GetFileName(rutaPdf) + \"\"");
68         builder.AppendLine("Content-Transfer-Encoding: base64");
69         builder.AppendLine("Content-Disposition: attachment; filename=\"\" + Path.GetFileName(rutaPdf) + \"\"");
70         builder.AppendLine();
71         builder.AppendLine(Base64Lines(File.ReadAllBytes(rutaPdf)));
72         builder.AppendLine("--" + boundary + "--");
73
74         File.WriteAllText(path, builder.ToString(), Encoding.UTF8);
75         auditTrailService.RegistrarAccion("Comprobantes", "Crear borrador de correo", destinatario.Trim());
76         return path;
77     }
78
79     private static bool IsValidEmail(string email)
80     {
81         string value = email.Trim();
82         int at = value.IndexOf('@');
83         return at > 0 && at < value.Length - 3 && value.LastIndexOf('.') > at + 1;
84     }
85
86     private static string ToBase64Lines(byte[] bytes)
87     {
88         string base64 = Convert.ToBase64String(bytes);
89         StringBuilder builder = new StringBuilder();
90         for (int i = 0; i < base64.Length; i += 76)
91         {
92             int length = Math.Min(76, base64.Length - i);
93             builder.AppendLine(base64.Substring(i, length));
94         }
95
96         return builder.ToString();
97     }
98 }
99 }
```

Services/EmployeeHistoryService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using System.Globalization;
5 using SistemaGestionNomina.Data;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8
9 namespace SistemaGestionNomina.Services
10 {
11     public sealed class EmployeeHistoryService
12     {
13         private readonly EmployeeHistoryRepository historyRepository = new EmployeeHistoryRepository();
14         private readonly EmpleadoRepository employeeRepository = new EmpleadoRepository();
15         private readonly NominaRepository payrollRepository = new NominaRepository();
16         private readonly AuditRepository auditRepository = new AuditRepository();
17         private readonly AuthorizationService authorizationService = new AuthorizationService();
18         private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
19
20         public List<HistorialEmpleado> Search(EmployeeHistoryQuery query)
21         {
22             authorizationService.DemandPermission(Permissions.EmployeeHistoryView);
23             if (query != null && query.EmployeeId.HasValue)
24                 employeeScopeService.DemandEmployeeInScope(query.EmployeeId.Value);
25             if (query != null && query.From.HasValue && query.To.HasValue && query.From.Value.Date > query.To.Value.Date)
26                 throw new ArgumentException("La fecha inicial no puede superar la fecha final.");
27             return historyRepository.Search(query, employeeScopeService.GetDepartmentScope());
28         }
29
30         internal EmployeeSaveResult SaveEmployee(Empleado candidate, EmployeeChangeContext context)
31         {
32             Empleado original = employeeRepository.GetById(candidate.IdEmpleado);
33             if (original == null) throw new InvalidOperationException("El empleado no existe.");
34             employeeScopeService.DemandEmployeeInScope(candidate.IdEmpleado);
35
36             List<HistorialEmpleado> changes = BuildChanges(original, candidate);
37             if (changes.Count == 0)
38             {
39                 candidate.FechaEfectivaLaboral = original.FechaEfectivaLaboral ?? original.FechaIngreso;
40                 employeeRepository.Update(candidate);
41                 return new EmployeeSaveResult
42                 {
43                     EmployeeId = candidate.IdEmpleado,
44                     AppliedImmediately = true
45                 };
46             }
47
48             if (context == null || string.IsNullOrEmpty(context.Reason))
49                 throw new ArgumentException("Debe indicar la fecha efectiva y el motivo del cambio laboral.");
50
51             DateTime effectiveDate = context.EffectiveDate.Date;
52             if (effectiveDate == DateTime.MinValue.Date)
53                 throw new ArgumentException("La fecha efectiva del cambio no es válida.");
54             if (effectiveDate < original.FechaIngreso.Date)
55                 throw new ArgumentException("La fecha efectiva no puede ser anterior al ingreso del empleado.");
56             if (effectiveDate > DateTime.Today)
57                 authorizationService.DemandPermission(Permissions.EmployeeChangesSchedule);
58             if (payrollRepository.HasClosedPeriodAffectedByEmployeeChange(effectiveDate))
59                 throw new InvalidOperationException("El cambio laboral afectaría un período de nómina cerrado.");
60 }
```

Services/EmployeeHistoryService.cs (continuacion)

```
61         bool applyNow = effectiveDate <= DateTime.Today;
62         string user = SessionContext.IsAuthenticated ? SessionContext.Username : "sistema";
63         DateTime now = DateTime.Now;
64         for (int i = 0; i < changes.Count; i++)
65         {
66             changes[i].FechaCambio = now;
67             changes[i].FechaEfectiva = effectiveDate;
68             changes[i].UsuarioResponsable = user;
69             changes[i].Motivo = context.Reason.Trim();
70             changes[i].Aplicado = applyNow;
71             changes[i].FechaAplicacion = applyNow ? (DateTime?)now : null;
72         }
73
74         Empleado valueToPersist = applyNow ? candidate : PreserveTrackedValues(candidate, original);
75         valueToPersist.FechaEfectivaLaboral = applyNow
76             ? effectiveDate : original.FechaEfectivaLaboral ?? original.FechaIngreso;
77
78         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
79         using (SQLiteTransaction transaction = connection.BeginTransaction())
80         {
81             try
82             {
83                 // Historial, dato vigente y auditoria se guardan juntos para no dejar cambios parciales.
84                 for (int i = 0; i < changes.Count; i++) historyRepository.Add(connection, transaction, changes[i]);
85                 employeeRepository.Update(connection, transaction, valueToPersist);
86                 auditRepository.Add(connection, transaction, new AuditRecord
87                 {
88                     Usuario = user,
89                     Modulo = "Empleados",
90                     Accion = applyNow ? "Aplicar cambio laboral" : "Programar cambio laboral",
91                     Detalle = "IdEmpleado=" + candidate.IdEmpleado + ", Campos=" + changes.Count +
92                         ", FechaEfectiva=" + effectiveDate.ToString("yyyy-MM-dd"),
93                     Fecha = now
94                 });
95                 transaction.Commit();
96             }
97             catch
98             {
99                 transaction.Rollback();
100                 throw;
101             }
102         }
103
104         return new EmployeeSaveResult
105         {
106             EmployeeId = candidate.IdEmpleado,
107             HistoryRecords = changes.Count,
108             AppliedImmediately = applyNow,
109             HasScheduledChanges = !applyNow
110         };
111     }
112
113     public int ApplyDueChanges(DateTime effectiveThrough)
114     {
115         int applied = 0;
116         DateTime now = DateTime.Now;
117         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
118         using (SQLiteTransaction transaction = connection.BeginTransaction())
119         {
120             try
```

Services/EmployeeHistoryService.cs (continuacion)

```
121         {
122             List<HistorialEmpleado> pending = historyRepository.GetPendingDue(
123                 connection, transaction, effectiveThrough.Date);
124             // Los cambios futuros se aplican por fecha efectiva, no por la fecha en que fueron creados.
125             for (int i = 0; i < pending.Count; i++)
126             {
127                 HistorialEmpleado item = pending[i];
128                 if (payrollRepository.HasClosedPeriodAffectedByEmployeeChange(
129                     connection, transaction, item.FechaEfectiva))
130                 {
131                     auditRepository.Add(connection, transaction, new AuditRecord
132                     {
133                         Usuario = "sistema",
134                         Modulo = "Empleados",
135                         Accion = "Cambio programado bloqueado",
136                         Detalle = "IdHistorial=" + item.IdHistorial + ", período cerrado relacionado",
137                         Fecha = now
138                     });
139                     continue;
140                 }
141
142                 historyRepository.ApplyChange(connection, transaction, item, now);
143                 auditRepository.Add(connection, transaction, new AuditRecord
144                 {
145                     Usuario = "sistema",
146                     Modulo = "Empleados",
147                     Accion = "Aplicar cambio programado",
148                     Detalle = "IdHistorial=" + item.IdHistorial + ", IdEmpleado=" + item.IdEmpleado,
149                     Fecha = now
150                 });
151                 applied++;
152             }
153             transaction.Commit();
154         }
155         catch
156         {
157             transaction.Rollback();
158             throw;
159         }
160     }
161     return applied;
162 }
163
164 private List<HistorialEmpleado> BuildChanges(Empleado original, Empleado candidate)
165 {
166     List<HistorialEmpleado> changes = new List<HistorialEmpleado>();
167     if (original.SalarioBase != candidate.SalarioBase)
168     {
169         changes.Add(NewChange(candidate.IdEmpleado, EmployeeHistoryFields.BaseSalary,
170             "B/. " + original.SalarioBase.ToString("0.00"),
171             "B/. " + candidate.SalarioBase.ToString("0.00"),
172             original.SalarioBase.ToString(CultureInfo.InvariantCulture),
173             candidate.SalarioBase.ToString(CultureInfo.InvariantCulture)));
174     }
175     if (!string.Equals(original.Cargo, candidate.Cargo, StringComparison.Ordinal))
176     {
177         changes.Add(NewChange(candidate.IdEmpleado, EmployeeHistoryFields.Position,
178             original.Cargo, candidate.Cargo, original.Cargo, candidate.Cargo));
179     }
180     if (original.IdDepartamento != candidate.IdDepartamento)
```


Services/EmployeeHistoryService.cs (continuacion)

```
181         {
182             changes.Add(NewChange(candidate.IdEmpleado, EmployeeHistoryFields.Department,
183                 original.DepartamentoNombre, employeeRepository.GetDepartmentName(candidate.IdDepartamento),
184                 original.IdDepartamento.ToString(CultureInfo.InvariantCulture),
185                 candidate.IdDepartamento.ToString(CultureInfo.InvariantCulture)));
186         }
187         if (!string.Equals(original.Estado, candidate.Estado, StringComparison.OrdinalIgnoreCase))
188         {
189             changes.Add(NewChange(candidate.IdEmpleado, EmployeeHistoryFields.Status,
190                 original.Estado, candidate.Estado, original.Estado, candidate.Estado));
191         }
192         return changes;
193     }
194
195     private static HistorialEmpleado NewChange(int employeeId, string field,
196         string oldDisplay, string newDisplay, string oldTechnical, string newTechnical)
197     {
198         return new HistorialEmpleado
199         {
200             IdEmpleado = employeeId,
201             CampoModificado = field,
202             ValorAnterior = oldDisplay,
203             ValorNuevo = newDisplay,
204             ValorAnteriorTecnico = oldTechnical,
205             ValorNuevoTecnico = newTechnical
206         };
207     }
208
209     private static Empleado PreserveTrackedValues(Empleado candidate, Empleado original)
210     {
211         return new Empleado
212         {
213             IdEmpleado = candidate.IdEmpleado,
214            Codigo = candidate.Codigo,
215             Nombre = candidate.Nombre,
216             Apellido = candidate.Apellido,
217             Cedula = candidate.Cedula,
218             Cargo = original.Cargo,
219             IdDepartamento = original.IdDepartamento,
220             DepartamentoNombre = original.DepartamentoNombre,
221             SalarioBase = original.SalarioBase,
222             Estado = original.Estado,
223             FechaIngreso = candidate.FechaIngreso,
224             FechaEfectivaLaboral = original.FechaEfectivaLaboral
225         };
226     }
227 }
228
229 public sealed class EmployeeEffectiveDataService
230 {
231     private readonly EmployeeHistoryRepository historyRepository = new EmployeeHistoryRepository();
232     private readonly EmpleadoRepository employeeRepository = new EmpleadoRepository();
233     private readonly EmployeeHistoryService historyService = new EmployeeHistoryService();
234
235     public List<Empleado> GetActiveEmployeesAsOf(DateTime effectiveDate, int? departmentId)
236     {
237         historyService.ApplyDueChanges(DateTime.Today);
238         List<Empleado> source = employeeRepository.GetAllForEffectiveDate(null);
239         List<Empleado> result = new List<Empleado>();
240         for (int i = 0; i < source.Count; i++)
```

Services/EmployeeHistoryService.cs (continuacion)

```
241         {
242             Empleado resolved = ResolveAsOf(source[i], effectiveDate.Date);
243             if (!string.Equals(resolved.Estado, "Activo", StringComparison.OrdinalIgnoreCase)) continue;
244             if (departmentId.HasValue && resolved.IdDepartamento != departmentId.Value) continue;
245             result.Add(resolved);
246         }
247         return result;
248     }
249
250     public Empleado GetEmployeeAsOf(int employeeId, DateTime effectiveDate)
251     {
252         historyService.ApplyDueChanges(DateTime.Today);
253         Empleado employee = employeeRepository.GetById(employeeId);
254         return employee == null ? null : ResolveAsOf(employee, effectiveDate.Date);
255     }
256
257     private Empleado ResolveAsOf(Empleado current, DateTime date)
258     {
259         Empleado result = Clone(current);
260         List<HistorialEmpleado> history = historyRepository.GetByEmployee(current.IdEmpleado);
261
262         for (int i = history.Count - 1; i >= 0; i--)
263         {
264             HistorialEmpleado item = history[i];
265             if (item.Aplicado && item.FechaEfectiva.Date > date)
266                 ApplyValue(result, item.CampoModificado, item.ValorAnteriorTecnico);
267         }
268         for (int i = 0; i < history.Count; i++)
269         {
270             HistorialEmpleado item = history[i];
271             if (!item.Aplicado && item.FechaEfectiva.Date <= date)
272                 ApplyValue(result, item.CampoModificado, item.ValorNuevoTecnico);
273         }
274         result.DepartamentoNombre = employeeRepository.GetDepartmentName(result.IdDepartamento);
275         return result;
276     }
277
278     private static void ApplyValue(Empleado employee, string field, string value)
279     {
280         if (field == EmployeeHistoryFields.BaseSalary)
281             employee.SalarioBase = decimal.Parse(value, CultureInfo.InvariantCulture);
282         else if (field == EmployeeHistoryFields.Position) employee.Cargo = value;
283         else if (field == EmployeeHistoryFields.Department)
284             employee.IdDepartamento = int.Parse(value, CultureInfo.InvariantCulture);
285         else if (field == EmployeeHistoryFields.Status) employee.Estado = value;
286     }
287
288     private static Empleado Clone(Empleado source)
289     {
290         return new Empleado
291         {
292             IdEmpleado = source.IdEmpleado,
293             Codigo = source.Codigo,
294             Nombre = source.Nombre,
295             Apellido = source.Apellido,
296             Cedula = source.Cedula,
297             Cargo = source.Cargo,
298             IdDepartamento = source.IdDepartamento,
299             DepartamentoNombre = source.DepartamentoNombre,
300             SalarioBase = source.SalarioBase,
```

Services/EmployeeHistoryService.cs (continuación)

```
301         Estado = source.Estado,  
302         FechaIngreso = source.FechaIngreso,  
303         FechaEfectivaLaboral = source.FechaEfectivaLaboral  
304     };  
305 }  
306 }  
307 }
```

Services/EmployeePortalService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.IO;
4 using SistemaGestionNomina.Data;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7
8 namespace SistemaGestionNomina.Services
9 {
10     public sealed class EmployeePortalService
11     {
12         private readonly AuthorizationService authorizationService = new AuthorizationService();
13         private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
14         private readonly EmpleadoRepository employeeRepository = new EmpleadoRepository();
15         private readonly AsistenciaRepository attendanceRepository = new AsistenciaRepository();
16         private readonly ComprobanteRepository payslipRepository = new ComprobanteRepository();
17         private readonly AuditTrailService auditTrailService = new AuditTrailService();
18         private readonly PdfExportService pdfExportService = new PdfExportService();
19
20         public EmployeePortalDashboardViewModel GetDashboard()
21         {
22             authorizationService.DemandPermission(Permissions.PortalView);
23             EmployeeProfileViewModel profile = GetProfileCore();
24             DateTime today = DateTime.Today;
25             List<Asistencia> monthAttendance = attendanceRepository.GetByEmployee(
26                 profile.IdEmpleado, new DateTime(today.Year, today.Month, 1), today, string.Empty);
27             List<Asistencia> recent = new List<Asistencia>();
28             for (int i = 0; i < monthAttendance.Count && i < 5; i++) recent.Add(monthAttendance[i]);
29
30             List<Comprobante> payslips = payslipRepository.GetByEmployee(profile.IdEmpleado, string.Empty);
31             auditTrailService.RegistrarAccion("Portal trabajador", "Consultar inicio", "Empleado=" + profile.IdEmpleado);
32             return new EmployeePortalDashboardViewModel
33             {
34                 Profile = profile,
35                 LatestPayslip = payslips.Count > 0 ? payslips[0] : null,
36                 RecentAttendance = recent,
37                 AttendanceSummary = SummarizeAttendance(monthAttendance)
38             };
39         }
40
41         public EmployeeProfileViewModel GetMyProfile()
42         {
43             authorizationService.DemandPermission(Permissions.OwnProfileView);
44             EmployeeProfileViewModel profile = GetProfileCore();
45             auditTrailService.RegistrarAccion("Portal trabajador", "Consultar perfil", "Empleado=" + profile.IdEmpleado);
46             return profile;
47         }
48
49         public List<Asistencia> GetMyAttendance(DateTime? start, DateTime? end, string status)
50         {
51             authorizationService.DemandPermission(Permissions.OwnAttendanceView);
52             if (start.HasValue && end.HasValue && start.Value.Date > end.Value.Date)
53             {
54                 throw new ArgumentException("La fecha inicial no puede ser posterior a la fecha final.");
55             }
56
57             string normalizedStatus = NormalizeAttendanceStatus(status);
58             int employeeId = employeeScopeService.RequireCurrentEmployeeId();
59             List<Asistencia> items = attendanceRepository.GetByEmployee(employeeId, start, end, normalizedStatus);
60             auditTrailService.RegistrarAccion("Portal trabajador", "Consultar asistencia",
```

Services/EmployeePortalService.cs (continuacion)

```
61         "Empleado=" + employeeId + ", Registros=" + items.Count);
62     return items;
63 }
64
65 public List<Comprobante> GetMyPayslips(string search)
66 {
67     authorizationService.DemandPermission(Permissions.OwnPayslipsView);
68     int employeeId = employeeScopeService.RequireCurrentEmployeeId();
69     List<Comprobante> items = payslipRepository.GetByEmployee(employeeId, (search ?? string.Empty).Trim());
70     auditTrailService.RegistrarAccion("Portal trabajador", "Consultar comprobantes",
71         "Empleado=" + employeeId + ", Registros=" + items.Count);
72     return items;
73 }
74
75 public Comprobante GetMyPayslipById(int payslipId)
76 {
77     authorizationService.DemandPermission(Permissions.OwnPayslipsView);
78     if (payslipId <= 0) throw new ArgumentException("Seleccione un comprobante valido.");
79
80     int employeeId = employeeScopeService.RequireCurrentEmployeeId();
81     Comprobante payslip = payslipRepository.GetByIdAndEmployee(payslipId, employeeId);
82     if (payslip == null)
83     {
84         auditTrailService.RegistrarAccion("Portal trabajador", "Acceso rechazado a comprobante",
85             "IdComprobante=" + payslipId);
86         throw new UnauthorizedAccessException("El comprobante no esta disponible para su cuenta.");
87     }
88
89     auditTrailService.RegistrarAccion("Portal trabajador", "Consultar comprobante",
90         "IdComprobante=" + payslipId);
91     return payslip;
92 }
93
94 public string DownloadMyPayslipPdf(int payslipId)
95 {
96     return DownloadMyPayslipPdfCore(payslipId, null);
97 }
98
99 internal string DownloadMyPayslipPdf(int payslipId, string destinationPath)
100 {
101     return DownloadMyPayslipPdfCore(payslipId, destinationPath);
102 }
103
104 public string GetExistingMyPayslipPdf(int payslipId)
105 {
106     Comprobante payslip = GetMyPayslipById(payslipId);
107     if (string.IsNullOrEmpty(payslip.RutaPdf) || !File.Exists(payslip.RutaPdf))
108     {
109         throw new FileNotFoundException("El comprobante no tiene un PDF existente. Descarguelo primero.");
110     }
111
112     auditTrailService.RegistrarAccion("Portal trabajador", "Abrir PDF propio",
113         "IdComprobante=" + payslipId);
114     return Path.GetFullPath(payslip.RutaPdf);
115 }
116
117 public Comprobante GetMyPayslipForPrint(int payslipId)
118 {
119     Comprobante payslip = GetMyPayslipById(payslipId);
120     auditTrailService.RegistrarAccion("Portal trabajador", "Imprimir comprobante propio",
```

Services/EmployeePortalService.cs (continuacion)

```
121         "IdComprobante=" + payslipId);
122     return payslip;
123 }
124
125 public static EmployeeAttendanceSummaryViewModel SummarizeAttendance(IEnumerable<Asistencia> attendance)
126 {
127     EmployeeAttendanceSummaryViewModel summary = new EmployeeAttendanceSummaryViewModel();
128     if (attendance == null) return summary;
129
130     foreach (Asistencia item in attendance)
131     {
132         if (string.Equals(item.Estado, "Puntual", StringComparison.OrdinalIgnoreCase)) summary.Puntuales++;
133         else if (string.Equals(item.Estado, "Tardanza", StringComparison.OrdinalIgnoreCase)) summary.Tardanzas++;
134         else if (string.Equals(item.Estado, "Falta", StringComparison.OrdinalIgnoreCase)) summary.Faltas++;
135         else if (string.Equals(item.Estado, "Permiso", StringComparison.OrdinalIgnoreCase)) summary.Permisos++;
136
137         summary.HorasTrabajadas += item.HorasTrabajadas;
138         if (item.HorasTrabajadas > 8) summary.HorasExtra += item.HorasTrabajadas - 8;
139     }
140
141     return summary;
142 }
143
144 private EmployeeProfileViewModel GetProfileCore()
145 {
146     int employeeId = employeeScopeService.RequireCurrentEmployeeId();
147     Empleado employee = employeeRepository.GetById(employeeId);
148     employeeScopeService.DemandCurrentEmployee(employee.IdEmpleado);
149     return new EmployeeProfileViewModel
150     {
151         IdEmpleado = employee.IdEmpleado,
152         Codigo = employee.Codigo,
153         NombreCompleto = employee.NombreCompleto,
154         Cedula = employee.Cedula,
155         Cargo = employee.Cargo,
156         Departamento = employee.DepartamentoNombre,
157         SalarioBase = employee.SalarioBase,
158         Estado = employee.Estado,
159         FechaIngreso = employee.FechaIngreso
160     };
161 }
162
163 private string DownloadMyPayslipPdfCore(int payslipId, string destinationPath)
164 {
165     authorizationService.DemandPermission(Permissions.OwnPayslipsDownload);
166     Comprobante payslip = GetMyPayslipById(payslipId);
167     int employeeId = employeeScopeService.RequireCurrentEmployeeId();
168     string path = destinationPath == null
169         ? pdfExportService.ExportarComprobantePersonal(payslip)
170         : pdfExportService.ExportarComprobantePersonal(payslip, destinationPath);
171     if (string.IsNullOrEmpty(path)) return string.Empty;
172
173     if (!payslipRepository.UpdateRutaPdfForEmployee(payslipId, employeeId, path))
174     {
175         throw new InvalidOperationException("No se pudo asociar el PDF al comprobante.");
176     }
177
178     auditTrailService.RegistrarAccion("Portal trabajador", "Descargar PDF propio",
179         "IdComprobante=" + payslipId + ", Archivo=" + Path.GetFileName(path));
180     return path;
181 }
```

Services/EmployeePortalService.cs (continuacion)

```
181     }
182
183     private static string NormalizeAttendanceStatus(string status)
184     {
185         string normalized = (status ?? string.Empty).Trim();
186         if (normalized.Length == 0 || string.Equals(normalized, "Todos", StringComparison.OrdinalIgnoreCase))
187         {
188             return string.Empty;
189         }
190
191         string[] allowed = { "Puntual", "Tardanza", "Falta", "Permiso" };
192         for (int i = 0; i < allowed.Length; i++)
193         {
194             if (string.Equals(normalized, allowed[i], StringComparison.OrdinalIgnoreCase)) return allowed[i];
195         }
196
197         throw new ArgumentException("El estado de asistencia no es valido.");
198     }
199 }
200 }
```

Services/ExcelExportService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using ClosedXML.Excel;
4 using SistemaGestionNomina.Helpers;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7
8 namespace SistemaGestionNomina.Services
9 {
10     public class ExcelExportService
11     {
12         private readonly AuthorizationService authorizationService = new AuthorizationService();
13         private readonly AuditTrailService auditTrailService = new AuditTrailService();
14
15         public string ExportarEmpleados(List<Empleado> empleados)
16         {
17             authorizationService.DemandAny(Permissions.EmployeesExport, Permissions.ReportsPersonal);
18             string path = BuildPath("Empleados");
19             if (string.IsNullOrEmpty(path)) return string.Empty;
20
21             using (XLWorkbook workbook = new XLWorkbook())
22             {
23                 IXLWorksheet sheet = workbook.Worksheets.Add("Empleados");
24                 WriteHeader(sheet, "Código", "Nombre", "Cédula", "Cargo", "Departamento", "Salario Base", "Estado");
25                 for (int i = 0; i < empleados.Count; i++)
26                 {
27                     Empleado e = empleados[i];
28                     int row = i + 2;
29                     sheet.Cell(row, 1).Value = e.Codigo;
30                     sheet.Cell(row, 2).Value = e.NombreCompleto;
31                     sheet.Cell(row, 3).Value = e.Cedula;
32                     sheet.Cell(row, 4).Value = e.Cargo;
33                     sheet.Cell(row, 5).Value = e.DepartamentoNombre;
34                     sheet.Cell(row, 6).Value = e.SalarioBase;
35                     sheet.Cell(row, 7).Value = e.Estado;
36                 }
37                 Finish(sheet);
38                 workbook.SaveAs(path);
39             }
40
41             auditTrailService.RegistrarAccion("Exportaciones", "Excel empleados", System.IO.Path.GetFileName(path));
42             return path;
43         }
44
45         public string ExportarAsistencias(List<Asistencia> asistencias)
46         {
47             authorizationService.DemandPermission(Permissions.AttendanceExport);
48             string path = BuildPath("Asistencia");
49             if (string.IsNullOrEmpty(path)) return string.Empty;
50
51             using (XLWorkbook workbook = new XLWorkbook())
52             {
53                 IXLWorksheet sheet = workbook.Worksheets.Add("Asistencia");
54                 WriteHeader(sheet, "Empleado", "Fecha", "Entrada", "Salida", "Horas", "Estado");
55                 for (int i = 0; i < asistencias.Count; i++)
56                 {
57                     Asistencia a = asistencias[i];
58                     int row = i + 2;
59                     sheet.Cell(row, 1).Value = a.EmpleadoNombre;
60                     sheet.Cell(row, 2).Value = a.Fecha.ToString("dd/MM/yyyy");
```


Services/ExcelExportService.cs (continuacion)

```
61         sheet.Cell(row, 3).Value = a.HoraEntrada.HasValue ? a.HoraEntrada.Value.ToString(@"hh:mm") : "--";
62         sheet.Cell(row, 4).Value = a.HoraSalida.HasValue ? a.HoraSalida.Value.ToString(@"hh:mm") : "--";
63         sheet.Cell(row, 5).Value = a.HorasTrabajadas;
64         sheet.Cell(row, 6).Value = a.Estado;
65     }
66     Finish(sheet);
67     workbook.SaveAs(path);
68 }
69
70 auditTrailService.RegistrarAccion("Exportaciones", "Excel asistencia", System.IO.Path.GetFileName(path));
71 return path;
72 }
73
74 public string ExportarHistorialEmpleados(List<HistorialEmpleado> history)
75 {
76     authorizationService.DemandPermission(Permissions.EmployeeHistoryExport);
77     string path = BuildPath("Historial_Laboral");
78     if (string.IsNullOrEmpty(path)) return string.Empty;
79     using (XLWorkbook workbook = new XLWorkbook())
80     {
81         IXLWorksheet sheet = workbook.Worksheets.Add("Historial laboral");
82         WriteHeader(sheet, "Código", "Empleado", "Campo", "Valor anterior", "Valor nuevo",
83             "Fecha efectiva", "Registrado", "Usuario", "Motivo", "Aplicado");
84         for (int i = 0; i < history.Count; i++)
85         {
86             HistorialEmpleado item = history[i];
87             int row = i + 2;
88             sheet.Cell(row, 1).Value = item.CodigoEmpleado;
89             sheet.Cell(row, 2).Value = item.EmpleadoNombre;
90             sheet.Cell(row, 3).Value = item.CampoModificado;
91             sheet.Cell(row, 4).Value = item.ValorAnterior;
92             sheet.Cell(row, 5).Value = item.ValorNuevo;
93             sheet.Cell(row, 6).Value = item.FechaEfectiva.ToString("dd/MM/yyyy");
94             sheet.Cell(row, 7).Value = item.FechaCambio.ToString("dd/MM/yyyy HH:mm");
95             sheet.Cell(row, 8).Value = item.UsuarioResponsable;
96             sheet.Cell(row, 9).Value = item.Motivo;
97             sheet.Cell(row, 10).Value = item.Aplicado ? "Sí" : "Programado";
98         }
99         Finish(sheet);
100        workbook.SaveAs(path);
101    }
102    auditTrailService.RegistrarAccion("Exportaciones", "Excel historial laboral",
103        System.IO.Path.GetFileName(path));
104    return path;
105 }
106
107 public string ExportarAusencias(List<SolicitudAusencia> requests)
108 {
109     authorizationService.DemandPermission(Permissions.AbsencesExport);
110     string path = BuildPath("Solicitudes_Ausencia");
111     if (string.IsNullOrEmpty(path)) return string.Empty;
112     using (XLWorkbook workbook = new XLWorkbook())
113     {
114         IXLWorksheet sheet = workbook.Worksheets.Add("Ausencias");
115         WriteHeader(sheet, "Código", "Empleado", "Departamento", "Tipo", "Desde", "Hasta",
116             "Estado", "Solicitante", "Resolución", "Motivo");
117         for (int i = 0; i < requests.Count; i++)
118         {
119             SolicitudAusencia item = requests[i];
120             int row = i + 2;
```

Services/ExcelExportService.cs (continuacion)

```
121         sheet.Cell(row, 1).Value = item.CodigoEmpleado;
122         sheet.Cell(row, 2).Value = item.EmpleadoNombre;
123         sheet.Cell(row, 3).Value = item.DepartamentoNombre;
124         sheet.Cell(row, 4).Value = item.Tipo;
125         sheet.Cell(row, 5).Value = item.FechaInicio.ToString("dd/MM/yyyy");
126         sheet.Cell(row, 6).Value = item.FechaFin.ToString("dd/MM/yyyy");
127         sheet.Cell(row, 7).Value = item.Estado;
128         sheet.Cell(row, 8).Value = item.UsuarioSolicitante;
129         sheet.Cell(row, 9).Value = item.ObservacionResolucion;
130         sheet.Cell(row, 10).Value = item.Motivo;
131     }
132     Finish(sheet);
133     workbook.SaveAs(path);
134 }
135 auditTrailService.RegistrarAccion("Exportaciones", "Excel ausencias",
136     System.IO.Path.GetFileName(path));
137 return path;
138 }
139
140 public string ExportarNomina(Nomina nomina)
141 {
142     authorizationService.DemandPermission(Permissions.PayrollExport);
143     string path = BuildPath("Nomina");
144     if (string.IsNullOrEmpty(path)) return string.Empty;
145
146     using (XLWorkbook workbook = new XLWorkbook())
147     {
148         IXLWorksheet sheet = workbook.Worksheets.Add("Nómina");
149         sheet.Cell(1, 1).Value = nomina.PeriodoNombre;
150         sheet.Range(1, 1, 1, 9).Merge().Style.Font.Bold = true;
151         WriteHeader(sheet, 3, "Código", "Empleado", "Departamento", "Sueldo Base", "Bonos", "Horas Extra", "Monto H.E.", "Deducciones", "Neto");
152         for (int i = 0; i < nomina.Detalles.Count; i++)
153         {
154             NominaDetalle d = nomina.Detalles[i];
155             int row = i + 4;
156             sheet.Cell(row, 1).Value = d.CodigoEmpleado;
157             sheet.Cell(row, 2).Value = d.EmpleadoNombre;
158             sheet.Cell(row, 3).Value = d.Departamento;
159             sheet.Cell(row, 4).Value = d.SueldoBase;
160             sheet.Cell(row, 5).Value = d.Bonos;
161             sheet.Cell(row, 6).Value = d.HorasExtra;
162             sheet.Cell(row, 7).Value = d.MontoHorasExtra;
163             sheet.Cell(row, 8).Value = d.TotalDeducciones;
164             sheet.Cell(row, 9).Value = d.NetoPagar;
165         }
166         int totalRow = nomina.Detalles.Count + 5;
167         sheet.Cell(totalRow, 7).Value = "Totales";
168         sheet.Cell(totalRow, 8).Value = nomina.TotalDeducciones;
169         sheet.Cell(totalRow, 9).Value = nomina.TotalNeto;
170         Finish(sheet);
171         workbook.SaveAs(path);
172     }
173
174     auditTrailService.RegistrarAccion("Exportaciones", "Excel nómina", System.IO.Path.GetFileName(path));
175     return path;
176 }
177
178 public string ExportarComprobantes(List<Comprobante> comprobantes)
179 {
180     authorizationService.DemandPermission(Permissions.PayslipsExport);
```

Services/ExcelExportService.cs (continuacion)

```
181         string path = BuildPath("Comprobantes");
182         if (string.IsNullOrEmpty(path)) return string.Empty;
183
184         using (XLWorkbook workbook = new XLWorkbook())
185         {
186             IXLWorksheet sheet = workbook.Worksheets.Add("Comprobantes");
187             WriteHeader(sheet, "Número", "Empleado", "Periodo", "Fecha", "Ingresos", "Deducciones", "Neto");
188             for (int i = 0; i < comprobantes.Count; i++)
189             {
190                 Comprobante c = comprobantes[i];
191                 int row = i + 2;
192                 sheet.Cell(row, 1).Value = c.NumeroComprobante;
193                 sheet.Cell(row, 2).Value = c.EmpleadoNombre;
194                 sheet.Cell(row, 3).Value = c.PeriodoNombre;
195                 sheet.Cell(row, 4).Value = c.FechaGeneracion.ToString("dd/MM/yyyy HH:mm");
196                 sheet.Cell(row, 5).Value = c.TotalIngresos;
197                 sheet.Cell(row, 6).Value = c.TotalDeducciones;
198                 sheet.Cell(row, 7).Value = c.Netopagar;
199             }
200             Finish(sheet);
201             workbook.SaveAs(path);
202         }
203
204         auditTrailService.RegistrarAccion("Exportaciones", "Excel comprobantes", System.IO.Path.GetFileName(path));
205         return path;
206     }
207
208     public string ExportarReportes(List<ReporteGenerado> reportes)
209     {
210         authorizationService.DemandPermission(Permissions.ReportsExport);
211         string path = BuildPath("Reportes");
212         if (string.IsNullOrEmpty(path)) return string.Empty;
213
214         using (XLWorkbook workbook = new XLWorkbook())
215         {
216             IXLWorksheet sheet = workbook.Worksheets.Add("Reportes");
217             WriteHeader(sheet, "Nombre", "Tipo", "Generado por", "Fecha", "Ruta");
218             for (int i = 0; i < reportes.Count; i++)
219             {
220                 ReporteGenerado r = reportes[i];
221                 int row = i + 2;
222                 sheet.Cell(row, 1).Value = r.NombreReporte;
223                 sheet.Cell(row, 2).Value = r.Tipo;
224                 sheet.Cell(row, 3).Value = r.GeneradoPor;
225                 sheet.Cell(row, 4).Value = r.FechaGeneracion.ToString("dd/MM/yyyy HH:mm");
226                 sheet.Cell(row, 5).Value = r.RutaArchivo;
227             }
228             Finish(sheet);
229             workbook.SaveAs(path);
230         }
231
232         auditTrailService.RegistrarAccion("Exportaciones", "Excel reportes", System.IO.Path.GetFileName(path));
233         return path;
234     }
235
236     private static string BuildPath(string prefix)
237     {
238         // El usuario elige la ubicacion final; la aplicacion no guarda exportaciones en rutas fijas.
239         return PathHelper.RequestExportPath(prefix, ".xlsx", "Libro de Excel (*.xlsx)|*.xlsx");
240     }
```

Services/ExcelExportService.cs (continuacion)

```
241
242     private static void WriteHeader(IXLWorksheet sheet, params string[] headers)
243     {
244         WriteHeader(sheet, 1, headers);
245     }
246
247     private static void WriteHeader(IXLWorksheet sheet, int row, params string[] headers)
248     {
249         for (int i = 0; i < headers.Length; i++)
250         {
251             sheet.Cell(row, i + 1).Value = headers[i];
252             sheet.Cell(row, i + 1).Style.Font.Bold = true;
253             sheet.Cell(row, i + 1).Style.Fill.BackgroundColor = XLColor.FromHtml("#A78BFA");
254         }
255     }
256
257     private static void Finish(IXLWorksheet sheet)
258     {
259         sheet.Columns().AdjustToContents();
260     }
261 }
262 }
```

Services/PayrollLifecycleService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Data.SQLite;
4 using SistemaGestionNomina.Data;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7
8 namespace SistemaGestionNomina.Services
9 {
10     public class PayrollLifecycleService
11     {
12         private readonly NominaVersionRepository versionRepository = new NominaVersionRepository();
13         private readonly AuditRepository auditRepository = new AuditRepository();
14         private readonly AuthorizationService authorizationService = new AuthorizationService();
15         private readonly PayrollPeriodPolicyService periodPolicyService = new PayrollPeriodPolicyService();
16
17         public int Confirmar(Nomina nominaCalculada, DateTime fechaInicio, DateTime fechaFin)
18         {
19             authorizationService.DemandPermission(Permissions.PayrollConfirm);
20             ValidateCalculatedPayroll(nominaCalculada);
21             periodPolicyService.VerificarFechasAbiertas(fechaInicio, fechaFin);
22
23             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
24             using (SQLiteTransaction transaction = connection.BeginTransaction())
25             {
26                 try
27                 {
28                     // Periodo, nomina, detalles, comprobantes y auditoria deben persistirse juntos.
29                     int idNomina = CreateConfirmedPayroll(connection, transaction, nominaCalculada,
30                                                         fechaInicio.Date, fechaFin.Date, CurrentUsername(), DateTime.Now);
31                     transaction.Commit();
32                     return idNomina;
33                 }
34                 catch
35                 {
36                     transaction.Rollback();
37                     throw;
38                 }
39             }
40         }
41
42         public void Pagar(int idNomina)
43         {
44             authorizationService.DemandPermission(Permissions.PayrollPay);
45             string user = CurrentUsername();
46             DateTime now = DateTime.Now;
47
48             using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
49             using (SQLiteTransaction transaction = connection.BeginTransaction())
50             {
51                 try
52                 {
53                     Nomina payroll = GetPayroll(connection, transaction, idNomina);
54                     if (payroll == null) throw new InvalidOperationException("La nómina no existe.");
55                     PayrollStateMachine.DemandTransition(payroll.Estado, PayrollStates.Paid);
56
57                     // La condicion por estado evita registrar un pago duplicado ante acciones simultaneas.
58                     using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Nominas
59                                                                     SET Estado = @nuevo, FechaPago = @fecha, PagadaPor = @usuario
60                                                                     WHERE IdNomina = @id AND Estado = @anterior;", connection, transaction))
```

Services/PayrollLifecycleService.cs (continuacion)

```
61         {
62             command.Parameters.AddWithValue("@nuevo", PayrollStates.Paid);
63             command.Parameters.AddWithValue("@fecha", now.ToString("yyyy-MM-dd HH:mm:ss"));
64             command.Parameters.AddWithValue("@usuario", user);
65             command.Parameters.AddWithValue("@id", idNomina);
66             command.Parameters.AddWithValue("@anterior", PayrollStates.Confirmed);
67             if (command.ExecuteNonQuery() != 1)
68                 throw new InvalidOperationException("La nómina cambió de estado antes de registrar el pago.");
69         }
70         using (SQLiteCommand command = new SQLiteCommand(@"UPDATE PeriodosNomina
71             SET Estado = @estado WHERE IdPeriodo = @id;", connection, transaction))
72         {
73             command.Parameters.AddWithValue("@estado", PayrollPeriodStates.Paid);
74             command.Parameters.AddWithValue("@id", payroll.IdPeriodo);
75             command.ExecuteNonQuery();
76         }
77         AddAudit(connection, transaction, user, "Nómina", "Pagar", "IdNomina=" + idNomina, now);
78         transaction.Commit();
79     }
80     catch
81     {
82         transaction.Rollback();
83         throw;
84     }
85 }
86 }
87
88 public void Anular(int idNomina, string motivo)
89 {
90     authorizationService.DemandPermission(Permissions.PayrollAnnul);
91     if (string.IsNullOrEmpty(motivo))
92         throw new InvalidOperationException("Debe indicar un motivo de anulación.");
93
94     string user = CurrentUsername();
95     DateTime now = DateTime.Now;
96     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
97     using (SQLiteTransaction transaction = connection.BeginTransaction())
98     {
99         try
100         {
101             Nomina payroll = GetPayroll(connection, transaction, idNomina);
102             if (payroll == null) throw new InvalidOperationException("La nómina no existe.");
103             PayrollStateMachine.DemandTransition(payroll.Estado, PayrollStates.Anulled);
104
105             // La anulacion conserva la nomina original para mantener la trazabilidad financiera.
106             using (SQLiteCommand command = new SQLiteCommand(@"UPDATE Nominas
107                 SET Estado = @nuevo, FechaAnulacion = @fecha, AnuladaPor = @usuario,
108                     MotivoAnulacion = @motivo
109                 WHERE IdNomina = @id AND Estado = @anterior;", connection, transaction))
110             {
111                 command.Parameters.AddWithValue("@nuevo", PayrollStates.Anulled);
112                 command.Parameters.AddWithValue("@fecha", now.ToString("yyyy-MM-dd HH:mm:ss"));
113                 command.Parameters.AddWithValue("@usuario", user);
114                 command.Parameters.AddWithValue("@motivo", motivo.Trim());
115                 command.Parameters.AddWithValue("@id", idNomina);
116                 command.Parameters.AddWithValue("@anterior", payroll.Estado);
117                 if (command.ExecuteNonQuery() != 1)
118                     throw new InvalidOperationException("La nómina cambió de estado antes de anularse.");
119             }
120             using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO NominaVersiones
```

Services/PayrollLifecycleService.cs (continuacion)

```

121         (IdNominaOriginal, IdNominaNueva, MotivoCambio, UsuarioResponsable, FechaCambio)
122         VALUES (@original, NULL, @motivo, @usuario, @fecha);", connection, transaction))
123     {
124         command.Parameters.AddWithValue("@original", idNomina);
125         command.Parameters.AddWithValue("@motivo", motivo.Trim());
126         command.Parameters.AddWithValue("@usuario", user);
127         command.Parameters.AddWithValue("@fecha", now.ToString("yyyy-MM-dd HH:mm:ss"));
128         command.ExecuteNonQuery();
129     }
130     using (SQLiteCommand command = new SQLiteCommand(@"UPDATE PeriodosNomina
131     SET Estado = @estado, Cerrado = 0, FechaCierre = NULL, CerradoPor = NULL
132     WHERE IdPeriodo = @id;", connection, transaction))
133     {
134         command.Parameters.AddWithValue("@estado", PayrollPeriodStates.Reopened);
135         command.Parameters.AddWithValue("@id", payroll.IdPeriodo);
136         command.ExecuteNonQuery();
137     }
138     AddAudit(connection, transaction, user, "Nómina", "Anular",
139         "IdNomina=" + idNomina + ", Motivo=" + motivo.Trim(), now);
140     transaction.Commit();
141 }
142 catch
143 {
144     transaction.Rollback();
145     throw;
146 }
147 }
148 }
149
150 public int Recalcular(int idNominaAnulada, DateTime fechaInicio, DateTime fechaFin,
151     Nomina nominaRecalculada)
152 {
153     authorizationService.DemandPermission(Permissions.PayrollRecalculate);
154     ValidateCalculatedPayroll(nominaRecalculada);
155     periodPolicyService.VerificarFechasAbiertas(fechaInicio, fechaFin);
156
157     string user = CurrentUsername();
158     DateTime now = DateTime.Now;
159     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
160     using (SQLiteTransaction transaction = connection.BeginTransaction())
161     {
162         try
163         {
164             Nomina original = GetPayroll(connection, transaction, idNominaAnulada);
165             if (original == null) throw new InvalidOperationException("La nómina original no existe.");
166             if (!string.Equals(original.Estado, PayrollStates.Anulled, StringComparison.OrdinalIgnoreCase))
167                 throw new InvalidOperationException("Solo se puede recalcular a partir de una nómina anulada.");
168
169             int pendingVersionId = GetPendingVersionId(connection, transaction, idNominaAnulada);
170             if (pendingVersionId <= 0)
171                 throw new InvalidOperationException("No hay una versión pendiente para la nómina anulada.");
172
173             int newPayrollId = CreateConfirmedPayroll(connection, transaction, nominaRecalculada,
174                 fechaInicio.Date, fechaFin.Date, user, now);
175
176             // La nueva nomina queda enlazada con la anulada como una version corregida.
177             using (SQLiteCommand command = new SQLiteCommand(@"UPDATE NominaVersiones
178             SET IdNominaNueva = @nueva
179             WHERE IdVersion = @id AND IdNominaNueva IS NULL;", connection, transaction))
180             {

```

Services/PayrollLifecycleService.cs (continuacion)

```
181         command.Parameters.AddWithValue("@nueva", newPayrollId);
182         command.Parameters.AddWithValue("@id", pendingVersionId);
183         if (command.ExecuteNonQuery() != 1)
184             throw new InvalidOperationException("No se pudo vincular la nueva versión de nómina.");
185     }
186
187     AddAudit(connection, transaction, user, "Nómina", "Recalcular",
188         "IdNominaOriginal=" + idNominaAnulada + ", IdNominaNueva=" + newPayrollId, now);
189     transaction.Commit();
190     return newPayrollId;
191 }
192 catch
193 {
194     transaction.Rollback();
195     throw;
196 }
197 }
198 }
199
200 public List<NominaVersion> ObtenerHistorial(int idNomina)
201 {
202     authorizationService.DemandPermission(Permissions.PayrollHistoryView);
203     return versionRepository.GetHistorialChain(idNomina);
204 }
205
206 private int CreateConfirmedPayroll(SQLiteConnection connection, SQLiteTransaction transaction,
207     Nomina calculated, DateTime start, DateTime end, string user, DateTime now)
208 {
209     int periodId;
210     using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO PeriodosNomina
211         (Nombre, FechaInicio, FechaFin, Estado, Cerrado, FechaCierre, CerradoPor)
212         VALUES (@nombre, @inicio, @fin, @estado, 1, @cierre, @usuario);
213         SELECT last_insert_rowid();", connection, transaction))
214     {
215         command.Parameters.AddWithValue("@nombre", calculated.PeriodoNombre);
216         command.Parameters.AddWithValue("@inicio", start.ToString("yyyy-MM-dd"));
217         command.Parameters.AddWithValue("@fin", end.ToString("yyyy-MM-dd"));
218         command.Parameters.AddWithValue("@estado", PayrollPeriodStates.Confirmed);
219         command.Parameters.AddWithValue("@cierre", now.ToString("yyyy-MM-dd HH:mm:ss"));
220         command.Parameters.AddWithValue("@usuario", user);
221         periodId = Convert.ToInt32(command.ExecuteScalar());
222     }
223
224     int payrollId;
225     using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Nominas
226         (IdPeriodo, FechaCalculo, TotalIngresos, TotalDeducciones, TotalNeto, Estado,
227         FechaConfirmacion, ConfirmadaPor)
228         VALUES (@periodo, @fecha, @ingresos, @deducciones, @neto, @estado,
229         @confirmacion, @usuario);
230         SELECT last_insert_rowid();", connection, transaction))
231     {
232         command.Parameters.AddWithValue("@periodo", periodId);
233         command.Parameters.AddWithValue("@fecha", calculated.FechaCalculo.ToString("yyyy-MM-dd HH:mm:ss"));
234         command.Parameters.AddWithValue("@ingresos", calculated.TotalIngresos);
235         command.Parameters.AddWithValue("@deducciones", calculated.TotalDeducciones);
236         command.Parameters.AddWithValue("@neto", calculated.TotalNeto);
237         command.Parameters.AddWithValue("@estado", PayrollStates.Confirmed);
238         command.Parameters.AddWithValue("@confirmacion", now.ToString("yyyy-MM-dd HH:mm:ss"));
239         command.Parameters.AddWithValue("@usuario", user);
240         payrollId = Convert.ToInt32(command.ExecuteScalar());
```


Services/PayrollLifecycleService.cs (continuacion)

```

241     }
242
243     for (int i = 0; i < calculated.Detalles.Count; i++)
244     {
245         NominaDetalle detail = calculated.Detalles[i];
246         // El snapshot evita que cambios futuros del empleado alteren comprobantes historicos.
247         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO NominaDetalle
248             (IdNomina, IdEmpleado, SueldoBase, Bonos, HorasExtra, MontoHorasExtra,
249             TotalIngresos, TotalDeducciones, NetoPagar, CódigoEmpleadoSnapshot,
250             NombreEmpleadoSnapshot, CargoEmpleadoSnapshot, DepartamentoSnapshot)
251             VALUES (@nomina, @empleado, @sueldo, @bonos, @horas, @montoHoras,
252             @ingresos, @deducciones, @neto, @codigo, @nombre, @cargo, @departamento);",
253             connection, transaction))
254         {
255             command.Parameters.AddWithValue("@nomina", payrollId);
256             command.Parameters.AddWithValue("@empleado", detail.IdEmpleado);
257             command.Parameters.AddWithValue("@sueldo", detail.SueldoBase);
258             command.Parameters.AddWithValue("@bonos", detail.Bonos);
259             command.Parameters.AddWithValue("@horas", detail.HorasExtra);
260             command.Parameters.AddWithValue("@montoHoras", detail.MontoHorasExtra);
261             command.Parameters.AddWithValue("@ingresos", detail.TotalIngresos);
262             command.Parameters.AddWithValue("@deducciones", detail.TotalDeducciones);
263             command.Parameters.AddWithValue("@neto", detail.NetoPagar);
264             command.Parameters.AddWithValue("@codigo", detail.CódigoEmpleado ?? string.Empty);
265             command.Parameters.AddWithValue("@nombre", detail.EmpleadoNombre ?? string.Empty);
266             command.Parameters.AddWithValue("@cargo", detail.CargoEmpleado ?? string.Empty);
267             command.Parameters.AddWithValue("@departamento", detail.Departamento ?? string.Empty);
268             command.ExecuteNonQuery();
269         }
270         using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO Comprobantes
271             (IdNomina, IdEmpleado, NumeroComprobante, FechaGeneracion, RutaPdf)
272             VALUES (@nomina, @empleado, @numero, @fecha, '');", connection, transaction))
273         {
274             command.Parameters.AddWithValue("@nomina", payrollId);
275             command.Parameters.AddWithValue("@empleado", detail.IdEmpleado);
276             command.Parameters.AddWithValue("@numero", "COMP-" + payrollId.ToString("0000") + "-" + detail.IdEmpleado.ToString("0000"));
277             command.Parameters.AddWithValue("@fecha", now.ToString("yyyy-MM-dd HH:mm:ss"));
278             command.ExecuteNonQuery();
279         }
280     }
281
282     AddAudit(connection, transaction, user, "Nómina", "Confirmar",
283         "IdNomina=" + payrollId + ", Periodo=" + periodId + ", Empleados=" + calculated.Detalles.Count, now);
284     AddAudit(connection, transaction, user, "Comprobantes", "Generar",
285         "IdNomina=" + payrollId + ", Cantidad=" + calculated.Detalles.Count, now);
286     return payrollId;
287 }
288
289 private static Nomina GetPayroll(SQLiteConnection connection, SQLiteTransaction transaction, int payrollId)
290 {
291     using (SQLiteCommand command = new SQLiteCommand(@"SELECT n.*, p.Nombre AS PeriodoNombre
292         FROM Nominas n INNER JOIN PeriodosNomina p ON p.IdPeriodo = n.IdPeriodo
293         WHERE n.IdNomina = @id LIMIT 1;", connection, transaction))
294     {
295         command.Parameters.AddWithValue("@id", payrollId);
296         using (SQLiteDataReader reader = command.ExecuteReader())
297         {
298             if (!reader.Read()) return null;
299             return new Nomina
300             {

```

Services/PayrollLifecycleService.cs (continuacion)

```
301         IdNomina = Convert.ToInt32(reader["IdNomina"]),
302         IdPeriodo = Convert.ToInt32(reader["IdPeriodo"]),
303         PeriodoNombre = Convert.ToString(reader["PeriodoNombre"]),
304         FechaCalculo = DateTime.Parse(Convert.ToString(reader["FechaCalculo"])),
305         TotalIngresos = Convert.ToDecimal(reader["TotalIngresos"]),
306         TotalDeducciones = Convert.ToDecimal(reader["TotalDeducciones"]),
307         TotalNeto = Convert.ToDecimal(reader["TotalNeto"]),
308         Estado = Convert.ToString(reader["Estado"]);
309     };
310 }
311 }
312 }
313
314 private static int GetPendingVersionId(SQLiteConnection connection, SQLiteTransaction transaction,
315     int originalPayrollId)
316 {
317     using (SQLiteCommand command = new SQLiteCommand(@"SELECT IdVersion FROM NominaVersiones
318         WHERE IdNominaOriginal = @id AND IdNominaNueva IS NULL
319         ORDER BY IdVersion DESC LIMIT 1;", connection, transaction))
320     {
321         command.Parameters.AddWithValue("@id", originalPayrollId);
322         object value = command.ExecuteScalar();
323         return value == null || value == DBNull.Value ? 0 : Convert.ToInt32(value);
324     }
325 }
326
327 private void AddAudit(SQLiteConnection connection, SQLiteTransaction transaction, string user,
328     string module, string action, string detail, DateTime date)
329 {
330     auditRepository.Add(connection, transaction, new AuditRecord
331     {
332         Usuario = user,
333         Modulo = module,
334         Accion = action,
335         Detalle = detail,
336         Fecha = date
337     });
338 }
339
340 private static void ValidateCalculatedPayroll(Nomina payroll)
341 {
342     if (payroll == null || payroll.Detalles == null || payroll.Detalles.Count == 0)
343         throw new InvalidOperationException("No hay una nómina calculada para confirmar.");
344     PayrollStateMachine.DemandTransition(payroll.Estado, PayrollStates.Confirmed);
345 }
346
347 private static string CurrentUsername()
348 {
349     if (!SessionContext.IsAuthenticated || string.IsNullOrEmpty(SessionContext.Username))
350         throw new UnauthorizedAccessException("Debe iniciar sesión para gestionar la nómina.");
351     return SessionContext.Username;
352 }
353 }
354 }
```

Services/PayrollPeriodPolicyService.cs

```
1 using System;
2 using SistemaGestionNomina.Data;
3 using SistemaGestionNomina.Models;
4
5 namespace SistemaGestionNomina.Services
6 {
7     public class PayrollPeriodPolicyService
8     {
9         private readonly NominaRepository nominaRepository = new NominaRepository();
10
11         public bool EstaAbierto(DateTime fecha)
12         {
13             return nominaRepository.GetOverlappingProtectedPeriod(fecha.Date, fecha.Date) == null;
14         }
15
16         public bool EstaAbierto(DateTime fechaInicio, DateTime fechaFin)
17         {
18             if (fechaInicio.Date > fechaFin.Date) return false;
19             return nominaRepository.GetOverlappingProtectedPeriod(fechaInicio.Date, fechaFin.Date) == null;
20         }
21
22         public void VerificarPeriodoAbierto(int idPeriodo)
23         {
24             PeriodoNomina periodo = nominaRepository.GetPeriodoById(idPeriodo);
25             if (periodo == null)
26                 throw new InvalidOperationException("El período no existe.");
27             if (periodo.Cerrado)
28                 throw new InvalidOperationException("El período ya está cerrado. No se pueden realizar cambios.");
29         }
30
31         public void VerificarFechasAbiertas(DateTime fechaInicio, DateTime fechaFin)
32         {
33             if (fechaInicio.Date > fechaFin.Date)
34                 throw new ArgumentException("La fecha inicial no puede superar la fecha final.");
35             if (!EstaAbierto(fechaInicio, fechaFin))
36                 throw new InvalidOperationException("El rango seleccionado se solapa con un período cerrado o confirmado.");
37         }
38
39         public void VerificarCambioLaboralPermitido(DateTime fechaEfectiva)
40         {
41             if (nominaRepository.HasClosedPeriodAffectedByEmployeeChange(fechaEfectiva.Date))
42                 throw new InvalidOperationException("La fecha efectiva del cambio afectaría una nómina cerrada.");
43         }
44     }
45 }
```

Services/PdfExportService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using PdfSharp.Drawing;
4 using PdfSharp.Pdf;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8
9 namespace SistemaGestionNomina.Services
10 {
11     public class PdfExportService
12     {
13         private readonly AuthorizationService authorizationService = new AuthorizationService();
14         private readonly AuditTrailService auditTrailService = new AuditTrailService();
15         private readonly EmployeeScopeService employeeScopeService = new EmployeeScopeService();
16
17         public string ExportarEmpleados(List<Empleado> empleados)
18         {
19             authorizationService.DemandAny(Permissions.EmployeesExport, Permissions.ReportsPersonal);
20             string path = BuildPath("Empleados");
21             if (string.IsNullOrEmpty(path)) return string.Empty;
22
23             PdfDocument document = CreateDocument("Empleados");
24             // Cada reporte crea un documento independiente para que pueda guardarse donde indique el usuario.
25             PdfPage page = document.AddPage();
26             XGraphics gfx = XGraphics.FromPdfPage(page);
27             DrawTitle(gfx, "Reporte de Empleados");
28             int y = 80;
29             for (int i = 0; i < empleados.Count; i++)
30             {
31                 Empleado e = empleados[i];
32                 DrawLine(gfx, 40, y, e.Codigo + " | " + e.NombreCompleto + " | " + e.DepartamentoNombre + " | B/. " + e.SalarioBase.ToString("0.00") + " \
33 | " + e.Estado);
34                 y = NextLine(document, ref page, ref gfx, y);
35             }
36             document.Save(path);
37             auditTrailService.RegistrarAccion("Exportaciones", "PDF empleados", System.IO.Path.GetFileName(path));
38             return path;
39         }
40
41         public string ExportarAsistencias(List<Asistencia> asistencias)
42         {
43             authorizationService.DemandPermission(Permissions.AttendanceExport);
44             string path = BuildPath("Asistencia");
45             if (string.IsNullOrEmpty(path)) return string.Empty;
46
47             PdfDocument document = CreateDocument("Asistencia");
48             PdfPage page = document.AddPage();
49             XGraphics gfx = XGraphics.FromPdfPage(page);
50             DrawTitle(gfx, "Reporte de Asistencia");
51             int y = 80;
52             for (int i = 0; i < asistencias.Count; i++)
53             {
54                 Asistencia a = asistencias[i];
55                 DrawLine(gfx, 40, y, a.Fecha.ToString("dd/MM/yyyy") + " | " + a.EmpleadoNombre + " | " + a.HorasTrabajadas.ToString("0.00") + " h | " + \
56 a.Estado);
57                 y = NextLine(document, ref page, ref gfx, y);
58             }
59             document.Save(path);
60             auditTrailService.RegistrarAccion("Exportaciones", "PDF asistencia", System.IO.Path.GetFileName(path));
61         }
62     }
63 }
```

Services/PdfExportService.cs (continuacion)

```
59         return path;
60     }
61
62     public string ExportarHistorialEmpleados(List<HistorialEmpleado> history)
63     {
64         authorizationService.DemandPermission(Permissions.EmployeeHistoryExport);
65         string path = BuildPath("Historial_Laboral");
66         if (string.IsNullOrEmpty(path)) return string.Empty;
67         PdfDocument document = CreateDocument("Historial laboral");
68         PdfPage page = document.AddPage();
69         XGraphics gfx = XGraphics.FromPdfPage(page);
70         DrawTitle(gfx, "Historial Laboral");
71         int y = 80;
72         for (int i = 0; i < history.Count; i++)
73         {
74             HistorialEmpleado item = history[i];
75             DrawLine(gfx, 40, y, item.FechaEfectiva.ToString("dd/MM/yyyy") + " | " +
76                 item.CodigoEmpleado + " | " + item.CampoModificado + " | " +
77                 item.ValorAnterior + " -> " + item.ValorNuevo + " | " + item.UsuarioResponsable);
78             y = NextLine(document, ref page, ref gfx, y);
79         }
80         document.Save(path);
81         auditTrailService.RegistrarAccion("Exportaciones", "PDF historial laboral",
82             System.IO.Path.GetFileName(path));
83         return path;
84     }
85
86     public string ExportarAusencias(List<SolicitudAusencia> requests)
87     {
88         authorizationService.DemandPermission(Permissions.AbsencesExport);
89         string path = BuildPath("Solicitudes_Ausencia");
90         if (string.IsNullOrEmpty(path)) return string.Empty;
91         PdfDocument document = CreateDocument("Solicitudes de ausencia");
92         PdfPage page = document.AddPage();
93         XGraphics gfx = XGraphics.FromPdfPage(page);
94         DrawTitle(gfx, "Solicitudes de Ausencia");
95         int y = 80;
96         for (int i = 0; i < requests.Count; i++)
97         {
98             SolicitudAusencia item = requests[i];
99             DrawLine(gfx, 40, y, item.CodigoEmpleado + " | " + item.EmpleadoNombre + " | " +
100                 item.Tipo + " | " + item.FechaInicio.ToString("dd/MM") + " - " +
101                 item.FechaFin.ToString("dd/MM/yyyy") + " | " + item.Estado);
102             y = NextLine(document, ref page, ref gfx, y);
103         }
104         document.Save(path);
105         auditTrailService.RegistrarAccion("Exportaciones", "PDF ausencias",
106             System.IO.Path.GetFileName(path));
107         return path;
108     }
109
110     public string ExportarNomina(Nomina nomina)
111     {
112         authorizationService.DemandPermission(Permissions.PayrollExport);
113         string path = BuildPath("Nomina");
114         if (string.IsNullOrEmpty(path)) return string.Empty;
115
116         PdfDocument document = CreateDocument("Nómina");
117         PdfPage page = document.AddPage();
118         XGraphics gfx = XGraphics.FromPdfPage(page);
```

Services/PdfExportService.cs (continuacion)

```
119         DrawTitle(gfx, nomina.PeriodoNombre);
120         int y = 80;
121         for (int i = 0; i < nomina.Detalles.Count; i++)
122         {
123             NominaDetalle d = nomina.Detalles[i];
124             DrawLine(gfx, 40, y, d.CodigoEmpleado + " | " + d.EmpleadoNombre + " | Ingresos B/. " + d.TotalIngresos.ToString("0.00") + " | \
Deducciones B/. " + d.TotalDeducciones.ToString("0.00") + " | Neto B/. " + d.NetoPagar.ToString("0.00"));
125             y = NextLine(document, ref page, ref gfx, y);
126         }
127         y += 20;
128         DrawLine(gfx, 40, y, "Total neto: B/. " + nomina.TotalNeto.ToString("0.00"));
129         document.Save(path);
130         auditTrailService.RegistrarAccion("Exportaciones", "PDF nómina", System.IO.Path.GetFileName(path));
131         return path;
132     }
133
134     public string ExportarComprobante(Comprobante comprobante)
135     {
136         authorizationService.DemandPermission(Permissions.PayslipsExport);
137         string path = PathHelper.RequestExportPath(comprobante.NumeroComprobante, ".pdf", "Documento PDF (*.pdf)|*.pdf");
138         if (string.IsNullOrEmpty(path)) return string.Empty;
139
140         SaveComprobantePdf(comprobante, path);
141         auditTrailService.RegistrarAccion("Exportaciones", "PDF comprobante", System.IO.Path.GetFileName(path));
142         return path;
143     }
144
145     public string ExportarComprobantePersonal(Comprobante comprobante)
146     {
147         authorizationService.DemandPermission(Permissions.OwnPayslipsDownload);
148         if (comprobante == null) throw new ArgumentNullException("comprobante");
149         employeeScopeService.DemandCurrentEmployee(comprobante.IdEmpleado);
150         string path = PathHelper.RequestExportPath(comprobante.NumeroComprobante, ".pdf", "Documento PDF (*.pdf)|*.pdf");
151         return ExportarComprobantePersonal(comprobante, path);
152     }
153
154     internal string ExportarComprobantePersonal(Comprobante comprobante, string path)
155     {
156         authorizationService.DemandPermission(Permissions.OwnPayslipsDownload);
157         if (comprobante == null) throw new ArgumentNullException("comprobante");
158         employeeScopeService.DemandCurrentEmployee(comprobante.IdEmpleado);
159         if (string.IsNullOrEmpty(path)) return string.Empty;
160         if (!string.Equals(System.IO.Path.GetExtension(path), ".pdf", StringComparison.OrdinalIgnoreCase))
161         {
162             throw new ArgumentException("La ubicacion seleccionada debe ser un archivo PDF.");
163         }
164
165         string fullPath = System.IO.Path.GetFullPath(path);
166         SaveComprobantePdf(comprobante, fullPath);
167         return fullPath;
168     }
169
170     public string ExportarReportes(List<ReporteGenerado> reportes)
171     {
172         authorizationService.DemandPermission(Permissions.ReportsExport);
173         string path = BuildPath("Reportes");
174         if (string.IsNullOrEmpty(path)) return string.Empty;
175
176         PdfDocument document = CreateDocument("Reportes");
177         PdfPage page = document.AddPage();
```

Services/PdfExportService.cs (continuacion)

```
178         XGraphics gfx = XGraphics.FromPdfPage(page);
179         DrawTitle(gfx, "Reportes Generados");
180         int y = 80;
181         for (int i = 0; i < reportes.Count; i++)
182         {
183             ReporteGenerado r = reportes[i];
184             DrawLine(gfx, 40, y, r.NombreReporte + " | " + r.Tipo + " | " + r.FechaGeneracion.ToString("dd/MM/yyyy HH:mm"));
185             y = NextLine(document, ref page, ref gfx, y);
186         }
187         document.Save(path);
188         auditTrailService.RegistrarAccion("Exportaciones", "PDF reportes", System.IO.Path.GetFileName(path));
189         return path;
190     }
191
192     private static PdfDocument CreateDocument(string title)
193     {
194         PdfDocument document = new PdfDocument();
195         document.Info.Title = title;
196         return document;
197     }
198
199     private static void SaveComprobantePdf(Comprobante comprobante, string path)
200     {
201         PdfDocument document = CreateDocument("Comprobante");
202         PdfPage page = document.AddPage();
203         XGraphics gfx = XGraphics.FromPdfPage(page);
204         DrawTitle(gfx, "ProyFinal_LPI_Eq01_NomiCore");
205         DrawLine(gfx, 40, 82, "COMPROBANTE DE PAGO");
206         DrawLine(gfx, 40, 120, "Numero: " + comprobante.NumeroComprobante);
207         DrawLine(gfx, 40, 145, "Empleado: " + comprobante.EmpleadoNombre + " (" + comprobante.CodigoEmpleado + ")");
208         DrawLine(gfx, 40, 170, "Periodo: " + comprobante.PeriodoNombre);
209         DrawLine(gfx, 40, 220, "Total ingresos: B/. " + comprobante.TotalIngresos.ToString("0.00"));
210         DrawLine(gfx, 40, 245, "Total deducciones: B/. " + comprobante.TotalDeducciones.ToString("0.00"));
211         DrawLine(gfx, 40, 280, "Neto a pagar: B/. " + comprobante.Netopagar.ToString("0.00"));
212         DrawLine(gfx, 40, 340, "Documento generado para fines academicos.");
213         document.Save(path);
214     }
215
216     private static string BuildPath(string prefix)
217     {
218         return PathHelper.RequestExportPath(prefix, ".pdf", "Documento PDF (*.pdf)|*.pdf");
219     }
220
221     private static void DrawTitle(XGraphics gfx, string title)
222     {
223         XFont font = new XFont("Segoe UI", 18, XFontStyleEx.Bold);
224         gfx.DrawString(title, font, XBrushes.Black, new XRect(40, 35, 520, 30), XStringFormats.TopLeft);
225     }
226
227     private static void DrawLine(XGraphics gfx, int x, int y, string text)
228     {
229         XFont font = new XFont("Segoe UI", 10, XFontStyleEx.Regular);
230         gfx.DrawString(text, font, XBrushes.Black, new XRect(x, y, 520, 18), XStringFormats.TopLeft);
231     }
232
233     private static int NextLine(PdfDocument document, ref PdfPage page, ref XGraphics gfx, int currentY)
234     {
235         int next = currentY + 22;
236         if (next > 760)
237         {
```

Services/PdfExportService.cs (continuacion)

```
238         page = document.AddPage();
239         gfx = XGraphics.FromPdfPage(page);
240         return 45;
241     }
242
243     return next;
244 }
245 }
246 }
```


Services/PdfFontResolver.cs

```
1 using System.Collections.Generic;
2 using System.IO;
3 using PdfSharp.Fonts;
4
5 namespace SistemaGestionNomina.Services
6 {
7     public class PdfFontResolver : IFontResolver
8     {
9         private const string RegularFace = "SegoeUI#Regular";
10        private const string BoldFace = "SegoeUI#Bold";
11        private static bool registered;
12
13        public static void EnsureRegistered()
14        {
15            if (!registered && GlobalFontSettings.FontResolver == null)
16            {
17                GlobalFontSettings.FontResolver = new PdfFontResolver();
18                registered = true;
19            }
20        }
21
22        public FontResolverInfo ResolveTypeface(string familyName, bool isBold, bool isItalic)
23        {
24            return new FontResolverInfo(isBold ? BoldFace : RegularFace);
25        }
26
27        public byte[] GetFont(string faceName)
28        {
29            Dictionary<string, string[]> candidates = new Dictionary<string, string[]>();
30            candidates[RegularFace] = new[] { "segoeui.ttf", "arial.ttf" };
31            candidates[BoldFace] = new[] { "segoeuib.ttf", "arialbd.ttf", "arial.ttf" };
32
33            string windowsFonts = Path.Combine(System.Environment.GetFolderPath(System.Environment.SpecialFolder.Windows), "Fonts");
34            string[] files = candidates.ContainsKey(faceName) ? candidates[faceName] : candidates[RegularFace];
35            for (int i = 0; i < files.Length; i++)
36            {
37                string path = Path.Combine(windowsFonts, files[i]);
38                if (File.Exists(path))
39                {
40                    return File.ReadAllBytes(path);
41                }
42            }
43
44            throw new FileNotFoundException("No se encontró una fuente compatible para generar el PDF.");
45        }
46    }
47 }
```

Services/RolePermissionService.cs

```
1 using System;
2 using System.Collections.Generic;
3 using SistemaGestionNomina.Security;
4
5 namespace SistemaGestionNomina.Services
6 {
7     /// <summary>
8     /// Servicio reservado para permisos avanzados por rol.
9     /// </summary>
10    public class RolePermissionService
11    {
12        private readonly Dictionary<string, HashSet<string>> permisosPorRol;
13
14        public RolePermissionService()
15        {
16            permisosPorRol = new Dictionary<string, HashSet<string>>(StringComparer.OrdinalIgnoreCase);
17            permisosPorRol[Roles.Admin] = new HashSet<string>(StringComparer.OrdinalIgnoreCase) { "" };
18            permisosPorRol[Roles.RRHH] = new HashSet<string>(StringComparer.OrdinalIgnoreCase)
19            {
20                Permissions.EmployeesView, Permissions.EmployeesCreate, Permissions.EmployeesEdit,
21                Permissions.EmployeesDeactivate, Permissions.EmployeesExport,
22                Permissions.EmployeeHistoryView, Permissions.EmployeeHistoryExport,
23                Permissions.EmployeeChangesSchedule,
24                Permissions.AttendanceView, Permissions.AttendanceRegister, Permissions.AttendanceImport,
25                Permissions.AttendanceExport, Permissions.PayslipsView, Permissions.PayslipsExport,
26                Permissions.PayslipsPrint, Permissions.PayslipsEmail,
27                Permissions.ReportsView, Permissions.ReportsPersonal, Permissions.ReportsExport,
28                Permissions.PayrollHistoryView,
29                Permissions.AbsencesView, Permissions.AbsencesCreate, Permissions.AbsencesApprove,
30                Permissions.AbsencesReject, Permissions.AbsencesCancel, Permissions.AbsencesExport
31            };
32            permisosPorRol[Roles.Contabilidad] = new HashSet<string>(StringComparer.OrdinalIgnoreCase)
33            {
34                Permissions.PayrollView, Permissions.PayrollCalculate, Permissions.PayrollConfirm,
35                Permissions.PayrollExport, Permissions.PayslipsView, Permissions.PayslipsExport,
36                Permissions.PayslipsPrint, Permissions.PayslipsEmail,
37                Permissions.ReportsView, Permissions.ReportsFinancial, Permissions.ReportsExport,
38                Permissions.PayrollPay, Permissions.PayrollAnnul, Permissions.PayrollRecalculate,
39                Permissions.PayrollHistoryView, Permissions.AbsencesView
40            };
41            permisosPorRol[Roles.Supervisor] = new HashSet<string>(StringComparer.OrdinalIgnoreCase)
42            {
43                Permissions.AttendanceView, Permissions.AttendanceRegister, Permissions.AttendanceExport,
44                Permissions.AbsencesView, Permissions.AbsencesApprove, Permissions.AbsencesReject
45            };
46            permisosPorRol[Roles.Trabajador] = new HashSet<string>(StringComparer.OrdinalIgnoreCase)
47            {
48                Permissions.PortalView, Permissions.OwnerProfileView, Permissions.OwnerAttendanceView,
49                Permissions.OwnerPayslipsView, Permissions.OwnerPayslipsDownload, Permissions.OwnerPasswordChange,
50                Permissions.OwnerAbsencesView, Permissions.OwnerAbsencesCreate, Permissions.OwnerAbsencesCancel
51            };
52
53            HashSet<string> admin = permisosPorRol[Roles.Admin];
54            admin.Add(Permissions.DashboardView);
55        }
56
57        /// <summary>
58        /// Indica si un rol tiene permiso para ejecutar una acción.
59        /// </summary>
60        public bool TienePermiso(string rol, string accion)
```

Services/RolePermissionService.cs (continuacion)

```
61     {
62         if (string.IsNullOrEmpty(rol) || string.IsNullOrEmpty(accion))
63         {
64             return false;
65         }
66
67         HashSet<string> permisos;
68         if (!permisosPorRol.TryGetValue(rol.Trim(), out permisos))
69         {
70             return false;
71         }
72
73         return permisos.Contains("*") || permisos.Contains(accion.Trim());
74     }
75
76     /// <summary>
77     /// Indica si un rol puede ejecutar una accion especifica en un modulo.
78     /// </summary>
79     public bool TienePermiso(string rol, string modulo, string accion)
80     {
81         if (string.IsNullOrEmpty(rol))
82         {
83             return false;
84         }
85
86         HashSet<string> permisos;
87         if (!permisosPorRol.TryGetValue(rol.Trim(), out permisos))
88         {
89             return false;
90         }
91
92         if (permisos.Contains("*"))
93         {
94             return true;
95         }
96
97         string key = (modulo ?? string.Empty).Trim().ToLowerInvariant() + "." + (accion ?? string.Empty).Trim().ToLowerInvariant();
98         return permisos.Contains(key);
99     }
100
101     public bool EsRolValido(string rol)
102     {
103         return !string.IsNullOrEmpty(rol) && permisosPorRol.ContainsKey(rol.Trim());
104     }
105
106     /// <summary>
107     /// Devuelve las acciones permitidas para mostrar u ocultar opciones de interfaz.
108     /// </summary>
109     public List<string> ObtenerPermisos(string rol)
110     {
111         List<string> result = new List<string>();
112         if (string.IsNullOrEmpty(rol))
113         {
114             return result;
115         }
116
117         HashSet<string> permisos;
118         if (permisosPorRol.TryGetValue(rol.Trim(), out permisos))
119         {
120             foreach (string permiso in permisos)
```

Services/RolePermissionService.cs (continuacion)

```
121         {  
122             result.Add(permiso);  
123         }  
124     }  
125  
126     return result;  
127 }  
128 }  
129 }
```

Tools/Fase2PdfSmokeHost.cs

```
1 using System;
2 using System.IO;
3 using System.Reflection;
4 using SistemaGestionNomina.Data;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Security;
7 using SistemaGestionNomina.Services;
8
9 internal static class Fase2PdfSmokeHost
10 {
11     private static int Main(string[] args)
12     {
13         if (args.Length != 6) return 2;
14         try
15         {
16             string dbPath = args[0];
17             string pdfPath = args[1];
18             int ownPayslipId = int.Parse(args[2]);
19             int foreignPayslipId = int.Parse(args[3]);
20             int userId = int.Parse(args[4]);
21             string username = args[5];
22             Environment.SetEnvironmentVariable("NOMINA_DB_PATH", dbPath);
23             DatabaseInitializer.Initialize();
24             SessionContext.Begin(new Usuario
25             {
26                 IdUsuario = userId,
27                 NombreUsuario = username,
28                 Rol = Roles.Trabajador,
29                 Estado = "Activo",
30                 IdEmpleado = 1
31             });
32
33             EmployeePortalService service = new EmployeePortalService();
34             MethodInfo download = null;
35             MethodInfo[] methods = typeof(EmployeePortalService).GetMethods(BindingFlags.Instance | BindingFlags.NonPublic);
36             for (int i = 0; i < methods.Length; i++)
37             {
38                 if (methods[i].Name == "DownloadMyPayslipPdf" && methods[i].GetParameters().Length == 2)
39                 {
40                     download = methods[i];
41                     break;
42                 }
43             }
44
45             if (download == null) throw new InvalidOperationException("No se encontro la descarga interna de prueba.");
46             string generated = Convert.ToString(download.Invoke(service, new object[] { ownPayslipId, pdfPath }));
47             if (!File.Exists(generated)) throw new InvalidOperationException("No se genero el PDF personal.");
48             byte[] bytes = File.ReadAllBytes(generated);
49             if (bytes.Length < 4 || bytes[0] != 37 || bytes[1] != 80 || bytes[2] != 68 || bytes[3] != 70)
50             {
51                 throw new InvalidOperationException("El archivo generado no tiene cabecera PDF.");
52             }
53
54             bool rejected = false;
55             try
56             {
57                 download.Invoke(service, new object[] { foreignPayslipId, Path.Combine(Path.GetDirectoryName(pdfPath), "ajeno.pdf") });
58             }
59             catch (TargetInvocationException ex)
60             {
61             }
```

Tools/Fase2PdfSmokeHost.cs (continuacion)

```
61         rejected = ex.InnerException is UnauthorizedAccessException;
62     }
63
64     if (!rejected) throw new InvalidOperationException("La descarga permitio un comprobante ajeno.");
65     SessionContext.Clear();
66     Console.WriteLine("PDF personal y rechazo de archivo ajeno OK");
67     return 0;
68 }
69 catch (Exception ex)
70 {
71     Console.Error.WriteLine(ex);
72     return 1;
73 }
74 }
75 }
```

Tools/Fase3NominaSmokeHost.cs

```

1  using System;
2  using System.Data.SQLite;
3  using SistemaGestionNomina.Data;
4  using SistemaGestionNomina.Models;
5  using SistemaGestionNomina.Security;
6  using SistemaGestionNomina.Services;
7
8  internal static class Fase3NominaSmokeHost
9  {
10     private static int Main(string[] args)
11     {
12         try
13         {
14             if (args.Length != 1) throw new ArgumentException("Debe indicar la base temporal.");
15             Environment.SetEnvironmentVariable("NOMINA_DB_PATH", args[0]);
16             DatabaseInitializer.Initialize();
17             DatabaseInitializer.Initialize();
18             Assert(ScalarInt("PRAGMA user_version;") == DatabaseMigrationRunner.LatestVersion,
19                 "Las migraciones no alcanzaron la última versión.");
20             Assert(ScalarInt("SELECT COUNT(1) FROM MigracionesLog WHERE Version IN (4,5,6);") == 3,
21                 "No se registraron las migraciones 4, 5 y 6.");
22
23             SessionContext.Begin(new Usuario
24             {
25                 IdUsuario = 1,
26                 NombreUsuario = "admin_smoke_f3",
27                 Rol = Roles.Admin,
28                 Estado = "Activo"
29             });
30
31             DateTime januaryStart = new DateTime(2034, 1, 1);
32             DateTime januaryEnd = new DateTime(2034, 1, 15);
33             AsistenciaService attendanceService = new AsistenciaService();
34             attendanceService.Register(new Asistencia
35             {
36                 IdEmpleado = 1,
37                 Fecha = new DateTime(2034, 1, 2),
38                 HoraEntrada = new TimeSpan(8, 0, 0),
39                 HoraSalida = new TimeSpan(17, 0, 0),
40                 Estado = "Puntual"
41             });
42
43             NominaService payrollService = new NominaService();
44             Nomina january = payrollService.CalcularNomina(januaryStart, januaryEnd, null);
45             int januaryId = payrollService.ConfirmarPago(january, januaryStart, januaryEnd);
46             Assert(ScalarInt("SELECT COUNT(1) FROM PeriodosNomina WHERE Cerrado=1 AND FechaInicio='2034-01-01' AND FechaFin='2034-01-15';") == 1,
47                 "La confirmación no cerró el período.");
48
49             ExpectFailure(delegate
50             {
51                 attendanceService.Register(new Asistencia
52                 {
53                     IdEmpleado = 1, Fecha = new DateTime(2034, 1, 3),
54                     HoraEntrada = new TimeSpan(8, 0, 0), HoraSalida = new TimeSpan(17, 0, 0), Estado = "Puntual"
55                 });
56             }, "El servicio permitió asistencia en período cerrado.");
57             ExpectSqlFailure("INSERT INTO Asistencias (IdEmpleado, Fecha, HorasTrabajadas, Estado) VALUES (1, '2034-01-04', 0, 'Falta');",
58                 "El trigger permitió insertar asistencia en período cerrado.");
59             ExpectSqlFailure("UPDATE Asistencias SET Estado='Tardanza' WHERE IdEmpleado=1 AND Fecha='2034-01-02';",
60                 "El trigger permitió actualizar asistencia en período cerrado.");

```

Tools/Fase3NominaSmokeHost.cs (continuacion)

```

61         ExpectFailure(delegate { payrollService.CalcularNomina(new DateTime(2034, 1, 5), new DateTime(2034, 1, 20), null); },
62             "Se permitió calcular un rango solapado.");
63         ExpectSqlFailure(@"INSERT INTO PeriodosNomina
64             (Nombre, FechaInicio, FechaFin, Estado, Cerrado) VALUES ('Solapado', '2034-01-10', '2034-01-20', 'Confirmado', 1);",
65             "SQLite permitió un periodo protegido solapado.");
66
67         payrollService.PagarNomina(januaryId);
68         Assert(ScalarString("SELECT Estado FROM Nominas WHERE IdNomina=" + januaryId + ";") == PayrollStates.Paid,
69             "La nómina confirmada no pasó a Pagada.");
70         ExpectFailure(delegate { payrollService.PagarNomina(januaryId); }, "Se pagó dos veces la misma nómina.");
71         payrollService.AnularNomina(januaryId, "Corrección integral de smoke test");
72         Assert(ScalarString("SELECT Estado FROM Nominas WHERE IdNomina=" + januaryId + ";") == PayrollStates.Anulled,
73             "La nómina pagada no se anuló con trazabilidad.");
74         Assert(ScalarInt("SELECT COUNT(1) FROM NominaDetalle WHERE IdNomina=" + januaryId + ";") > 0 &&
75             ScalarInt("SELECT COUNT(1) FROM Comprobantes WHERE IdNomina=" + januaryId + ";") > 0,
76             "La anulación eliminó detalles o comprobantes.");
77         ExpectFailure(delegate { payrollService.PagarNomina(januaryId); }, "Se permitió pagar una nómina anulada.");
78
79         Nomina januaryRecalculated = payrollService.CalcularNomina(januaryStart, januaryEnd, null);
80         int januaryNewId = payrollService.RecalcularNomina(januaryId, januaryStart, januaryEnd, januaryRecalculated);
81         Assert(ScalarInt("SELECT COUNT(1) FROM NominaVersiones WHERE IdNominaOriginal=" + januaryId +
82             " AND IdNominaNueva=" + januaryNewId + ";") == 1, "El recálculo no vinculó la versión nueva.");
83
84         DateTime februaryStart = new DateTime(2034, 2, 1);
85         DateTime februaryEnd = new DateTime(2034, 2, 15);
86         Nomina february = payrollService.CalcularNomina(februaryStart, februaryEnd, null);
87         int februaryId = payrollService.ConfirmarPago(february, februaryStart, februaryEnd);
88         payrollService.AnularNomina(februaryId, "Preparar prueba de rollback");
89         Nomina februaryRecalculated = payrollService.CalcularNomina(februaryStart, februaryEnd, null);
90         Execute(@"CREATE TRIGGER Smoke_Fallar_Version BEFORE UPDATE ON NominaVersiones
91             BEGIN SELECT RAISE(ABORT, 'Fallo controlado de version'); END;");
92         int periodCount = ScalarInt("SELECT COUNT(1) FROM PeriodosNomina;");
93         int payrollCount = ScalarInt("SELECT COUNT(1) FROM Nominas;");
94         int detailCount = ScalarInt("SELECT COUNT(1) FROM NominaDetalle;");
95         int payslipCount = ScalarInt("SELECT COUNT(1) FROM Comprobantes;");
96         int auditCount = ScalarInt("SELECT COUNT(1) FROM Auditoria;");
97         ExpectFailure(delegate
98         {
99             payrollService.RecalcularNomina(februaryId, februaryStart, februaryEnd, februaryRecalculated);
100         }, "El recálculo no propagó el fallo controlado.");
101         Execute("DROP TRIGGER Smoke_Fallar_Version;");
102         Assert(periodCount == ScalarInt("SELECT COUNT(1) FROM PeriodosNomina;") &&
103             payrollCount == ScalarInt("SELECT COUNT(1) FROM Nominas;") &&
104             detailCount == ScalarInt("SELECT COUNT(1) FROM NominaDetalle;") &&
105             payslipCount == ScalarInt("SELECT COUNT(1) FROM Comprobantes;") &&
106             auditCount == ScalarInt("SELECT COUNT(1) FROM Auditoria;"),
107             "El recálculo fallido dejó datos parciales.");
108
109         EmpleadoService employeeService = new EmpleadoService();
110         Empleado blocked = employeeService.GetById(1);
111         blocked.SalarioBase += 10m;
112         ExpectFailure(delegate
113         {
114             employeeService.Save(blocked, new EmployeeChangeContext
115             {
116                 EffectiveDate = new DateTime(2034, 1, 10), Reason = "Cambio que afectaría período cerrado"
117             });
118         }, "Se permitió un cambio laboral que afecta nómina cerrada.");
119
120         Empleado scheduled = employeeService.GetById(1);

```


Tools/Fase3NominaSmokeHost.cs (continuacion)

```
121         decimal originalSalary = scheduled.SalarioBase;
122         string originalDepartment = scheduled.DepartamentoNombre;
123         scheduled.SalarioBase += 175m;
124         scheduled.Cargo = scheduled.Cargo + " II";
125         scheduled.IdDepartamento = 2;
126         scheduled.Estado = "Inactivo";
127         EmployeeSaveResult saveResult = employeeService.Save(scheduled, new EmployeeChangeContext
128         {
129             EffectiveDate = new DateTime(2034, 1, 20), Reason = "Prueba de cambios laborales programados"
130         });
131         Assert(saveResult.HistoryRecords == 4 && saveResult.HasScheduledChanges,
132             "No se registraron los cuatro cambios programados.");
133         Assert(employeeService.GetById(1).SalarioBase == originalSalary,
134             "Un cambio futuro se aplicó antes de su fecha.");
135         new EmployeeHistoryService().ApplyDueChanges(new DateTime(2034, 1, 20));
136         Assert(employeeService.GetById(1).SalarioBase == originalSalary + 175m &&
137             employeeService.GetById(1).Estado == "Inactivo", "Los cambios vencidos no se aplicaron.");
138         Assert(ScalarInt("SELECT COUNT(1) FROM HistorialEmpleado WHERE IdEmpleado=1;") == 4,
139             "El historial laboral no conservó todos los campos.");
140         Assert(ScalarDecimal("SELECT SueldoBase FROM NominaDetalle WHERE IdNomina=" + januaryNewId + " AND IdEmpleado=1;") == originalSalary &&
141             ScalarString("SELECT DepartamentoSnapshot FROM NominaDetalle WHERE IdNomina=" + januaryNewId + " AND IdEmpleado=1;") == originalDepartment,
142             "Los snapshots históricos cambiaron después de modificar el empleado.");
143         Assert(ScalarInt("SELECT COUNT(1) FROM Auditoria WHERE Usuario='admin_smoke_f3';") > 0,
144             "La auditoría no registró al usuario real.");
145
146         SessionContext.Clear();
147         SQLiteConnection.ClearAllPools();
148         Console.WriteLine("Migraciones, periodos, estados, rollback, historial y snapshots OK");
149         return 0;
150     }
151     catch (Exception ex)
152     {
153         Console.Error.WriteLine(ex.ToString());
154         try { SessionContext.Clear(); SQLiteConnection.ClearAllPools(); } catch { }
155         return 1;
156     }
157 }
158
159 private static void Assert(bool condition, string message)
160 {
161     if (!condition) throw new InvalidOperationException(message);
162 }
163
164 private static void ExpectFailure(Action action, string message)
165 {
166     try { action(); }
167     catch { return; }
168     throw new InvalidOperationException(message);
169 }
170
171 private static void ExpectSqlFailure(string sql, string message)
172 {
173     try { Execute(sql); }
174     catch (SQLiteException) { return; }
175     throw new InvalidOperationException(message);
176 }
177
178 private static void Execute(string sql)
179 {
180     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
```

Tools/Fase3NominaSmokeHost.cs (continuacion)

```
181         using (SQLiteCommand command = new SQLiteCommand(sql, connection)) command.ExecuteNonQuery();
182     }
183
184     private static object Scalar(string sql)
185     {
186         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
187         using (SQLiteCommand command = new SQLiteCommand(sql, connection)) return command.ExecuteScalar();
188     }
189
190     private static int ScalarInt(string sql) { return Convert.ToInt32(Scalar(sql)); }
191     private static decimal ScalarDecimal(string sql) { return Convert.ToDecimal(Scalar(sql)); }
192     private static string ScalarString(string sql) { return Convert.ToString(Scalar(sql)); }
193 }
```

Tools/Fase4RrhhAvanzadoSmokeHost.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Data.SQLite;
4  using SistemaGestionNomina.Data;
5  using SistemaGestionNomina.Models;
6  using SistemaGestionNomina.Security;
7  using SistemaGestionNomina.Services;
8
9  internal static class Fase4RrhhAvanzadoSmokeHost
10 {
11     private static int Main(string[] args)
12     {
13         try
14         {
15             if (args.Length != 1) throw new ArgumentException("Debe indicar la base temporal.");
16             Environment.SetEnvironmentVariable("NOMINA_DB_PATH", args[0]);
17             DatabaseInitializer.Initialize();
18             Assert(ScalarInt("PRAGMA user_version;") == 6, "La base temporal no llegó a la migración v6.");
19             Assert(ScalarInt("SELECT COUNT(1) FROM pragma_table_info('Asistencias') WHERE name='IdSolicitudAusencia';") == 1,
20                 "Falta la relación entre asistencia y solicitud.");
21
22             RolePermissionService roles = new RolePermissionService();
23             Assert(roles.TienePermiso(Roles.RRHH, Permissions.AbsencesApprove), "RRHH no recibió gestión de ausencias.");
24             Assert(roles.TienePermiso(Roles.Supervisor, Permissions.AbsencesApprove), "Supervisor no recibió aprobación departamental.");
25             Assert(!roles.TienePermiso(Roles.Contabilidad, Permissions.AbsencesApprove) &&
26                 roles.TienePermiso(Roles.Contabilidad, Permissions.AbsencesView), "Contabilidad tiene una matriz incorrecta.");
27             Assert(roles.TienePermiso(Roles.Trabajador, Permissions.OwnAbsencesCreate) &&
28                 !roles.TienePermiso(Roles.Trabajador, Permissions.AbsencesView), "Trabajador recibió alcance administrativo.");
29
30             DateTime monday = NextMonday(new DateTime(2035, 3, 1));
31             AbsenceRequestService service = new AbsenceRequestService();
32             BeginSession(40, "trabajador_f4", Roles.Trabajador, 1);
33             int ownId = service.CreateOwn(new OwnAbsenceRequest
34             {
35                 Type = AbsenceTypes.PaidLeave,
36                 StartDate = monday,
37                 EndDate = monday.AddDays(1),
38                 Reason = "Gestión personal de prueba"
39             });
40             List<SolicitudAusencia> ownItems = service.Search(new AbsenceQuery());
41             Assert(ownItems.Count == 1 && ownItems[0].IdEmpleado == 1, "El portal no aisló solicitudes propias.");
42
43             BeginSession(1, "admin_f4", Roles.Admin, null);
44             int employeeTwoId = service.CreateForEmployee(NewRequest(2, AbsenceTypes.Vacation,
45                 monday, monday.AddDays(4), "Vacaciones de prueba"));
46             BeginSession(40, "trabajador_f4", Roles.Trabajador, 1);
47             ExpectUnauthorized(delegate { service.GetById(employeeTwoId); }, "El trabajador consultó una solicitud ajena.");
48             service.Cancel(ownId, "Cancelación personal de prueba");
49             Assert(ScalarString("SELECT Estado FROM SolicitudesAusencia WHERE IdSolicitud=" + ownId + ";") == AbsenceStates.Cancelled,
50                 "El trabajador no pudo cancelar su solicitud pendiente.");
51
52             BeginSession(1, "admin_f4", Roles.Admin, null);
53             service.Approve(employeeTwoId, "Aprobada por RRHH de prueba");
54             Assert(ScalarInt("SELECT COUNT(1) FROM Asistencias WHERE IdSolicitudAusencia=" + employeeTwoId + ";") == 5,
55                 "La aprobación no creó exactamente cinco días laborables.");
56             Assert(ScalarInt("SELECT COUNT(1) FROM Asistencias WHERE IdSolicitudAusencia=" + employeeTwoId + " AND Estado='Permiso';") == 5,
57                 "Vacaciones no se reflejaron como Permiso.");
58
59             int overlapId = service.CreateForEmployee(NewRequest(2, AbsenceTypes.MedicalLeave,
60                 monday.AddDays(2), monday.AddDays(4), "Solicitud solapada"));

```

Tools/Fase4RrhhAvanzadoSmokeHost.cs (continuacion)

```

61     ExpectFailure(delegate { service.Approve(overlapId, "No debe aprobar"); },
62         "Se aprobó una ausencia solapada.");
63     Assert(ScalarString("SELECT Estado FROM SolicitudesAusencia WHERE IdSolicitud=" + overlapId + ";") == AbsenceStates.Pending,
64         "La aprobación fallida cambió el estado de la solicitud.");
65
66     service.Cancel(employeeTwoId, "Cancelar ausencia aprobada");
67     Assert(ScalarInt("SELECT COUNT(1) FROM Asistencias WHERE IdSolicitudAusencia=" + employeeTwoId + ";") == 0,
68         "La cancelación no retiró las asistencias generadas.");
69
70     DateTime conflictDate = monday.AddDays(14);
71     Execute("INSERT INTO Asistencias (IdEmpleado, Fecha, HoraEntrada, HoraSalida, HorasTrabajadas, Estado) VALUES (2, ' +
72         conflictDate.ToString("yyyy-MM-dd") + "', '08:00', '17:00', 9, 'Puntual');");
73     int conflictId = service.CreateForEmployee(NewRequest(2, AbsenceTypes.UnpaidLeave,
74         conflictDate, conflictDate, "Permiso con asistencia existente"));
75     ExpectFailure(delegate { service.Approve(conflictId, "Debe fallar sin sobrescribir"); },
76         "Se sobrescribió una asistencia existente.");
77     Assert(ScalarInt("SELECT COUNT(1) FROM Asistencias WHERE IdEmpleado=2 AND Fecha=' +
78         conflictDate.ToString("yyyy-MM-dd") + ';'") == 1 &&
79         ScalarString("SELECT Estado FROM SolicitudesAusencia WHERE IdSolicitud=" + conflictId + ";") == AbsenceStates.Pending,
80         "El conflicto de asistencia dejó una operación parcial.");
81
82     DateTime suspensionDate = monday.AddDays(21);
83     int suspensionId = service.CreateForEmployee(NewRequest(3, AbsenceTypes.Suspension,
84         suspensionDate, suspensionDate, "Suspensión académica"));
85     service.Approve(suspensionId, "Aprobada");
86     Assert(ScalarString("SELECT Estado FROM Asistencias WHERE IdSolicitudAusencia=" + suspensionId + ";") == "Falta",
87         "La suspensión no se reflejó como Falta.");
88
89     DateTime supervisorDate = monday.AddDays(28);
90     int departmentRequestId = service.CreateForEmployee(NewRequest(4, AbsenceTypes.License,
91         supervisorDate, supervisorDate.AddDays(1), "Solicitud de Operaciones"));
92     int foreignDepartmentRequestId = service.CreateForEmployee(NewRequest(2, AbsenceTypes.PaidLeave,
93         supervisorDate, supervisorDate, "Solicitud de Ventas"));
94     BeginSession(50, "supervisor_f4", Roles.Supervisor, 4);
95     service.Approve(departmentRequestId, "Aprobación departamental");
96     ExpectUnauthorized(delegate { service.Reject(foreignDepartmentRequestId, "Fuera de alcance"); },
97         "El supervisor resolvió una solicitud de otro departamento.");
98     List<SolicitudAusencia> scoped = service.Search(new AbsenceQuery());
99     for (int i = 0; i < scoped.Count; i++)
100         Assert(scoped[i].DepartamentoNombre == "Operaciones", "La consulta del supervisor filtró otro departamento.");
101
102     BeginSession(60, "contabilidad_f4", Roles.Contabilidad, null);
103     Assert(service.Search(new AbsenceQuery()).Count > 0, "Contabilidad no pudo consultar ausencias.");
104     ExpectUnauthorized(delegate { service.Approve(foreignDepartmentRequestId, "No autorizado"); },
105         "Contabilidad aprobó una solicitud.");
106
107     BeginSession(1, "admin_f4", Roles.Admin, null);
108     DateTime pendingClosedDate = monday.AddDays(70);
109     int pendingClosedId = service.CreateForEmployee(NewRequest(3, AbsenceTypes.JustifiedAbsence,
110         pendingClosedDate, pendingClosedDate, "Pendiente antes del cierre"));
111     InsertClosedPeriod(pendingClosedDate, pendingClosedDate, "Cierre posterior");
112     ExpectFailure(delegate { service.Reject(pendingClosedId, "No puede alterar período cerrado"); },
113         "Se modificó una ausencia después de cerrar el período.");
114     ExpectSqlFailure("UPDATE SolicitudesAusencia SET Estado='Rechazada' WHERE IdSolicitud=" + pendingClosedId + ";",
115         "SQLite permitió modificar una ausencia de período cerrado.");
116
117     DateTime closedStart = monday.AddDays(84);
118     InsertClosedPeriod(closedStart, closedStart.AddDays(4), "Período cerrado de ausencia");
119     ExpectFailure(delegate
120     {

```

Tools/Fase4RrhhAvanzadoSmokeHost.cs (continuacion)

```

121         service.CreateForEmployee(NewRequest(2, AbsenceTypes.Vacation,
122             closedStart, closedStart.AddDays(1), "Debe bloquearse"));
123     }, "Se creó una solicitud dentro de un período cerrado.");
124     ExpectSqlFailure(@"INSERT INTO SolicitudesAusencia
125         (IdEmpleado,Tipo,FechaInicio,FechaFin,Estado,Motivo,UsuarioSolicitante,FechaCreacion)
126         VALUES (2,'Vacaciones','" + closedStart.ToString("yyyy-MM-dd") + "','" +
127             closedStart.AddDays(1).ToString("yyyy-MM-dd") + "','Pendiente','Directa','sql','2035-01-01');",
128         "SQLite permitió insertar una ausencia dentro de un período cerrado.");
129
130     Assert(ScalarInt("SELECT COUNT(1) FROM Auditoria WHERE Modulo='Ausencias' AND Usuario IN ('admin_f4','trabajador_f4','supervisor_f4');") > 0,
131         "La auditoría no registró a los usuarios reales de ausencias.");
132     SessionContext.Clear();
133     SQLiteConnection.ClearAllPools();
134     Console.WriteLine("Permisos, alcance, ausencias, asistencia, cancelación y periodos cerrados OK");
135     return 0;
136 }
137 catch (Exception ex)
138 {
139     Console.Error.WriteLine(ex.ToString());
140     try { SessionContext.Clear(); SQLiteConnection.ClearAllPools(); } catch { }
141     return 1;
142 }
143 }
144
145 private static SolicitudAusencia NewRequest(int employeeId, string type, DateTime start, DateTime end, string reason)
146 {
147     return new SolicitudAusencia { IdEmpleado = employeeId, Tipo = type, FechaInicio = start, FechaFin = end, Motivo = reason };
148 }
149
150 private static void BeginSession(int id, string username, string role, int? employeeId)
151 {
152     SessionContext.Begin(new Usuario
153     {
154         IdUsuario = id, NombreUsuario = username, Rol = role, Estado = "Activo", IdEmpleado = employeeId
155     });
156 }
157
158 private static DateTime NextMonday(DateTime date)
159 {
160     DateTime result = date.Date;
161     while (result.DayOfWeek != DayOfWeek.Monday) result = result.AddDays(1);
162     return result;
163 }
164
165 private static void InsertClosedPeriod(DateTime start, DateTime end, string name)
166 {
167     using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
168     using (SQLiteCommand command = new SQLiteCommand(@"INSERT INTO PeriodosNomina
169         (Nombre,FechaInicio,FechaFin,Estado,Cerrado,FechaCierre,CerradoPor)
170         VALUES (@nombre,@inicio,@fin,'Confirmado',1,@cierre,'admin_f4');", connection))
171     {
172         command.Parameters.AddWithValue("@nombre", name);
173         command.Parameters.AddWithValue("@inicio", start.ToString("yyyy-MM-dd"));
174         command.Parameters.AddWithValue("@fin", end.ToString("yyyy-MM-dd"));
175         command.Parameters.AddWithValue("@cierre", DateTime.Now.ToString("yyyy-MM-dd HH:mm:ss"));
176         command.ExecuteNonQuery();
177     }
178 }
179
180 private static void Assert(bool condition, string message) { if (!condition) throw new InvalidOperationException(message); }

```

Tools/Fase4RrhhAvanzadoSmokeHost.cs (continuacion)

```
181     private static void ExpectFailure(Action action, string message) { try { action(); } catch { return; } throw new InvalidOperationException(message); }
182     private static void ExpectUnauthorized(Action action, string message)
183     {
184         try { action(); }
185         catch (UnauthorizedAccessException) { return; }
186         throw new InvalidOperationException(message);
187     }
188     private static void ExpectSqlFailure(string sql, string message)
189     {
190         try { Execute(sql); } catch (SQLiteException) { return; }
191         throw new InvalidOperationException(message);
192     }
193     private static void Execute(string sql)
194     {
195         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
196         using (SQLiteCommand command = new SQLiteCommand(sql, connection)) command.ExecuteNonQuery();
197     }
198     private static object Scalar(string sql)
199     {
200         using (SQLiteConnection connection = SQLiteConnectionFactory.CreateConnection())
201         using (SQLiteCommand command = new SQLiteCommand(sql, connection)) return command.ExecuteScalar();
202     }
203     private static int ScalarInt(string sql) { return Convert.ToInt32(Scalar(sql)); }
204     private static string ScalarString(string sql) { return Convert.ToString(Scalar(sql)); }
205 }
```

UI/Controls/RoundedButton.cs

```
1 using System.Drawing;
2 using System.Drawing.Drawing2D;
3 using System.Windows.Forms;
4 using SistemaGestionNomina.Helpers;
5
6 namespace SistemaGestionNomina.UI.Controls
7 {
8     public class RoundedButton : Button
9     {
10         public int Radius { get; set; }
11         public Color BorderColor { get; set; }
12
13         public RoundedButton()
14         {
15             Radius = 8;
16             BorderColor = ThemeHelper.Border;
17             FlatStyle = FlatStyle.Flat;
18             FlatAppearance.BorderSize = 0;
19             BackColor = ThemeHelper.Accent;
20             ForeColor = Color.FromArgb(32, 18, 67);
21             Font = ThemeHelper.BodyFont(9.5F, FontStyle.Bold);
22             Cursor = Cursors.Hand;
23             Height = 38;
24         }
25
26         protected override void OnPaint(PaintEventArgs pevent)
27         {
28             pevent.Graphics.SmoothingMode = SmoothingMode.AntiAlias;
29             Rectangle rect = new Rectangle(0, 0, Width - 1, Height - 1);
30             using (GraphicsPath path = CreateRoundPath(rect, Radius))
31             using (SolidBrush brush = new SolidBrush(BackColor))
32             using (Pen pen = new Pen(BorderColor, 1))
33             using (SolidBrush textBrush = new SolidBrush(ForeColor))
34             {
35                 pevent.Graphics.FillPath(brush, path);
36                 pevent.Graphics.DrawPath(pen, path);
37                 TextRenderer.DrawText(pevent.Graphics, Text, Font, rect, ForeColor,
38                     TextFormatFlags.HorizontalCenter | TextFormatFlags.VerticalCenter | TextFormatFlags.EndEllipsis);
39             }
40         }
41
42         private static GraphicsPath CreateRoundPath(Rectangle bounds, int radius)
43         {
44             int diameter = radius * 2;
45             GraphicsPath path = new GraphicsPath();
46             path.AddArc(bounds.X, bounds.Y, diameter, diameter, 180, 90);
47             path.AddArc(bounds.Right - diameter, bounds.Y, diameter, diameter, 270, 90);
48             path.AddArc(bounds.Right - diameter, bounds.Bottom - diameter, diameter, diameter, 0, 90);
49             path.AddArc(bounds.X, bounds.Bottom - diameter, diameter, diameter, 90, 90);
50             path.CloseFigure();
51             return path;
52         }
53     }
54 }
```

UI/Controls/RoundedPanel.cs

```
1 using System.Drawing;
2 using System.Drawing.Drawing2D;
3 using System.Windows.Forms;
4 using SistemaGestionNomina.Helpers;
5
6 namespace SistemaGestionNomina.UI.Controls
7 {
8     public class RoundedPanel : Panel
9     {
10         public int Radius { get; set; }
11         public Color BorderColor { get; set; }
12
13         public RoundedPanel()
14         {
15             Radius = 12;
16             BorderColor = ThemeHelper.Border;
17             BackColor = ThemeHelper.Card;
18             DoubleBuffered = true;
19             Padding = new Padding(16);
20         }
21
22         protected override void OnPaint(PaintEventArgs e)
23         {
24             base.OnPaint(e);
25             e.Graphics.SmoothingMode = SmoothingMode.AntiAlias;
26             using (GraphicsPath path = CreateRoundPath(ClientRectangle, Radius))
27             using (Pen pen = new Pen(BorderColor, 1))
28             {
29                 Region = new Region(path);
30                 e.Graphics.DrawPath(pen, path);
31             }
32         }
33
34         private static GraphicsPath CreateRoundPath(Rectangle bounds, int radius)
35         {
36             int diameter = radius * 2;
37             Rectangle rect = new Rectangle(bounds.X, bounds.Y, bounds.Width - 1, bounds.Height - 1);
38             GraphicsPath path = new GraphicsPath();
39             path.AddArc(rect.X, rect.Y, diameter, diameter, 180, 90);
40             path.AddArc(rect.Right - diameter, rect.Y, diameter, diameter, 270, 90);
41             path.AddArc(rect.Right - diameter, rect.Bottom - diameter, diameter, diameter, 0, 90);
42             path.AddArc(rect.X, rect.Bottom - diameter, diameter, diameter, 90, 90);
43             path.CloseFigure();
44             return path;
45         }
46     }
47 }
```


UI/FrmAcercaDe.cs

```
1 using System;
2 using System.Windows.Forms;
3 using SistemaGestionNomina.Helpers;
4
5 namespace SistemaGestionNomina.UI
6 {
7     public partial class FrmAcercaDe : Form
8     {
9         public FrmAcercaDe()
10         {
11             InitializeComponent();
12             ControlStyleHelper.ApplyModernForm(this);
13         }
14
15         private void btnCerrar_Click(object sender, EventArgs e)
16         {
17             Close();
18         }
19
20         private void lblIntegrantes_Click(object sender, EventArgs e)
21         {
22         }
23     }
24 }
25 }
```

UI/FrmAcercaDe.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmAcercaDe
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Panel panelInfo;
7         private System.Windows.Forms.Label lblTitulo;
8         private System.Windows.Forms.Label lblProyecto;
9         private System.Windows.Forms.Label lblCurso;
10        private System.Windows.Forms.Label lblIntegrantes;
11        private System.Windows.Forms.Label lblDescripcion;
12        private System.Windows.Forms.Button btnCerrar;
13
14        protected override void Dispose(bool disposing)
15        {
16            if (disposing && (components != null)) components.Dispose();
17            base.Dispose(disposing);
18        }
19
20        private void InitializeComponent()
21        {
22            this.panelInfo = new System.Windows.Forms.Panel();
23            this.lblTitulo = new System.Windows.Forms.Label();
24            this.lblProyecto = new System.Windows.Forms.Label();
25            this.lblCurso = new System.Windows.Forms.Label();
26            this.lblIntegrantes = new System.Windows.Forms.Label();
27            this.lblDescripcion = new System.Windows.Forms.Label();
28            this.btnCerrar = new System.Windows.Forms.Button();
29            this.panelInfo.SuspendLayout();
30            this.SuspendLayout();
31            //
32            // panelInfo
33            //
34            this.panelInfo.BackColor = System.Drawing.Color.Black;
35            this.panelInfo.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
36            this.panelInfo.Controls.Add(this.lblTitulo);
37            this.panelInfo.Controls.Add(this.lblProyecto);
38            this.panelInfo.Controls.Add(this.lblCurso);
39            this.panelInfo.Controls.Add(this.lblIntegrantes);
40            this.panelInfo.Controls.Add(this.lblDescripcion);
41            this.panelInfo.Controls.Add(this.btnCerrar);
42            this.panelInfo.Location = new System.Drawing.Point(120, 108);
43            this.panelInfo.Name = "panelInfo";
44            this.panelInfo.Size = new System.Drawing.Size(780, 460);
45            this.panelInfo.TabIndex = 0;
46            //
47            // lblTitulo
48            //
49            this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
50            this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)((byte)(167))), ((int)((byte)(139))), ((int)((byte)(250))));
51            this.lblTitulo.Location = new System.Drawing.Point(34, 28);
52            this.lblTitulo.Name = "lblTitulo";
53            this.lblTitulo.Size = new System.Drawing.Size(650, 44);
54            this.lblTitulo.TabIndex = 0;
55            this.lblTitulo.Text = "ProyFinal_LPI_Eq01_NomiCore";
56            //
57            // lblProyecto
58            //
59            this.lblProyecto.ForeColor = System.Drawing.Color.FromArgb(((int)((byte)(229))), ((int)((byte)(228))), ((int)((byte)(245))));
60            this.lblProyecto.Location = new System.Drawing.Point(36, 104);
```

UI/FrmAcercaDe.Designer.cs (continuacion)

```

61         this.lblProyecto.Name = "lblProyecto";
62         this.lblProyecto.Size = new System.Drawing.Size(680, 90);
63         this.lblProyecto.TabIndex = 1;
64         this.lblProyecto.Text = "Proyecto N° 2\r\nTema H: Manejo de los campos en las clases + PrintDocument\r\nAplica" +
65 "ción: ProyFinal_LPI_Eq01_NomiCore\r\nProblema: control de empleados, a" +
66 "sistencia, nómina y comprobantes";
67         //
68         // lblCurso
69         //
70         this.lblCurso.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
71         this.lblCurso.Location = new System.Drawing.Point(36, 208);
72         this.lblCurso.Name = "lblCurso";
73         this.lblCurso.Size = new System.Drawing.Size(680, 82);
74         this.lblCurso.TabIndex = 2;
75         this.lblCurso.Text = "Universidad Tecnológica de Panamá\r\nFacultad de Ingeniería de Sistemas Computacion" +
76 "ales\r\nLenguaje de Programación I\r\nProfesora: Anna Araba de Ruiz | Grupo: 1SF121";
77         //
78         // lblIntegrantes
79         //
80         this.lblIntegrantes.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
81         this.lblIntegrantes.Location = new System.Drawing.Point(36, 304);
82         this.lblIntegrantes.Name = "lblIntegrantes";
83         this.lblIntegrantes.Size = new System.Drawing.Size(680, 50);
84         this.lblIntegrantes.TabIndex = 3;
85         this.lblIntegrantes.Text = "Equipo: Eq01\r\nIntegrantes: Jonathan Romero, Jose Martinez, Karen Wen, Davis Batis" +
86 "ta";
87         this.lblIntegrantes.Click += new System.EventHandler(this.lblIntegrantes_Click);
88         //
89         // lblDescripcion
90         //
91         this.lblDescripcion.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
92         this.lblDescripcion.Location = new System.Drawing.Point(36, 366);
93         this.lblDescripcion.Name = "lblDescripcion";
94         this.lblDescripcion.Size = new System.Drawing.Size(520, 72);
95         this.lblDescripcion.TabIndex = 4;
96         this.lblDescripcion.Text = "Windows Forms App (.NET Framework 4.8)\r\nVisual Studio 2022 | Versión académica 1." +
97 "0 | Año 2026\r\nIncluye SQLite, servicios, repositorios, exportación e impresión.";
98         //
99         // btnCerrar
100        //
101        this.btnCerrar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(124)))), ((int)(((byte)(58)))), ((int)(((byte)(237))))));
102        this.btnCerrar.FlatAppearance.BorderSize = 0;
103        this.btnCerrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
104        this.btnCerrar.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
105        this.btnCerrar.ForeColor = System.Drawing.Color.White;
106        this.btnCerrar.Location = new System.Drawing.Point(600, 386);
107        this.btnCerrar.Name = "btnCerrar";
108        this.btnCerrar.Size = new System.Drawing.Size(116, 38);
109        this.btnCerrar.TabIndex = 5;
110        this.btnCerrar.Text = "Cerrar";
111        this.btnCerrar.UseVisualStyleBackColor = false;
112        this.btnCerrar.Click += new System.EventHandler(this.btnCerrar_Click);
113        //
114        // FrmAcercaDe
115        //
116        this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F);
117        this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
118        this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
119        this.ClientSize = new System.Drawing.Size(1020, 742);
120        this.Controls.Add(this.panelInfo);

```

UI/FrmAcercaDe.Designer.cs (continuacion)

```
121         this.Font = new System.Drawing.Font("Segoe UI", 9F);
122         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
123         this.Name = "FrmAcercaDe";
124         this.Text = "Acerca de";
125         this.panelInfo.ResumeLayout(false);
126         this.ResumeLayout(false);
127
128     }
129 }
130 }
```

UI\FrmAnularNomina.cs

```
1 using System;
2 using System.Windows.Forms;
3 using SistemaGestionNomina.Models;
4
5 namespace SistemaGestionNomina.UI
6 {
7     public partial class FrmAnularNomina : Form
8     {
9         private TextBox txtMotivo;
10        private Label lblAdvertencia;
11        private Label lblResumen;
12        private Button btnConfirmar;
13        private Button btnCancelar;
14
15        public string Motivo { get; private set; }
16
17        public FrmAnularNomina(Nomina nomina)
18        {
19            InitializeComponent();
20            lblResumen.Text = "Nómina: " + (nomina.PeriodoNombre ?? "Sin nombre") +
21                "\nEstado actual: " + (nomina.Estado ?? "Desconocido") +
22                "\nTotal neto: B/. " + nomina.TotalNeto.ToString("0.00") +
23                "\nEmpleados: " + nomina.Detalles.Count;
24        }
25
26        private void InitializeComponent()
27        {
28            this.lblResumen = new Label();
29            this.lblAdvertencia = new Label();
30            this.txtMotivo = new TextBox();
31            this.btnConfirmar = new Button();
32            this.btnCancelar = new Button();
33            this.SuspendLayout();
34
35            this.lblResumen.AutoSize = true;
36            this.lblResumen.Font = new System.Drawing.Font("Segoe UI", 10F);
37            this.lblResumen.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
38            this.lblResumen.Location = new System.Drawing.Point(16, 16);
39            this.lblResumen.Size = new System.Drawing.Size(50, 19);
40            this.lblResumen.Text = "Resumen";
41
42            this.lblAdvertencia.AutoSize = true;
43            this.lblAdvertencia.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
44            this.lblAdvertencia.ForeColor = System.Drawing.Color.FromArgb(239, 68, 68);
45            this.lblAdvertencia.Location = new System.Drawing.Point(16, 96);
46            this.lblAdvertencia.Size = new System.Drawing.Size(80, 15);
47            this.lblAdvertencia.Text = "La nómina anterior seguirá existiendo como registro histórico.";
48
49            this.txtMotivo.Location = new System.Drawing.Point(16, 124);
50            this.txtMotivo.Multiline = true;
51            this.txtMotivo.Size = new System.Drawing.Size(400, 80);
52            this.txtMotivo.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
53            this.txtMotivo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
54            this.txtMotivo.BorderStyle = BorderStyle.FixedSingle;
55
56            this.btnConfirmar.BackColor = System.Drawing.Color.FromArgb(239, 68, 68);
57            this.btnConfirmar.FlatStyle = FlatStyle.Flat;
58            this.btnConfirmar.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
59            this.btnConfirmar.ForeColor = System.Drawing.Color.White;
60            this.btnConfirmar.Location = new System.Drawing.Point(220, 220);
```

UI/FrmAnularNomina.cs (continuacion)

```
61         this.btnConfirmar.Size = new System.Drawing.Size(100, 36);
62         this.btnConfirmar.Text = "Anular";
63         this.btnConfirmar.UseVisualStyleBackColor = false;
64         this.btnConfirmar.Click += new EventHandler(this.btnConfirmar_Click);
65
66         this.btnCancelar.BackColor = System.Drawing.Color.Black;
67         this.btnCancelar.FlatStyle = FlatStyle.Flat;
68         this.btnCancelar.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
69         this.btnCancelar.Location = new System.Drawing.Point(336, 220);
70         this.btnCancelar.Size = new System.Drawing.Size(80, 36);
71         this.btnCancelar.Text = "Cancelar";
72         this.btnCancelar.UseVisualStyleBackColor = false;
73         this.btnCancelar.Click += new EventHandler(this.btnCancelar_Click);
74
75         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
76         this.AutoScaleMode = AutoScaleMode.Font;
77         this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
78         this.ClientSize = new System.Drawing.Size(434, 271);
79         this.Controls.Add(this.lblResumen);
80         this.Controls.Add(this.lblAdvertencia);
81         this.Controls.Add(this.txtMotivo);
82         this.Controls.Add(this.btnConfirmar);
83         this.Controls.Add(this.btnCancelar);
84         this.Font = new System.Drawing.Font("Segoe UI", 9F);
85         this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
86         this.FormBorderStyle = FormBorderStyle.FixedDialog;
87         this.MaximizeBox = false;
88         this.MinimizeBox = false;
89         this.StartPosition = FormStartPosition.CenterParent;
90         this.Text = "Anular nómina";
91         this.ResumeLayout(false);
92         this.PerformLayout();
93     }
94
95     private void btnConfirmar_Click(object sender, EventArgs e)
96     {
97         if (string.IsNullOrEmpty(txtMotivo.Text))
98         {
99             MessageBox.Show("Debe indicar el motivo de la anulación.", "Anular",
100                 MessageBoxButtons.OK, MessageBoxIcon.Warning);
101             return;
102         }
103
104         Motivo = txtMotivo.Text.Trim();
105         DialogResult = DialogResult.OK;
106         Close();
107     }
108
109     private void btnCancelar_Click(object sender, EventArgs e)
110     {
111         DialogResult = DialogResult.Cancel;
112         Close();
113     }
114 }
115 }
```

UI/FrmAsistencia.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Services;
8 using SistemaGestionNomina.Security;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmAsistencia : Form
13     {
14         private readonly EmpleadoService empleadoService = new EmpleadoService();
15         private readonly AsistenciaService asistenciaService = new AsistenciaService();
16         private readonly ExcelExportService excelExportService = new ExcelExportService();
17         private readonly PdfExportService pdfExportService = new PdfExportService();
18         private readonly AttendanceDeviceImportService attendanceDeviceImportService = new AttendanceDeviceImportService();
19         private readonly AuthorizationService authorizationService = new AuthorizationService();
20         private List<Asistencia> currentItems = new List<Asistencia>();
21
22         public FrmAsistencia()
23         {
24             InitializeComponent();
25             ControlStyleHelper.ApplyModernForm(this);
26         }
27
28         private void FrmAsistencia_Load(object sender, EventArgs e)
29         {
30             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
31             btnRegistrar.Visible = authorizationService.HasPermission(Permissions.AttendanceRegister);
32             btnCargarReloj.Visible = authorizationService.HasPermission(Permissions.AttendanceImport);
33             btnExportarExcel.Visible = authorizationService.HasPermission(Permissions.AttendanceExport);
34             btnExportarPdf.Visible = authorizationService.HasPermission(Permissions.AttendanceExport);
35             LoadEmployees();
36             LoadAttendance();
37         }
38
39         private void LoadEmployees()
40         {
41             List<Empleado> employees = empleadoService.GetActive(null);
42             cmbEmpleado.DataSource = new List<Empleado>(employees);
43             cmbEmpleado.DisplayMember = "NombreCompleto";
44             cmbEmpleado.ValueMember = "IdEmpleado";
45
46             List<Empleado> filters = new List<Empleado>();
47             filters.Add(new Empleado { IdEmpleado = 0, Nombre = "Todos", Apellido = "los empleados" });
48             filters.AddRange(employees);
49             cmbFiltroEmpleado.DataSource = filters;
50             cmbFiltroEmpleado.DisplayMember = "NombreCompleto";
51             cmbFiltroEmpleado.ValueMember = "IdEmpleado";
52         }
53
54         private void LoadAttendance()
55         {
56             if (dtpFiltroInicio.Value.Date > dtpFiltroFin.Value.Date)
57             {
58                 MessageBox.Show("La fecha inicial no puede ser mayor que la fecha final.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Warning);
59                 return;
60             }
61         }
62     }
63 }
```

UI/FrmAsistencia.cs (continuacion)

```
61
62     int? employeeId = null;
63     if (cmbFiltroEmpleado.SelectedValue is int && (int)cmbFiltroEmpleado.SelectedValue > 0)
64     {
65         employeeId = (int)cmbFiltroEmpleado.SelectedValue;
66     }
67
68     string status = cmbFiltroEstado.SelectedItem == null || cmbFiltroEstado.SelectedItem.ToString() == "Todos"
69         ? string.Empty
70         : cmbFiltroEstado.SelectedItem.ToString();
71
72     currentItems = asistenciaService.GetAll(dtpFiltroInicio.Value.Date, dtpFiltroFin.Value.Date, employeeId, status);
73     dgvAsistencia.Rows.Clear();
74     foreach (Asistencia a in currentItems)
75     {
76         dgvAsistencia.Rows.Add(a.EmpleadoNombre, a.Fecha.ToString("dd/MM/yyyy"),
77             a.HoraEntrada.HasValue ? a.HoraEntrada.Value.ToString(@"hh\:mm") : "--",
78             a.HoraSalida.HasValue ? a.HoraSalida.Value.ToString(@"hh\:mm") : "--",
79             a.HorasTrabajadas.ToString("0.00"), a.Estado);
80     }
81
82     UpdateSummaryCards();
83 }
84
85 private void UpdateSummaryCards()
86 {
87     int presentes = 0;
88     int ausentes = 0;
89     int tardanzas = 0;
90
91     foreach (Asistencia asistencia in currentItems)
92     {
93         if (asistencia.Estado == "Puntual")
94         {
95             presentes++;
96         }
97         else if (asistencia.Estado == "Tardanza")
98         {
99             presentes++;
100             tardanzas++;
101         }
102         else if (asistencia.Estado == "Falta")
103         {
104             ausentes++;
105         }
106     }
107
108     int total = currentItems.Count;
109     decimal porcentaje = total == 0 ? 0 : Math.Round((presentes * 100m) / total, 0);
110     lblPresentesValor.Text = presentes.ToString();
111     lblAusentesValor.Text = ausentes.ToString();
112     lblTardanzasValor.Text = tardanzas.ToString();
113     lblPorcentajeValor.Text = porcentaje.ToString("0") + "%";
114 }
115
116 private void btnRegistrar_Click(object sender, EventArgs e)
117 {
118     if (cmbEmpleado.SelectedValue == null)
119     {
120         MessageBox.Show("Seleccione un empleado.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Warning);
```


UI/FrmAsistencia.cs (continuacion)

```
121         return;
122     }
123
124     if (cmbEstado.SelectedItem == null)
125     {
126         MessageBox.Show("Seleccione el estado de asistencia.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Warning);
127         return;
128     }
129
130     try
131     {
132         Asistencia asistencia = new Asistencia();
133         asistencia.IdEmpleado = Convert.ToInt32(cmbEmpleado.SelectedValue);
134         asistencia.Fecha = dtpFecha.Value.Date;
135         asistencia.HoraEntrada = dtpHoraEntrada.Value.TimeOfDay;
136         asistencia.HoraSalida = dtpHoraSalida.Value.TimeOfDay;
137         asistencia.Estado = cmbEstado.SelectedItem.ToString();
138         asistenciaService.Register(asistencia);
139         MessageBox.Show("Asistencia registrada correctamente.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Information);
140         LoadAttendance();
141     }
142     catch (Exception ex)
143     {
144         MessageBox.Show(ex.Message, "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Warning);
145     }
146 }
147
148 private void btnFiltrar_Click(object sender, EventArgs e)
149 {
150     LoadAttendance();
151 }
152
153 private void btnExportarExcel_Click(object sender, EventArgs e)
154 {
155     if (currentItems.Count == 0)
156     {
157         MessageBox.Show("No hay asistencias para exportar.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Information);
158         return;
159     }
160
161     UiFactory.ShowExported(excelExportService.ExportarAsistencias(currentItems));
162 }
163
164 private void btnExportarPdf_Click(object sender, EventArgs e)
165 {
166     if (currentItems.Count == 0)
167     {
168         MessageBox.Show("No hay asistencias para exportar.", "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Information);
169         return;
170     }
171
172     UiFactory.ShowExported(pdfExportService.ExportarAsistencias(currentItems));
173 }
174
175 private void btnCargarReloj_Click(object sender, EventArgs e)
176 {
177     using (OpenFileDialog dialog = new OpenFileDialog())
178     {
179         dialog.Filter = "Archivos CSV (*.csv)|*.csv|Todos los archivos (*.*)|*.*";
180         dialog.Title = "Importar asistencia desde archivo";
```

UI/FrmAsistencia.cs (continuacion)

```
181         if (dialog.ShowDialog(this) != DialogResult.OK)
182         {
183             return;
184         }
185
186         try
187         {
188             int imported = attendanceDeviceImportService.ImportarDesdeArchivo(dialog.FileName);
189             MessageBox.Show("Importación completada. Registros procesados: " + imported,
190                 "Asistencia", MessageBoxButtons.OK, MessageBoxIcon.Information);
191             LoadAttendance();
192         }
193         catch (Exception ex)
194         {
195             MessageBox.Show(ex.Message, "Importar asistencia", MessageBoxButtons.OK, MessageBoxIcon.Warning);
196         }
197     }
198 }
199
200 }
```

UI\FrmAsistencia.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmAsistencia
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelRegistro;
8         private System.Windows.Forms.ComboBox cmbEmpleado;
9         private System.Windows.Forms.Label lblEmpleado;
10        private System.Windows.Forms.Label lblFecha;
11        private System.Windows.Forms.Label lblEntrada;
12        private System.Windows.Forms.Label lblSalida;
13        private System.Windows.Forms.Label lblEstado;
14        private System.Windows.Forms.DateTimePicker dtpFecha;
15        private System.Windows.Forms.DateTimePicker dtpHoraEntrada;
16        private System.Windows.Forms.DateTimePicker dtpHoraSalida;
17        private System.Windows.Forms.ComboBox cmbEstado;
18        private System.Windows.Forms.Button btnRegistrar;
19        private System.Windows.Forms.Button btnCargarReloj;
20        private System.Windows.Forms.Panel panelFiltros;
21        private System.Windows.Forms.DateTimePicker dtpFiltroInicio;
22        private System.Windows.Forms.DateTimePicker dtpFiltroFin;
23        private System.Windows.Forms.ComboBox cmbFiltroEmpleado;
24        private System.Windows.Forms.ComboBox cmbFiltroEstado;
25        private System.Windows.Forms.Button btnFiltrar;
26        private System.Windows.Forms.Button btnExportarExcel;
27        private System.Windows.Forms.Button btnExportarPdf;
28        private System.Windows.Forms.Panel cardPresentes;
29        private System.Windows.Forms.Panel cardAusentes;
30        private System.Windows.Forms.Panel cardTardanzas;
31        private System.Windows.Forms.Panel cardPorcentaje;
32        private System.Windows.Forms.Label lblPresentesTitulo;
33        private System.Windows.Forms.Label lblAusentesTitulo;
34        private System.Windows.Forms.Label lblTardanzasTitulo;
35        private System.Windows.Forms.Label lblPorcentajeTitulo;
36        private System.Windows.Forms.Label lblPresentesValor;
37        private System.Windows.Forms.Label lblAusentesValor;
38        private System.Windows.Forms.Label lblTardanzasValor;
39        private System.Windows.Forms.Label lblPorcentajeValor;
40        private System.Windows.Forms.DataGridView dgvAsistencia;
41
42        protected override void Dispose(bool disposing)
43        {
44            if (disposing && (components != null)) components.Dispose();
45            base.Dispose(disposing);
46        }
47
48        private void InitializeComponent()
49        {
50            this.lblTitulo = new System.Windows.Forms.Label();
51            this.panelRegistro = new System.Windows.Forms.Panel();
52            this.lblEmpleado = new System.Windows.Forms.Label();
53            this.lblFecha = new System.Windows.Forms.Label();
54            this.lblEntrada = new System.Windows.Forms.Label();
55            this.lblSalida = new System.Windows.Forms.Label();
56            this.lblEstado = new System.Windows.Forms.Label();
57            this.cmbEmpleado = new System.Windows.Forms.ComboBox();
58            this.dtpFecha = new System.Windows.Forms.DateTimePicker();
59            this.dtpHoraEntrada = new System.Windows.Forms.DateTimePicker();
60            this.dtpHoraSalida = new System.Windows.Forms.DateTimePicker();
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
61         this.cmbEstado = new System.Windows.Forms.ComboBox();
62         this.btnRegistrar = new System.Windows.Forms.Button();
63         this.btnCargarReloj = new System.Windows.Forms.Button();
64         this.panelFiltros = new System.Windows.Forms.Panel();
65         this.dtpFiltroInicio = new System.Windows.Forms.DateTimePicker();
66         this.dtpFiltroFin = new System.Windows.Forms.DateTimePicker();
67         this.cmbFiltroEmpleado = new System.Windows.Forms.ComboBox();
68         this.cmbFiltroEstado = new System.Windows.Forms.ComboBox();
69         this.btnFiltrar = new System.Windows.Forms.Button();
70         this.btnExportarExcel = new System.Windows.Forms.Button();
71         this.btnExportarPdf = new System.Windows.Forms.Button();
72         this.cardPresentes = new System.Windows.Forms.Panel();
73         this.lblPresentesTitulo = new System.Windows.Forms.Label();
74         this.lblPresentesValor = new System.Windows.Forms.Label();
75         this.cardAusentes = new System.Windows.Forms.Panel();
76         this.lblAusentesTitulo = new System.Windows.Forms.Label();
77         this.lblAusentesValor = new System.Windows.Forms.Label();
78         this.cardTardanzas = new System.Windows.Forms.Panel();
79         this.lblTardanzasTitulo = new System.Windows.Forms.Label();
80         this.lblTardanzasValor = new System.Windows.Forms.Label();
81         this.cardPorcentaje = new System.Windows.Forms.Panel();
82         this.lblPorcentajeTitulo = new System.Windows.Forms.Label();
83         this.lblPorcentajeValor = new System.Windows.Forms.Label();
84         this.dgvAsistencia = new System.Windows.Forms.DataGridView();
85         this.dataGridViewTextBoxColumn1 = new System.Windows.Forms.DataGridViewTextBoxColumn();
86         this.dataGridViewTextBoxColumn2 = new System.Windows.Forms.DataGridViewTextBoxColumn();
87         this.dataGridViewTextBoxColumn3 = new System.Windows.Forms.DataGridViewTextBoxColumn();
88         this.dataGridViewTextBoxColumn4 = new System.Windows.Forms.DataGridViewTextBoxColumn();
89         this.dataGridViewTextBoxColumn5 = new System.Windows.Forms.DataGridViewTextBoxColumn();
90         this.dataGridViewTextBoxColumn6 = new System.Windows.Forms.DataGridViewTextBoxColumn();
91         this.panelRegistro.SuspendLayout();
92         this.panelFiltros.SuspendLayout();
93         this.cardPresentes.SuspendLayout();
94         this.cardAusentes.SuspendLayout();
95         this.cardTardanzas.SuspendLayout();
96         this.cardPorcentaje.SuspendLayout();
97         ((System.ComponentModel.ISupportInitialize)(this.dgvAsistencia)).BeginInit();
98         this.SuspendLayout();
99         //
100        // lblTitulo
101        //
102        this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
103        this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
104        this.lblTitulo.Location = new System.Drawing.Point(34, 24);
105        this.lblTitulo.Name = "lblTitulo";
106        this.lblTitulo.Size = new System.Drawing.Size(500, 42);
107        this.lblTitulo.TabIndex = 0;
108        this.lblTitulo.Text = "Control de Asistencia";
109        //
110        // panelRegistro
111        //
112        this.panelRegistro.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(17)))), ((int)(((byte)(18)))), ((int)(((byte)(26))))));
113        this.panelRegistro.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
114        this.panelRegistro.Controls.Add(this.lblEmpleado);
115        this.panelRegistro.Controls.Add(this.lblFecha);
116        this.panelRegistro.Controls.Add(this.lblEntrada);
117        this.panelRegistro.Controls.Add(this.lblSalida);
118        this.panelRegistro.Controls.Add(this.lblEstado);
119        this.panelRegistro.Controls.Add(this.cmbEmpleado);
120        this.panelRegistro.Controls.Add(this.dtpFecha);
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
121         this.panelRegistro.Controls.Add(this.dtpHoraEntrada);
122         this.panelRegistro.Controls.Add(this.dtpHoraSalida);
123         this.panelRegistro.Controls.Add(this.cmbEstado);
124         this.panelRegistro.Controls.Add(this.btnRegistrar);
125         this.panelRegistro.Controls.Add(this.btnCargarReloj);
126         this.panelRegistro.Location = new System.Drawing.Point(36, 92);
127         this.panelRegistro.Name = "panelRegistro";
128         this.panelRegistro.Size = new System.Drawing.Size(948, 92);
129         this.panelRegistro.TabIndex = 1;
130         //
131         // lblEmpleado
132         //
133         this.lblEmpleado.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
134         this.lblEmpleado.Location = new System.Drawing.Point(18, 22);
135         this.lblEmpleado.Name = "lblEmpleado";
136         this.lblEmpleado.Size = new System.Drawing.Size(120, 18);
137         this.lblEmpleado.TabIndex = 0;
138         this.lblEmpleado.Text = "Empleado";
139         //
140         // lblFecha
141         //
142         this.lblFecha.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
143         this.lblFecha.Location = new System.Drawing.Point(270, 22);
144         this.lblFecha.Name = "lblFecha";
145         this.lblFecha.Size = new System.Drawing.Size(80, 18);
146         this.lblFecha.TabIndex = 1;
147         this.lblFecha.Text = "Fecha";
148         //
149         // lblEntrada
150         //
151         this.lblEntrada.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
152         this.lblEntrada.Location = new System.Drawing.Point(392, 22);
153         this.lblEntrada.Name = "lblEntrada";
154         this.lblEntrada.Size = new System.Drawing.Size(90, 18);
155         this.lblEntrada.TabIndex = 2;
156         this.lblEntrada.Text = "Entrada";
157         //
158         // lblSalida
159         //
160         this.lblSalida.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
161         this.lblSalida.Location = new System.Drawing.Point(504, 22);
162         this.lblSalida.Name = "lblSalida";
163         this.lblSalida.Size = new System.Drawing.Size(90, 18);
164         this.lblSalida.TabIndex = 3;
165         this.lblSalida.Text = "Salida";
166         //
167         // lblEstado
168         //
169         this.lblEstado.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
170         this.lblEstado.Location = new System.Drawing.Point(616, 22);
171         this.lblEstado.Name = "lblEstado";
172         this.lblEstado.Size = new System.Drawing.Size(90, 18);
173         this.lblEstado.TabIndex = 4;
174         this.lblEstado.Text = "Estado";
175         //
176         // cmbEmpleado
177         //
178         this.cmbEmpleado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
179         this.cmbEmpleado.Location = new System.Drawing.Point(18, 44);
180         this.cmbEmpleado.Name = "cmbEmpleado";
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
181         this.cmbEmpleado.Size = new System.Drawing.Size(240, 28);
182         this.cmbEmpleado.TabIndex = 0;
183         //
184         // dtpFecha
185         //
186         this.dtpFecha.Format = System.Windows.Forms.DateTimePickerFormat.Short;
187         this.dtpFecha.Location = new System.Drawing.Point(270, 44);
188         this.dtpFecha.Name = "dtpFecha";
189         this.dtpFecha.Size = new System.Drawing.Size(110, 27);
190         this.dtpFecha.TabIndex = 1;
191         //
192         // dtpHoraEntrada
193         //
194         this.dtpHoraEntrada.Format = System.Windows.Forms.DateTimePickerFormat.Time;
195         this.dtpHoraEntrada.Location = new System.Drawing.Point(392, 44);
196         this.dtpHoraEntrada.Name = "dtpHoraEntrada";
197         this.dtpHoraEntrada.ShowUpDown = true;
198         this.dtpHoraEntrada.Size = new System.Drawing.Size(100, 27);
199         this.dtpHoraEntrada.TabIndex = 2;
200         //
201         // dtpHoraSalida
202         //
203         this.dtpHoraSalida.Format = System.Windows.Forms.DateTimePickerFormat.Time;
204         this.dtpHoraSalida.Location = new System.Drawing.Point(504, 44);
205         this.dtpHoraSalida.Name = "dtpHoraSalida";
206         this.dtpHoraSalida.ShowUpDown = true;
207         this.dtpHoraSalida.Size = new System.Drawing.Size(100, 27);
208         this.dtpHoraSalida.TabIndex = 3;
209         //
210         // cmbEstado
211         //
212         this.cmbEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
213         this.cmbEstado.Items.AddRange(new object[] {
214             "Puntual",
215             "Tardanza",
216             "Falta",
217             "Permiso"});
218         this.cmbEstado.Location = new System.Drawing.Point(616, 44);
219         this.cmbEstado.Name = "cmbEstado";
220         this.cmbEstado.Size = new System.Drawing.Size(120, 28);
221         this.cmbEstado.TabIndex = 4;
222         //
223         // btnRegistrar
224         //
225         this.btnRegistrar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
226         this.btnRegistrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
227         this.btnRegistrar.Location = new System.Drawing.Point(750, 38);
228         this.btnRegistrar.Name = "btnRegistrar";
229         this.btnRegistrar.Size = new System.Drawing.Size(90, 30);
230         this.btnRegistrar.TabIndex = 5;
231         this.btnRegistrar.Text = "Registrar";
232         this.btnRegistrar.UseVisualStyleBackColor = false;
233         this.btnRegistrar.Click += new System.EventHandler(this.btnRegistrar_Click);
234         //
235         // btnCargarReloj
236         //
237         this.btnCargarReloj.BackColor = System.Drawing.Color.Black;
238         this.btnCargarReloj.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
239         this.btnCargarReloj.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
240         this.btnCargarReloj.Location = new System.Drawing.Point(846, 38);
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
241         this.btnCargarReloj.Name = "btnCargarReloj";
242         this.btnCargarReloj.Size = new System.Drawing.Size(86, 30);
243         this.btnCargarReloj.TabIndex = 6;
244         this.btnCargarReloj.Text = "Reloj";
245         this.btnCargarReloj.UseVisualStyleBackColor = false;
246         this.btnCargarReloj.Click += new System.EventHandler(this.btnCargarReloj_Click);
247         //
248         // panelFiltros
249         //
250         this.panelFiltros.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(17)))), ((int)(((byte)(18)))), ((int)(((byte)(26))))));
251         this.panelFiltros.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
252         this.panelFiltros.Controls.Add(this.dtpFiltroInicio);
253         this.panelFiltros.Controls.Add(this.dtpFiltroFin);
254         this.panelFiltros.Controls.Add(this.cmbFiltroEmpleado);
255         this.panelFiltros.Controls.Add(this.cmbFiltroEstado);
256         this.panelFiltros.Controls.Add(this.btnFiltrar);
257         this.panelFiltros.Controls.Add(this.btnExportarExcel);
258         this.panelFiltros.Controls.Add(this.btnExportarPdf);
259         this.panelFiltros.Location = new System.Drawing.Point(36, 206);
260         this.panelFiltros.Name = "panelFiltros";
261         this.panelFiltros.Size = new System.Drawing.Size(948, 80);
262         this.panelFiltros.TabIndex = 2;
263         //
264         // dtpFiltroInicio
265         //
266         this.dtpFiltroInicio.Format = System.Windows.Forms.DateTimePickerFormat.Short;
267         this.dtpFiltroInicio.Location = new System.Drawing.Point(18, 28);
268         this.dtpFiltroInicio.Name = "dtpFiltroInicio";
269         this.dtpFiltroInicio.Size = new System.Drawing.Size(110, 27);
270         this.dtpFiltroInicio.TabIndex = 0;
271         //
272         // dtpFiltroFin
273         //
274         this.dtpFiltroFin.Format = System.Windows.Forms.DateTimePickerFormat.Short;
275         this.dtpFiltroFin.Location = new System.Drawing.Point(140, 28);
276         this.dtpFiltroFin.Name = "dtpFiltroFin";
277         this.dtpFiltroFin.Size = new System.Drawing.Size(110, 27);
278         this.dtpFiltroFin.TabIndex = 1;
279         //
280         // cmbFiltroEmpleado
281         //
282         this.cmbFiltroEmpleado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
283         this.cmbFiltroEmpleado.Location = new System.Drawing.Point(262, 28);
284         this.cmbFiltroEmpleado.Name = "cmbFiltroEmpleado";
285         this.cmbFiltroEmpleado.Size = new System.Drawing.Size(220, 28);
286         this.cmbFiltroEmpleado.TabIndex = 2;
287         //
288         // cmbFiltroEstado
289         //
290         this.cmbFiltroEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
291         this.cmbFiltroEstado.Items.AddRange(new object[] {
292             "Todos",
293             "Puntual",
294             "Tardanza",
295             "Falta",
296             "Permiso"});
297         this.cmbFiltroEstado.Location = new System.Drawing.Point(494, 28);
298         this.cmbFiltroEstado.Name = "cmbFiltroEstado";
299         this.cmbFiltroEstado.Size = new System.Drawing.Size(120, 28);
300         this.cmbFiltroEstado.TabIndex = 3;
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
301         //
302         // btnFiltrar
303         //
304         this.btnFiltrar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167))))), ((int)(((byte)(139))))), ((int)(((byte)(250)))));
305         this.btnFiltrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
306         this.btnFiltrar.Location = new System.Drawing.Point(630, 24);
307         this.btnFiltrar.Name = "btnFiltrar";
308         this.btnFiltrar.Size = new System.Drawing.Size(80, 30);
309         this.btnFiltrar.TabIndex = 4;
310         this.btnFiltrar.Text = "Filtrar";
311         this.btnFiltrar.UseVisualStyleBackColor = false;
312         this.btnFiltrar.Click += new System.EventHandler(this.btnFiltrar_Click);
313         //
314         // btnExportarExcel
315         //
316         this.btnExportarExcel.BackColor = System.Drawing.Color.Black;
317         this.btnExportarExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
318         this.btnExportarExcel.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229))))), ((int)(((byte)(228))))), ((int)(((byte)(245)))));
319         this.btnExportarExcel.Location = new System.Drawing.Point(724, 24);
320         this.btnExportarExcel.Name = "btnExportarExcel";
321         this.btnExportarExcel.Size = new System.Drawing.Size(80, 30);
322         this.btnExportarExcel.TabIndex = 5;
323         this.btnExportarExcel.Text = "Excel";
324         this.btnExportarExcel.UseVisualStyleBackColor = false;
325         this.btnExportarExcel.Click += new System.EventHandler(this.btnExportarExcel_Click);
326         //
327         // btnExportarPdf
328         //
329         this.btnExportarPdf.BackColor = System.Drawing.Color.Black;
330         this.btnExportarPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
331         this.btnExportarPdf.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229))))), ((int)(((byte)(228))))), ((int)(((byte)(245)))));
332         this.btnExportarPdf.Location = new System.Drawing.Point(816, 24);
333         this.btnExportarPdf.Name = "btnExportarPdf";
334         this.btnExportarPdf.Size = new System.Drawing.Size(80, 30);
335         this.btnExportarPdf.TabIndex = 6;
336         this.btnExportarPdf.Text = "PDF";
337         this.btnExportarPdf.UseVisualStyleBackColor = false;
338         this.btnExportarPdf.Click += new System.EventHandler(this.btnExportarPdf_Click);
339         //
340         // cardPresentes
341         //
342         this.cardPresentes.BackColor = System.Drawing.Color.Black;
343         this.cardPresentes.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
344         this.cardPresentes.Controls.Add(this.lblPresentesTitulo);
345         this.cardPresentes.Controls.Add(this.lblPresentesValor);
346         this.cardPresentes.Location = new System.Drawing.Point(36, 304);
347         this.cardPresentes.Name = "cardPresentes";
348         this.cardPresentes.Size = new System.Drawing.Size(222, 64);
349         this.cardPresentes.TabIndex = 3;
350         //
351         // lblPresentesTitulo
352         //
353         this.lblPresentesTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170))))), ((int)(((byte)(170))))), ((int)(((byte)(185)))));
354         this.lblPresentesTitulo.Location = new System.Drawing.Point(16, 10);
355         this.lblPresentesTitulo.Name = "lblPresentesTitulo";
356         this.lblPresentesTitulo.Size = new System.Drawing.Size(160, 18);
357         this.lblPresentesTitulo.TabIndex = 0;
358         this.lblPresentesTitulo.Text = "Presentes";
359         //
360         // lblPresentesValor
```


UI/FrmAsistencia.Designer.cs (continuacion)

```
361        //
362        this.lblPresentesValor.Font = new System.Drawing.Font("Segoe UI", 16F, System.Drawing.FontStyle.Bold);
363        this.lblPresentesValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(34)))), ((int)(((byte)(197)))), ((int)(((byte)(94))))));
364        this.lblPresentesValor.Location = new System.Drawing.Point(16, 28);
365        this.lblPresentesValor.Name = "lblPresentesValor";
366        this.lblPresentesValor.Size = new System.Drawing.Size(170, 35);
367        this.lblPresentesValor.TabIndex = 1;
368        this.lblPresentesValor.Text = "0";
369        //
370        // cardAusentes
371        //
372        this.cardAusentes.BackColor = System.Drawing.Color.Black;
373        this.cardAusentes.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
374        this.cardAusentes.Controls.Add(this.lblAusentesTitulo);
375        this.cardAusentes.Controls.Add(this.lblAusentesValor);
376        this.cardAusentes.Location = new System.Drawing.Point(278, 304);
377        this.cardAusentes.Name = "cardAusentes";
378        this.cardAusentes.Size = new System.Drawing.Size(222, 64);
379        this.cardAusentes.TabIndex = 4;
380        //
381        // lblAusentesTitulo
382        //
383        this.lblAusentesTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
384        this.lblAusentesTitulo.Location = new System.Drawing.Point(16, 10);
385        this.lblAusentesTitulo.Name = "lblAusentesTitulo";
386        this.lblAusentesTitulo.Size = new System.Drawing.Size(160, 18);
387        this.lblAusentesTitulo.TabIndex = 0;
388        this.lblAusentesTitulo.Text = "Ausentes";
389        //
390        // lblAusentesValor
391        //
392        this.lblAusentesValor.Font = new System.Drawing.Font("Segoe UI", 16F, System.Drawing.FontStyle.Bold);
393        this.lblAusentesValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(239)))), ((int)(((byte)(68)))), ((int)(((byte)(68))))));
394        this.lblAusentesValor.Location = new System.Drawing.Point(16, 28);
395        this.lblAusentesValor.Name = "lblAusentesValor";
396        this.lblAusentesValor.Size = new System.Drawing.Size(170, 34);
397        this.lblAusentesValor.TabIndex = 1;
398        this.lblAusentesValor.Text = "0";
399        //
400        // cardTardanzas
401        //
402        this.cardTardanzas.BackColor = System.Drawing.Color.Black;
403        this.cardTardanzas.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
404        this.cardTardanzas.Controls.Add(this.lblTardanzasTitulo);
405        this.cardTardanzas.Controls.Add(this.lblTardanzasValor);
406        this.cardTardanzas.Location = new System.Drawing.Point(520, 304);
407        this.cardTardanzas.Name = "cardTardanzas";
408        this.cardTardanzas.Size = new System.Drawing.Size(222, 64);
409        this.cardTardanzas.TabIndex = 5;
410        //
411        // lblTardanzasTitulo
412        //
413        this.lblTardanzasTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
414        this.lblTardanzasTitulo.Location = new System.Drawing.Point(16, 10);
415        this.lblTardanzasTitulo.Name = "lblTardanzasTitulo";
416        this.lblTardanzasTitulo.Size = new System.Drawing.Size(160, 18);
417        this.lblTardanzasTitulo.TabIndex = 0;
418        this.lblTardanzasTitulo.Text = "Tardanzas";
419        //
420        // lblTardanzasValor
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
421        //
422        this.lblTardanzasValor.Font = new System.Drawing.Font("Segoe UI", 16F, System.Drawing.FontStyle.Bold);
423        this.lblTardanzasValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(251))))), ((int)(((byte)(191))))), ((int)(((byte)(36))))));
424        this.lblTardanzasValor.Location = new System.Drawing.Point(16, 28);
425        this.lblTardanzasValor.Name = "lblTardanzasValor";
426        this.lblTardanzasValor.Size = new System.Drawing.Size(170, 34);
427        this.lblTardanzasValor.TabIndex = 1;
428        this.lblTardanzasValor.Text = "0";
429        //
430        // cardPorcentaje
431        //
432        this.cardPorcentaje.BackColor = System.Drawing.Color.Black;
433        this.cardPorcentaje.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
434        this.cardPorcentaje.Controls.Add(this.lblPorcentajeTitulo);
435        this.cardPorcentaje.Controls.Add(this.lblPorcentajeValor);
436        this.cardPorcentaje.Location = new System.Drawing.Point(762, 304);
437        this.cardPorcentaje.Name = "cardPorcentaje";
438        this.cardPorcentaje.Size = new System.Drawing.Size(222, 64);
439        this.cardPorcentaje.TabIndex = 6;
440        //
441        // lblPorcentajeTitulo
442        //
443        this.lblPorcentajeTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170))))), ((int)(((byte)(170))))), ((int)(((byte)(185))))));
444        this.lblPorcentajeTitulo.Location = new System.Drawing.Point(16, 10);
445        this.lblPorcentajeTitulo.Name = "lblPorcentajeTitulo";
446        this.lblPorcentajeTitulo.Size = new System.Drawing.Size(160, 18);
447        this.lblPorcentajeTitulo.TabIndex = 0;
448        this.lblPorcentajeTitulo.Text = "Asistencia";
449        //
450        // lblPorcentajeValor
451        //
452        this.lblPorcentajeValor.Font = new System.Drawing.Font("Segoe UI", 16F, System.Drawing.FontStyle.Bold);
453        this.lblPorcentajeValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(139))))), ((int)(((byte)(92))))), ((int)(((byte)(246))))));
454        this.lblPorcentajeValor.Location = new System.Drawing.Point(16, 28);
455        this.lblPorcentajeValor.Name = "lblPorcentajeValor";
456        this.lblPorcentajeValor.Size = new System.Drawing.Size(170, 34);
457        this.lblPorcentajeValor.TabIndex = 1;
458        this.lblPorcentajeValor.Text = "0%";
459        //
460        // dgvAsistencia
461        //
462        this.dgvAsistencia.AllowUserToAddRows = false;
463        this.dgvAsistencia.AllowUserToDeleteRows = false;
464        this.dgvAsistencia.BackgroundColor = System.Drawing.Color.Black;
465        this.dgvAsistencia.ColumnHeadersHeightSizeMode = System.Windows.Forms.DataGridViewColumnHeadersHeightSizeMode.AutoSize;
466        this.dgvAsistencia.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] {
467            this.dataGridViewTextBoxColumn1,
468            this.dataGridViewTextBoxColumn2,
469            this.dataGridViewTextBoxColumn3,
470            this.dataGridViewTextBoxColumn4,
471            this.dataGridViewTextBoxColumn5,
472            this.dataGridViewTextBoxColumn6});
473        this.dgvAsistencia.Location = new System.Drawing.Point(36, 392);
474        this.dgvAsistencia.Name = "dgvAsistencia";
475        this.dgvAsistencia.ReadOnly = true;
476        this.dgvAsistencia.RowHeadersVisible = false;
477        this.dgvAsistencia.RowHeadersWidth = 51;
478        this.dgvAsistencia.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect;
479        this.dgvAsistencia.Size = new System.Drawing.Size(948, 286);
480        this.dgvAsistencia.TabIndex = 7;
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
481         //
482         // dataGridViewTextBoxColumn1
483         //
484         this.dataGridViewTextBoxColumn1.HeaderText = "Empleado";
485         this.dataGridViewTextBoxColumn1.MinimumWidth = 6;
486         this.dataGridViewTextBoxColumn1.Name = "dataGridViewTextBoxColumn1";
487         this.dataGridViewTextBoxColumn1.ReadOnly = true;
488         this.dataGridViewTextBoxColumn1.Width = 125;
489         //
490         // dataGridViewTextBoxColumn2
491         //
492         this.dataGridViewTextBoxColumn2.HeaderText = "Fecha";
493         this.dataGridViewTextBoxColumn2.MinimumWidth = 6;
494         this.dataGridViewTextBoxColumn2.Name = "dataGridViewTextBoxColumn2";
495         this.dataGridViewTextBoxColumn2.ReadOnly = true;
496         this.dataGridViewTextBoxColumn2.Width = 125;
497         //
498         // dataGridViewTextBoxColumn3
499         //
500         this.dataGridViewTextBoxColumn3.HeaderText = "Entrada";
501         this.dataGridViewTextBoxColumn3.MinimumWidth = 6;
502         this.dataGridViewTextBoxColumn3.Name = "dataGridViewTextBoxColumn3";
503         this.dataGridViewTextBoxColumn3.ReadOnly = true;
504         this.dataGridViewTextBoxColumn3.Width = 125;
505         //
506         // dataGridViewTextBoxColumn4
507         //
508         this.dataGridViewTextBoxColumn4.HeaderText = "Salida";
509         this.dataGridViewTextBoxColumn4.MinimumWidth = 6;
510         this.dataGridViewTextBoxColumn4.Name = "dataGridViewTextBoxColumn4";
511         this.dataGridViewTextBoxColumn4.ReadOnly = true;
512         this.dataGridViewTextBoxColumn4.Width = 125;
513         //
514         // dataGridViewTextBoxColumn5
515         //
516         this.dataGridViewTextBoxColumn5.HeaderText = "Horas";
517         this.dataGridViewTextBoxColumn5.MinimumWidth = 6;
518         this.dataGridViewTextBoxColumn5.Name = "dataGridViewTextBoxColumn5";
519         this.dataGridViewTextBoxColumn5.ReadOnly = true;
520         this.dataGridViewTextBoxColumn5.Width = 125;
521         //
522         // dataGridViewTextBoxColumn6
523         //
524         this.dataGridViewTextBoxColumn6.HeaderText = "Estado";
525         this.dataGridViewTextBoxColumn6.MinimumWidth = 6;
526         this.dataGridViewTextBoxColumn6.Name = "dataGridViewTextBoxColumn6";
527         this.dataGridViewTextBoxColumn6.ReadOnly = true;
528         this.dataGridViewTextBoxColumn6.Width = 125;
529         //
530         // FrmAsistencia
531         //
532         this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F);
533         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
534         this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
535         this.ClientSize = new System.Drawing.Size(1020, 742);
536         this.Controls.Add(this.dgvAsistencia);
537         this.Controls.Add(this.cardPorcentaje);
538         this.Controls.Add(this.cardTardanzas);
539         this.Controls.Add(this.cardAusentes);
540         this.Controls.Add(this.cardPresentes);
```

UI/FrmAsistencia.Designer.cs (continuacion)

```
541         this.Controls.Add(this.panelFiltros);
542         this.Controls.Add(this.panelRegistro);
543         this.Controls.Add(this.lblTitulo);
544         this.Font = new System.Drawing.Font("Segoe UI", 9F);
545         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
546         this.Name = "FrmAsistencia";
547         this.Text = "Asistencia";
548         this.Load += new System.EventHandler(this.FrmAsistencia_Load);
549         this.panelRegistro.ResumeLayout(false);
550         this.panelFiltros.ResumeLayout(false);
551         this.cardPresentes.ResumeLayout(false);
552         this.cardAusentes.ResumeLayout(false);
553         this.cardTardanzas.ResumeLayout(false);
554         this.cardPorcentaje.ResumeLayout(false);
555         ((System.ComponentModel.ISupportInitialize)(this.dgvAsistencia)).EndInit();
556         this.ResumeLayout(false);
557
558     }
559
560     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn1;
561     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn2;
562     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn3;
563     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn4;
564     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn5;
565     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn6;
566 }
567 }
```

UI/FrmAuditoria.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmAuditoria : Form
13     {
14         private const int PageSize = 100;
15         private readonly AuditTrailService auditTrailService = new AuditTrailService();
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         private int pageIndex;
18
19         public FrmAuditoria()
20         {
21             InitializeComponent();
22             ControlStyleHelper.ApplyModernForm(this);
23         }
24
25         private void FrmAuditoria_Load(object sender, EventArgs e)
26         {
27             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
28             try
29             {
30                 authorizationService.DemandPermission(Permissions.AuditView);
31                 LoadFilters();
32                 LoadRecords();
33             }
34             catch (Exception ex)
35             {
36                 MessageBox.Show(ex.Message, "Auditoría", MessageBoxButtons.OK, MessageBoxIcon.Warning);
37             }
38         }
39
40         private void LoadFilters()
41         {
42             cmbModulo.Items.Clear();
43             cmbModulo.Items.Add("Todos");
44             foreach (string module in auditTrailService.ObtenerModulos()) cmbModulo.Items.Add(module);
45             cmbModulo.SelectedIndex = 0;
46
47             cmbAccion.Items.Clear();
48             cmbAccion.Items.Add("Todos");
49             foreach (string action in auditTrailService.ObtenerAcciones()) cmbAccion.Items.Add(action);
50             cmbAccion.SelectedIndex = 0;
51         }
52
53         private void LoadRecords()
54         {
55             AuditQuery query = new AuditQuery
56             {
57                 Usuario = txtUsuario.Text.Trim(),
58                 Modulo = SelectedFilter(cmbModulo),
59                 Accion = SelectedFilter(cmbAccion),
60                 Detalle = txtDetalle.Text.Trim(),
```

UI/FrmAuditoria.cs (continuacion)

```
61         FechaDesde = dtpDesde.Checked ? dtpDesde.Value.Date : (DateTime?)null,
62         FechaHasta = dtpHasta.Checked ? dtpHasta.Value.Date : (DateTime?)null,
63         Offset = pageIndex * PageSize,
64         Limit = PageSize
65     };
66
67     if (query.FechaDesde.HasValue && query.FechaHasta.HasValue &&
68         query.FechaDesde.Value > query.FechaHasta.Value)
69     {
70         throw new ArgumentException("La fecha inicial no puede ser mayor que la fecha final.");
71     }
72
73     List<AuditRecord> records = auditTrailService.Buscar(query);
74     dgvAuditoria.Rows.Clear();
75     foreach (AuditRecord record in records)
76     {
77         dgvAuditoria.Rows.Add(record.Fecha.ToString("dd/MM/yyyy HH:mm:ss"), record.Usuario,
78             record.Modulo, record.Accion, record.Detalle);
79     }
80
81     lblPagina.Text = "Página " + (pageIndex + 1) + " . " + records.Count + " registros";
82     btnAnterior.Enabled = pageIndex > 0;
83     btnSiguiente.Enabled = records.Count == PageSize;
84 }
85
86 private static string SelectedFilter(ComboBox combo)
87 {
88     return combo.SelectedItem == null || combo.SelectedItem.ToString() == "Todos"
89         ? string.Empty : combo.SelectedItem.ToString();
90 }
91
92 private void btnBuscar_Click(object sender, EventArgs e)
93 {
94     pageIndex = 0;
95     TryLoadRecords();
96 }
97
98 private void btnLimpiar_Click(object sender, EventArgs e)
99 {
100     txtUsuario.Clear();
101     txtDetalle.Clear();
102     cmbModulo.SelectedIndex = 0;
103     cmbAccion.SelectedIndex = 0;
104     dtpDesde.Checked = false;
105     dtpHasta.Checked = false;
106     pageIndex = 0;
107     TryLoadRecords();
108 }
109
110 private void btnAnterior_Click(object sender, EventArgs e)
111 {
112     if (pageIndex > 0) pageIndex--;
113     TryLoadRecords();
114 }
115
116 private void btnSiguiente_Click(object sender, EventArgs e)
117 {
118     pageIndex++;
119     TryLoadRecords();
120 }
```

UI/FrmAuditoria.cs (continuacion)

```
121
122     private void TryLoadRecords()
123     {
124         try
125         {
126             LoadRecords();
127         }
128         catch (Exception ex)
129         {
130             MessageBox.Show(ex.Message, "Auditoría", MessageBoxButtons.OK, MessageBoxIcon.Warning);
131         }
132     }
133 }
134 }
```

UI\FrmAuditoria.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmAuditoria
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelFiltros;
8         private System.Windows.Forms.Label lblUsuario;
9         private System.Windows.Forms.TextBox txtUsuario;
10        private System.Windows.Forms.Label lblModulo;
11        private System.Windows.Forms.ComboBox cmbModulo;
12        private System.Windows.Forms.Label lblAccion;
13        private System.Windows.Forms.ComboBox cmbAccion;
14        private System.Windows.Forms.Label lblDetalle;
15        private System.Windows.Forms.TextBox txtDetalle;
16        private System.Windows.Forms.Label lblDesde;
17        private System.Windows.Forms.DateTimePicker dtpDesde;
18        private System.Windows.Forms.Label lblHasta;
19        private System.Windows.Forms.DateTimePicker dtpHasta;
20        private System.Windows.Forms.Button btnBuscar;
21        private System.Windows.Forms.Button btnLimpiar;
22        private System.Windows.Forms.DataGridView dgvAuditoria;
23        private System.Windows.Forms.DataGridViewTextBoxColumn colFecha;
24        private System.Windows.Forms.DataGridViewTextBoxColumn colUsuario;
25        private System.Windows.Forms.DataGridViewTextBoxColumn colModulo;
26        private System.Windows.Forms.DataGridViewTextBoxColumn colAccion;
27        private System.Windows.Forms.DataGridViewTextBoxColumn colDetalle;
28        private System.Windows.Forms.Button btnAnterior;
29        private System.Windows.Forms.Button btnSiguiente;
30        private System.Windows.Forms.Label lblPagina;
31
32        protected override void Dispose(bool disposing)
33        {
34            if (disposing && (components != null)) components.Dispose();
35            base.Dispose(disposing);
36        }
37
38        private void InitializeComponent()
39        {
40            System.Windows.Forms.DataGridViewCellStyle headerStyle = new System.Windows.Forms.DataGridViewCellStyle();
41            System.Windows.Forms.DataGridViewCellStyle cellStyle = new System.Windows.Forms.DataGridViewCellStyle();
42            this.lblTitulo = new System.Windows.Forms.Label();
43            this.panelFiltros = new System.Windows.Forms.Panel();
44            this.lblUsuario = new System.Windows.Forms.Label();
45            this.txtUsuario = new System.Windows.Forms.TextBox();
46            this.lblModulo = new System.Windows.Forms.Label();
47            this.cmbModulo = new System.Windows.Forms.ComboBox();
48            this.lblAccion = new System.Windows.Forms.Label();
49            this.cmbAccion = new System.Windows.Forms.ComboBox();
50            this.lblDetalle = new System.Windows.Forms.Label();
51            this.txtDetalle = new System.Windows.Forms.TextBox();
52            this.lblDesde = new System.Windows.Forms.Label();
53            this.dtpDesde = new System.Windows.Forms.DateTimePicker();
54            this.lblHasta = new System.Windows.Forms.Label();
55            this.dtpHasta = new System.Windows.Forms.DateTimePicker();
56            this.btnBuscar = new System.Windows.Forms.Button();
57            this.btnLimpiar = new System.Windows.Forms.Button();
58            this.dgvAuditoria = new System.Windows.Forms.DataGridView();
59            this.colFecha = new System.Windows.Forms.DataGridViewTextBoxColumn();
60            this.colUsuario = new System.Windows.Forms.DataGridViewTextBoxColumn();
```


UI/FrmAuditoria.Designer.cs (continuacion)

```

61         this.colModulo = new System.Windows.Forms.DataGridViewTextBoxColumn();
62         this.colAccion = new System.Windows.Forms.DataGridViewTextBoxColumn();
63         this.colDetalle = new System.Windows.Forms.DataGridViewTextBoxColumn();
64         this.btnAnterior = new System.Windows.Forms.Button();
65         this.btnSiguiente = new System.Windows.Forms.Button();
66         this.lblPagina = new System.Windows.Forms.Label();
67         this.panelFiltros.SuspendLayout();
68         ((System.ComponentModel.ISupportInitialize)(this.dgvAuditoria)).BeginInit();
69         this.SuspendLayout();
70         // lblTitulo
71         this.lblTitulo.AutoSize = true;
72         this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
73         this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
74         this.lblTitulo.Location = new System.Drawing.Point(28, 20);
75         this.lblTitulo.Name = "lblTitulo";
76         this.lblTitulo.Size = new System.Drawing.Size(329, 50);
77         this.lblTitulo.Text = "Auditoría del sistema";
78         // panelFiltros
79         this.panelFiltros.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Left) | System.Windows.Forms.AnchorStyles.Right));
80         this.panelFiltros.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
81         this.panelFiltros.Controls.Add(this.lblUsuario);
82         this.panelFiltros.Controls.Add(this.txtUsuario);
83         this.panelFiltros.Controls.Add(this.lblModulo);
84         this.panelFiltros.Controls.Add(this.cmbModulo);
85         this.panelFiltros.Controls.Add(this.lblAccion);
86         this.panelFiltros.Controls.Add(this.cmbAccion);
87         this.panelFiltros.Controls.Add(this.lblDetalle);
88         this.panelFiltros.Controls.Add(this.txtDetalle);
89         this.panelFiltros.Controls.Add(this.lblDesde);
90         this.panelFiltros.Controls.Add(this.dtpDesde);
91         this.panelFiltros.Controls.Add(this.lblHasta);
92         this.panelFiltros.Controls.Add(this.dtpHasta);
93         this.panelFiltros.Controls.Add(this.btnBuscar);
94         this.panelFiltros.Controls.Add(this.btnLimpiar);
95         this.panelFiltros.Location = new System.Drawing.Point(32, 82);
96         this.panelFiltros.Name = "panelFiltros";
97         this.panelFiltros.Size = new System.Drawing.Size(956, 136);
98         // labels
99         this.lblUsuario.AutoSize = true; this.lblUsuario.Location = new System.Drawing.Point(18, 14); this.lblUsuario.Text = "Usuario";
100        this.lblModulo.AutoSize = true; this.lblModulo.Location = new System.Drawing.Point(206, 14); this.lblModulo.Text = "Módulo";
101        this.lblAccion.AutoSize = true; this.lblAccion.Location = new System.Drawing.Point(394, 14); this.lblAccion.Text = "Acción";
102        this.lblDetalle.AutoSize = true; this.lblDetalle.Location = new System.Drawing.Point(582, 14); this.lblDetalle.Text = "Detalle";
103        this.lblDesde.AutoSize = true; this.lblDesde.Location = new System.Drawing.Point(18, 75); this.lblDesde.Text = "Desde";
104        this.lblHasta.AutoSize = true; this.lblHasta.Location = new System.Drawing.Point(238, 75); this.lblHasta.Text = "Hasta";
105        // inputs
106        this.txtUsuario.Location = new System.Drawing.Point(18, 36); this.txtUsuario.Size = new System.Drawing.Size(170, 27);
107        this.cmbModulo.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbModulo.Location = new System.Drawing.Point(206, 36); \
this.cmbModulo.Size = new System.Drawing.Size(170, 28);
108        this.cmbAccion.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbAccion.Location = new System.Drawing.Point(394, 36); \
this.cmbAccion.Size = new System.Drawing.Size(170, 28);
109        this.txtDetalle.Location = new System.Drawing.Point(582, 36); this.txtDetalle.Size = new System.Drawing.Size(350, 27);
110        this.dtpDesde.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpDesde.Location = new System.Drawing.Point(18, 96); \
this.dtpDesde.ShowCheckBox = true; this.dtpDesde.Size = new System.Drawing.Size(202, 27);
111        this.dtpHasta.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpHasta.Location = new System.Drawing.Point(238, 96); \
this.dtpHasta.ShowCheckBox = true; this.dtpHasta.Size = new System.Drawing.Size(202, 27);
112        // buttons
113        this.btnBuscar.BackColor = System.Drawing.Color.FromArgb(124, 58, 237); this.btnBuscar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnBuscar.ForeColor = System.Drawing.Color.White; this.btnBuscar.Location = new System.Drawing.Point(674, 88); this.btnBuscar.Size = new \
System.Drawing.Size(122, 36); this.btnBuscar.Text = "Buscar"; this.btnBuscar.UseVisualStyleBackColor = false; this.btnBuscar.Click += new \

```

UI/FrmAuditoria.Designer.cs (continuacion)

```

114     System.EventHandler(this.btnBuscar_Click);
        this.btnLimpiar.BackColor = System.Drawing.Color.FromArgb(42, 43, 54); this.btnLimpiar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
        this.btnLimpiar.ForeColor = System.Drawing.Color.White; this.btnLimpiar.Location = new System.Drawing.Point(810, 88); this.btnLimpiar.Size = new \
        System.Drawing.Size(122, 36); this.btnLimpiar.Text = "Limpiar"; this.btnLimpiar.UseVisualStyleBackColor = false; this.btnLimpiar.Click += new \
        System.EventHandler(this.btnLimpiar_Click);
115     // dgvAuditoria
116     this.dgvAuditoria.AllowUserToAddRows = false;
117     this.dgvAuditoria.AllowUserToDeleteRows = false;
118     this.dgvAuditoria.AllowUserToResizeRows = false;
119     this.dgvAuditoria.Anchor = ((System.Windows.Forms.AnchorStyles)((((System.Windows.Forms.AnchorStyles.Top | \
        System.Windows.Forms.AnchorStyles.Bottom) | System.Windows.Forms.AnchorStyles.Left) | System.Windows.Forms.AnchorStyles.Right)));
120     this.dgvAuditoria.AutoSizeColumnsMode = System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill;
121     this.dgvAuditoria.BackgroundColor = System.Drawing.Color.FromArgb(18, 19, 27);
122     this.dgvAuditoria.BorderStyle = System.Windows.Forms.BorderStyle.None;
123     headerStyle.BackColor = System.Drawing.Color.FromArgb(42, 43, 54); headerStyle.ForeColor = System.Drawing.Color.White; \
        headerStyle.SelectionBackColor = System.Drawing.Color.FromArgb(42, 43, 54); headerStyle.Font = new System.Drawing.Font("Segoe UI", 9F, \
        System.Drawing.FontStyle.Bold);
124     this.dgvAuditoria.ColumnHeadersDefaultCellStyle = headerStyle;
125     this.dgvAuditoria.ColumnHeadersHeight = 40;
126     this.dgvAuditoria.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] { this.colFecha, this.colUsuario, this.colModulo, \
        this.colAccion, this.colDetalle });
127     cellStyle.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); cellStyle.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); \
        cellStyle.SelectionBackColor = System.Drawing.Color.FromArgb(76, 59, 118); cellStyle.SelectionForeColor = System.Drawing.Color.White;
128     this.dgvAuditoria.DefaultCellStyle = cellStyle;
129     this.dgvAuditoria.EnableHeadersVisualStyles = false;
130     this.dgvAuditoria.Location = new System.Drawing.Point(32, 236);
131     this.dgvAuditoria.MultiSelect = false;
132     this.dgvAuditoria.ReadOnly = true;
133     this.dgvAuditoria.RowHeadersVisible = false;
134     this.dgvAuditoria.RowTemplate.Height = 32;
135     this.dgvAuditoria.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect;
136     this.dgvAuditoria.Size = new System.Drawing.Size(956, 418);
137     this.colFecha.HeaderText = "Fecha"; this.colFecha.FillWeight = 95F; this.colFecha.ReadOnly = true;
138     this.colUsuario.HeaderText = "Usuario"; this.colUsuario.FillWeight = 75F; this.colUsuario.ReadOnly = true;
139     this.colModulo.HeaderText = "Módulo"; this.colModulo.FillWeight = 75F; this.colModulo.ReadOnly = true;
140     this.colAccion.HeaderText = "Acción"; this.colAccion.FillWeight = 95F; this.colAccion.ReadOnly = true;
141     this.colDetalle.HeaderText = "Detalle"; this.colDetalle.FillWeight = 190F; this.colDetalle.ReadOnly = true;
142     // pagination
143     this.btnAnterior.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Bottom | \
        System.Windows.Forms.AnchorStyles.Right))); this.btnAnterior.Location = new System.Drawing.Point(764, 670); this.btnAnterior.Size = new \
        System.Drawing.Size(100, 34); this.btnAnterior.Text = "Anterior"; this.btnAnterior.Click += new System.EventHandler(this.btnAnterior_Click);
144     this.btnSiguiente.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Bottom | \
        System.Windows.Forms.AnchorStyles.Right))); this.btnSiguiente.Location = new System.Drawing.Point(888, 670); this.btnSiguiente.Size = new \
        System.Drawing.Size(100, 34); this.btnSiguiente.Text = "Siguiente"; this.btnSiguiente.Click += new System.EventHandler(this.btnSiguiente_Click);
145     this.lblPagina.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Bottom | \
        System.Windows.Forms.AnchorStyles.Left))); this.lblPagina.Location = new System.Drawing.Point(32, 675); this.lblPagina.Size = new \
        System.Drawing.Size(300, 24); this.lblPagina.Text = "Página 1";
146     // FrmAuditoria
147     this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F);
148     this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
149     this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
150     this.ClientSize = new System.Drawing.Size(1020, 720);
151     this.Controls.Add(this.lblPagina);
152     this.Controls.Add(this.btnAnterior);
153     this.Controls.Add(this.btnSiguiente);
154     this.Controls.Add(this.dgvAuditoria);
155     this.Controls.Add(this.panelFiltros);
156     this.Controls.Add(this.lblTitulo);
157     this.Font = new System.Drawing.Font("Segoe UI", 9F);
158     this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);

```

UI/FrmAuditoria.Designer.cs (continuación)

```
159         this.Name = "FrmAuditoria";
160         this.Text = "Auditoria";
161         this.Load += new System.EventHandler(this.FrmAuditoria_Load);
162         this.panelFiltros.ResumeLayout(false);
163         this.panelFiltros.PerformLayout();
164         ((System.ComponentModel.ISupportInitialize)(this.dgvAuditoria)).EndInit();
165         this.ResumeLayout(false);
166         this.PerformLayout();
167     }
168 }
169 }
```

UI/FrmCambiarPassword.cs

```
1 using System.Windows.Forms;
2 using SistemaGestionNomina.Helpers;
3 using SistemaGestionNomina.Services;
4
5 namespace SistemaGestionNomina.UI
6 {
7     public partial class FrmCambiarPassword : Form
8     {
9         private readonly AccountService accountService = new AccountService();
10
11         public FrmCambiarPassword()
12         {
13             InitializeComponent();
14             ControlStyleHelper.ApplyModernForm(this);
15         }
16
17         private void ClearFields()
18         {
19             txtActual.Clear();
20             txtNueva.Clear();
21             txtConfirmar.Clear();
22             txtActual.Focus();
23         }
24
25         private void btnGuardar_Click(object sender, System.EventArgs e)
26         {
27             try
28             {
29                 accountService.ChangeOwnPassword(txtActual.Text, txtNueva.Text, txtConfirmar.Text);
30                 MessageBox.Show("La contraseña se actualizó correctamente.", "Cuenta",
31                     MessageBoxButtons.OK, MessageBoxIcon.Information);
32                 ClearFields();
33             }
34             catch (System.Exception ex)
35             {
36                 MessageBox.Show(ex.Message, "Cambiar contraseña", MessageBoxButtons.OK, MessageBoxIcon.Warning);
37             }
38         }
39     }
40 }
```

UI/FrmCambiarPassword.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmCambiarPassword
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Label lblSubtitulo;
8         private System.Windows.Forms.Panel panelCuenta;
9         private System.Windows.Forms.Label lblActual;
10        private System.Windows.Forms.TextBox txtActual;
11        private System.Windows.Forms.Label lblNueva;
12        private System.Windows.Forms.TextBox txtNueva;
13        private System.Windows.Forms.Label lblConfirmar;
14        private System.Windows.Forms.TextBox txtConfirmar;
15        private System.Windows.Forms.Label lblPolitica;
16        private System.Windows.Forms.Button btnGuardar;
17
18        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
19
20        private void InitializeComponent()
21        {
22            this.lblTitulo = new System.Windows.Forms.Label(); this.lblSubtitulo = new System.Windows.Forms.Label(); this.panelCuenta = new \
System.Windows.Forms.Panel(); this.lblActual = new System.Windows.Forms.Label(); this.txtActual = new System.Windows.Forms.TextBox(); this.lblNueva = \
new System.Windows.Forms.Label(); this.txtNueva = new System.Windows.Forms.TextBox(); this.lblConfirmar = new System.Windows.Forms.Label(); \
this.txtConfirmar = new System.Windows.Forms.TextBox(); this.lblPolitica = new System.Windows.Forms.Label(); this.btnGuardar = new \
System.Windows.Forms.Button(); this.panelCuenta.SuspendLayout(); this.SuspendLayout();
23            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 23F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(30, 24); this.lblTitulo.Text = "Cambiar contraseña";
24            this.lblSubtitulo.AutoSize = true; this.lblSubtitulo.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblSubtitulo.Location = \
new System.Drawing.Point(34, 78); this.lblSubtitulo.Text = "Actualiza la clave utilizada para iniciar sesión.";
25            this.panelCuenta.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); this.panelCuenta.Controls.Add(this.lblActual); \
this.panelCuenta.Controls.Add(this.txtActual); this.panelCuenta.Controls.Add(this.lblNueva); this.panelCuenta.Controls.Add(this.txtNueva); \
this.panelCuenta.Controls.Add(this.lblConfirmar); this.panelCuenta.Controls.Add(this.txtConfirmar); this.panelCuenta.Controls.Add(this.lblPolitica); \
this.panelCuenta.Controls.Add(this.btnGuardar); this.panelCuenta.Location = new System.Drawing.Point(38, 126); this.panelCuenta.Name = "panelCuenta"; \
this.panelCuenta.Size = new System.Drawing.Size(560, 444);
26            this.lblActual.AutoSize = true; this.lblActual.Location = new System.Drawing.Point(28, 28); this.lblActual.Text = "Contraseña actual";
27            this.txtActual.Location = new System.Drawing.Point(28, 56); this.txtActual.Name = "txtActual"; this.txtActual.Size = new \
System.Drawing.Size(500, 27); this.txtActual.UseSystemPasswordChar = true;
28            this.lblNueva.AutoSize = true; this.lblNueva.Location = new System.Drawing.Point(28, 110); this.lblNueva.Text = "Nueva contraseña";
29            this.txtNueva.Location = new System.Drawing.Point(28, 138); this.txtNueva.Name = "txtNueva"; this.txtNueva.Size = new \
System.Drawing.Size(500, 27); this.txtNueva.UseSystemPasswordChar = true;
30            this.lblConfirmar.AutoSize = true; this.lblConfirmar.Location = new System.Drawing.Point(28, 192); this.lblConfirmar.Text = "Confirmar nueva \
contraseña";
31            this.txtConfirmar.Location = new System.Drawing.Point(28, 220); this.txtConfirmar.Name = "txtConfirmar"; this.txtConfirmar.Size = new \
System.Drawing.Size(500, 27); this.txtConfirmar.UseSystemPasswordChar = true;
32            this.lblPolitica.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblPolitica.Location = new System.Drawing.Point(28, 274); \
this.lblPolitica.Name = "lblPolitica"; this.lblPolitica.Size = new System.Drawing.Size(500, 52); this.lblPolitica.Text = "Mínimo 8 caracteres, con \
mayúscula y número.";
33            this.btnGuardar.Location = new System.Drawing.Point(28, 354); this.btnGuardar.Name = "btnGuardar"; this.btnGuardar.Size = new \
System.Drawing.Size(500, 44); this.btnGuardar.Text = "Guardar nueva contraseña"; this.btnGuardar.Click += new \
System.EventHandler(this.btnGuardar_Click);
34            this.AcceptButton = this.btnGuardar; this.AutoScaleMode = new System.Drawing.SizeF(8F, 20F); this.AutoScaleMode = \
System.Windows.Forms.AutoScaleMode.Font; this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20); this.ClientSize = new System.Drawing.Size(1020, \
742); this.Controls.Add(this.panelCuenta); this.Controls.Add(this.lblSubtitulo); this.Controls.Add(this.lblTitulo); this.Font = new \
System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.Name = "FrmCambiarPassword"; this.Text = \
"Cambiar contraseña"; this.panelCuenta.ResumeLayout(false); this.panelCuenta.PerformLayout(); this.ResumeLayout(false); this.PerformLayout();
35        }
36    }
37 }

```

UI/FrmCambioLaboral.cs

```
1 using System;
2 using System.Windows.Forms;
3 using SistemaGestionNomina.Helpers;
4 using SistemaGestionNomina.Models;
5
6 namespace SistemaGestionNomina.UI
7 {
8     public partial class FrmCambioLaboral : Form
9     {
10         public EmployeeChangeContext ChangeContext { get; private set; }
11
12         public FrmCambioLaboral() : this(null) { }
13
14         public FrmCambioLaboral(Empleado employee)
15         {
16             InitializeComponent();
17             ControlStyleHelper.ApplyModernForm(this);
18             if (employee != null)
19                 lblEmpleado.Text = employee.Codigo + " - " + employee.NombreCompleto;
20         }
21
22         private void btnAceptar_Click(object sender, EventArgs e)
23         {
24             if (string.IsNullOrEmpty(txtMotivo.Text))
25             {
26                 MessageBox.Show("Indique el motivo del cambio laboral.", "Cambio laboral",
27                     MessageBoxButtons.OK, MessageBoxIcon.Warning);
28                 txtMotivo.Focus();
29                 return;
30             }
31             ChangeContext = new EmployeeChangeContext
32             {
33                 EffectiveDate = dtpFechaEfectiva.Value.Date,
34                 Reason = txtMotivo.Text.Trim()
35             };
36             DialogResult = DialogResult.OK;
37             Close();
38         }
39
40         private void btnCancelar_Click(object sender, EventArgs e)
41         {
42             DialogResult = DialogResult.Cancel;
43             Close();
44         }
45     }
46 }
```

UI/FrmCambioLaboral.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmCambioLaboral
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Label lblEmpleado;
8         private System.Windows.Forms.Label lblFecha;
9         private System.Windows.Forms.DateTimePicker dtpFechaEfectiva;
10        private System.Windows.Forms.Label lblMotivo;
11        private System.Windows.Forms.TextBox txtMotivo;
12        private System.Windows.Forms.Button btnAceptar;
13        private System.Windows.Forms.Button btnCancelar;
14
15        protected override void Dispose(bool disposing)
16        {
17            if (disposing && (components != null)) components.Dispose();
18            base.Dispose(disposing);
19        }
20
21        private void InitializeComponent()
22        {
23            this.lblTitulo = new System.Windows.Forms.Label();
24            this.lblEmpleado = new System.Windows.Forms.Label();
25            this.lblFecha = new System.Windows.Forms.Label();
26            this.dtpFechaEfectiva = new System.Windows.Forms.DateTimePicker();
27            this.lblMotivo = new System.Windows.Forms.Label();
28            this.txtMotivo = new System.Windows.Forms.TextBox();
29            this.btnAceptar = new System.Windows.Forms.Button();
30            this.btnCancelar = new System.Windows.Forms.Button();
31            this.SuspendLayout();
32            this.BackColor = System.Drawing.Color.FromArgb(18, 18, 27);
33            this.ClientSize = new System.Drawing.Size(520, 330);
34            this.Font = new System.Drawing.Font("Segoe UI", 9F);
35            this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
36            this.FormBorderStyle = System.Windows.Forms.FormBorderStyle.FixedDialog;
37            this.MaximizeBox = false;
38            this.MinimizeBox = false;
39            this.Name = "FrmCambioLaboral";
40            this.StartPosition = System.Windows.Forms.FormStartPosition.CenterParent;
41            this.Text = "Cambio laboral";
42            this.lblTitulo.AutoSize = true;
43            this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 18F, System.Drawing.FontStyle.Bold);
44            this.lblTitulo.Location = new System.Drawing.Point(28, 24);
45            this.lblTitulo.Text = "Fecha efectiva del cambio";
46            this.lblEmpleado.AutoSize = true;
47            this.lblEmpleado.ForeColor = System.Drawing.Color.FromArgb(167, 139, 250);
48            this.lblEmpleado.Location = new System.Drawing.Point(31, 72);
49            this.lblEmpleado.Text = "Empleado seleccionado";
50            this.lblFecha.AutoSize = true;
51            this.lblFecha.Location = new System.Drawing.Point(31, 112);
52            this.lblFecha.Text = "Fecha efectiva";
53            this.dtpFechaEfectiva.Location = new System.Drawing.Point(34, 134);
54            this.dtpFechaEfectiva.Size = new System.Drawing.Size(210, 23);
55            this.lblMotivo.AutoSize = true;
56            this.lblMotivo.Location = new System.Drawing.Point(31, 176);
57            this.lblMotivo.Text = "Motivo obligatorio";
58            this.txtMotivo.Location = new System.Drawing.Point(34, 198);
59            this.txtMotivo.Multiline = true;
60            this.txtMotivo.Size = new System.Drawing.Size(450, 68);
```

UI/FrmCambioLaboral.Designer.cs (continuacion)

```
61         this.txtMotivo.MaxLength = 300;
62         this.btnAceptar.BackColor = System.Drawing.Color.FromArgb(124, 58, 237);
63         this.btnAceptar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
64         this.btnAceptar.Location = new System.Drawing.Point(294, 282);
65         this.btnAceptar.Size = new System.Drawing.Size(92, 32);
66         this.btnAceptar.Text = "Aceptar";
67         this.btnAceptar.UseVisualStyleBackColor = false;
68         this.btnAceptar.Click += new System.EventHandler(this.btnAceptar_Click);
69         this.btnCancel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
70         this.btnCancel.Location = new System.Drawing.Point(392, 282);
71         this.btnCancel.Size = new System.Drawing.Size(92, 32);
72         this.btnCancel.Text = "Cancelar";
73         this.btnCancel.Click += new System.EventHandler(this.btnCancel_Click);
74         this.Controls.Add(this.lblTitulo);
75         this.Controls.Add(this.lblEmpleado);
76         this.Controls.Add(this.lblFecha);
77         this.Controls.Add(this.dtpFechaEfectiva);
78         this.Controls.Add(this.lblMotivo);
79         this.Controls.Add(this.txtMotivo);
80         this.Controls.Add(this.btnAceptar);
81         this.Controls.Add(this.btnCancel);
82         this.ResumeLayout(false);
83         this.PerformLayout();
84     }
85 }
86 }
```


UI/FrmComprobantes.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Drawing;
5 using System.Drawing.Printing;
6 using System.IO;
7 using System.Windows.Forms;
8 using SistemaGestionNomina.Helpers;
9 using SistemaGestionNomina.Models;
10 using SistemaGestionNomina.Services;
11 using SistemaGestionNomina.Security;
12
13 namespace SistemaGestionNomina.UI
14 {
15     public partial class FrmComprobantes : Form
16     {
17         private readonly ComprobanteService comprobanteService = new ComprobanteService();
18         private readonly ExcelExportService excelExportService = new ExcelExportService();
19         private readonly PdfExportService pdfExportService = new PdfExportService();
20         private readonly EmailService emailService = new EmailService();
21         private readonly AuthorizationService authorizationService = new AuthorizationService();
22         private readonly PrintDocument printDocument = new PrintDocument();
23         private Comprobante selectedComprobante;
24         private List<Comprobante> currentItems = new List<Comprobante>();
25
26         public FrmComprobantes()
27         {
28             InitializeComponent();
29             ControlStyleHelper.ApplyModernForm(this);
30             printDocument.PrintPage += PrintDocument_PrintPage;
31         }
32
33         private void FrmComprobantes_Load(object sender, EventArgs e)
34         {
35             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime)
36             {
37                 btnExportarExcel.Visible = authorizationService.HasPermission(Permissions.PayslipsExport);
38                 btnExportarPdf.Visible = authorizationService.HasPermission(Permissions.PayslipsExport);
39                 btnImprimir.Visible = authorizationService.HasPermission(Permissions.PayslipsPrint);
40                 btnEnviarEmail.Visible = authorizationService.HasPermission(Permissions.PayslipsEmail);
41                 LoadReceipts();
42             }
43         }
44
45         private void LoadReceipts()
46         {
47             currentItems = comprobanteService.GetAll(txtBuscar.Text);
48             dgvComprobantes.Rows.Clear();
49             foreach (Comprobante c in currentItems)
50             {
51                 dgvComprobantes.Rows.Add(c.IdComprobante, c.NumeroComprobante, c.EmpleadoNombre, c.PeriodoNombre,
52                     "B/. " + c.NetoPagar.ToString("0.00"));
53             }
54         }
55
56         private void txtBuscar_TextChanged(object sender, EventArgs e)
57         {
58             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime)
59             {
60                 LoadReceipts();
61             }
62         }
63     }
64 }
```

UI/FrmComprobantes.cs (continuación)

```
61     }
62 }
63
64 private void dgvComprobantes_SelectionChanged(object sender, EventArgs e)
65 {
66     if (dgvComprobantes.CurrentRow == null || dgvComprobantes.CurrentRow.Cells["IdComprobante"].Value == null)
67     {
68         return;
69     }
70
71     int id = Convert.ToInt32(dgvComprobantes.CurrentRow.Cells["IdComprobante"].Value);
72     selectedComprobante = comprobanteService.GetById(id);
73     RenderPreview();
74 }
75
76 private void RenderPreview()
77 {
78     if (selectedComprobante == null) return;
79     lblVistaPrevia.Text =
80         "SISTEMA DE GESTIÓN DE NÓMINA\n" +
81         "COMPROBANTE DE PAGO\n\n" +
82         "-----\n" +
83         "Documento:      " + selectedComprobante.NumeroComprobante + "\n" +
84         "Fecha emisión:   " + selectedComprobante.FechaGeneracion.ToString("dd/MM/yyyy HH:mm") + "\n" +
85         "Periodo:         " + selectedComprobante.PeriodoNombre + "\n" +
86         "-----\n" +
87         "Empleado:        " + selectedComprobante.EmpleadoNombre + "\n" +
88         "Código:          " + selectedComprobante.CodigoEmpleado + "\n" +
89         "-----\n" +
90         "Ingresos:         B/. " + selectedComprobante.TotalIngresos.ToString("0.00") + "\n" +
91         "Deducciones:      B/. " + selectedComprobante.TotalDeducciones.ToString("0.00") + "\n" +
92         "-----\n" +
93         "NETO A PAGAR:     B/. " + selectedComprobante.Netopagar.ToString("0.00") + "\n" +
94         "-----\n\n" +
95         "Firma autorizada: _____";
96 }
97
98 private void btnExportarPdf_Click(object sender, EventArgs e)
99 {
100     if (!EnsureSelected()) return;
101     string path = pdfExportService.ExportarComprobante(selectedComprobante);
102     if (string.IsNullOrEmpty(path)) return;
103
104     comprobanteService.SaveRutaPdf(selectedComprobante.IdComprobante, path);
105     UiFactory.ShowExported(path);
106 }
107
108 private void btnExportarExcel_Click(object sender, EventArgs e)
109 {
110     if (currentItems.Count == 0)
111     {
112         MessageBox.Show("No hay comprobantes para exportar.", "Comprobantes", MessageBoxButtons.OK, MessageBoxIcon.Information);
113         return;
114     }
115
116     UiFactory.ShowExported(excelExportService.ExportarComprobantes(currentItems));
117 }
118
119 private void btnImprimir_Click(object sender, EventArgs e)
120 {
```

UI/FrmComprobantes.cs (continuacion)

```
121         authorizationService.DemandPermission(Permissions.PayslipsPrint);
122         if (!EnsureSelected()) return;
123
124         try
125         {
126             using (PrintPreviewDialog preview = new PrintPreviewDialog())
127             {
128                 preview.Document = printDocument;
129                 preview.Width = 900;
130                 preview.Height = 700;
131                 preview.StartPosition = FormStartPosition.CenterParent;
132                 preview.ShowDialog(this);
133             }
134         }
135         catch (Exception ex)
136         {
137             MessageBox.Show("No se pudo abrir la vista previa de impresión.\n\n" + ex.Message,
138                 "Imprimir", MessageBoxButtons.OK, MessageBoxIcon.Warning);
139         }
140     }
141
142     private void PrintDocument_PrintPage(object sender, PrintPageEventArgs e)
143     {
144         if (selectedComprobante == null)
145         {
146             e.HasMorePages = false;
147             return;
148         }
149
150         ComprobantePrintRenderer.Draw(e.Graphics, e.MarginBounds, selectedComprobante);
151         e.HasMorePages = false;
152     }
153
154     private void btnEnviarEmail_Click(object sender, EventArgs e)
155     {
156         if (!EnsureSelected()) return;
157
158         string email = PromptEmail();
159         if (string.IsNullOrEmpty(email))
160         {
161             return;
162         }
163
164         try
165         {
166             string pdfPath = selectedComprobante.RutaPdf;
167             if (string.IsNullOrEmpty(pdfPath) || !File.Exists(pdfPath))
168             {
169                 pdfPath = pdfExportService.ExportarComprobante(selectedComprobante);
170                 if (string.IsNullOrEmpty(pdfPath)) return;
171
172                 comprobanteService.SaveRutaPdf(selectedComprobante.IdComprobante, pdfPath);
173                 selectedComprobante.RutaPdf = pdfPath;
174             }
175
176             string emlPath = emailService.CrearBorradorComprobante(email, pdfPath,
177                 "Comprobante de pago " + selectedComprobante.NumeroComprobante);
178             UiFactory.ShowExported(emlPath);
179         }
180         catch (Exception ex)
```

UI/FrmComprobantes.cs (continuacion)

```
181         {
182             MessageBox.Show(ex.Message, "Enviar email", MessageBoxButtons.OK, MessageBoxIcon.Warning);
183         }
184     }
185
186     private bool EnsureSelected()
187     {
188         if (selectedComprobante != null) return true;
189         MessageBox.Show("Seleccione un comprobante.", "Comprobantes", MessageBoxButtons.OK, MessageBoxIcon.Information);
190         return false;
191     }
192
193     private string PromptEmail()
194     {
195         using (Form prompt = new Form())
196         using (Label label = new Label())
197         using (TextBox input = new TextBox())
198         using (Button okButton = new Button())
199         using (Button cancelButton = new Button())
200         {
201             prompt.Text = "Enviar comprobante";
202             prompt.StartPosition = FormStartPosition.CenterParent;
203             prompt.FormBorderStyle = FormBorderStyle.FixedDialog;
204             prompt.MinimizeBox = false;
205             prompt.MaximizeBox = false;
206             prompt.ClientSize = new Size(420, 140);
207
208             label.Text = "Correo del destinatario:";
209             label.Location = new Point(18, 18);
210             label.Size = new Size(360, 24);
211
212             input.Location = new Point(18, 48);
213             input.Size = new Size(380, 24);
214
215             okButton.Text = "Crear email";
216             okButton.DialogResult = DialogResult.OK;
217             okButton.Location = new Point(210, 92);
218             okButton.Size = new Size(92, 30);
219
220             cancelButton.Text = "Cancelar";
221             cancelButton.DialogResult = DialogResult.Cancel;
222             cancelButton.Location = new Point(306, 92);
223             cancelButton.Size = new Size(92, 30);
224
225             prompt.Controls.Add(label);
226             prompt.Controls.Add(input);
227             prompt.Controls.Add(okButton);
228             prompt.Controls.Add(cancelButton);
229             prompt.AcceptButton = okButton;
230             prompt.CancelButton = cancelButton;
231
232             return prompt.ShowDialog(this) == DialogResult.OK ? input.Text.Trim() : string.Empty;
233         }
234     }
235 }
236 }
```

UI/FrmComprobantes.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmComprobantes
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.TextBox txtBuscar;
8         private System.Windows.Forms.DataGridView dgvComprobantes;
9         private System.Windows.Forms.Panel panelVistaPrevia;
10        private System.Windows.Forms.Label lblVistaPrevia;
11        private System.Windows.Forms.Button btnExportarPdf;
12        private System.Windows.Forms.Button btnExportarExcel;
13        private System.Windows.Forms.Button btnImprimir;
14        private System.Windows.Forms.Button btnEnviarEmail;
15
16        protected override void Dispose(bool disposing)
17        {
18            if (disposing && (components != null)) components.Dispose();
19            base.Dispose(disposing);
20        }
21
22        private void InitializeComponent()
23        {
24            this.lblTitulo = new System.Windows.Forms.Label();
25            this.txtBuscar = new System.Windows.Forms.TextBox();
26            this.dgvComprobantes = new System.Windows.Forms.DataGridView();
27            this.panelVistaPrevia = new System.Windows.Forms.Panel();
28            this.lblVistaPrevia = new System.Windows.Forms.Label();
29            this.btnExportarPdf = new System.Windows.Forms.Button();
30            this.btnExportarExcel = new System.Windows.Forms.Button();
31            this.btnImprimir = new System.Windows.Forms.Button();
32            this.btnEnviarEmail = new System.Windows.Forms.Button();
33            ((System.ComponentModel.ISupportInitialize)(this.dgvComprobantes)).BeginInit();
34            this.panelVistaPrevia.SuspendLayout();
35            this.SuspendLayout();
36            //
37            // lblTitulo
38            //
39            this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
40            this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
41            this.lblTitulo.Location = new System.Drawing.Point(34, 24);
42            this.lblTitulo.Name = "lblTitulo";
43            this.lblTitulo.Size = new System.Drawing.Size(420, 42);
44            this.lblTitulo.TabIndex = 0;
45            this.lblTitulo.Text = "Comprobantes de pago";
46            //
47            // txtBuscar
48            //
49            this.txtBuscar.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
50            this.txtBuscar.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
51            this.txtBuscar.Location = new System.Drawing.Point(36, 92);
52            this.txtBuscar.Name = "txtBuscar";
53            this.txtBuscar.Size = new System.Drawing.Size(320, 23);
54            this.txtBuscar.TabIndex = 1;
55            this.txtBuscar.TextChanged += new System.EventHandler(this.txtBuscar_TextChanged);
56            //
57            // dgvComprobantes
58            //
59            this.dgvComprobantes.AllowUserToAddRows = false;
60            this.dgvComprobantes.AllowUserToDeleteRows = false;
```

UI/FrmComprobantes.Designer.cs (continuacion)

```
61         this.dgvComprobantes.BackgroundColor = System.Drawing.Color.Black;
62         this.dgvComprobantes.ColumnHeadersHeightSizeMode = System.Windows.Forms.DataGridViewColumnHeadersHeightSizeMode.AutoSize;
63         this.dgvComprobantes.Location = new System.Drawing.Point(36, 144);
64         this.dgvComprobantes.Name = "dgvComprobantes";
65         this.dgvComprobantes.ReadOnly = true;
66         this.dgvComprobantes.RowHeadersVisible = false;
67         this.dgvComprobantes.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect;
68         this.dgvComprobantes.Size = new System.Drawing.Size(480, 460);
69         this.dgvComprobantes.TabIndex = 2;
70         this.dgvComprobantes.Columns.Add("IdComprobante", "Id");
71         this.dgvComprobantes.Columns["IdComprobante"].Visible = false;
72         this.dgvComprobantes.Columns.Add("Numero", "Número");
73         this.dgvComprobantes.Columns.Add("Empleado", "Empleado");
74         this.dgvComprobantes.Columns.Add("Periodo", "Periodo");
75         this.dgvComprobantes.Columns.Add("Neto", "Neto");
76         this.dgvComprobantes.SelectionChanged += new System.EventHandler(this.dgvComprobantes_SelectionChanged);
77         //
78         // panelVistaPrevia
79         //
80         this.panelVistaPrevia.BackColor = System.Drawing.Color.FromArgb(243, 244, 246);
81         this.panelVistaPrevia.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
82         this.panelVistaPrevia.Controls.Add(this.lblVistaPrevia);
83         this.panelVistaPrevia.Location = new System.Drawing.Point(548, 144);
84         this.panelVistaPrevia.Name = "panelVistaPrevia";
85         this.panelVistaPrevia.Size = new System.Drawing.Size(436, 460);
86         this.panelVistaPrevia.TabIndex = 3;
87         //
88         // lblVistaPrevia
89         //
90         this.lblVistaPrevia.Font = new System.Drawing.Font("Consolas", 10F);
91         this.lblVistaPrevia.ForeColor = System.Drawing.Color.FromArgb(17, 24, 39);
92         this.lblVistaPrevia.Location = new System.Drawing.Point(22, 22);
93         this.lblVistaPrevia.Name = "lblVistaPrevia";
94         this.lblVistaPrevia.Size = new System.Drawing.Size(390, 410);
95         this.lblVistaPrevia.TabIndex = 0;
96         this.lblVistaPrevia.Text = "Seleccione un comprobante para ver el detalle.";
97         //
98         // buttons
99         //
100        this.btnExportarPdf.BackColor = System.Drawing.Color.Black;
101        this.btnExportarPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
102        this.btnExportarPdf.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
103        this.btnExportarPdf.Location = new System.Drawing.Point(568, 86);
104        this.btnExportarPdf.Name = "btnExportarPdf";
105        this.btnExportarPdf.Size = new System.Drawing.Size(100, 34);
106        this.btnExportarPdf.TabIndex = 4;
107        this.btnExportarPdf.Text = "PDF";
108        this.btnExportarPdf.UseVisualStyleBackColor = false;
109        this.btnExportarPdf.Click += new System.EventHandler(this.btnExportarPdf_Click);
110        this.btnExportarExcel.BackColor = System.Drawing.Color.Black;
111        this.btnExportarExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
112        this.btnExportarExcel.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
113        this.btnExportarExcel.Location = new System.Drawing.Point(678, 86);
114        this.btnExportarExcel.Name = "btnExportarExcel";
115        this.btnExportarExcel.Size = new System.Drawing.Size(110, 34);
116        this.btnExportarExcel.TabIndex = 5;
117        this.btnExportarExcel.Text = "Excel";
118        this.btnExportarExcel.UseVisualStyleBackColor = false;
119        this.btnExportarExcel.Click += new System.EventHandler(this.btnExportarExcel_Click);
120        this.btnImprimir.BackColor = System.Drawing.Color.Black;
```

UI/FrmComprobantes.Designer.cs (continuacion)

```
121         this.btnImprimir.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
122         this.btnImprimir.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
123         this.btnImprimir.Location = new System.Drawing.Point(798, 86);
124         this.btnImprimir.Name = "btnImprimir";
125         this.btnImprimir.Size = new System.Drawing.Size(80, 34);
126         this.btnImprimir.TabIndex = 6;
127         this.btnImprimir.Text = "Imprimir";
128         this.btnImprimir.UseVisualStyleBackColor = false;
129         this.btnImprimir.Click += new System.EventHandler(this.btnImprimir_Click);
130         this.btnEnviarEmail.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
131         this.btnEnviarEmail.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
132         this.btnEnviarEmail.Location = new System.Drawing.Point(888, 86);
133         this.btnEnviarEmail.Name = "btnEnviarEmail";
134         this.btnEnviarEmail.Size = new System.Drawing.Size(96, 34);
135         this.btnEnviarEmail.TabIndex = 7;
136         this.btnEnviarEmail.Text = "Enviar Email";
137         this.btnEnviarEmail.UseVisualStyleBackColor = false;
138         this.btnEnviarEmail.Click += new System.EventHandler(this.btnEnviarEmail_Click);
139         //
140         // FrmComprobantes
141         //
142         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
143         this.AutoScaleModeMode = System.Windows.Forms.AutoScaleMode.Font;
144         this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
145         this.ClientSize = new System.Drawing.Size(1020, 742);
146         this.Controls.Add(this.btnEnviarEmail);
147         this.Controls.Add(this.btnImprimir);
148         this.Controls.Add(this.btnExportarExcel);
149         this.Controls.Add(this.btnExportarPdf);
150         this.Controls.Add(this.panelVistaPrevia);
151         this.Controls.Add(this.dgvComprobantes);
152         this.Controls.Add(this.txtBuscar);
153         this.Controls.Add(this.lblTitulo);
154         this.Font = new System.Drawing.Font("Segoe UI", 9F);
155         this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
156         this.Name = "FrmComprobantes";
157         this.Text = "Comprobantes";
158         this.Load += new System.EventHandler(this.FrmComprobantes_Load);
159         ((System.ComponentModel.ISupportInitialize)(this.dgvComprobantes)).EndInit();
160         this.panelVistaPrevia.ResumeLayout(false);
161         this.ResumeLayout(false);
162         this.PerformLayout();
163     }
164 }
165 }
```

UI/FrmConfiguracion.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Services;
7
8 namespace SistemaGestionNomina.UI
9 {
10     public partial class FrmConfiguracion : Form
11     {
12         private readonly ConfiguracionService configuracionService = new ConfiguracionService();
13         private readonly BackupService backupService = new BackupService();
14
15         public FrmConfiguracion()
16         {
17             InitializeComponent();
18             ControlStyleHelper.ApplyModernForm(this);
19         }
20
21         private void FrmConfiguracion_Load(object sender, EventArgs e)
22         {
23             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime)
24             {
25                 LoadSettings();
26             }
27         }
28
29         private void LoadSettings()
30         {
31             nudSeguroSocial.Value = Clamp(configuracionService.GetValue("SeguroSocial", 9.75m), nudSeguroSocial);
32             nudISR.Value = Clamp(configuracionService.GetValue("ISR", 10m), nudISR);
33             nudSeguroEducativo.Value = Clamp(configuracionService.GetValue("SeguroEducativo", 1.25m), nudSeguroEducativo);
34             nudRecargoHoraExtra.Value = Clamp(configuracionService.GetValue("RecargoHoraExtra", 1.25m), nudRecargoHoraExtra);
35             nudHorasMensualesBase.Value = Clamp(configuracionService.GetValue("HorasMensualesBase", 160m), nudHorasMensualesBase);
36         }
37
38         private void btnGuardar_Click(object sender, EventArgs e)
39         {
40             Dictionary<string, decimal> values = new Dictionary<string, decimal>();
41             values["SeguroSocial"] = nudSeguroSocial.Value;
42             values["ISR"] = nudISR.Value;
43             values["SeguroEducativo"] = nudSeguroEducativo.Value;
44             values["RecargoHoraExtra"] = nudRecargoHoraExtra.Value;
45             values["HorasMensualesBase"] = nudHorasMensualesBase.Value;
46             configuracionService.SaveDefaults(values);
47             MessageBox.Show("Configuración guardada correctamente.", "Configuración", MessageBoxButtons.OK, MessageBoxIcon.Information);
48         }
49
50         private void btnDescartar_Click(object sender, EventArgs e)
51         {
52             LoadSettings();
53         }
54
55         private void btnBackup_Click(object sender, EventArgs e)
56         {
57             using (FolderBrowserDialog dialog = new FolderBrowserDialog())
58             {
59                 dialog.Description = "Seleccione la carpeta donde se guardará el backup.";
60                 if (dialog.ShowDialog(this) != DialogResult.OK)
```


UI/FrmConfiguracion.cs (continuacion)

```
61         {
62             return;
63         }
64
65         try
66         {
67             string path = backupService.CrearBackup(dialog.SelectedPath);
68             bool verified = backupService.VerificarBackup(path);
69             MessageBox.Show("Backup generado correctamente:\n" + path + "\n\nVerificación SHA-256: " +
70                 (verified ? "correcta" : "no disponible"), "Backup", MessageBoxButtons.OK, MessageBoxIcon.Information);
71         }
72         catch (Exception ex)
73         {
74             MessageBox.Show(ex.Message, "Backup", MessageBoxButtons.OK, MessageBoxIcon.Warning);
75         }
76     }
77 }
78
79 private static decimal Clamp(decimal value, NumericUpDown control)
80 {
81     if (value < control.Minimum) return control.Minimum;
82     if (value > control.Maximum) return control.Maximum;
83     return value;
84 }
85 }
86 }
```

UI\FrmConfiguracion.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmConfiguracion
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelConfiguracion;
8         private System.Windows.Forms.NumericUpDown nudSeguroSocial;
9         private System.Windows.Forms.NumericUpDown nudISR;
10        private System.Windows.Forms.NumericUpDown nudSeguroEducativo;
11        private System.Windows.Forms.NumericUpDown nudRecargoHoraExtra;
12        private System.Windows.Forms.NumericUpDown nudHorasMensualesBase;
13        private System.Windows.Forms.Label lblSeguroSocial;
14        private System.Windows.Forms.Label lblISR;
15        private System.Windows.Forms.Label lblSeguroEducativo;
16        private System.Windows.Forms.Label lblRecargoHoraExtra;
17        private System.Windows.Forms.Label lblHorasMensualesBase;
18        private System.Windows.Forms.Label lblDescripcion;
19        private System.Windows.Forms.Button btnGuardar;
20        private System.Windows.Forms.Button btnDescartar;
21        private System.Windows.Forms.Button btnBackup;
22
23        protected override void Dispose(bool disposing)
24        {
25            if (disposing && (components != null)) components.Dispose();
26            base.Dispose(disposing);
27        }
28
29        private void InitializeComponent()
30        {
31            this.lblTitulo = new System.Windows.Forms.Label();
32            this.panelConfiguracion = new System.Windows.Forms.Panel();
33            this.nudSeguroSocial = new System.Windows.Forms.NumericUpDown();
34            this.nudISR = new System.Windows.Forms.NumericUpDown();
35            this.nudSeguroEducativo = new System.Windows.Forms.NumericUpDown();
36            this.nudRecargoHoraExtra = new System.Windows.Forms.NumericUpDown();
37            this.nudHorasMensualesBase = new System.Windows.Forms.NumericUpDown();
38            this.lblSeguroSocial = new System.Windows.Forms.Label();
39            this.lblISR = new System.Windows.Forms.Label();
40            this.lblSeguroEducativo = new System.Windows.Forms.Label();
41            this.lblRecargoHoraExtra = new System.Windows.Forms.Label();
42            this.lblHorasMensualesBase = new System.Windows.Forms.Label();
43            this.lblDescripcion = new System.Windows.Forms.Label();
44            this.btnGuardar = new System.Windows.Forms.Button();
45            this.btnDescartar = new System.Windows.Forms.Button();
46            this.btnBackup = new System.Windows.Forms.Button();
47            this.panelConfiguracion.SuspendLayout();
48            ((System.ComponentModel.ISupportInitialize)(this.nudSeguroSocial)).BeginInit();
49            ((System.ComponentModel.ISupportInitialize)(this.nudISR)).BeginInit();
50            ((System.ComponentModel.ISupportInitialize)(this.nudSeguroEducativo)).BeginInit();
51            ((System.ComponentModel.ISupportInitialize)(this.nudRecargoHoraExtra)).BeginInit();
52            ((System.ComponentModel.ISupportInitialize)(this.nudHorasMensualesBase)).BeginInit();
53            this.SuspendLayout();
54            //
55            // lblTitulo
56            //
57            this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
58            this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
59            this.lblTitulo.Location = new System.Drawing.Point(34, 24);
60            this.lblTitulo.Name = "lblTitulo";
```

UI/FrmConfiguracion.Designer.cs (continuacion)

```
61         this.lblTitulo.Size = new System.Drawing.Size(520, 42);
62         this.lblTitulo.TabIndex = 0;
63         this.lblTitulo.Text = "Configuración del Sistema";
64         //
65         // panelConfiguracion
66         //
67         this.panelConfiguracion.BackColor = System.Drawing.Color.Black;
68         this.panelConfiguracion.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
69         this.panelConfiguracion.Controls.Add(this.lblSeguroSocial);
70         this.panelConfiguracion.Controls.Add(this.lblISR);
71         this.panelConfiguracion.Controls.Add(this.lblSeguroEducativo);
72         this.panelConfiguracion.Controls.Add(this.lblRecargoHoraExtra);
73         this.panelConfiguracion.Controls.Add(this.lblHorasMensualesBase);
74         this.panelConfiguracion.Controls.Add(this.lblDescripcion);
75         this.panelConfiguracion.Controls.Add(this.nudSeguroSocial);
76         this.panelConfiguracion.Controls.Add(this.nudISR);
77         this.panelConfiguracion.Controls.Add(this.nudSeguroEducativo);
78         this.panelConfiguracion.Controls.Add(this.nudRecargoHoraExtra);
79         this.panelConfiguracion.Controls.Add(this.nudHorasMensualesBase);
80         this.panelConfiguracion.Controls.Add(this.btnGuardar);
81         this.panelConfiguracion.Controls.Add(this.btnDescartar);
82         this.panelConfiguracion.Controls.Add(this.btnBackup);
83         this.panelConfiguracion.Location = new System.Drawing.Point(210, 116);
84         this.panelConfiguracion.Name = "panelConfiguracion";
85         this.panelConfiguracion.Size = new System.Drawing.Size(600, 390);
86         this.panelConfiguracion.TabIndex = 1;
87         //
88         // numeric controls
89         //
90         this.nudSeguroSocial.DecimalPlaces = 2;
91         this.nudSeguroSocial.Location = new System.Drawing.Point(42, 54);
92         this.nudSeguroSocial.Maximum = new decimal(new int[] { 100, 0, 0, 0 });
93         this.nudSeguroSocial.Name = "nudSeguroSocial";
94         this.nudSeguroSocial.Size = new System.Drawing.Size(210, 23);
95         this.nudSeguroSocial.TabIndex = 0;
96         this.nudISR.DecimalPlaces = 2;
97         this.nudISR.Location = new System.Drawing.Point(320, 54);
98         this.nudISR.Maximum = new decimal(new int[] { 100, 0, 0, 0 });
99         this.nudISR.Name = "nudISR";
100        this.nudISR.Size = new System.Drawing.Size(210, 23);
101        this.nudISR.TabIndex = 1;
102        this.nudSeguroEducativo.DecimalPlaces = 2;
103        this.nudSeguroEducativo.Location = new System.Drawing.Point(42, 132);
104        this.nudSeguroEducativo.Maximum = new decimal(new int[] { 100, 0, 0, 0 });
105        this.nudSeguroEducativo.Name = "nudSeguroEducativo";
106        this.nudSeguroEducativo.Size = new System.Drawing.Size(210, 23);
107        this.nudSeguroEducativo.TabIndex = 2;
108        this.nudRecargoHoraExtra.DecimalPlaces = 2;
109        this.nudRecargoHoraExtra.Location = new System.Drawing.Point(320, 132);
110        this.nudRecargoHoraExtra.Maximum = new decimal(new int[] { 10, 0, 0, 0 });
111        this.nudRecargoHoraExtra.Name = "nudRecargoHoraExtra";
112        this.nudRecargoHoraExtra.Size = new System.Drawing.Size(210, 23);
113        this.nudRecargoHoraExtra.TabIndex = 3;
114        this.nudHorasMensualesBase.Location = new System.Drawing.Point(42, 210);
115        this.nudHorasMensualesBase.Maximum = new decimal(new int[] { 400, 0, 0, 0 });
116        this.nudHorasMensualesBase.Minimum = new decimal(new int[] { 1, 0, 0, 0 });
117        this.nudHorasMensualesBase.Name = "nudHorasMensualesBase";
118        this.nudHorasMensualesBase.Size = new System.Drawing.Size(210, 23);
119        this.nudHorasMensualesBase.TabIndex = 4;
120        this.nudHorasMensualesBase.Value = new decimal(new int[] { 160, 0, 0, 0 });
```

UI/FrmConfiguracion.Designer.cs (continuacion)

```
121         //
122         // labels
123         //
124         this.lblDescripcion.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
125         this.lblDescripcion.Location = new System.Drawing.Point(42, 18);
126         this.lblDescripcion.Name = "lblDescripcion";
127         this.lblDescripcion.Size = new System.Drawing.Size(500, 24);
128         this.lblDescripcion.TabIndex = 7;
129         this.lblDescripcion.Text = "Valores académicos configurables para el cálculo de nómina.";
130         this.lblSeguroSocial.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
131         this.lblSeguroSocial.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
132         this.lblSeguroSocial.Location = new System.Drawing.Point(42, 34);
133         this.lblSeguroSocial.Name = "lblSeguroSocial";
134         this.lblSeguroSocial.Size = new System.Drawing.Size(210, 18);
135         this.lblSeguroSocial.TabIndex = 8;
136         this.lblSeguroSocial.Text = "Seguro Social (%)";
137         this.lblISR.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
138         this.lblISR.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
139         this.lblISR.Location = new System.Drawing.Point(320, 34);
140         this.lblISR.Name = "lblISR";
141         this.lblISR.Size = new System.Drawing.Size(210, 18);
142         this.lblISR.TabIndex = 9;
143         this.lblISR.Text = "ISR (%)";
144         this.lblSeguroEducativo.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
145         this.lblSeguroEducativo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
146         this.lblSeguroEducativo.Location = new System.Drawing.Point(42, 112);
147         this.lblSeguroEducativo.Name = "lblSeguroEducativo";
148         this.lblSeguroEducativo.Size = new System.Drawing.Size(210, 18);
149         this.lblSeguroEducativo.TabIndex = 10;
150         this.lblSeguroEducativo.Text = "Seguro Educativo (%)";
151         this.lblRecargoHoraExtra.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
152         this.lblRecargoHoraExtra.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
153         this.lblRecargoHoraExtra.Location = new System.Drawing.Point(320, 112);
154         this.lblRecargoHoraExtra.Name = "lblRecargoHoraExtra";
155         this.lblRecargoHoraExtra.Size = new System.Drawing.Size(210, 18);
156         this.lblRecargoHoraExtra.TabIndex = 11;
157         this.lblRecargoHoraExtra.Text = "Recargo de horas extra";
158         this.lblHorasMensualesBase.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
159         this.lblHorasMensualesBase.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
160         this.lblHorasMensualesBase.Location = new System.Drawing.Point(42, 190);
161         this.lblHorasMensualesBase.Name = "lblHorasMensualesBase";
162         this.lblHorasMensualesBase.Size = new System.Drawing.Size(210, 18);
163         this.lblHorasMensualesBase.TabIndex = 12;
164         this.lblHorasMensualesBase.Text = "Horas mensuales base";
165         //
166         // buttons
167         //
168         this.btnGuardar.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
169         this.btnGuardar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
170         this.btnGuardar.Location = new System.Drawing.Point(320, 304);
171         this.btnGuardar.Name = "btnGuardar";
172         this.btnGuardar.Size = new System.Drawing.Size(130, 34);
173         this.btnGuardar.TabIndex = 5;
174         this.btnGuardar.Text = "Guardar cambios";
175         this.btnGuardar.UseVisualStyleBackColor = false;
176         this.btnGuardar.Click += new System.EventHandler(this.btnGuardar_Click);
177         this.btnDescartar.BackColor = System.Drawing.Color.Black;
178         this.btnDescartar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
179         this.btnDescartar.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
180         this.btnDescartar.Location = new System.Drawing.Point(462, 304);
```

UI/FrmConfiguracion.Designer.cs (continuacion)

```
181         this.btnDescartar.Name = "btnDescartar";
182         this.btnDescartar.Size = new System.Drawing.Size(92, 34);
183         this.btnDescartar.TabIndex = 6;
184         this.btnDescartar.Text = "Descartar";
185         this.btnDescartar.UseVisualStyleBackColor = false;
186         this.btnDescartar.Click += new System.EventHandler(this.btnDescartar_Click);
187         this.btnBackup.BackColor = System.Drawing.Color.Black;
188         this.btnBackup.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
189         this.btnBackup.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
190         this.btnBackup.Location = new System.Drawing.Point(42, 304);
191         this.btnBackup.Name = "btnBackup";
192         this.btnBackup.Size = new System.Drawing.Size(150, 34);
193         this.btnBackup.TabIndex = 7;
194         this.btnBackup.Text = "Crear backup";
195         this.btnBackup.UseVisualStyleBackColor = false;
196         this.btnBackup.Click += new System.EventHandler(this.btnBackup_Click);
197         //
198         // FrmConfiguracion
199         //
200         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
201         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
202         this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
203         this.ClientSize = new System.Drawing.Size(1020, 742);
204         this.Controls.Add(this.panelConfiguracion);
205         this.Controls.Add(this.lblTitulo);
206         this.Font = new System.Drawing.Font("Segoe UI", 9F);
207         this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
208         this.Name = "FrmConfiguracion";
209         this.Text = "Configuración";
210         this.Load += new System.EventHandler(this.FrmConfiguracion_Load);
211         this.panelConfiguracion.ResumeLayout(false);
212         ((System.ComponentModel.ISupportInitialize)(this.nudSeguroSocial)).EndInit();
213         ((System.ComponentModel.ISupportInitialize)(this.nudISR)).EndInit();
214         ((System.ComponentModel.ISupportInitialize)(this.nudSeguroEducativo)).EndInit();
215         ((System.ComponentModel.ISupportInitialize)(this.nudRecargoHoraExtra)).EndInit();
216         ((System.ComponentModel.ISupportInitialize)(this.nudHorasMensualesBase)).EndInit();
217         this.ResumeLayout(false);
218     }
219 }
220 }
```

UI/FrmDashboard.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Drawing;
5 using System.Windows.Forms;
6 using SistemaGestionNomina.Helpers;
7 using SistemaGestionNomina.Models;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmDashboard : Form
13     {
14         private readonly Action openEmployees;
15         private readonly Action openPayroll;
16         private readonly EmpleadoService empleadoService = new EmpleadoService();
17         private readonly NominaService nominaService = new NominaService();
18         private readonly ComprobanteService comprobanteService = new ComprobanteService();
19
20         public FrmDashboard()
21             : this(null, null)
22         {
23         }
24
25         public FrmDashboard(Action openEmployeesAction, Action openPayrollAction)
26         {
27             openEmployees = openEmployeesAction;
28             openPayroll = openPayrollAction;
29             InitializeComponent();
30             ControlStyleHelper.ApplyModernForm(this);
31         }
32
33         private void FrmDashboard_Load(object sender, EventArgs e)
34         {
35             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime)
36             {
37                 RefreshDashboard();
38             }
39         }
40
41         private void btnRegistrarEmpleado_Click(object sender, EventArgs e)
42         {
43             if (openEmployees != null)
44             {
45                 openEmployees();
46             }
47         }
48
49         private void btnProcesarNomina_Click(object sender, EventArgs e)
50         {
51             if (openPayroll != null)
52             {
53                 openPayroll();
54             }
55         }
56
57         private void RefreshDashboard()
58         {
59             List<Empleado> activos = empleadoService.GetActive(null);
60             List<Nomina> nominas = nominaService.GetNominas();
```

UI/FrmDashboard.cs (continuacion)

```
61         List<Comprobante> comprobantes = comprobanteService.GetAll(string.Empty);
62         decimal totalNomina = nominas.Count > 0 ? nominas[0].TotalNeto : 0;
63         decimal totalDeducciones = nominas.Count > 0 ? nominas[0].TotalDeducciones : 0;
64
65         lblEmpleadosActivosValor.Text = activos.Count.ToString();
66         SetMoneyLabel(lblNominaActualValor, totalNomina);
67         lblComprobantesValor.Text = comprobantes.Count.ToString();
68         SetMoneyLabel(lblAlertasValor, totalDeducciones);
69
70         chartPagos.Series["Neto"].Points.Clear();
71         string[] months = { "Ene", "Feb", "Mar", "Abr", "May", "Jun" };
72         decimal baseValue = totalNomina > 0 ? totalNomina : 8500;
73         for (int i = 0; i < months.Length; i++)
74         {
75             chartPagos.Series["Neto"].Points.AddXY(months[i], Convert.ToDouble(baseValue + (i * 420)));
76         }
77     }
78
79     private static void SetMoneyLabel(Label label, decimal amount)
80     {
81         string value = "B/. " + amount.ToString("#,##0.00");
82         float size = value.Length > 13 ? 15F : value.Length > 11 ? 17F : 20F;
83
84         label.Font = new Font("Segoe UI", size, FontStyle.Bold);
85         label.Text = value;
86     }
87 }
88 }
```

UI/FrmDashboard.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmDashboard
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Label lblSubtitulo;
8         private System.Windows.Forms.Button btnRegistrarEmpleado;
9         private System.Windows.Forms.Button btnProcesarNomina;
10        private System.Windows.Forms.Panel cardEmpleados;
11        private System.Windows.Forms.Panel cardNomina;
12        private System.Windows.Forms.Panel cardComprobantes;
13        private System.Windows.Forms.Panel cardAlertas;
14        private System.Windows.Forms.Label lblEmpleadosActivosValor;
15        private System.Windows.Forms.Label lblNominaActualValor;
16        private System.Windows.Forms.Label lblComprobantesValor;
17        private System.Windows.Forms.Label lblAlertasValor;
18        private System.Windows.Forms.Label lblCardEmpleados;
19        private System.Windows.Forms.Label lblCardNomina;
20        private System.Windows.Forms.Label lblCardComprobantes;
21        private System.Windows.Forms.Label lblCardAlertas;
22        private System.Windows.Forms.Label lblActividadTitulo;
23        private System.Windows.Forms.Label lblActividadUno;
24        private System.Windows.Forms.Label lblActividadDos;
25        private System.Windows.Forms.Label lblActividadTres;
26        private System.Windows.Forms.Panel panelGrafico;
27        private System.Windows.Forms.Panel panelActividad;
28        private System.Windows.Forms.DataVisualization.Charting.Chart chartPagos;
29
30        protected override void Dispose(bool disposing)
31        {
32            if (disposing && (components != null))
33            {
34                components.Dispose();
35            }
36            base.Dispose(disposing);
37        }
38
39        private void InitializeComponent()
40        {
41            System.Windows.Forms.DataVisualization.Charting.ChartArea chartArea1 = new System.Windows.Forms.DataVisualization.Charting.ChartArea();
42            System.Windows.Forms.DataVisualization.Charting.Series series1 = new System.Windows.Forms.DataVisualization.Charting.Series();
43            this.lblTitulo = new System.Windows.Forms.Label();
44            this.lblSubtitulo = new System.Windows.Forms.Label();
45            this.btnRegistrarEmpleado = new System.Windows.Forms.Button();
46            this.btnProcesarNomina = new System.Windows.Forms.Button();
47            this.cardEmpleados = new System.Windows.Forms.Panel();
48            this.lblCardEmpleados = new System.Windows.Forms.Label();
49            this.lblEmpleadosActivosValor = new System.Windows.Forms.Label();
50            this.cardNomina = new System.Windows.Forms.Panel();
51            this.lblCardNomina = new System.Windows.Forms.Label();
52            this.lblNominaActualValor = new System.Windows.Forms.Label();
53            this.cardComprobantes = new System.Windows.Forms.Panel();
54            this.lblCardComprobantes = new System.Windows.Forms.Label();
55            this.lblComprobantesValor = new System.Windows.Forms.Label();
56            this.cardAlertas = new System.Windows.Forms.Panel();
57            this.lblCardAlertas = new System.Windows.Forms.Label();
58            this.lblAlertasValor = new System.Windows.Forms.Label();
59            this.panelGrafico = new System.Windows.Forms.Panel();
60            this.chartPagos = new System.Windows.Forms.DataVisualization.Charting.Chart();
```


UI/FrmDashboard.Designer.cs (continuacion)

```
61         this.panelActividad = new System.Windows.Forms.Panel();
62         this.lblActividadTitulo = new System.Windows.Forms.Label();
63         this.lblActividadUno = new System.Windows.Forms.Label();
64         this.lblActividadDos = new System.Windows.Forms.Label();
65         this.lblActividadTres = new System.Windows.Forms.Label();
66         this.cardEmpleados.SuspendLayout();
67         this.cardNomina.SuspendLayout();
68         this.cardComprobantes.SuspendLayout();
69         this.cardAlertas.SuspendLayout();
70         this.panelGrafico.SuspendLayout();
71         ((System.ComponentModel.ISupportInitialize)(this.chartPagos)).BeginInit();
72         this.SuspendLayout();
73         //
74         // lblTitulo
75         //
76         this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
77         this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
78         this.lblTitulo.Location = new System.Drawing.Point(34, 28);
79         this.lblTitulo.Name = "lblTitulo";
80         this.lblTitulo.Size = new System.Drawing.Size(420, 42);
81         this.lblTitulo.TabIndex = 0;
82         this.lblTitulo.Text = "Panel Ejecutivo";
83         //
84         // lblSubtitulo
85         //
86         this.lblSubtitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
87         this.lblSubtitulo.Location = new System.Drawing.Point(38, 70);
88         this.lblSubtitulo.Name = "lblSubtitulo";
89         this.lblSubtitulo.Size = new System.Drawing.Size(520, 25);
90         this.lblSubtitulo.TabIndex = 1;
91         this.lblSubtitulo.Text = "Resumen ejecutivo del periodo actual de nómina.";
92         //
93         // btnRegistrarEmpleado
94         //
95         this.btnRegistrarEmpleado.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right)));
96         this.btnRegistrarEmpleado.BackColor = System.Drawing.Color.Black;
97         this.btnRegistrarEmpleado.FlatAppearance.BorderColor = System.Drawing.Color.FromArgb(((int)(((byte)(42)))), ((int)(((byte)(43)))), \
((int)(((byte)(56))))));
98         this.btnRegistrarEmpleado.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
99         this.btnRegistrarEmpleado.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
100        this.btnRegistrarEmpleado.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
101        this.btnRegistrarEmpleado.Location = new System.Drawing.Point(650, 42);
102        this.btnRegistrarEmpleado.Name = "btnRegistrarEmpleado";
103        this.btnRegistrarEmpleado.Size = new System.Drawing.Size(170, 38);
104        this.btnRegistrarEmpleado.TabIndex = 2;
105        this.btnRegistrarEmpleado.Text = "+ Nuevo empleado";
106        this.btnRegistrarEmpleado.UseVisualStyleBackColor = false;
107        this.btnRegistrarEmpleado.Click += new System.EventHandler(this.btnRegistrarEmpleado_Click);
108        //
109        // btnProcesarNomina
110        //
111        this.btnProcesarNomina.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right)));
112        this.btnProcesarNomina.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
113        this.btnProcesarNomina.FlatAppearance.BorderSize = 0;
114        this.btnProcesarNomina.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
115        this.btnProcesarNomina.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
116        this.btnProcesarNomina.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(48)))), ((int)(((byte)(25)))), ((int)(((byte)(92))))));
117        this.btnProcesarNomina.Location = new System.Drawing.Point(836, 42);
```

UI\FrmDashboard.Designer.cs (continuacion)

```
118         this.btnProcesarNomina.Name = "btnProcesarNomina";
119         this.btnProcesarNomina.Size = new System.Drawing.Size(170, 38);
120         this.btnProcesarNomina.TabIndex = 3;
121         this.btnProcesarNomina.Text = "Procesar nómina";
122         this.btnProcesarNomina.UseVisualStyleBackColor = false;
123         this.btnProcesarNomina.Click += new System.EventHandler(this.btnProcesarNomina_Click);
124         //
125         // cardEmpleados
126         //
127         this.cardEmpleados.BackColor = System.Drawing.Color.Black;
128         this.cardEmpleados.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
129         this.cardEmpleados.Controls.Add(this.lblCardEmpleados);
130         this.cardEmpleados.Controls.Add(this.lblEmpleadosActivosValor);
131         this.cardEmpleados.Location = new System.Drawing.Point(36, 120);
132         this.cardEmpleados.Name = "cardEmpleados";
133         this.cardEmpleados.Size = new System.Drawing.Size(230, 130);
134         this.cardEmpleados.TabIndex = 4;
135         //
136         // lblCardEmpleados
137         //
138         this.lblCardEmpleados.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
139         this.lblCardEmpleados.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
140         this.lblCardEmpleados.Location = new System.Drawing.Point(18, 18);
141         this.lblCardEmpleados.Name = "lblCardEmpleados";
142         this.lblCardEmpleados.Size = new System.Drawing.Size(180, 24);
143         this.lblCardEmpleados.TabIndex = 0;
144         this.lblCardEmpleados.Text = "Empleados activos";
145         //
146         // lblEmpleadosActivosValor
147         //
148         this.lblEmpleadosActivosValor.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
149         this.lblEmpleadosActivosValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(59)))), ((int)(((byte)(130)))), ((int)(((byte)(246))))));
150         this.lblEmpleadosActivosValor.Location = new System.Drawing.Point(18, 56);
151         this.lblEmpleadosActivosValor.Name = "lblEmpleadosActivosValor";
152         this.lblEmpleadosActivosValor.Size = new System.Drawing.Size(170, 44);
153         this.lblEmpleadosActivosValor.TabIndex = 1;
154         this.lblEmpleadosActivosValor.Text = "0";
155         //
156         // cardNomina
157         //
158         this.cardNomina.BackColor = System.Drawing.Color.Black;
159         this.cardNomina.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
160         this.cardNomina.Controls.Add(this.lblCardNomina);
161         this.cardNomina.Controls.Add(this.lblNominaActualValor);
162         this.cardNomina.Location = new System.Drawing.Point(282, 120);
163         this.cardNomina.Name = "cardNomina";
164         this.cardNomina.Size = new System.Drawing.Size(230, 130);
165         this.cardNomina.TabIndex = 5;
166         //
167         // lblCardNomina
168         //
169         this.lblCardNomina.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
170         this.lblCardNomina.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
171         this.lblCardNomina.Location = new System.Drawing.Point(18, 18);
172         this.lblCardNomina.Name = "lblCardNomina";
173         this.lblCardNomina.Size = new System.Drawing.Size(180, 24);
174         this.lblCardNomina.TabIndex = 0;
175         this.lblCardNomina.Text = "Nómina actual";
176         //
177         // lblNominaActualValor
```

UI/FrmDashboard.Designer.cs (continuacion)

```
178        //
179        this.lblNominaActualValor.Font = new System.Drawing.Font("Segoe UI", 17F, System.Drawing.FontStyle.Bold);
180        this.lblNominaActualValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(34)))), ((int)(((byte)(197)))), ((int)(((byte)(94))))));
181        this.lblNominaActualValor.Location = new System.Drawing.Point(18, 56);
182        this.lblNominaActualValor.Name = "lblNominaActualValor";
183        this.lblNominaActualValor.Size = new System.Drawing.Size(206, 44);
184        this.lblNominaActualValor.TabIndex = 1;
185        this.lblNominaActualValor.Text = "B/. 0.00";
186        //
187        // cardComprobantes
188        //
189        this.cardComprobantes.BackColor = System.Drawing.Color.Black;
190        this.cardComprobantes.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
191        this.cardComprobantes.Controls.Add(this.lblCardComprobantes);
192        this.cardComprobantes.Controls.Add(this.lblComprobantesValor);
193        this.cardComprobantes.Location = new System.Drawing.Point(528, 120);
194        this.cardComprobantes.Name = "cardComprobantes";
195        this.cardComprobantes.Size = new System.Drawing.Size(230, 130);
196        this.cardComprobantes.TabIndex = 6;
197        //
198        // lblCardComprobantes
199        //
200        this.lblCardComprobantes.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
201        this.lblCardComprobantes.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
202        this.lblCardComprobantes.Location = new System.Drawing.Point(18, 18);
203        this.lblCardComprobantes.Name = "lblCardComprobantes";
204        this.lblCardComprobantes.Size = new System.Drawing.Size(190, 24);
205        this.lblCardComprobantes.TabIndex = 0;
206        this.lblCardComprobantes.Text = "Comprobantes generados";
207        //
208        // lblComprobantesValor
209        //
210        this.lblComprobantesValor.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
211        this.lblComprobantesValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(139)))), ((int)(((byte)(92)))), ((int)(((byte)(246))))));
212        this.lblComprobantesValor.Location = new System.Drawing.Point(18, 56);
213        this.lblComprobantesValor.Name = "lblComprobantesValor";
214        this.lblComprobantesValor.Size = new System.Drawing.Size(170, 44);
215        this.lblComprobantesValor.TabIndex = 1;
216        this.lblComprobantesValor.Text = "0";
217        //
218        // cardAlertas
219        //
220        this.cardAlertas.BackColor = System.Drawing.Color.Black;
221        this.cardAlertas.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
222        this.cardAlertas.Controls.Add(this.lblCardAlertas);
223        this.cardAlertas.Controls.Add(this.lblAlertasValor);
224        this.cardAlertas.Location = new System.Drawing.Point(774, 120);
225        this.cardAlertas.Name = "cardAlertas";
226        this.cardAlertas.Size = new System.Drawing.Size(230, 130);
227        this.cardAlertas.TabIndex = 7;
228        //
229        // lblCardAlertas
230        //
231        this.lblCardAlertas.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
232        this.lblCardAlertas.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
233        this.lblCardAlertas.Location = new System.Drawing.Point(18, 18);
234        this.lblCardAlertas.Name = "lblCardAlertas";
235        this.lblCardAlertas.Size = new System.Drawing.Size(180, 24);
236        this.lblCardAlertas.TabIndex = 0;
237        this.lblCardAlertas.Text = "Deducciones del mes";
```

UI/FrmDashboard.Designer.cs (continuacion)

```
238        //
239        // lblAlertasValor
240        //
241        this.lblAlertasValor.Font = new System.Drawing.Font("Segoe UI", 17F, System.Drawing.FontStyle.Bold);
242        this.lblAlertasValor.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(249)))), ((int)(((byte)(115)))), ((int)(((byte)(134))))));
243        this.lblAlertasValor.Location = new System.Drawing.Point(18, 56);
244        this.lblAlertasValor.Name = "lblAlertasValor";
245        this.lblAlertasValor.Size = new System.Drawing.Size(206, 44);
246        this.lblAlertasValor.TabIndex = 1;
247        this.lblAlertasValor.Text = "B/. 0.00";
248        //
249        // panelGrafico
250        //
251        this.panelGrafico.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
252        this.panelGrafico.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
253        this.panelGrafico.Controls.Add(this.chartPagos);
254        this.panelGrafico.Location = new System.Drawing.Point(36, 290);
255        this.panelGrafico.Name = "panelGrafico";
256        this.panelGrafico.Size = new System.Drawing.Size(650, 330);
257        this.panelGrafico.TabIndex = 8;
258        //
259        // chartPagos
260        //
261        this.chartPagos.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
262        chartArea1.AxisX.LabelStyle.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
263        chartArea1.AxisX.MajorGrid.LineColor = System.Drawing.Color.FromArgb(((int)(((byte)(42)))), ((int)(((byte)(43)))), ((int)(((byte)(56))))));
264        chartArea1.AxisY.LabelStyle.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
265        chartArea1.AxisY.MajorGrid.LineColor = System.Drawing.Color.FromArgb(((int)(((byte)(42)))), ((int)(((byte)(43)))), ((int)(((byte)(56))))));
266        chartArea1.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
267        chartArea1.Name = "Pagos";
268        this.chartPagos.ChartAreas.Add(chartArea1);
269        this.chartPagos.Location = new System.Drawing.Point(20, 42);
270        this.chartPagos.Name = "chartPagos";
271        series1.BorderColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
272        series1.BorderWidth = 3;
273        series1.ChartArea = "Pagos";
274        series1.ChartType = System.Windows.Forms.DataVisualization.Charting.SeriesChartType.SplineArea;
275        series1.Color = System.Drawing.Color.FromArgb(((int)(((byte)(120)))), ((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
276        series1.Name = "Neto";
277        this.chartPagos.Series.Add(series1);
278        this.chartPagos.Size = new System.Drawing.Size(600, 260);
279        this.chartPagos.TabIndex = 0;
280        this.chartPagos.Text = "Tendencia de pagos";
281        //
282        // panelActividad
283        //
284        this.panelActividad.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
285        this.panelActividad.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
286        this.panelActividad.Controls.Add(this.lblActividadTitulo);
287        this.panelActividad.Controls.Add(this.lblActividadUno);
288        this.panelActividad.Controls.Add(this.lblActividadDos);
289        this.panelActividad.Controls.Add(this.lblActividadTres);
290        this.panelActividad.Location = new System.Drawing.Point(710, 290);
291        this.panelActividad.Name = "panelActividad";
292        this.panelActividad.Size = new System.Drawing.Size(294, 330);
293        this.panelActividad.TabIndex = 9;
294        //
295        // lblActividadTitulo
296        //
297        this.lblActividadTitulo.Font = new System.Drawing.Font("Segoe UI", 15F, System.Drawing.FontStyle.Bold);
```

UI/FrmDashboard.Designer.cs (continuacion)

```
298         this.lblActividadTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
299         this.lblActividadTitulo.Location = new System.Drawing.Point(20, 22);
300         this.lblActividadTitulo.Name = "lblActividadTitulo";
301         this.lblActividadTitulo.Size = new System.Drawing.Size(240, 30);
302         this.lblActividadTitulo.TabIndex = 0;
303         this.lblActividadTitulo.Text = "Actividad reciente";
304         //
305         // lblActividadUno
306         //
307         this.lblActividadUno.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
308         this.lblActividadUno.Location = new System.Drawing.Point(24, 82);
309         this.lblActividadUno.Name = "lblActividadUno";
310         this.lblActividadUno.Size = new System.Drawing.Size(230, 42);
311         this.lblActividadUno.TabIndex = 1;
312         this.lblActividadUno.Text = "Nómina lista para revisión\r\nHace 2 horas";
313         //
314         // lblActividadDos
315         //
316         this.lblActividadDos.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
317         this.lblActividadDos.Location = new System.Drawing.Point(24, 146);
318         this.lblActividadDos.Name = "lblActividadDos";
319         this.lblActividadDos.Size = new System.Drawing.Size(230, 42);
320         this.lblActividadDos.TabIndex = 2;
321         this.lblActividadDos.Text = "Nuevo empleado registrado\r\nDatos iniciales cargados";
322         //
323         // lblActividadTres
324         //
325         this.lblActividadTres.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(249)))), ((int)(((byte)(115)))), ((int)(((byte)(134))))));
326         this.lblActividadTres.Location = new System.Drawing.Point(24, 210);
327         this.lblActividadTres.Name = "lblActividadTres";
328         this.lblActividadTres.Size = new System.Drawing.Size(230, 42);
329         this.lblActividadTres.TabIndex = 3;
330         this.lblActividadTres.Text = "Validar asistencia\r\nRevisar tardanzas";
331         //
332         // FrmDashboard
333         //
334         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
335         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
336         this.AutoScroll = true;
337         this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
338         this.ClientSize = new System.Drawing.Size(1020, 742);
339         this.Controls.Add(this.panelActividad);
340         this.Controls.Add(this.panelGrafico);
341         this.Controls.Add(this.cardAlertas);
342         this.Controls.Add(this.cardComprobantes);
343         this.Controls.Add(this.cardNomina);
344         this.Controls.Add(this.cardEmpleados);
345         this.Controls.Add(this.btnProcesarNomina);
346         this.Controls.Add(this.btnRegistrarEmpleado);
347         this.Controls.Add(this.lblSubtitulo);
348         this.Controls.Add(this.lblTitulo);
349         this.Font = new System.Drawing.Font("Segoe UI", 9F);
350         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
351         this.Name = "FrmDashboard";
352         this.Text = "Dashboard";
353         this.Load += new System.EventHandler(this.FrmDashboard_Load);
354         this.cardEmpleados.ResumeLayout(false);
355         this.cardNomina.ResumeLayout(false);
356         this.cardComprobantes.ResumeLayout(false);
357         this.cardAlertas.ResumeLayout(false);
```

UI/FrmDashboard.Designer.cs (continuacion)

```
358         this.panelGrafico.ResumeLayout(false);
359         ((System.ComponentModel.ISupportInitialize)(this.chartPagos)).EndInit();
360         this.ResumeLayout(false);
361     }
362 }
363 }
```

UI/FrmEmpleados.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Services;
8 using SistemaGestionNomina.Security;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmEmpleados : Form
13     {
14         private readonly EmpleadoService empleadoService = new EmpleadoService();
15         private readonly ExcelExportService excelExportService = new ExcelExportService();
16         private readonly PdfExportService pdfExportService = new PdfExportService();
17         private readonly AuthorizationService authorizationService = new AuthorizationService();
18         private int selectedId;
19         private List<Empleado> currentEmployees = new List<Empleado>();
20
21         public FrmEmpleados()
22         {
23             InitializeComponent();
24             ControlStyleHelper.ApplyModernForm(this);
25         }
26
27         private void FrmEmpleados_Load(object sender, EventArgs e)
28         {
29             if (LicenseManager.UsageMode == LicenseUsageMode.DesignTime)
30             {
31                 return;
32             }
33
34             btnNuevo.Visible = authorizationService.HasPermission(Permissions.EmployeesCreate);
35             btnEditar.Visible = authorizationService.HasPermission(Permissions.EmployeesEdit);
36             btnDesactivar.Visible = authorizationService.HasPermission(Permissions.EmployeesDeactivate);
37             btnExportarExcel.Visible = authorizationService.HasPermission(Permissions.EmployeesExport);
38             btnExportarPdf.Visible = authorizationService.HasPermission(Permissions.EmployeesExport);
39             btnHistorialLaboral.Visible = authorizationService.HasPermission(Permissions.EmployeeHistoryView);
40             LoadDepartments();
41             LoadEmployees();
42         }
43
44         private void LoadDepartments()
45         {
46             List<Departamento> departments = empleadoService.GetDepartments();
47             cmbDepartamento.DataSource = new List<Departamento>(departments);
48             cmbDepartamento.DisplayMember = "Nombre";
49             cmbDepartamento.ValueMember = "IdDepartamento";
50
51             List<Departamento> filters = new List<Departamento>();
52             filters.Add(new Departamento { IdDepartamento = 0, Nombre = "Todos los departamentos" });
53             filters.AddRange(departments);
54             cmbFiltroDepartamento.DataSource = filters;
55             cmbFiltroDepartamento.DisplayMember = "Nombre";
56             cmbFiltroDepartamento.ValueMember = "IdDepartamento";
57         }
58
59         private void LoadEmployees()
60         {
```

UI/FrmEmpleados.cs (continuacion)

```
61         int? department = null;
62         if (cmbFiltroDepartamento.SelectedValue is int && (int)cmbFiltroDepartamento.SelectedValue > 0)
63         {
64             department = (int)cmbFiltroDepartamento.SelectedValue;
65         }
66
67         string status = cmbFiltroEstado.SelectedItem == null || cmbFiltroEstado.SelectedItem.ToString() == "Todos"
68             ? string.Empty
69             : cmbFiltroEstado.SelectedItem.ToString();
70
71         currentEmployees = empleadoService.GetAll(txtBuscar.Text, department, status);
72         dgvEmpleados.Rows.Clear();
73         foreach (Empleado e in currentEmployees)
74         {
75             dgvEmpleados.Rows.Add(e.IdEmpleado, e.Codigo, e.NombreCompleto, e.Cedula, e.Cargo,
76                 e.DepartamentoNombre, "B/. " + e.SalarioBase.ToString("0.00"), e.Estado, "Editar");
77         }
78     }
79
80     private void dgvEmpleados_SelectionChanged(object sender, EventArgs e)
81     {
82         if (dgvEmpleados.CurrentRow == null || dgvEmpleados.CurrentRow.Cells["IdEmpleado"].Value == null)
83         {
84             return;
85         }
86
87         selectedId = Convert.ToInt32(dgvEmpleados.CurrentRow.Cells["IdEmpleado"].Value);
88         Empleado empleado = empleadoService.GetById(selectedId);
89         if (empleado == null)
90         {
91             return;
92         }
93
94         txtCodigo.Text = empleado.Codigo;
95         txtNombre.Text = empleado.Nombre;
96         txtApellido.Text = empleado.Apellido;
97         txtCedula.Text = empleado.Cedula;
98         txtCargo.Text = empleado.Cargo;
99         cmbDepartamento.SelectedValue = empleado.IdDepartamento;
100        txtSalario.Text = empleado.SalarioBase.ToString("0.00");
101        cmbEstado.SelectedItem = empleado.Estado;
102        dtpIngreso.Value = empleado.FechaIngreso;
103    }
104
105     private void btnNuevo_Click(object sender, EventArgs e)
106     {
107         ClearForm();
108     }
109
110     private void btnEditar_Click(object sender, EventArgs e)
111     {
112         decimal salary;
113         if (!ValidationHelper.RequireText(txtCodigo, "Código") ||
114             !ValidationHelper.RequireText(txtNombre, "Nombre") ||
115             !ValidationHelper.RequireText(txtApellido, "Apellido") ||
116             !ValidationHelper.RequireText(txtCedula, "Cédula") ||
117             !ValidationHelper.RequireText(txtCargo, "Cargo") ||
118             !ValidationHelper.TryPositiveDecimal(txtSalario, "Salario base", out salary))
119         {
120             return;
```


UI/FrmEmpleados.cs (continuación)

```
121     }
122
123     if (cmbDepartamento.SelectedValue == null)
124     {
125         MessageBox.Show("Seleccione un departamento.", "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Warning);
126         return;
127     }
128
129     if (cmbEstado.SelectedItem == null)
130     {
131         MessageBox.Show("Seleccione el estado del empleado.", "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Warning);
132         return;
133     }
134
135     try
136     {
137         Empleado empleado = new Empleado();
138         empleado.IdEmpleado = selectedId;
139         empleado.Codigo = txtCodigo.Text.Trim();
140         empleado.Nombre = txtNombre.Text.Trim();
141         empleado.Apellido = txtApellido.Text.Trim();
142         empleado.Cedula = txtCedula.Text.Trim();
143         empleado.Cargo = txtCargo.Text.Trim();
144         empleado.IdDepartamento = Convert.ToInt32(cmbDepartamento.SelectedValue);
145         empleado.SalarioBase = salary;
146         empleado.Estado = cmbEstado.SelectedItem.ToString();
147         empleado.FechaIngreso = dtpIngreso.Value.Date;
148         EmployeeChangeContext changeContext = null;
149         if (selectedId > 0)
150         {
151             Empleado original = empleadoService.GetById(selectedId);
152             if (HasTrackedChanges(original, empleado))
153             {
154                 using (FrmCambioLaboral dialog = new FrmCambioLaboral(original))
155                 {
156                     if (dialog.ShowDialog() != DialogResult.OK) return;
157                     changeContext = dialog.ChangeContext;
158                 }
159             }
160         }
161         EmployeeSaveResult result = empleadoService.Save(empleado, changeContext);
162         string message = result.HasScheduledChanges
163             ? "El cambio laboral quedó programado para la fecha indicada."
164             : "Empleado guardado correctamente.";
165         MessageBox.Show(message, "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Information);
166         ClearForm();
167         LoadEmployees();
168     }
169     catch (Exception ex)
170     {
171         MessageBox.Show(ex.Message, "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Warning);
172     }
173 }
174
175 private void btnDesactivar_Click(object sender, EventArgs e)
176 {
177     if (selectedId == 0)
178     {
179         MessageBox.Show("Seleccione un empleado.", "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Information);
180         return;
181     }
182 }
```

UI/FrmEmpleados.cs (continuacion)

```
181         }
182
183         if (MessageBox.Show("¿Desea desactivar el empleado seleccionado?", "Confirmar",
184             MessageBoxButtons.YesNo, MessageBoxIcon.Question) == DialogResult.Yes)
185         {
186             try
187             {
188                 Empleado employee = empleadoService.GetById(selectedId);
189                 using (FrmCambioLaboral dialog = new FrmCambioLaboral(employee))
190                 {
191                     if (dialog.ShowDialog() != DialogResult.OK) return;
192                     EmployeeSaveResult result = empleadoService.Deactivate(selectedId, dialog.ChangeContext);
193                     MessageBox.Show(result.HasScheduledChanges
194                         ? "La desactivación quedó programada." : "Empleado desactivado correctamente.",
195                         "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Information);
196                 }
197                 ClearForm();
198                 LoadEmployees();
199             }
200             catch (Exception ex)
201             {
202                 MessageBox.Show(ex.Message, "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Warning);
203             }
204         }
205     }
206
207     private void btnExportarExcel_Click(object sender, EventArgs e)
208     {
209         if (currentEmployees.Count == 0)
210         {
211             MessageBox.Show("No hay empleados para exportar.", "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Information);
212             return;
213         }
214
215         UiFactory.ShowExported(excelExportService.ExportarEmpleados(currentEmployees));
216     }
217
218     private void btnExportarPdf_Click(object sender, EventArgs e)
219     {
220         if (currentEmployees.Count == 0)
221         {
222             MessageBox.Show("No hay empleados para exportar.", "Empleados", MessageBoxButtons.OK, MessageBoxIcon.Information);
223             return;
224         }
225
226         UiFactory.ShowExported(pdfExportService.ExportarEmpleados(currentEmployees));
227     }
228
229     private void FilterChanged(object sender, EventArgs e)
230     {
231         if (LicenseManager.UsageMode != LicenseUsageMode.DesignTime && cmbFiltroDepartamento.DataSource != null)
232         {
233             LoadEmployees();
234         }
235     }
236
237     private void btnHistorialLaboral_Click(object sender, EventArgs e)
238     {
239         using (FrmHistorialEmpleado form = new FrmHistorialEmpleado(selectedId > 0 ? (int?)selectedId : null))
240             form.ShowDialog();
241     }
```

UI/FrmEmpleados.cs (continuacion)

```
241     }
242
243     private static bool HasTrackedChanges(Empleado original, Empleado updated)
244     {
245         return original != null && (original.SalarioBase != updated.SalarioBase ||
246             original.IdDepartamento != updated.IdDepartamento ||
247             !string.Equals(original.Cargo, updated.Cargo, StringComparison.Ordinal) ||
248             !string.Equals(original.Estado, updated.Estado, StringComparison.OrdinalIgnoreCase));
249     }
250
251     private void ClearForm()
252     {
253         selectedId = 0;
254         txtCodigo.Clear();
255         txtNombre.Clear();
256         txtApellido.Clear();
257         txtCedula.Clear();
258         txtCargo.Clear();
259         txtSalario.Clear();
260         if (cmbDepartamento.Items.Count > 0)
261         {
262             cmbDepartamento.SelectedIndex = 0;
263         }
264         cmbEstado.SelectedIndex = 0;
265         dtpIngreso.Value = DateTime.Today;
266     }
267 }
268 }
```

UI\FrmEmpleados.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmEmpleados
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelFormulario;
8         private System.Windows.Forms.Panel panelFiltros;
9         private System.Windows.Forms.TextBox txtBuscar;
10        private System.Windows.Forms.ComboBox cmbFiltroDepartamento;
11        private System.Windows.Forms.ComboBox cmbFiltroEstado;
12        private System.Windows.Forms.Button btnNuevo;
13        private System.Windows.Forms.Button btnEditar;
14        private System.Windows.Forms.Button btnDesactivar;
15        private System.Windows.Forms.Button btnExportarExcel;
16        private System.Windows.Forms.Button btnExportarPdf;
17        private System.Windows.Forms.Button btnHistorialLaboral;
18        private System.Windows.Forms.TextBox txtCodigo;
19        private System.Windows.Forms.TextBox txtNombre;
20        private System.Windows.Forms.TextBox txtApellido;
21        private System.Windows.Forms.TextBox txtCedula;
22        private System.Windows.Forms.TextBox txtCargo;
23        private System.Windows.Forms.TextBox txtSalario;
24        private System.Windows.Forms.Label lblCodigo;
25        private System.Windows.Forms.Label lblNombre;
26        private System.Windows.Forms.Label lblApellido;
27        private System.Windows.Forms.Label lblCedula;
28        private System.Windows.Forms.Label lblCargo;
29        private System.Windows.Forms.Label lblDepartamento;
30        private System.Windows.Forms.Label lblSalario;
31        private System.Windows.Forms.ComboBox cmbDepartamento;
32        private System.Windows.Forms.ComboBox cmbEstado;
33        private System.Windows.Forms.DateTimePicker dtpIngreso;
34        private System.Windows.Forms.DataGridView dgvEmpleados;
35        private System.Windows.Forms.DataGridViewTextBoxColumn colIdEmpleado;
36        private System.Windows.Forms.DataGridViewTextBoxColumn colCodigo;
37        private System.Windows.Forms.DataGridViewTextBoxColumn colNombre;
38        private System.Windows.Forms.DataGridViewTextBoxColumn colCedula;
39        private System.Windows.Forms.DataGridViewTextBoxColumn colCargo;
40        private System.Windows.Forms.DataGridViewTextBoxColumn colDepartamento;
41        private System.Windows.Forms.DataGridViewTextBoxColumn colSalarioBase;
42        private System.Windows.Forms.DataGridViewTextBoxColumn colEstado;
43        private System.Windows.Forms.DataGridViewTextBoxColumn colAcciones;
44
45        protected override void Dispose(bool disposing)
46        {
47            if (disposing && (components != null)) components.Dispose();
48            base.Dispose(disposing);
49        }
50
51        private void InitializeComponent()
52        {
53            this.lblTitulo = new System.Windows.Forms.Label();
54            this.panelFormulario = new System.Windows.Forms.Panel();
55            this.txtCodigo = new System.Windows.Forms.TextBox();
56            this.txtNombre = new System.Windows.Forms.TextBox();
57            this.txtApellido = new System.Windows.Forms.TextBox();
58            this.txtCedula = new System.Windows.Forms.TextBox();
59            this.txtCargo = new System.Windows.Forms.TextBox();
60            this.lblCodigo = new System.Windows.Forms.Label();
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
61         this.lblNombre = new System.Windows.Forms.Label();
62         this.lblApellido = new System.Windows.Forms.Label();
63         this.lblCedula = new System.Windows.Forms.Label();
64         this.lblCargo = new System.Windows.Forms.Label();
65         this.lblDepartamento = new System.Windows.Forms.Label();
66         this.lblSalario = new System.Windows.Forms.Label();
67         this.cmbDepartamento = new System.Windows.Forms.ComboBox();
68         this.txtSalario = new System.Windows.Forms.TextBox();
69         this.cmbEstado = new System.Windows.Forms.ComboBox();
70         this.dtpIngreso = new System.Windows.Forms.DateTimePicker();
71         this.panelFiltros = new System.Windows.Forms.Panel();
72         this.txtBuscar = new System.Windows.Forms.TextBox();
73         this.cmbFiltroDepartamento = new System.Windows.Forms.ComboBox();
74         this.cmbFiltroEstado = new System.Windows.Forms.ComboBox();
75         this.btnNuevo = new System.Windows.Forms.Button();
76         this.btnEditar = new System.Windows.Forms.Button();
77         this.btnDesactivar = new System.Windows.Forms.Button();
78         this.btnExportarExcel = new System.Windows.Forms.Button();
79         this.btnExportarPdf = new System.Windows.Forms.Button();
80         this.btnHistorialLaboral = new System.Windows.Forms.Button();
81         this.dgvEmpleados = new System.Windows.Forms.DataGridView();
82         this.colIdEmpleado = new System.Windows.Forms.DataGridViewTextBoxColumn();
83         this.colCodigo = new System.Windows.Forms.DataGridViewTextBoxColumn();
84         this.colNombre = new System.Windows.Forms.DataGridViewTextBoxColumn();
85         this.colCedula = new System.Windows.Forms.DataGridViewTextBoxColumn();
86         this.colCargo = new System.Windows.Forms.DataGridViewTextBoxColumn();
87         this.colDepartamento = new System.Windows.Forms.DataGridViewTextBoxColumn();
88         this.colSalarioBase = new System.Windows.Forms.DataGridViewTextBoxColumn();
89         this.colEstado = new System.Windows.Forms.DataGridViewTextBoxColumn();
90         this.colAcciones = new System.Windows.Forms.DataGridViewTextBoxColumn();
91         this.panelFormulario.SuspendLayout();
92         this.panelFiltros.SuspendLayout();
93         ((System.ComponentModel.ISupportInitialize)(this.dgvEmpleados)).BeginInit();
94         this.SuspendLayout();
95         //
96         // lblTitulo
97         //
98         this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
99         this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
100        this.lblTitulo.Location = new System.Drawing.Point(34, 24);
101        this.lblTitulo.Name = "lblTitulo";
102        this.lblTitulo.Size = new System.Drawing.Size(420, 42);
103        this.lblTitulo.TabIndex = 0;
104        this.lblTitulo.Text = "Empleados";
105        //
106        // btnHistorialLaboral
107        //
108        this.btnHistorialLaboral.BackColor = System.Drawing.Color.FromArgb(17, 18, 26);
109        this.btnHistorialLaboral.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
110        this.btnHistorialLaboral.Location = new System.Drawing.Point(824, 28);
111        this.btnHistorialLaboral.Name = "btnHistorialLaboral";
112        this.btnHistorialLaboral.Size = new System.Drawing.Size(160, 34);
113        this.btnHistorialLaboral.TabIndex = 20;
114        this.btnHistorialLaboral.Text = "Historial laboral";
115        this.btnHistorialLaboral.UseVisualStyleBackColor = false;
116        this.btnHistorialLaboral.Click += new System.EventHandler(this.btnHistorialLaboral_Click);
117        //
118        // panelFormulario
119        //
120        this.panelFormulario.BackColor = System.Drawing.Color.FromArgb(17, 18, 26);
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
121         this.panelFormulario.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
122         this.panelFormulario.Controls.Add(this.lblCodigo);
123         this.panelFormulario.Controls.Add(this.lblNombre);
124         this.panelFormulario.Controls.Add(this.lblApellido);
125         this.panelFormulario.Controls.Add(this.lblCedula);
126         this.panelFormulario.Controls.Add(this.lblCargo);
127         this.panelFormulario.Controls.Add(this.lblDepartamento);
128         this.panelFormulario.Controls.Add(this.lblSalario);
129         this.panelFormulario.Controls.Add(this.txtCodigo);
130         this.panelFormulario.Controls.Add(this.txtNombre);
131         this.panelFormulario.Controls.Add(this.txtApellido);
132         this.panelFormulario.Controls.Add(this.txtCedula);
133         this.panelFormulario.Controls.Add(this.txtCargo);
134         this.panelFormulario.Controls.Add(this.cmbDepartamento);
135         this.panelFormulario.Controls.Add(this.txtSalario);
136         this.panelFormulario.Controls.Add(this.cmbEstado);
137         this.panelFormulario.Controls.Add(this.dtpIngreso);
138         this.panelFormulario.Location = new System.Drawing.Point(36, 86);
139         this.panelFormulario.Name = "panelFormulario";
140         this.panelFormulario.Size = new System.Drawing.Size(948, 118);
141         this.panelFormulario.TabIndex = 1;
142         //
143         // txtCodigo
144         //
145         this.txtCodigo.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
146         this.txtCodigo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
147         this.txtCodigo.Location = new System.Drawing.Point(18, 36);
148         this.txtCodigo.Name = "txtCodigo";
149         this.txtCodigo.Size = new System.Drawing.Size(100, 23);
150         this.txtCodigo.TabIndex = 0;
151         this.txtCodigo.Text = "EMP-";
152         //
153         // txtNombre
154         //
155         this.txtNombre.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
156         this.txtNombre.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
157         this.txtNombre.Location = new System.Drawing.Point(128, 36);
158         this.txtNombre.Name = "txtNombre";
159         this.txtNombre.Size = new System.Drawing.Size(120, 23);
160         this.txtNombre.TabIndex = 1;
161         //
162         // txtApellido
163         //
164         this.txtApellido.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
165         this.txtApellido.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
166         this.txtApellido.Location = new System.Drawing.Point(258, 36);
167         this.txtApellido.Name = "txtApellido";
168         this.txtApellido.Size = new System.Drawing.Size(120, 23);
169         this.txtApellido.TabIndex = 2;
170         //
171         // txtCedula
172         //
173         this.txtCedula.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
174         this.txtCedula.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
175         this.txtCedula.Location = new System.Drawing.Point(388, 36);
176         this.txtCedula.Name = "txtCedula";
177         this.txtCedula.Size = new System.Drawing.Size(120, 23);
178         this.txtCedula.TabIndex = 3;
179         //
180         // txtCargo
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
181        //
182        this.txtCargo.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
183        this.txtCargo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
184        this.txtCargo.Location = new System.Drawing.Point(518, 36);
185        this.txtCargo.Name = "txtCargo";
186        this.txtCargo.Size = new System.Drawing.Size(150, 23);
187        this.txtCargo.TabIndex = 4;
188        //
189        // cmbDepartamento
190        //
191        this.cmbDepartamento.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
192        this.cmbDepartamento.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
193        this.cmbDepartamento.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
194        this.cmbDepartamento.Location = new System.Drawing.Point(678, 36);
195        this.cmbDepartamento.Name = "cmbDepartamento";
196        this.cmbDepartamento.Size = new System.Drawing.Size(150, 23);
197        this.cmbDepartamento.TabIndex = 5;
198        //
199        // txtSalario
200        //
201        this.txtSalario.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
202        this.txtSalario.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
203        this.txtSalario.Location = new System.Drawing.Point(18, 84);
204        this.txtSalario.Name = "txtSalario";
205        this.txtSalario.Size = new System.Drawing.Size(120, 23);
206        this.txtSalario.TabIndex = 6;
207        //
208        // input labels
209        //
210        this.lblCodigo.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
211        this.lblCodigo.Location = new System.Drawing.Point(18, 16);
212        this.lblCodigo.Name = "lblCodigo";
213        this.lblCodigo.Size = new System.Drawing.Size(100, 18);
214        this.lblCodigo.TabIndex = 20;
215        this.lblCodigo.Text = "Código";
216        this.lblNombre.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
217        this.lblNombre.Location = new System.Drawing.Point(128, 16);
218        this.lblNombre.Name = "lblNombre";
219        this.lblNombre.Size = new System.Drawing.Size(100, 18);
220        this.lblNombre.TabIndex = 21;
221        this.lblNombre.Text = "Nombre";
222        this.lblApellido.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
223        this.lblApellido.Location = new System.Drawing.Point(258, 16);
224        this.lblApellido.Name = "lblApellido";
225        this.lblApellido.Size = new System.Drawing.Size(100, 18);
226        this.lblApellido.TabIndex = 22;
227        this.lblApellido.Text = "Apellido";
228        this.lblCedula.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
229        this.lblCedula.Location = new System.Drawing.Point(388, 16);
230        this.lblCedula.Name = "lblCedula";
231        this.lblCedula.Size = new System.Drawing.Size(100, 18);
232        this.lblCedula.TabIndex = 23;
233        this.lblCedula.Text = "Cédula";
234        this.lblCargo.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
235        this.lblCargo.Location = new System.Drawing.Point(518, 16);
236        this.lblCargo.Name = "lblCargo";
237        this.lblCargo.Size = new System.Drawing.Size(100, 18);
238        this.lblCargo.TabIndex = 24;
239        this.lblCargo.Text = "Cargo";
240        this.lblDepartamento.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
241         this.lblDepartamento.Location = new System.Drawing.Point(678, 16);
242         this.lblDepartamento.Name = "lblDepartamento";
243         this.lblDepartamento.Size = new System.Drawing.Size(120, 18);
244         this.lblDepartamento.TabIndex = 25;
245         this.lblDepartamento.Text = "Departamento";
246         this.lblSalario.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
247         this.lblSalario.Location = new System.Drawing.Point(18, 64);
248         this.lblSalario.Name = "lblSalario";
249         this.lblSalario.Size = new System.Drawing.Size(100, 18);
250         this.lblSalario.TabIndex = 26;
251         this.lblSalario.Text = "Salario base";
252         //
253         // cmbEstado
254         //
255         this.cmbEstado.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
256         this.cmbEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
257         this.cmbEstado.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
258         this.cmbEstado.Items.AddRange(new object[] { "Activo", "Inactivo" });
259         this.cmbEstado.Location = new System.Drawing.Point(150, 70);
260         this.cmbEstado.Name = "cmbEstado";
261         this.cmbEstado.SelectedIndex = 0;
262         this.cmbEstado.Size = new System.Drawing.Size(120, 23);
263         this.cmbEstado.TabIndex = 7;
264         //
265         // dtpIngreso
266         //
267         this.dtpIngreso.Format = System.Windows.Forms.DateTimePickerFormat.Short;
268         this.dtpIngreso.Location = new System.Drawing.Point(282, 70);
269         this.dtpIngreso.Name = "dtpIngreso";
270         this.dtpIngreso.Size = new System.Drawing.Size(120, 23);
271         this.dtpIngreso.TabIndex = 8;
272         //
273         // panelFiltros
274         //
275         this.panelFiltros.BackColor = System.Drawing.Color.FromArgb(17, 18, 26);
276         this.panelFiltros.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
277         this.panelFiltros.Controls.Add(this.txtBuscar);
278         this.panelFiltros.Controls.Add(this.cmbFiltroDepartamento);
279         this.panelFiltros.Controls.Add(this.cmbFiltroEstado);
280         this.panelFiltros.Controls.Add(this.btnNuevo);
281         this.panelFiltros.Controls.Add(this.btnEditar);
282         this.panelFiltros.Controls.Add(this.btnDesactivar);
283         this.panelFiltros.Controls.Add(this.btnExportarExcel);
284         this.panelFiltros.Controls.Add(this.btnExportarPdf);
285         this.panelFiltros.Location = new System.Drawing.Point(36, 224);
286         this.panelFiltros.Name = "panelFiltros";
287         this.panelFiltros.Size = new System.Drawing.Size(948, 70);
288         this.panelFiltros.TabIndex = 2;
289         //
290         // txtBuscar
291         //
292         this.txtBuscar.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
293         this.txtBuscar.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
294         this.txtBuscar.Location = new System.Drawing.Point(18, 24);
295         this.txtBuscar.Name = "txtBuscar";
296         this.txtBuscar.Size = new System.Drawing.Size(190, 23);
297         this.txtBuscar.TabIndex = 0;
298         this.txtBuscar.TextChanged += new System.EventHandler(this.FilterChanged);
299         //
300         // cmbFiltroDepartamento
```


UI/FrmEmpleados.Designer.cs (continuacion)

```
301        //
302        this.cmbFiltroDepartamento.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
303        this.cmbFiltroDepartamento.Location = new System.Drawing.Point(222, 24);
304        this.cmbFiltroDepartamento.Name = "cmbFiltroDepartamento";
305        this.cmbFiltroDepartamento.Size = new System.Drawing.Size(180, 23);
306        this.cmbFiltroDepartamento.TabIndex = 1;
307        this.cmbFiltroDepartamento.SelectedIndexChanged += new System.EventHandler(this.FilterChanged);
308        //
309        // cmbFiltroEstado
310        //
311        this.cmbFiltroEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
312        this.cmbFiltroEstado.Items.AddRange(new object[] { "Todos", "Activo", "Inactivo" });
313        this.cmbFiltroEstado.Location = new System.Drawing.Point(414, 24);
314        this.cmbFiltroEstado.Name = "cmbFiltroEstado";
315        this.cmbFiltroEstado.SelectedIndex = 0;
316        this.cmbFiltroEstado.Size = new System.Drawing.Size(110, 23);
317        this.cmbFiltroEstado.TabIndex = 2;
318        this.cmbFiltroEstado.SelectedIndexChanged += new System.EventHandler(this.FilterChanged);
319        //
320        // btnNuevo
321        //
322        this.btnNuevo.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
323        this.btnNuevo.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
324        this.btnNuevo.Location = new System.Drawing.Point(540, 20);
325        this.btnNuevo.Name = "btnNuevo";
326        this.btnNuevo.Size = new System.Drawing.Size(80, 30);
327        this.btnNuevo.TabIndex = 3;
328        this.btnNuevo.Text = "Nuevo";
329        this.btnNuevo.UseVisualStyleBackColor = false;
330        this.btnNuevo.Click += new System.EventHandler(this.btnNuevo_Click);
331        //
332        // btnEditar
333        //
334        this.btnEditar.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
335        this.btnEditar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
336        this.btnEditar.Location = new System.Drawing.Point(626, 20);
337        this.btnEditar.Name = "btnEditar";
338        this.btnEditar.Size = new System.Drawing.Size(80, 30);
339        this.btnEditar.TabIndex = 4;
340        this.btnEditar.Text = "Guardar";
341        this.btnEditar.UseVisualStyleBackColor = false;
342        this.btnEditar.Click += new System.EventHandler(this.btnEditar_Click);
343        //
344        // btnDesactivar
345        //
346        this.btnDesactivar.BackColor = System.Drawing.Color.Black;
347        this.btnDesactivar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
348        this.btnDesactivar.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
349        this.btnDesactivar.Location = new System.Drawing.Point(712, 20);
350        this.btnDesactivar.Name = "btnDesactivar";
351        this.btnDesactivar.Size = new System.Drawing.Size(90, 30);
352        this.btnDesactivar.TabIndex = 5;
353        this.btnDesactivar.Text = "Desactivar";
354        this.btnDesactivar.UseVisualStyleBackColor = false;
355        this.btnDesactivar.Click += new System.EventHandler(this.btnDesactivar_Click);
356        //
357        // btnExportarExcel
358        //
359        this.btnExportarExcel.BackColor = System.Drawing.Color.Black;
360        this.btnExportarExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
361         this.btnExportarExcel.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
362         this.btnExportarExcel.Location = new System.Drawing.Point(808, 20);
363         this.btnExportarExcel.Name = "btnExportarExcel";
364         this.btnExportarExcel.Size = new System.Drawing.Size(60, 30);
365         this.btnExportarExcel.TabIndex = 6;
366         this.btnExportarExcel.Text = "Excel";
367         this.btnExportarExcel.UseVisualStyleBackColor = false;
368         this.btnExportarExcel.Click += new System.EventHandler(this.btnExportarExcel_Click);
369         //
370         // btnExportarPdf
371         //
372         this.btnExportarPdf.BackColor = System.Drawing.Color.Black;
373         this.btnExportarPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
374         this.btnExportarPdf.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
375         this.btnExportarPdf.Location = new System.Drawing.Point(874, 20);
376         this.btnExportarPdf.Name = "btnExportarPdf";
377         this.btnExportarPdf.Size = new System.Drawing.Size(55, 30);
378         this.btnExportarPdf.TabIndex = 7;
379         this.btnExportarPdf.Text = "PDF";
380         this.btnExportarPdf.UseVisualStyleBackColor = false;
381         this.btnExportarPdf.Click += new System.EventHandler(this.btnExportarPdf_Click);
382         //
383         // dgvEmpleados
384         //
385         this.dgvEmpleados.AllowUserToAddRows = false;
386         this.dgvEmpleados.AllowUserToDeleteRows = false;
387         this.dgvEmpleados.BackgroundColor = System.Drawing.Color.Black;
388         this.dgvEmpleados.ColumnHeadersHeightSizeMode = System.Windows.Forms.DataGridViewColumnHeadersHeightSizeMode.AutoSize;
389         this.dgvEmpleados.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] {
390             this.colIdEmpleado,
391             this.colCodigo,
392             this.colNombre,
393             this.colCedula,
394             this.colCargo,
395             this.colDepartamento,
396             this.colSalarioBase,
397             this.colEstado,
398             this.colAcciones});
399         this.dgvEmpleados.Location = new System.Drawing.Point(36, 318);
400         this.dgvEmpleados.MultiSelect = false;
401         this.dgvEmpleados.Name = "dgvEmpleados";
402         this.dgvEmpleados.ReadOnly = true;
403         this.dgvEmpleados.RowHeadersVisible = false;
404         this.dgvEmpleados.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect;
405         this.dgvEmpleados.Size = new System.Drawing.Size(948, 360);
406         this.dgvEmpleados.TabIndex = 3;
407         this.dgvEmpleados.SelectionChanged += new System.EventHandler(this.dgvEmpleados_SelectionChanged);
408         //
409         // columns
410         //
411         this.colIdEmpleado.HeaderText = "Id";
412         this.colIdEmpleado.Name = "IdEmpleado";
413         this.colIdEmpleado.ReadOnly = true;
414         this.colIdEmpleado.Visible = false;
415         this.colCodigo.HeaderText = "Código";
416         this.colCodigo.Name = "Codigo";
417         this.colCodigo.ReadOnly = true;
418         this.colNombre.HeaderText = "Nombre";
419         this.colNombre.Name = "Nombre";
420         this.colNombre.ReadOnly = true;
```

UI/FrmEmpleados.Designer.cs (continuacion)

```
421         this.colCedula.HeaderText = "Cédula";
422         this.colCedula.Name = "Cedula";
423         this.colCedula.ReadOnly = true;
424         this.colCargo.HeaderText = "Cargo";
425         this.colCargo.Name = "Cargo";
426         this.colCargo.ReadOnly = true;
427         this.colDepartamento.HeaderText = "Departamento";
428         this.colDepartamento.Name = "Departamento";
429         this.colDepartamento.ReadOnly = true;
430         this.colSalarioBase.HeaderText = "Salario Base";
431         this.colSalarioBase.Name = "SalarioBase";
432         this.colSalarioBase.ReadOnly = true;
433         this.colEstado.HeaderText = "Estado";
434         this.colEstado.Name = "Estado";
435         this.colEstado.ReadOnly = true;
436         this.colAcciones.HeaderText = "Acciones";
437         this.colAcciones.Name = "Acciones";
438         this.colAcciones.ReadOnly = true;
439         //
440         // FrmEmpleados
441         //
442         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
443         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
444         this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
445         this.ClientSize = new System.Drawing.Size(1020, 742);
446         this.Controls.Add(this.dgvEmpleados);
447         this.Controls.Add(this.panelFiltros);
448         this.Controls.Add(this.panelFormulario);
449         this.Controls.Add(this.btnHistorialLaboral);
450         this.Controls.Add(this.lblTitulo);
451         this.Font = new System.Drawing.Font("Segoe UI", 9F);
452         this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
453         this.Name = "FrmEmpleados";
454         this.Text = "Empleados";
455         this.Load += new System.EventHandler(this.FrmEmpleados_Load);
456         this.panelFormulario.ResumeLayout(false);
457         this.panelFormulario.PerformLayout();
458         this.panelFiltros.ResumeLayout(false);
459         this.panelFiltros.PerformLayout();
460         ((System.ComponentModel.ISupportInitialize)(this.dgvEmpleados)).EndInit();
461         this.ResumeLayout(false);
462     }
463 }
464 }
```

UI/FrmGestionAusencias.cs

```

1  using System;
2  using System.ComponentModel;
3  using System.Windows.Forms;
4  using SistemaGestionNomina.Helpers;
5  using SistemaGestionNomina.Models;
6  using SistemaGestionNomina.Security;
7  using SistemaGestionNomina.Services;
8
9  namespace SistemaGestionNomina.UI
10 {
11     public partial class FrmGestionAusencias : Form
12     {
13         private readonly int requestId;
14         private readonly bool cancelOnly;
15         private readonly AbsenceRequestService service = new AbsenceRequestService();
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         public bool RequestChanged { get; private set; }
18
19         public FrmGestionAusencias() : this(0, false) { }
20         public FrmGestionAusencias(int id, bool showCancelOnly)
21         {
22             requestId = id; cancelOnly = showCancelOnly; InitializeComponent(); ControlStyleHelper.ApplyModernForm(this);
23         }
24
25         private void FrmGestionAusencias_Load(object sender, EventArgs e)
26         {
27             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime || requestId <= 0) return;
28             try
29             {
30                 SolicitudAusencia item = service.GetById(requestId);
31                 lblSolicitud.Text = item.CodigoEmpleado + " - " + item.EmpleadoNombre;
32                 lblDetalle.Text = item.Tipo + " | " + item.FechaInicio.ToString("dd/MM/yyyy") + " al " + item.FechaFin.ToString("dd/MM/yyyy") + " | " + \
item.Estado;
33                 txtMotivo.Text = item.Motivo;
34                 btnAprobar.Visible = !cancelOnly && authorizationService.HasPermission(Permissions.AbsencesApprove) && item.Estado == AbsenceStates.Pending;
35                 btnRechazar.Visible = !cancelOnly && authorizationService.HasPermission(Permissions.AbsencesReject) && item.Estado == AbsenceStates.Pending;
36                 btnCancelarSolicitud.Visible = (authorizationService.HasPermission(Permissions.AbsencesCancel) || \
authorizationService.HasPermission(Permissions.OwnAbsencesCancel)) &&
37                     (item.Estado == AbsenceStates.Pending || item.Estado == AbsenceStates.Approved);
38             }
39             catch (Exception ex) { MessageBox.Show(ex.Message, "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Warning); }
40         }
41
42         private void btnAprobar_Click(object sender, EventArgs e) { Execute("aprobar", delegate { service.Approve(requestId, txtObservacion.Text); }); }
43         private void btnRechazar_Click(object sender, EventArgs e) { Execute("rechazar", delegate { service.Reject(requestId, txtObservacion.Text); }); }
44         private void btnCancelarSolicitud_Click(object sender, EventArgs e) { Execute("cancelar", delegate { service.Cancel(requestId, \
txtObservacion.Text); }); }
45         private void btnCerrar_Click(object sender, EventArgs e) { Close(); }
46
47         private void Execute(string action, Action operation)
48         {
49             if (MessageBox.Show("¿Confirma que desea " + action + " esta solicitud?", "Ausencias", MessageBoxButtons.YesNo, MessageBoxIcon.Question) != \
DialogResult.Yes) return;
50             try { operation(); RequestChanged = true; DialogResult = DialogResult.OK; Close(); }
51             catch (Exception ex) { MessageBox.Show(ex.Message, "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Warning); }
52         }
53     }
54 }

```

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmGestionAusencias
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo, lblSolicitud, lblDetalle, lblMotivo, lblObservacion;
7         private System.Windows.Forms.TextBox txtMotivo, txtObservacion;
8         private System.Windows.Forms.Button btnAprobar, btnRechazar, btnCancelarSolicitud, btnCerrar;
9         protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
10        private void InitializeComponent()
11        {
12            this.lblTitulo = new System.Windows.Forms.Label(); this.lblSolicitud = new System.Windows.Forms.Label(); this.lblDetalle = new \
System.Windows.Forms.Label(); this.lblMotivo = new System.Windows.Forms.Label(); this.lblObservacion = new System.Windows.Forms.Label(); \
this.txtMotivo = new System.Windows.Forms.TextBox(); this.txtObservacion = new System.Windows.Forms.TextBox(); this.btnAprobar = new \
System.Windows.Forms.Button(); this.btnRechazar = new System.Windows.Forms.Button(); this.btnCancelarSolicitud = new System.Windows.Forms.Button(); \
this.btnCerrar = new System.Windows.Forms.Button(); this.SuspendLayout();
13            this.BackColor = System.Drawing.Color.FromArgb(18, 18, 27); this.ClientSize = new System.Drawing.Size(660, 460); this.Font = new \
System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.FormBorderStyle = \
System.Windows.Forms.FormBorderStyle.FixedDialog; this.MaximizeBox = false; this.MinimizeBox = false; this.Name = "FrmGestionAusencias"; \
this.StartPosition = System.Windows.Forms.FormStartPosition.CenterParent; this.Text = "Gestión de ausencia"; this.Load += new \
System.EventHandler(this.FrmGestionAusencias_Load);
14            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 20F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Resolver solicitud";
15            this.lblSolicitud.AutoSize = true; this.lblSolicitud.ForeColor = System.Drawing.Color.FromArgb(167, 139, 250); this.lblSolicitud.Location = \
new System.Drawing.Point(32, 72); this.lblSolicitud.Text = "Empleado";
16            this.lblDetalle.AutoSize = true; this.lblDetalle.Location = new System.Drawing.Point(32, 99); this.lblDetalle.Text = "Detalle de la solicitud";
17            this.lblMotivo.AutoSize = true; this.lblMotivo.Location = new System.Drawing.Point(32, 136); this.lblMotivo.Text = "Motivo del trabajador"; \
this.txtMotivo.Location = new System.Drawing.Point(35, 158); this.txtMotivo.Multiline = true; this.txtMotivo.ReadOnly = true; this.txtMotivo.Size = \
new System.Drawing.Size(590, 72);
18            this.lblObservacion.AutoSize = true; this.lblObservacion.Location = new System.Drawing.Point(32, 250); this.lblObservacion.Text = \
"Observación o motivo de resolución"; this.txtObservacion.Location = new System.Drawing.Point(35, 272); this.txtObservacion.MaxLength = 500; \
this.txtObservacion.Multiline = true; this.txtObservacion.Size = new System.Drawing.Size(590, 72);
19            this.btnAprobar.BackColor = System.Drawing.Color.FromArgb(34, 197, 94); this.btnAprobar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnAprobar.Location = new System.Drawing.Point(194, 380); this.btnAprobar.Size = new System.Drawing.Size(94, 34); this.btnAprobar.Text = \
"Aprobar"; this.btnAprobar.UseVisualStyleBackColor = false; this.btnAprobar.Click += new System.EventHandler(this.btnAprobar_Click);
20            this.btnRechazar.BackColor = System.Drawing.Color.FromArgb(239, 68, 68); this.btnRechazar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnRechazar.Location = new System.Drawing.Point(296, 380); this.btnRechazar.Size = new System.Drawing.Size(94, 34); this.btnRechazar.Text = \
"Rechazar"; this.btnRechazar.UseVisualStyleBackColor = false; this.btnRechazar.Click += new System.EventHandler(this.btnRechazar_Click);
21            this.btnCancelarSolicitud.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnCancelarSolicitud.Location = new System.Drawing.Point(398, \
380); this.btnCancelarSolicitud.Size = new System.Drawing.Size(108, 34); this.btnCancelarSolicitud.Text = "Cancelar solicitud"; \
this.btnCancelarSolicitud.Click += new System.EventHandler(this.btnCancelarSolicitud_Click);
22            this.btnCerrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnCerrar.Location = new System.Drawing.Point(514, 380); \
this.btnCerrar.Size = new System.Drawing.Size(110, 34); this.btnCerrar.Text = "Cerrar"; this.btnCerrar.Click += new \
System.EventHandler(this.btnCerrar_Click);
23            this.Controls.Add(this.lblTitulo); this.Controls.Add(this.lblSolicitud); this.Controls.Add(this.lblDetalle); \
this.Controls.Add(this.lblMotivo); this.Controls.Add(this.txtMotivo); this.Controls.Add(this.lblObservacion); this.Controls.Add(this.txtObservacion); \
this.Controls.Add(this.btnAprobar); this.Controls.Add(this.btnRechazar); this.Controls.Add(this.btnCancelarSolicitud); \
this.Controls.Add(this.btnCerrar); this.ResumeLayout(false); this.PerformLayout();
24        }
25    }
26 }

```

UI\FrmHistorialEmpleado.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmHistorialEmpleado : Form
13     {
14         private readonly int? initialEmployeeId;
15         private readonly EmployeeHistoryService historyService = new EmployeeHistoryService();
16         private readonly EmpleadoService empleadoService = new EmpleadoService();
17         private readonly ExcelExportService excelExportService = new ExcelExportService();
18         private readonly PdfExportService pdfExportService = new PdfExportService();
19         private readonly AuthorizationService authorizationService = new AuthorizationService();
20         private List<HistorialEmpleado> currentHistory = new List<HistorialEmpleado>();
21
22         public FrmHistorialEmpleado() : this(null) { }
23
24         public FrmHistorialEmpleado(int? employeeId)
25         {
26             initialEmployeeId = employeeId;
27             InitializeComponent();
28             ControlStyleHelper.ApplyModernForm(this);
29         }
30
31         private void FrmHistorialEmpleado_Load(object sender, EventArgs e)
32         {
33             if (LicenseManager.UsageMode == LicenseUsageMode.DesignTime) return;
34             List<Empleado> employees = new List<Empleado>();
35             employees.Add(new Empleado
36             {
37                 IdEmpleado = 0, Codigo = "TODOS", Nombre = "Todos los empleados", Apellido = string.Empty,
38                 Cedula = "N/A", Cargo = "N/A", IdDepartamento = 1, SalarioBase = 1,
39                 Estado = "Activo", FechaIngreso = DateTime.Today
40             });
41             employees.AddRange(empleadoService.GetAll(string.Empty, null, string.Empty));
42             cmbEmpleado.DataSource = employees;
43             cmbEmpleado.DisplayMember = "NombreCompleto";
44             cmbEmpleado.ValueMember = "IdEmpleado";
45             if (initialEmployeeId.HasValue) cmbEmpleado.SelectedValue = initialEmployeeId.Value;
46             cmbCampo.Items.Add("Todos");
47             cmbCampo.Items.Add(EmployeeHistoryFields.BaseSalary);
48             cmbCampo.Items.Add(EmployeeHistoryFields.Position);
49             cmbCampo.Items.Add(EmployeeHistoryFields.Department);
50             cmbCampo.Items.Add(EmployeeHistoryFields.Status);
51             cmbCampo.SelectedIndex = 0;
52             btnExcel.Visible = authorizationService.HasPermission(Permissions.EmployeeHistoryExport);
53             btnPdf.Visible = btnExcel.Visible;
54             LoadHistory();
55         }
56
57         private void btnBuscar_Click(object sender, EventArgs e) { LoadHistory(); }
58
59         private void LoadHistory()
60         {
```

UI/FrmHistorialEmpleado.cs (continuacion)

```
61         try
62         {
63             int employeeId = cmbEmpleado.SelectedValue is int ? (int)cmbEmpleado.SelectedValue : 0;
64             currentHistory = historyService.Search(new EmployeeHistoryQuery
65             {
66                 EmployeeId = employeeId > 0 ? (int?)employeeId : null,
67                 Field = cmbCampo.SelectedIndex > 0 ? Convert.ToString(cmbCampo.SelectedItem) : string.Empty,
68                 User = txtUsuario.Text.Trim(),
69                 From = dtpDesde.Checked ? (DateTime?)dtpDesde.Value.Date : null,
70                 To = dtpHasta.Checked ? (DateTime?)dtpHasta.Value.Date : null
71             });
72             dgvHistorial.Rows.Clear();
73             for (int i = 0; i < currentHistory.Count; i++)
74             {
75                 HistorialEmpleado item = currentHistory[i];
76                 dgvHistorial.Rows.Add(item.CodigoEmpleado, item.EmpleadoNombre, item.CampoModificado,
77                                     item.ValorAnterior, item.ValorNuevo, item.FechaEfectiva.ToString("dd/MM/yyyy"),
78                                     item.FechaCambio.ToString("dd/MM/yyyy HH:mm"), item.UsuarioResponsable,
79                                     item.Motivo, item.Aplicado ? "Aplicado" : "Programado");
80             }
81             lblResultado.Text = currentHistory.Count + " cambios encontrados";
82         }
83         catch (Exception ex)
84         {
85             MessageBox.Show(ex.Message, "Historial laboral", MessageBoxButtons.OK, MessageBoxIcon.Warning);
86         }
87     }
88
89     private void btnExcel_Click(object sender, EventArgs e)
90     {
91         if (currentHistory.Count == 0) return;
92         UiFactory.ShowExported(excelExportService.ExportarHistorialEmpleados(currentHistory));
93     }
94
95     private void btnPdf_Click(object sender, EventArgs e)
96     {
97         if (currentHistory.Count == 0) return;
98         UiFactory.ShowExported(pdfExportService.ExportarHistorialEmpleados(currentHistory));
99     }
100
101     private void btnCerrar_Click(object sender, EventArgs e) { Close(); }
102 }
103 }
```

UI/FrmHistorialEmpleado.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmHistorialEmpleado
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.ComboBox cmbEmpleado;
8         private System.Windows.Forms.ComboBox cmbCampo;
9         private System.Windows.Forms.TextBox txtUsuario;
10        private System.Windows.Forms.DateTimePicker dtpDesde;
11        private System.Windows.Forms.DateTimePicker dtpHasta;
12        private System.Windows.Forms.Button btnBuscar;
13        private System.Windows.Forms.Button btnExcel;
14        private System.Windows.Forms.Button btnPdf;
15        private System.Windows.Forms.Button btnCerrar;
16        private System.Windows.Forms.DataGridView dgvHistorial;
17        private System.Windows.Forms.Label lblResultado;
18
19        protected override void Dispose(bool disposing)
20        {
21            if (disposing && (components != null)) components.Dispose();
22            base.Dispose(disposing);
23        }
24
25        private void InitializeComponent()
26        {
27            this.lblTitulo = new System.Windows.Forms.Label(); this.cmbEmpleado = new System.Windows.Forms.ComboBox();
28            this.cmbCampo = new System.Windows.Forms.ComboBox(); this.txtUsuario = new System.Windows.Forms.TextBox();
29            this.dtpDesde = new System.Windows.Forms.DateTimePicker(); this.dtpHasta = new System.Windows.Forms.DateTimePicker();
30            this.btnBuscar = new System.Windows.Forms.Button(); this.btnExcel = new System.Windows.Forms.Button();
31            this.btnPdf = new System.Windows.Forms.Button(); this.btnCerrar = new System.Windows.Forms.Button();
32            this.dgvHistorial = new System.Windows.Forms.DataGridView(); this.lblResultado = new System.Windows.Forms.Label();
33            ((System.ComponentModel.ISupportInitialize)(this.dgvHistorial)).BeginInit(); this.SuspendLayout();
34            this.BackColor = System.Drawing.Color.FromArgb(12, 12, 18); this.ClientSize = new System.Drawing.Size(1120, 700);
35            this.Font = new System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
36            this.Name = "FrmHistorialEmpleado"; this.StartPosition = System.Windows.Forms.FormStartPosition.CenterParent;
37            this.Text = "Historial laboral"; this.Load += new System.EventHandler(this.FrmHistorialEmpleado_Load);
38            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
39            this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Historial laboral";
40            this.cmbEmpleado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbEmpleado.Location = new System.Drawing.Point(34, \
82); this.cmbEmpleado.Size = new System.Drawing.Size(220, 23);
41            this.cmbCampo.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbCampo.Location = new System.Drawing.Point(266, 82); \
this.cmbCampo.Size = new System.Drawing.Size(150, 23);
42            this.txtUsuario.Location = new System.Drawing.Point(428, 82); this.txtUsuario.Size = new System.Drawing.Size(130, 23);
43            this.dtpDesde.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpDesde.Location = new System.Drawing.Point(570, 82); \
this.dtpDesde.ShowCheckBox = true; this.dtpDesde.Size = new System.Drawing.Size(130, 23); this.dtpDesde.Checked = false;
44            this.dtpHasta.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpHasta.Location = new System.Drawing.Point(712, 82); \
this.dtpHasta.ShowCheckBox = true; this.dtpHasta.Size = new System.Drawing.Size(130, 23); this.dtpHasta.Checked = false;
45            this.btnBuscar.BackColor = System.Drawing.Color.FromArgb(124, 58, 237); this.btnBuscar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnBuscar.Location = new System.Drawing.Point(854, 78); this.btnBuscar.Size = new System.Drawing.Size(78, 31); this.btnBuscar.Text = "Buscar"; \
this.btnBuscar.UseVisualStyleBackColor = false; this.btnBuscar.Click += new System.EventHandler(this.btnBuscar_Click);
46            this.btnExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnExcel.Location = new System.Drawing.Point(938, 78); this.btnExcel.Size \
= new System.Drawing.Size(64, 31); this.btnExcel.Text = "Excel"; this.btnExcel.Click += new System.EventHandler(this.btnExcel_Click);
47            this.btnPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnPdf.Location = new System.Drawing.Point(1008, 78); this.btnPdf.Size = \
new System.Drawing.Size(64, 31); this.btnPdf.Text = "PDF"; this.btnPdf.Click += new System.EventHandler(this.btnPdf_Click);
48            this.dgvHistorial.AllowUserToAddRows = false; this.dgvHistorial.AllowUserToDeleteRows = false; this.dgvHistorial.AutoSizeColumnsMode = \
System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill; this.dgvHistorial.BackgroundColor = System.Drawing.Color.FromArgb(18, 18, 27); \
this.dgvHistorial.Location = new System.Drawing.Point(34, 130); this.dgvHistorial.ReadOnly = true; this.dgvHistorial.RowHeadersVisible = false; \
this.dgvHistorial.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect; this.dgvHistorial.Size = new \
System.Drawing.Size(1038, 500);

```


UI/FrmHistorialEmpleado.Designer.cs (continuacion)

```
49         this.dgvHistorial.Columns.Add("Codigo", "Código"); this.dgvHistorial.Columns.Add("Empleado", "Empleado"); \
this.dgvHistorial.Columns.Add("Campo", "Campo"); this.dgvHistorial.Columns.Add("Anterior", "Anterior"); this.dgvHistorial.Columns.Add("Nuevo", \
"Nuevo"); this.dgvHistorial.Columns.Add("Efectiva", "Fecha efectiva"); this.dgvHistorial.Columns.Add("Registro", "Registrado"); \
this.dgvHistorial.Columns.Add("Usuario", "Usuario"); this.dgvHistorial.Columns.Add("Motivo", "Motivo"); this.dgvHistorial.Columns.Add("Estado", \
"Aplicación");
50         this.lblResultado.AutoSize = true; this.lblResultado.Location = new System.Drawing.Point(34, 650); this.lblResultado.Text = "0 cambios \
encontrados";
51         this.btnCerrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnCerrar.Location = new System.Drawing.Point(976, 644); \
this.btnCerrar.Size = new System.Drawing.Size(96, 32); this.btnCerrar.Text = "Cerrar"; this.btnCerrar.Click += new \
System.EventHandler(this.btnCerrar_Click);
52         this.Controls.Add(this.lblTitulo); this.Controls.Add(this.cmbEmpleado); this.Controls.Add(this.cmbCampo); this.Controls.Add(this.txtUsuario); \
this.Controls.Add(this.dtpDesde); this.Controls.Add(this.dtpHasta); this.Controls.Add(this.btnBuscar); this.Controls.Add(this.btnExcel); \
this.Controls.Add(this.btnPdf); this.Controls.Add(this.dgvHistorial); this.Controls.Add(this.lblResultado); this.Controls.Add(this.btnCerrar);
53         ((System.ComponentModel.ISupportInitialize)(this.dgvHistorial)).EndInit(); this.ResumeLayout(false); this.PerformLayout();
54     }
55 }
56 }
```

UI\FrmHistorialNomina.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.Drawing;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Models;
6
7 namespace SistemaGestionNomina.UI
8 {
9     public partial class FrmHistorialNomina : Form
10     {
11         private DataGridView dgvHistorial;
12         private Button btnCerrar;
13
14         public FrmHistorialNomina(List<NominaVersion> versiones)
15         {
16             InitializeComponent();
17             CargarHistorial(versiones);
18         }
19
20         private void InitializeComponent()
21         {
22             this.dgvHistorial = new DataGridView();
23             this.btnCerrar = new Button();
24             ((System.ComponentModel.ISupportInitialize)(this.dgvHistorial)).BeginInit();
25             this.SuspendLayout();
26
27             this.dgvHistorial.AllowUserToAddRows = false;
28             this.dgvHistorial.AllowUserToDeleteRows = false;
29             this.dgvHistorial.BackgroundColor = Color.FromArgb(24, 25, 34);
30             this.dgvHistorial.BorderStyle = BorderStyle.None;
31             this.dgvHistorial.CellBorderStyle = DataGridViewCellBorderStyle.SingleHorizontal;
32             this.dgvHistorial.ColumnHeadersHeightSizeMode = DataGridViewColumnHeadersHeightSizeMode.AutoSize;
33             this.dgvHistorial.Location = new Point(16, 16);
34             this.dgvHistorial.Name = "dgvHistorial";
35             this.dgvHistorial.ReadOnly = true;
36             this.dgvHistorial.RowHeadersVisible = false;
37             this.dgvHistorial.Size = new Size(600, 300);
38             this.dgvHistorial.TabIndex = 0;
39             this.dgvHistorial.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
40             this.dgvHistorial.EnableHeadersVisualStyles = false;
41
42             this.dgvHistorial.DefaultCellStyle.BackColor = Color.FromArgb(13, 14, 20);
43             this.dgvHistorial.DefaultCellStyle.ForeColor = Color.FromArgb(229, 228, 245);
44             this.dgvHistorial.DefaultCellStyle.SelectionBackColor = Color.FromArgb(167, 139, 250);
45             this.dgvHistorial.DefaultCellStyle.SelectionForeColor = Color.Black;
46             this.dgvHistorial.DefaultCellStyle.Font = new Font("Segoe UI", 9F);
47
48             this.dgvHistorial.ColumnHeadersDefaultCellStyle.BackColor = Color.FromArgb(17, 18, 26);
49             this.dgvHistorial.ColumnHeadersDefaultCellStyle.ForeColor = Color.FromArgb(167, 139, 250);
50             this.dgvHistorial.ColumnHeadersDefaultCellStyle.Font = new Font("Segoe UI", 9F, FontStyle.Bold);
51             this.dgvHistorial.ColumnHeadersDefaultCellStyle.Alignment = DataGridViewContentAlignment.MiddleCenter;
52
53             this.dgvHistorial.AlternatingRowsDefaultCellStyle.BackColor = Color.FromArgb(20, 21, 30);
54
55             this.dgvHistorial.GridColor = Color.FromArgb(30, 31, 40);
56
57             this.btnCerrar.BackColor = Color.Black;
58             this.btnCerrar.FlatStyle = FlatStyle.Flat;
59             this.btnCerrar.ForeColor = Color.FromArgb(229, 228, 245);
60             this.btnCerrar.Location = new Point(520, 332);
```

UI/FrmHistorialNomina.cs (continuacion)

```
61         this.btnCerrar.Size = new Size(96, 32);
62         this.btnCerrar.Text = "Cerrar";
63         this.btnCerrar.UseVisualStyleBackColor = false;
64         this.btnCerrar.Click += new EventHandler(this.btnCerrar_Click);
65
66         this.AutoScaleDimensions = new SizeF(7F, 15F);
67         this.AutoScaleMode = AutoScaleMode.Font;
68         this.BackColor = Color.FromArgb(13, 14, 20);
69         this.ClientSize = new Size(634, 381);
70         this.Controls.Add(this.dgvHistorial);
71         this.Controls.Add(this.btnCerrar);
72         this.Font = new Font("Segoe UI", 9F);
73         this.ForeColor = Color.FromArgb(229, 228, 245);
74         this.FormBorderStyle = FormBorderStyle.FixedDialog;
75         this.MaximizeBox = false;
76         this.MinimizeBox = false;
77         this.StartPosition = FormStartPosition.CenterParent;
78         this.Text = "Historial de versiones";
79         ((System.ComponentModel.ISupportInitialize)(this.dgvHistorial)).EndInit();
80         this.ResumeLayout(false);
81     }
82
83     private void CargarHistorial(List<NominaVersion> versiones)
84     {
85         dgvHistorial.Columns.Clear();
86         dgvHistorial.Columns.Add("Version", "Versión");
87         dgvHistorial.Columns.Add("Direccion", "Original ? Nueva");
88         dgvHistorial.Columns.Add("Motivo", "Motivo");
89         dgvHistorial.Columns.Add("Usuario", "Usuario");
90         dgvHistorial.Columns.Add("Fecha", "Fecha");
91
92         for (int i = 0; i < versiones.Count; i++)
93         {
94             NominaVersion v = versiones[i];
95             string direccion = v.IdNominaOriginal.ToString() + " ? " +
96                 (v.IdNominaNueva.HasValue ? v.IdNominaNueva.Value.ToString() : "Pendiente");
97             dgvHistorial.Rows.Add(
98                 v.IdVersion.ToString(),
99                 direccion,
100                 v.MotivoCambio,
101                 v.UsuarioResponsable,
102                 v.FechaCambio.ToString("yyyy-MM-dd HH:mm")
103             );
104         }
105     }
106
107     private void btnCerrar_Click(object sender, EventArgs e)
108     {
109         Close();
110     }
111 }
112 }
```

UI\FrmLogin.cs

```
1 using System;
2 using System.ComponentModel;
3 using System.Drawing;
4 using System.Drawing.Drawing2D;
5 using System.Windows.Forms;
6 using SistemaGestionNomina.Helpers;
7 using SistemaGestionNomina.Models;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmLogin : Form
13     {
14         readonly AuthService authService = new AuthService();
15
16         public Usuario AuthenticatedUser { get; private set; }
17
18         public FrmLogin()
19         {
20             InitializeComponent();
21             ControlStyleHelper.ApplyModernForm(this);
22             string rememberedUsername = RememberedUsernameHelper.Load();
23             if (!string.IsNullOrEmpty(rememberedUsername))
24             {
25                 txtUsuario.Text = rememberedUsername;
26                 chkRecordar.Checked = true;
27                 txtPassword.Focus();
28             }
29         }
30
31         private void FrmLogin_Paint(object sender, PaintEventArgs e)
32         {
33             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime)
34             {
35                 return;
36             }
37
38             using (LinearGradientBrush brush = new LinearGradientBrush(
39                 ClientRectangle,
40                 Color.FromArgb(13, 14, 20),
41                 Color.FromArgb(24, 18, 45),
42                 LinearGradientMode.ForwardDiagonal))
43             {
44                 e.Graphics.FillRectangle(brush, ClientRectangle);
45             }
46
47             using (Pen pen = new Pen(Color.FromArgb(24, 29, 42)))
48             {
49                 for (int x = 0; x < Width; x += 34)
50                 {
51                     e.Graphics.DrawLine(pen, x, 0, x, Height);
52                 }
53
54                 for (int y = 0; y < Height; y += 34)
55                 {
56                     e.Graphics.DrawLine(pen, 0, y, Width, y);
57                 }
58             }
59         }
60     }
```

UI/FrmLogin.cs (continuacion)

```
61     private void btnIngresar_Click(object sender, EventArgs e)
62     {
63         if (string.IsNullOrEmpty(txtUsuario.Text) || string.IsNullOrEmpty(txtPassword.Text))
64         {
65             lblMensaje.Text = "Ingrese usuario y contraseña.";
66             lblMensaje.ForeColor = ThemeHelper.Error;
67             return;
68         }
69
70         try
71         {
72             AuthenticationResult result = authService.Authenticate(txtUsuario.Text.Trim(), txtPassword.Text);
73             if (!result.IsSuccess)
74             {
75                 lblMensaje.Text = result.Message;
76                 lblMensaje.ForeColor = ThemeHelper.Error;
77                 txtPassword.Clear();
78                 txtPassword.Focus();
79                 return;
80             }
81
82             if (chkRecordar.Checked)
83             {
84                 RememberedUsernameHelper.Save(result.User.NombreUsuario);
85             }
86             else
87             {
88                 RememberedUsernameHelper.Clear();
89             }
90
91             AuthenticatedUser = result.User;
92             DialogResult = DialogResult.OK;
93             Close();
94         }
95         catch (Exception)
96         {
97             MessageBox.Show("No se pudo iniciar sesión. Revise la conexión con la base de datos e intente nuevamente.",
98                 "Login", MessageBoxButtons.OK, MessageBoxIcon.Error);
99         }
100     }
101
102     private void foxBigLabel1_Click(object sender, EventArgs e)
103     {
104     }
105 }
106 }
107 }
```

UI\FrmLogin.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmLogin
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Panel panelLogin;
7         private System.Windows.Forms.Label lblSubtitulo;
8         private System.Windows.Forms.Label lblUsuario;
9         private System.Windows.Forms.Label lblPassword;
10        private System.Windows.Forms.TextBox txtUsuario;
11        private System.Windows.Forms.TextBox txtPassword;
12        private System.Windows.Forms.CheckBox chkRecordar;
13        private System.Windows.Forms.Button btnIngresar;
14        private System.Windows.Forms.Label lblMensaje;
15
16        protected override void Dispose(bool disposing)
17        {
18            if (disposing && (components != null))
19            {
20                components.Dispose();
21            }
22            base.Dispose(disposing);
23        }
24
25        private void InitializeComponent()
26        {
27            this.panelLogin = new System.Windows.Forms.Panel();
28            this.foxBigLabel1 = new RealTaiizor.Controls.FoxBigLabel();
29            this.lblSubtitulo = new System.Windows.Forms.Label();
30            this.lblUsuario = new System.Windows.Forms.Label();
31            this.lblPassword = new System.Windows.Forms.Label();
32            this.txtUsuario = new System.Windows.Forms.TextBox();
33            this.txtPassword = new System.Windows.Forms.TextBox();
34            this.chkRecordar = new System.Windows.Forms.CheckBox();
35            this.btnIngresar = new System.Windows.Forms.Button();
36            this.lblMensaje = new System.Windows.Forms.Label();
37            this.panelLogin.SuspendLayout();
38            this.SuspendLayout();
39            //
40            // panelLogin
41            //
42            this.panelLogin.Anchor = System.Windows.Forms.AnchorStyles.None;
43            this.panelLogin.BackColor = System.Drawing.Color.Black;
44            this.panelLogin.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
45            this.panelLogin.Controls.Add(this.foxBigLabel1);
46            this.panelLogin.Controls.Add(this.lblSubtitulo);
47            this.panelLogin.Controls.Add(this.lblUsuario);
48            this.panelLogin.Controls.Add(this.lblPassword);
49            this.panelLogin.Controls.Add(this.txtUsuario);
50            this.panelLogin.Controls.Add(this.txtPassword);
51            this.panelLogin.Controls.Add(this.chkRecordar);
52            this.panelLogin.Controls.Add(this.btnIngresar);
53            this.panelLogin.Controls.Add(this.lblMensaje);
54            this.panelLogin.Location = new System.Drawing.Point(275, 78);
55            this.panelLogin.Name = "panelLogin";
56            this.panelLogin.Size = new System.Drawing.Size(430, 520);
57            this.panelLogin.TabIndex = 0;
58            //
59            // foxBigLabel1
60            //
```

UI/FrmLogin.Designer.cs (continuación)

```
61         this.foxBigLabel1.BackColor = System.Drawing.Color.Transparent;
62         this.foxBigLabel1.Font = new System.Drawing.Font("Segoe UI Semibold", 20F);
63         this.foxBigLabel1.ForeColor = System.Drawing.Color.Indigo;
64         this.foxBigLabel1.Line = RealTaiizor.Controls.FoxBigLabel.Direction.Top;
65         this.foxBigLabel1.LineColor = System.Drawing.Color.Transparent;
66         this.foxBigLabel1.Location = new System.Drawing.Point(135, 81);
67         this.foxBigLabel1.Name = "foxBigLabel1";
68         this.foxBigLabel1.Size = new System.Drawing.Size(165, 41);
69         this.foxBigLabel1.TabIndex = 10;
70         this.foxBigLabel1.Text = "NomiCore";
71         this.foxBigLabel1.Click += new System.EventHandler(this.foxBigLabel1_Click);
72         //
73         // lblSubtitulo
74         //
75         this.lblSubtitulo.Font = new System.Drawing.Font("Segoe UI", 10F);
76         this.lblSubtitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
77         this.lblSubtitulo.Location = new System.Drawing.Point(40, 125);
78         this.lblSubtitulo.Name = "lblSubtitulo";
79         this.lblSubtitulo.Size = new System.Drawing.Size(350, 24);
80         this.lblSubtitulo.TabIndex = 2;
81         this.lblSubtitulo.Text = "Acceso corporativo seguro";
82         this.lblSubtitulo.TextAlign = System.Drawing.ContentAlignment.MiddleCenter;
83         //
84         // lblUsuario
85         //
86         this.lblUsuario.AutoSize = true;
87         this.lblUsuario.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
88         this.lblUsuario.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
89         this.lblUsuario.Location = new System.Drawing.Point(40, 224);
90         this.lblUsuario.Name = "lblUsuario";
91         this.lblUsuario.Size = new System.Drawing.Size(63, 20);
92         this.lblUsuario.TabIndex = 3;
93         this.lblUsuario.Text = "Usuario";
94         //
95         // lblPassword
96         //
97         this.lblPassword.AutoSize = true;
98         this.lblPassword.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
99         this.lblPassword.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
100        this.lblPassword.Location = new System.Drawing.Point(40, 302);
101        this.lblPassword.Name = "lblPassword";
102        this.lblPassword.Size = new System.Drawing.Size(88, 20);
103        this.lblPassword.TabIndex = 5;
104        this.lblPassword.Text = "Contraseña";
105        //
106        // txtUsuario
107        //
108        this.txtUsuario.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
109        this.txtUsuario.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
110        this.txtUsuario.Font = new System.Drawing.Font("Segoe UI", 10F);
111        this.txtUsuario.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
112        this.txtUsuario.Location = new System.Drawing.Point(40, 252);
113        this.txtUsuario.Name = "txtUsuario";
114        this.txtUsuario.Size = new System.Drawing.Size(350, 30);
115        this.txtUsuario.TabIndex = 4;
116        //
117        // txtPassword
118        //
119        this.txtPassword.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(24)))), ((int)(((byte)(25)))), ((int)(((byte)(34))))));
120        this.txtPassword.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
```

UI\FrmLogin.Designer.cs (continuación)

```
121         this.txtPassword.Font = new System.Drawing.Font("Segoe UI", 10F);
122         this.txtPassword.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
123         this.txtPassword.Location = new System.Drawing.Point(40, 330);
124         this.txtPassword.Name = "txtPassword";
125         this.txtPassword.Size = new System.Drawing.Size(350, 30);
126         this.txtPassword.TabIndex = 6;
127         this.txtPassword.UseSystemPasswordChar = true;
128         //
129         // chkRecordar
130         //
131         this.chkRecordar.AutoSize = true;
132         this.chkRecordar.BackColor = System.Drawing.Color.Transparent;
133         this.chkRecordar.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
134         this.chkRecordar.Location = new System.Drawing.Point(40, 378);
135         this.chkRecordar.Name = "chkRecordar";
136         this.chkRecordar.Size = new System.Drawing.Size(145, 24);
137         this.chkRecordar.TabIndex = 7;
138         this.chkRecordar.Text = "Recordar usuario";
139         this.chkRecordar.UseVisualStyleBackColor = false;
140         //
141         // btnIngresar
142         //
143         this.btnIngresar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
144         this.btnIngresar.FlatAppearance.BorderSize = 0;
145         this.btnIngresar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
146         this.btnIngresar.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
147         this.btnIngresar.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(48)))), ((int)(((byte)(25)))), ((int)(((byte)(92))))));
148         this.btnIngresar.Location = new System.Drawing.Point(40, 426);
149         this.btnIngresar.Name = "btnIngresar";
150         this.btnIngresar.Size = new System.Drawing.Size(350, 42);
151         this.btnIngresar.TabIndex = 8;
152         this.btnIngresar.Text = "Iniciar sesión";
153         this.btnIngresar.UseVisualStyleBackColor = false;
154         this.btnIngresar.Click += new System.EventHandler(this.btnIngresar_Click);
155         //
156         // lblMensaje
157         //
158         this.lblMensaje.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
159         this.lblMensaje.Location = new System.Drawing.Point(40, 476);
160         this.lblMensaje.Name = "lblMensaje";
161         this.lblMensaje.Size = new System.Drawing.Size(350, 24);
162         this.lblMensaje.TabIndex = 9;
163         this.lblMensaje.Text = "Ingrese sus credenciales para continuar.";
164         this.lblMensaje.TextAlign = System.Drawing.ContentAlignment.MiddleCenter;
165         //
166         // FrmLogin
167         //
168         this.AcceptButton = this.btnIngresar;
169         this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F);
170         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
171         this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
172         this.ClientSize = new System.Drawing.Size(980, 700);
173         this.Controls.Add(this.panelLogin);
174         this.Font = new System.Drawing.Font("Segoe UI", 9F);
175         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
176         this.FormBorderStyle = System.Windows.Forms.FormBorderStyle.FixedSingle;
177         this.MaximizeBox = false;
178         this.MinimumSize = new System.Drawing.Size(900, 640);
179         this.Name = "FrmLogin";
180         this.StartPosition = System.Windows.Forms.FormStartPosition.CenterScreen;
```


UI/FrmLogin.Designer.cs (continuacion)

```
181         this.Text = "ProyFinal_LPI_Eq01_NomiCore - Login";
182         this.Paint += new System.Windows.Forms.PaintEventHandler(this.FrmLogin_Paint);
183         this.panelLogin.ResumeLayout(false);
184         this.panelLogin.PerformLayout();
185         this.ResumeLayout(false);
186
187     }
188
189     private RealTaiizor.Controls.FoxBigLabel foxBigLabel1;
190 }
191 }
```

UI/FrmMain.cs

```
1 using System;
2 using System.ComponentModel;
3 using System.Windows.Forms;
4 using FontAwesome.Sharp;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmMain : Form
13     {
14         private Usuario currentUser;
15         private Form activeChild;
16         private readonly AuthorizationService authorizationService = new AuthorizationService();
17         private readonly AuditTrailService auditTrailService = new AuditTrailService();
18
19         public bool LogoutRequested { get; private set; }
20
21         public FrmMain()
22         {
23             InitializeComponent();
24             ControlStyleHelper.ApplyModernForm(this);
25         }
26
27         public FrmMain(Usuario usuario)
28             : this()
29         {
30             if (usuario == null) throw new ArgumentNullException("usuario");
31             currentUser = usuario;
32             if (!SessionContext.IsAuthenticated) SessionContext.Begin(usuario);
33             ConfigureRuntime();
34         }
35
36         public void OpenChildForm(Form childForm)
37         {
38             if (activeChild != null)
39             {
40                 activeChild.Close();
41                 activeChild.Dispose();
42             }
43
44             activeChild = childForm;
45             childForm.TopLevel = false;
46             childForm.FormBorderStyle = FormBorderStyle.None;
47             childForm.Dock = DockStyle.Fill;
48             panelContent.Controls.Clear();
49             panelContent.Controls.Add(childForm);
50             childForm.Show();
51         }
52
53         private void ConfigureRuntime()
54         {
55             lblUsuarioActual.Text = currentUser.NombreUsuario + " | " + currentUser.Rol;
56             ApplyPermissions();
57             if (!IsDesignTime())
58             {
59                 OpenInitialModule();
60             }
61         }
62     }
63 }
```

UI/FrmMain.cs (continuacion)

```
61     }
62
63     private void ApplyPermissions()
64     {
65         btnPortal.Visible = string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase) &&
66             authorizationService.HasPermission(Permissions.PortalView);
67         btnDashboard.Visible = authorizationService.HasPermission(Permissions.DashboardView);
68         btnEmpleados.Visible = authorizationService.HasPermission(Permissions.EmployeesView);
69         btnAsistencia.Visible = authorizationService.HasPermission(Permissions.AttendanceView);
70         btnNomina.Visible = authorizationService.HasPermission(Permissions.PayrollView);
71         btnComprobantes.Visible = authorizationService.HasPermission(Permissions.PayslipsView);
72         btnReportes.Visible = authorizationService.HasPermission(Permissions.ReportsView);
73         btnConfiguracion.Visible = authorizationService.HasPermission(Permissions.ConfigurationView);
74         btnAuditoria.Visible = authorizationService.HasPermission(Permissions.AuditView);
75         btnAusencias.Visible = authorizationService.HasPermission(Permissions.AbsencesView);
76     }
77
78     private void OpenInitialModule()
79     {
80         if (btnPortal.Visible) OpenPortal();
81         else if (btnDashboard.Visible) OpenDashboard();
82         else if (btnEmpleados.Visible) OpenAuthorized(Permissions.EmployeesView, btnEmpleados, new FrmEmpleados());
83         else if (btnNomina.Visible) OpenAuthorized(Permissions.PayrollView, btnNomina, new FrmNomina());
84         else if (btnAsistencia.Visible) OpenAuthorized(Permissions.AttendanceView, btnAsistencia, new FrmAsistencia());
85         else
86         {
87             SetActiveButton(btnAcercaDe);
88             OpenChildForm(new FrmAcercaDe());
89         }
90     }
91
92     private static bool IsDesignTime()
93     {
94         return LicenseManager.UsageMode == LicenseUsageMode.DesignTime;
95     }
96
97     private void btnDashboard_Click(object sender, EventArgs e)
98     {
99         SetActiveButton(btnDashboard);
100         OpenDashboard();
101     }
102
103     private void btnPortal_Click(object sender, EventArgs e)
104     {
105         OpenPortal();
106     }
107
108     private void btnEmpleados_Click(object sender, EventArgs e)
109     {
110         OpenAuthorized(Permissions.EmployeesView, btnEmpleados, new FrmEmpleados());
111     }
112
113     private void btnAsistencia_Click(object sender, EventArgs e)
114     {
115         OpenAuthorized(Permissions.AttendanceView, btnAsistencia, new FrmAsistencia());
116     }
117
118     private void btnNomina_Click(object sender, EventArgs e)
119     {
120         OpenAuthorized(Permissions.PayrollView, btnNomina, new FrmNomina());
```

UI/FrmMain.cs (continuacion)

```
121     }
122
123     private void btnComprobantes_Click(object sender, EventArgs e)
124     {
125         OpenAuthorized(Permissions.PayslipsView, btnComprobantes, new FrmComprobantes());
126     }
127
128     private void btnReportes_Click(object sender, EventArgs e)
129     {
130         OpenAuthorized(Permissions.ReportsView, btnReportes, new FrmReportes());
131     }
132
133     private void btnConfiguracion_Click(object sender, EventArgs e)
134     {
135         OpenAuthorized(Permissions.ConfigurationView, btnConfiguracion, new FrmConfiguracion());
136     }
137
138     private void btnAcercaDe_Click(object sender, EventArgs e)
139     {
140         SetActiveButton(btnAcercaDe);
141         OpenChildForm(new FrmAcercaDe());
142     }
143
144     private void btnCerrarSesion_Click(object sender, EventArgs e)
145     {
146         auditTrailService.RegistrarAccion("Seguridad", "Cerrar sesión", string.Empty);
147         SessionContext.Clear();
148         LogoutRequested = true;
149         Close();
150     }
151
152     private void btnSalir_Click(object sender, EventArgs e)
153     {
154         DialogResult result = MessageBox.Show("¿Desea salir de la aplicación?",
155         "Salir", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
156         if (result == DialogResult.Yes)
157         {
158             auditTrailService.RegistrarAccion("Seguridad", "Salir de la aplicación", string.Empty);
159             SessionContext.Clear();
160             Application.Exit();
161         }
162     }
163
164     private void btnAuditoria_Click(object sender, EventArgs e)
165     {
166         OpenAuthorized(Permissions.AuditView, btnAuditoria, new FrmAuditoria());
167     }
168
169     private void btnAusencias_Click(object sender, EventArgs e)
170     {
171         OpenAuthorized(Permissions.AbsencesView, btnAusencias, new FrmSolicitudesAusencia());
172     }
173
174     private void OpenDashboard()
175     {
176         authorizationService.DemandPermission(Permissions.DashboardView);
177         SetActiveButton(btnDashboard);
178         OpenChildForm(new FrmDashboard(
179             delegate { OpenChildForm(new FrmEmpleados()); },
180             delegate { OpenChildForm(new FrmNomina()); }));
```

UI/FrmMain.cs (continuacion)

```
181     }
182
183     private void OpenPortal()
184     {
185         authorizationService.DemandPermission(Permissions.PortalView);
186         SetActiveButton(btnPortal);
187         OpenChildForm(new FrmPortalTrabajador(
188             delegate { OpenPortalChild(Permissions.OwnProfileView, new FrmMiPerfil()); },
189             delegate { OpenPortalChild(Permissions.OwnAttendanceView, new FrmMisAsistencias()); },
190             delegate { OpenPortalChild(Permissions.OwnPayslipsView, new FrmMisComprobantes()); },
191             delegate { OpenPortalChild(Permissions.OwnPasswordChange, new FrmCambiarPassword()); },
192             delegate { OpenPortalChild(Permissions.OwnAbsencesView, new FrmSolicitudesAusencia()); }));
193     }
194
195     private void OpenPortalChild(string permission, Form form)
196     {
197         try
198         {
199             authorizationService.DemandPermission(permission);
200             SetActiveButton(btnPortal);
201             OpenChildForm(form);
202         }
203         catch (UnauthorizedAccessException ex)
204         {
205             form.Dispose();
206             MessageBox.Show(ex.Message, "Acceso restringido", MessageBoxButtons.OK, MessageBoxIcon.Warning);
207         }
208     }
209
210     private void OpenAuthorized(string permission, IconButton button, Form form)
211     {
212         try
213         {
214             authorizationService.DemandPermission(permission);
215             SetActiveButton(button);
216             OpenChildForm(form);
217         }
218         catch (UnauthorizedAccessException ex)
219         {
220             form.Dispose();
221             MessageBox.Show(ex.Message, "Acceso restringido", MessageBoxButtons.OK, MessageBoxIcon.Warning);
222         }
223     }
224
225     private void SetActiveButton(IconButton button)
226     {
227         ControlStyleHelper.SetActiveSidebarButton(panelSidebar, button);
228     }
229 }
230 }
```

UI/FrmMain.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmMain
4     {
5         private System.ComponentModel.IContainer components = null;
6         public System.Windows.Forms.Panel panelSidebar;
7         private System.Windows.Forms.Panel panelTopbar;
8         public System.Windows.Forms.Panel panelContent;
9         private System.Windows.Forms.Panel panelBrand;
10        private System.Windows.Forms.FlowLayoutPanel flowMenu;
11        private System.Windows.Forms.FlowLayoutPanel flowAccount;
12        private System.Windows.Forms.Panel panelAccountSeparator;
13        private System.Windows.Forms.Label lblMarca;
14        private System.Windows.Forms.Label lblSubmarca;
15        private System.Windows.Forms.Label lblUsuarioActual;
16        public FontAwesome.Sharp.IconButton btnPortal;
17        public FontAwesome.Sharp.IconButton btnDashboard;
18        public FontAwesome.Sharp.IconButton btnEmpleados;
19        public FontAwesome.Sharp.IconButton btnAsistencia;
20        public FontAwesome.Sharp.IconButton btnNomina;
21        public FontAwesome.Sharp.IconButton btnComprobantes;
22        public FontAwesome.Sharp.IconButton btnReportes;
23        public FontAwesome.Sharp.IconButton btnConfiguracion;
24        public FontAwesome.Sharp.IconButton btnAcercaDe;
25        public FontAwesome.Sharp.IconButton btnAuditoria;
26        public FontAwesome.Sharp.IconButton btnAusencias;
27        public FontAwesome.Sharp.IconButton btnSalir;
28        public FontAwesome.Sharp.IconButton btnCerrarSesion;
29
30        protected override void Dispose(bool disposing)
31        {
32            if (disposing && (components != null))
33            {
34                components.Dispose();
35            }
36            base.Dispose(disposing);
37        }
38
39        private void InitializeComponent()
40        {
41            this.panelSidebar = new System.Windows.Forms.Panel();
42            this.panelBrand = new System.Windows.Forms.Panel();
43            this.lblMarca = new System.Windows.Forms.Label();
44            this.lblSubmarca = new System.Windows.Forms.Label();
45            this.flowMenu = new System.Windows.Forms.FlowLayoutPanel();
46            this.btnPortal = new FontAwesome.Sharp.IconButton();
47            this.btnDashboard = new FontAwesome.Sharp.IconButton();
48            this.btnEmpleados = new FontAwesome.Sharp.IconButton();
49            this.btnAsistencia = new FontAwesome.Sharp.IconButton();
50            this.btnNomina = new FontAwesome.Sharp.IconButton();
51            this.btnComprobantes = new FontAwesome.Sharp.IconButton();
52            this.btnReportes = new FontAwesome.Sharp.IconButton();
53            this.btnConfiguracion = new FontAwesome.Sharp.IconButton();
54            this.btnAcercaDe = new FontAwesome.Sharp.IconButton();
55            this.btnAuditoria = new FontAwesome.Sharp.IconButton();
56            this.btnAusencias = new FontAwesome.Sharp.IconButton();
57            this.btnSalir = new FontAwesome.Sharp.IconButton();
58            this.btnCerrarSesion = new FontAwesome.Sharp.IconButton();
59            this.flowAccount = new System.Windows.Forms.FlowLayoutPanel();
60            this.panelAccountSeparator = new System.Windows.Forms.Panel();
```

UI/FrmMain.Designer.cs (continuacion)

```
61         this.panelTopbar = new System.Windows.Forms.Panel();
62         this.lblUsuarioActual = new System.Windows.Forms.Label();
63         this.panelContent = new System.Windows.Forms.Panel();
64         this.panelSidebar.SuspendLayout();
65         this.panelBrand.SuspendLayout();
66         this.flowMenu.SuspendLayout();
67         this.flowAccount.SuspendLayout();
68         this.panelTopbar.SuspendLayout();
69         this.SuspendLayout();
70         //
71         // panelSidebar
72         //
73         this.panelSidebar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(11)))), ((int)(((byte)(15)))), ((int)(((byte)(25))))));
74         this.panelSidebar.Controls.Add(this.flowMenu);
75         this.panelSidebar.Controls.Add(this.flowAccount);
76         this.panelSidebar.Controls.Add(this.panelBrand);
77         this.panelSidebar.Dock = System.Windows.Forms.DockStyle.Left;
78         this.panelSidebar.Location = new System.Drawing.Point(0, 0);
79         this.panelSidebar.Name = "panelSidebar";
80         this.panelSidebar.Size = new System.Drawing.Size(260, 820);
81         this.panelSidebar.TabIndex = 0;
82         //
83         // panelBrand
84         //
85         this.panelBrand.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(11)))), ((int)(((byte)(15)))), ((int)(((byte)(25))))));
86         this.panelBrand.Controls.Add(this.lblMarca);
87         this.panelBrand.Controls.Add(this.lblSubmarca);
88         this.panelBrand.Dock = System.Windows.Forms.DockStyle.Top;
89         this.panelBrand.Location = new System.Drawing.Point(0, 0);
90         this.panelBrand.Name = "panelBrand";
91         this.panelBrand.Size = new System.Drawing.Size(260, 112);
92         this.panelBrand.TabIndex = 0;
93         //
94         // lblMarca
95         //
96         this.lblMarca.Font = new System.Drawing.Font("Segoe UI", 16F, System.Drawing.FontStyle.Bold);
97         this.lblMarca.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(243)))), ((int)(((byte)(244)))), ((int)(((byte)(246))))));
98         this.lblMarca.Location = new System.Drawing.Point(24, 28);
99         this.lblMarca.Name = "lblMarca";
100        this.lblMarca.Size = new System.Drawing.Size(210, 32);
101        this.lblMarca.TabIndex = 0;
102        this.lblMarca.Text = "NomiCore";
103        //
104        // lblSubmarca
105        //
106        this.lblSubmarca.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
107        this.lblSubmarca.Location = new System.Drawing.Point(26, 63);
108        this.lblSubmarca.Name = "lblSubmarca";
109        this.lblSubmarca.Size = new System.Drawing.Size(210, 22);
110        this.lblSubmarca.TabIndex = 1;
111        this.lblSubmarca.Text = "Sistema de nómina";
112        //
113        // flowMenu
114        //
115        this.flowMenu.AutoScroll = true;
116        this.flowMenu.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(11)))), ((int)(((byte)(15)))), ((int)(((byte)(25))))));
117        this.flowMenu.Controls.Add(this.btnPortal);
118        this.flowMenu.Controls.Add(this.btnDashboard);
119        this.flowMenu.Controls.Add(this.btnEmpleados);
120        this.flowMenu.Controls.Add(this.btnAsistencia);
```

UI/FrmMain.Designer.cs (continuacion)

```
121         this.flowMenu.Controls.Add(this.btnNomina);
122         this.flowMenu.Controls.Add(this.btnComprobantes);
123         this.flowMenu.Controls.Add(this.btnReportes);
124         this.flowMenu.Controls.Add(this.btnAusencias);
125         this.flowMenu.Controls.Add(this.btnConfiguracion);
126         this.flowMenu.Controls.Add(this.btnAuditoria);
127         this.flowMenu.Controls.Add(this.btnAcercade);
128         this.flowMenu.Dock = System.Windows.Forms.DockStyle.Fill;
129         this.flowMenu.FlowDirection = System.Windows.Forms.FlowDirection.TopDown;
130         this.flowMenu.Location = new System.Drawing.Point(0, 112);
131         this.flowMenu.Name = "flowMenu";
132         this.flowMenu.Padding = new System.Windows.Forms.Padding(12, 20, 12, 12);
133         this.flowMenu.Size = new System.Drawing.Size(260, 592);
134         this.flowMenu.TabIndex = 1;
135         this.flowMenu.WrapContents = false;
136         //
137         // btnPortal
138         //
139         this.btnPortal.FlatAppearance.BorderSize = 0;
140         this.btnPortal.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
141         this.btnPortal.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
142         this.btnPortal.ForeColor = System.Drawing.Color.FromArgb((int)((byte)(229))), ((int)((byte)(228))), ((int)((byte)(245))));
143         this.btnPortal.IconChar = FontAwesome.Sharp.IconChar.User;
144         this.btnPortal.IconColor = System.Drawing.Color.FromArgb((int)((byte)(167))), ((int)((byte)(139))), ((int)((byte)(250))));
145         this.btnPortal.IconFont = FontAwesome.Sharp.IconFont.Auto;
146         this.btnPortal.IconSize = 20;
147         this.btnPortal.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
148         this.btnPortal.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
149         this.btnPortal.Name = "btnPortal";
150         this.btnPortal.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
151         this.btnPortal.Size = new System.Drawing.Size(228, 44);
152         this.btnPortal.TabIndex = 2;
153         this.btnPortal.Text = " Mi portal";
154         this.btnPortal.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
155         this.btnPortal.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
156         this.btnPortal.UseVisualStyleBackColor = true;
157         this.btnPortal.Visible = false;
158         this.btnPortal.Click += new System.EventHandler(this.btnPortal_Click);
159         //
160         // btnDashboard
161         //
162         this.btnDashboard.FlatAppearance.BorderSize = 0;
163         this.btnDashboard.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
164         this.btnDashboard.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
165         this.btnDashboard.ForeColor = System.Drawing.Color.FromArgb((int)((byte)(229))), ((int)((byte)(228))), ((int)((byte)(245))));
166         this.btnDashboard.IconChar = FontAwesome.Sharp.IconChar.BorderAll;
167         this.btnDashboard.IconColor = System.Drawing.Color.FromArgb((int)((byte)(167))), ((int)((byte)(139))), ((int)((byte)(250))));
168         this.btnDashboard.IconFont = FontAwesome.Sharp.IconFont.Auto;
169         this.btnDashboard.IconSize = 20;
170         this.btnDashboard.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
171         this.btnDashboard.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
172         this.btnDashboard.Name = "btnDashboard";
173         this.btnDashboard.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
174         this.btnDashboard.Size = new System.Drawing.Size(228, 44);
175         this.btnDashboard.TabIndex = 2;
176         this.btnDashboard.Text = " Dashboard";
177         this.btnDashboard.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
178         this.btnDashboard.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
179         this.btnDashboard.UseVisualStyleBackColor = true;
180         this.btnDashboard.Click += new System.EventHandler(this.btnDashboard_Click);
```


UI/FrmMain.Designer.cs (continuacion)

```
181         //
182         // btnEmpleados
183         //
184         this.btnEmpleados.FlatAppearance.BorderSize = 0;
185         this.btnEmpleados.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
186         this.btnEmpleados.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
187         this.btnEmpleados.ForeColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
188         this.btnEmpleados.IconChar = FontAwesome.Sharp.IconChar.Users;
189         this.btnEmpleados.IconColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
190         this.btnEmpleados.IconFont = FontAwesome.Sharp.IconFont.Auto;
191         this.btnEmpleados.IconSize = 20;
192         this.btnEmpleados.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
193         this.btnEmpleados.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
194         this.btnEmpleados.Name = "btnEmpleados";
195         this.btnEmpleados.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
196         this.btnEmpleados.Size = new System.Drawing.Size(228, 44);
197         this.btnEmpleados.TabIndex = 3;
198         this.btnEmpleados.Text = " Empleados";
199         this.btnEmpleados.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
200         this.btnEmpleados.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
201         this.btnEmpleados.UseVisualStyleBackColor = true;
202         this.btnEmpleados.Click += new System.EventHandler(this.btnEmpleados_Click);
203         //
204         // btnAsistencia
205         //
206         this.btnAsistencia.FlatAppearance.BorderSize = 0;
207         this.btnAsistencia.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
208         this.btnAsistencia.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
209         this.btnAsistencia.ForeColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
210         this.btnAsistencia.IconChar = FontAwesome.Sharp.IconChar.CalendarCheck;
211         this.btnAsistencia.IconColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
212         this.btnAsistencia.IconFont = FontAwesome.Sharp.IconFont.Auto;
213         this.btnAsistencia.IconSize = 20;
214         this.btnAsistencia.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
215         this.btnAsistencia.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
216         this.btnAsistencia.Name = "btnAsistencia";
217         this.btnAsistencia.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
218         this.btnAsistencia.Size = new System.Drawing.Size(228, 44);
219         this.btnAsistencia.TabIndex = 4;
220         this.btnAsistencia.Text = " Asistencia";
221         this.btnAsistencia.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
222         this.btnAsistencia.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
223         this.btnAsistencia.UseVisualStyleBackColor = true;
224         this.btnAsistencia.Click += new System.EventHandler(this.btnAsistencia_Click);
225         //
226         // btnNomina
227         //
228         this.btnNomina.FlatAppearance.BorderSize = 0;
229         this.btnNomina.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
230         this.btnNomina.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
231         this.btnNomina.ForeColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
232         this.btnNomina.IconChar = FontAwesome.Sharp.IconChar.MoneyCheckDollar;
233         this.btnNomina.IconColor = System.Drawing.Color.FromArgb((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
234         this.btnNomina.IconFont = FontAwesome.Sharp.IconFont.Auto;
235         this.btnNomina.IconSize = 20;
236         this.btnNomina.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
237         this.btnNomina.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
238         this.btnNomina.Name = "btnNomina";
239         this.btnNomina.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
240         this.btnNomina.Size = new System.Drawing.Size(228, 44);
```

UI/FrmMain.Designer.cs (continuacion)

```
241         this.btnNomina.TabIndex = 5;
242         this.btnNomina.Text = "  Nómina";
243         this.btnNomina.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
244         this.btnNomina.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
245         this.btnNomina.UseVisualStyleBackColor = true;
246         this.btnNomina.Click += new System.EventHandler(this.btnNomina_Click);
247         //
248         // btnComprobantes
249         //
250         this.btnComprobantes.FlatAppearance.BorderSize = 0;
251         this.btnComprobantes.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
252         this.btnComprobantes.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
253         this.btnComprobantes.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
254         this.btnComprobantes.IconChar = FontAwesome.Sharp.IconChar.Receipt;
255         this.btnComprobantes.IconColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
256         this.btnComprobantes.IconFont = FontAwesome.Sharp.IconFont.Auto;
257         this.btnComprobantes.IconSize = 20;
258         this.btnComprobantes.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
259         this.btnComprobantes.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
260         this.btnComprobantes.Name = "btnComprobantes";
261         this.btnComprobantes.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
262         this.btnComprobantes.Size = new System.Drawing.Size(228, 44);
263         this.btnComprobantes.TabIndex = 6;
264         this.btnComprobantes.Text = "  Comprobantes";
265         this.btnComprobantes.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
266         this.btnComprobantes.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
267         this.btnComprobantes.UseVisualStyleBackColor = true;
268         this.btnComprobantes.Click += new System.EventHandler(this.btnComprobantes_Click);
269         //
270         // btnReportes
271         //
272         this.btnReportes.FlatAppearance.BorderSize = 0;
273         this.btnReportes.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
274         this.btnReportes.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
275         this.btnReportes.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
276         this.btnReportes.IconChar = FontAwesome.Sharp.IconChar.ChartSimple;
277         this.btnReportes.IconColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
278         this.btnReportes.IconFont = FontAwesome.Sharp.IconFont.Auto;
279         this.btnReportes.IconSize = 20;
280         this.btnReportes.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
281         this.btnReportes.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
282         this.btnReportes.Name = "btnReportes";
283         this.btnReportes.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
284         this.btnReportes.Size = new System.Drawing.Size(228, 44);
285         this.btnReportes.TabIndex = 7;
286         this.btnReportes.Text = "  Reportes";
287         this.btnReportes.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
288         this.btnReportes.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
289         this.btnReportes.UseVisualStyleBackColor = true;
290         this.btnReportes.Click += new System.EventHandler(this.btnReportes_Click);
291         //
292         // btnConfiguracion
293         //
294         this.btnConfiguracion.FlatAppearance.BorderSize = 0;
295         this.btnConfiguracion.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
296         this.btnConfiguracion.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
297         this.btnConfiguracion.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
298         this.btnConfiguracion.IconChar = FontAwesome.Sharp.IconChar.Cog;
299         this.btnConfiguracion.IconColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
300         this.btnConfiguracion.IconFont = FontAwesome.Sharp.IconFont.Auto;
```

UI/FrmMain.Designer.cs (continuacion)

```
301         this.btnConfiguracion.IconSize = 20;
302         this.btnConfiguracion.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
303         this.btnConfiguracion.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
304         this.btnConfiguracion.Name = "btnConfiguracion";
305         this.btnConfiguracion.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
306         this.btnConfiguracion.Size = new System.Drawing.Size(228, 44);
307         this.btnConfiguracion.TabIndex = 8;
308         this.btnConfiguracion.Text = " Configuración";
309         this.btnConfiguracion.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
310         this.btnConfiguracion.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
311         this.btnConfiguracion.UseVisualStyleBackColor = true;
312         this.btnConfiguracion.Click += new System.EventHandler(this.btnConfiguracion_Click);
313         //
314         // btnAcercaDe
315         //
316         this.btnAcercaDe.FlatAppearance.BorderSize = 0;
317         this.btnAcercaDe.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
318         this.btnAcercaDe.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
319         this.btnAcercaDe.ForeColor = System.Drawing.Color.FromArgb(((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
320         this.btnAcercaDe.IconChar = FontAwesome.Sharp.IconChar.CircleInfo;
321         this.btnAcercaDe.IconColor = System.Drawing.Color.FromArgb(((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
322         this.btnAcercaDe.IconFont = FontAwesome.Sharp.IconFont.Auto;
323         this.btnAcercaDe.IconSize = 20;
324         this.btnAcercaDe.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
325         this.btnAcercaDe.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
326         this.btnAcercaDe.Name = "btnAcercaDe";
327         this.btnAcercaDe.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
328         this.btnAcercaDe.Size = new System.Drawing.Size(228, 44);
329         this.btnAcercaDe.TabIndex = 9;
330         this.btnAcercaDe.Text = " Acerca de";
331         this.btnAcercaDe.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
332         this.btnAcercaDe.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
333         this.btnAcercaDe.UseVisualStyleBackColor = true;
334         this.btnAcercaDe.Click += new System.EventHandler(this.btnAcercaDe_Click);
335         //
336         // btnAuditoria
337         //
338         this.btnAuditoria.FlatAppearance.BorderSize = 0;
339         this.btnAuditoria.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
340         this.btnAuditoria.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
341         this.btnAuditoria.ForeColor = System.Drawing.Color.FromArgb(((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
342         this.btnAuditoria.IconChar = FontAwesome.Sharp.IconChar.Clipboard;
343         this.btnAuditoria.IconColor = System.Drawing.Color.FromArgb(((int)((byte)(170))), ((int)((byte)(170))), ((int)((byte)(185))));
344         this.btnAuditoria.IconFont = FontAwesome.Sharp.IconFont.Auto;
345         this.btnAuditoria.IconSize = 20;
346         this.btnAuditoria.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
347         this.btnAuditoria.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
348         this.btnAuditoria.Name = "btnAuditoria";
349         this.btnAuditoria.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
350         this.btnAuditoria.Size = new System.Drawing.Size(228, 44);
351         this.btnAuditoria.TabIndex = 10;
352         this.btnAuditoria.Text = " Auditoria";
353         this.btnAuditoria.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
354         this.btnAuditoria.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
355         this.btnAuditoria.UseVisualStyleBackColor = true;
356         this.btnAuditoria.Click += new System.EventHandler(this.btnAuditoria_Click);
357         //
358         // btnAusencias
359         //
360         this.btnAusencias.FlatAppearance.BorderSize = 0;
```

UI/FrmMain.Designer.cs (continuacion)

```
361         this.btnAusencias.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
362         this.btnAusencias.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
363         this.btnAusencias.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
364         this.btnAusencias.IconChar = FontAwesome.Sharp.IconChar.CalendarMinus;
365         this.btnAusencias.IconColor = System.Drawing.Color.FromArgb(170, 170, 185);
366         this.btnAusencias.IconFont = FontAwesome.Sharp.IconFont.Auto;
367         this.btnAusencias.IconSize = 20;
368         this.btnAusencias.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
369         this.btnAusencias.Margin = new System.Windows.Forms.Padding(0, 0, 0, 8);
370         this.btnAusencias.Name = "btnAusencias";
371         this.btnAusencias.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
372         this.btnAusencias.Size = new System.Drawing.Size(228, 44);
373         this.btnAusencias.TabIndex = 11;
374         this.btnAusencias.Text = " Ausencias";
375         this.btnAusencias.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
376         this.btnAusencias.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
377         this.btnAusencias.UseVisualStyleBackColor = true;
378         this.btnAusencias.Click += new System.EventHandler(this.btnAusencias_Click);
379         //
380         // btnSalir
381         //
382         this.btnSalir.FlatAppearance.BorderSize = 0;
383         this.btnSalir.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
384         this.btnSalir.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
385         this.btnSalir.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
386         this.btnSalir.IconChar = FontAwesome.Sharp.IconChar.PowerOff;
387         this.btnSalir.IconColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
388         this.btnSalir.IconFont = FontAwesome.Sharp.IconFont.Auto;
389         this.btnSalir.IconSize = 20;
390         this.btnSalir.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
391         this.btnSalir.Margin = new System.Windows.Forms.Padding(0, 0, 0, 4);
392         this.btnSalir.Name = "btnSalir";
393         this.btnSalir.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
394         this.btnSalir.Size = new System.Drawing.Size(228, 44);
395         this.btnSalir.TabIndex = 10;
396         this.btnSalir.Text = " Salir";
397         this.btnSalir.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
398         this.btnSalir.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
399         this.btnSalir.UseVisualStyleBackColor = true;
400         this.btnSalir.Click += new System.EventHandler(this.btnSalir_Click);
401         //
402         // btnCerrarSesion
403         //
404         this.btnCerrarSesion.FlatAppearance.BorderSize = 0;
405         this.btnCerrarSesion.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
406         this.btnCerrarSesion.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
407         this.btnCerrarSesion.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
408         this.btnCerrarSesion.IconChar = FontAwesome.Sharp.IconChar.RightFromBracket;
409         this.btnCerrarSesion.IconColor = System.Drawing.Color.FromArgb(((int)(((byte)(170)))), ((int)(((byte)(170)))), ((int)(((byte)(185))))));
410         this.btnCerrarSesion.IconFont = FontAwesome.Sharp.IconFont.Auto;
411         this.btnCerrarSesion.IconSize = 20;
412         this.btnCerrarSesion.ImageAlign = System.Drawing.ContentAlignment.MiddleLeft;
413         this.btnCerrarSesion.Margin = new System.Windows.Forms.Padding(0, 0, 0, 4);
414         this.btnCerrarSesion.Name = "btnCerrarSesion";
415         this.btnCerrarSesion.Padding = new System.Windows.Forms.Padding(14, 0, 0, 0);
416         this.btnCerrarSesion.Size = new System.Drawing.Size(228, 44);
417         this.btnCerrarSesion.TabIndex = 11;
418         this.btnCerrarSesion.Text = " Cerrar sesión";
419         this.btnCerrarSesion.TextAlign = System.Drawing.ContentAlignment.MiddleLeft;
420         this.btnCerrarSesion.TextImageRelation = System.Windows.Forms.TextImageRelation.ImageBeforeText;
```

UI/FrmMain.Designer.cs (continuacion)

```
421         this.btnCerrarSesion.UseVisualStyleBackColor = true;
422         this.btnCerrarSesion.Click += new System.EventHandler(this.btnCerrarSesion_Click);
423         //
424         // flowAccount
425         //
426         this.flowAccount.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(11)))), ((int)(((byte)(15)))), ((int)(((byte)(25)))));
427         this.flowAccount.Controls.Add(this.panelAccountSeparator);
428         this.flowAccount.Controls.Add(this.btnCerrarSesion);
429         this.flowAccount.Controls.Add(this.btnSalir);
430         this.flowAccount.Dock = System.Windows.Forms.DockStyle.Bottom;
431         this.flowAccount.FlowDirection = System.Windows.Forms.FlowDirection.TopDown;
432         this.flowAccount.Location = new System.Drawing.Point(0, 704);
433         this.flowAccount.Name = "flowAccount";
434         this.flowAccount.Padding = new System.Windows.Forms.Padding(12, 8, 12, 8);
435         this.flowAccount.Size = new System.Drawing.Size(260, 116);
436         this.flowAccount.TabIndex = 2;
437         this.flowAccount.WrapContents = false;
438         //
439         // panelAccountSeparator
440         //
441         this.panelAccountSeparator.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(50)))), ((int)(((byte)(54)))), ((int)(((byte)(68)))));
442         this.panelAccountSeparator.Margin = new System.Windows.Forms.Padding(0, 0, 0, 7);
443         this.panelAccountSeparator.Name = "panelAccountSeparator";
444         this.panelAccountSeparator.Size = new System.Drawing.Size(228, 1);
445         this.panelAccountSeparator.TabIndex = 0;
446         //
447         // panelTopbar
448         //
449         this.panelTopbar.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(17)))), ((int)(((byte)(18)))), ((int)(((byte)(26)))));
450         this.panelTopbar.Controls.Add(this.lblUsuarioActual);
451         this.panelTopbar.Dock = System.Windows.Forms.DockStyle.Top;
452         this.panelTopbar.Location = new System.Drawing.Point(260, 0);
453         this.panelTopbar.Name = "panelTopbar";
454         this.panelTopbar.Size = new System.Drawing.Size(1020, 78);
455         this.panelTopbar.TabIndex = 1;
456         //
457         // lblUsuarioActual
458         //
459         this.lblUsuarioActual.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right)));
460         this.lblUsuarioActual.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
461         this.lblUsuarioActual.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(161)))), ((int)(((byte)(161)))), ((int)(((byte)(170)))));
462         this.lblUsuarioActual.Location = new System.Drawing.Point(708, 25);
463         this.lblUsuarioActual.Name = "lblUsuarioActual";
464         this.lblUsuarioActual.Size = new System.Drawing.Size(282, 24);
465         this.lblUsuarioActual.TabIndex = 1;
466         this.lblUsuarioActual.Text = "Sin sesión";
467         this.lblUsuarioActual.TextAlign = System.Drawing.ContentAlignment.MiddleRight;
468         //
469         // panelContent
470         //
471         this.panelContent.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20)))));
472         this.panelContent.Dock = System.Windows.Forms.DockStyle.Fill;
473         this.panelContent.Location = new System.Drawing.Point(260, 78);
474         this.panelContent.Name = "panelContent";
475         this.panelContent.Size = new System.Drawing.Size(1020, 742);
476         this.panelContent.TabIndex = 2;
477         //
478         // FrmMain
479         //
```

UI/FrmMain.Designer.cs (continuacion)

```
480         this.AutoScaleModeDimensions = new System.Drawing.SizeF(8F, 20F);
481         this.AutoScaleModeMode = System.Windows.Forms.AutoScaleMode.Font;
482         this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
483         this.ClientSize = new System.Drawing.Size(1280, 820);
484         this.Controls.Add(this.panelContent);
485         this.Controls.Add(this.panelTopbar);
486         this.Controls.Add(this.panelSidebar);
487         this.Font = new System.Drawing.Font("Segoe UI", 9F);
488         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
489         this.MinimumSize = new System.Drawing.Size(1120, 720);
490         this.Name = "FrmMain";
491         this.StartPosition = System.Windows.Forms.FormStartPosition.CenterScreen;
492         this.Text = "ProyFinal_LPI_Eq01_NomiCore";
493         this.panelSidebar.ResumeLayout(false);
494         this.panelBrand.ResumeLayout(false);
495         this.flowMenu.ResumeLayout(false);
496         this.flowAccount.ResumeLayout(false);
497         this.panelTopbar.ResumeLayout(false);
498         this.ResumeLayout(false);
499
500     }
501 }
502 }
```

UI/FrmMiPerfil.cs

```
1 using System;
2 using System.ComponentModel;
3 using System.Windows.Forms;
4 using SistemaGestionNomina.Helpers;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Services;
7
8 namespace SistemaGestionNomina.UI
9 {
10     public partial class FrmMiPerfil : Form
11     {
12         private readonly EmployeePortalService portalService = new EmployeePortalService();
13
14         public FrmMiPerfil()
15         {
16             InitializeComponent();
17             ControlStyleHelper.ApplyModernForm(this);
18         }
19
20         private void FrmMiPerfil_Load(object sender, EventArgs e)
21         {
22             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
23             try
24             {
25                 EmployeeProfileViewModel profile = portalService.GetMyProfile();
26                 lblNombreValor.Text = profile.NombreCompleto;
27                 lblCodigoValor.Text = profile.Codigo;
28                 lblCedulaValor.Text = profile.Cedula;
29                 lblCargoValor.Text = profile.Cargo;
30                 lblDepartamentoValor.Text = profile.Departamento;
31                 lblSalarioValor.Text = "B/. " + profile.SalarioBase.ToString("#,##0.00");
32                 lblEstadoValor.Text = profile.Estado;
33                 lblIngresoValor.Text = profile.FechaIngreso.ToString("dd/MM/yyyy");
34             }
35             catch (Exception ex)
36             {
37                 MessageBox.Show(ex.Message, "Mi perfil", MessageBoxButtons.OK, MessageBoxIcon.Warning);
38             }
39         }
40     }
41 }
```


UI/FrmMiPerfil.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmMiPerfil
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Label lblSubtitulo;
8         private System.Windows.Forms.TableLayoutPanel tablePerfil;
9         private System.Windows.Forms.Label lblNombre;
10        private System.Windows.Forms.Label lblNombreValor;
11        private System.Windows.Forms.Label lblCodigo;
12        private System.Windows.Forms.Label lblCodigoValor;
13        private System.Windows.Forms.Label lblCedula;
14        private System.Windows.Forms.Label lblCedulaValor;
15        private System.Windows.Forms.Label lblCargo;
16        private System.Windows.Forms.Label lblCargoValor;
17        private System.Windows.Forms.Label lblDepartamento;
18        private System.Windows.Forms.Label lblDepartamentoValor;
19        private System.Windows.Forms.Label lblSalario;
20        private System.Windows.Forms.Label lblSalarioValor;
21        private System.Windows.Forms.Label lblEstado;
22        private System.Windows.Forms.Label lblEstadoValor;
23        private System.Windows.Forms.Label lblIngreso;
24        private System.Windows.Forms.Label lblIngresoValor;
25
26        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
27
28        private void InitializeComponent()
29        {
30            this.lblTitulo = new System.Windows.Forms.Label(); this.lblSubtitulo = new System.Windows.Forms.Label(); this.tablePerfil = new \
System.Windows.Forms.TableLayoutPanel();
31            this.lblNombre = new System.Windows.Forms.Label(); this.lblNombreValor = new System.Windows.Forms.Label(); this.lblCodigo = new \
System.Windows.Forms.Label(); this.lblCodigoValor = new System.Windows.Forms.Label(); this.lblCedula = new System.Windows.Forms.Label(); \
this.lblCedulaValor = new System.Windows.Forms.Label(); this.lblCargo = new System.Windows.Forms.Label(); this.lblCargoValor = new \
System.Windows.Forms.Label(); this.lblDepartamento = new System.Windows.Forms.Label(); this.lblDepartamentoValor = new System.Windows.Forms.Label(); \
this.lblSalario = new System.Windows.Forms.Label(); this.lblSalarioValor = new System.Windows.Forms.Label(); this.lblEstado = new \
System.Windows.Forms.Label(); this.lblEstadoValor = new System.Windows.Forms.Label(); this.lblIngreso = new System.Windows.Forms.Label(); \
this.lblIngresoValor = new System.Windows.Forms.Label();
32            this.tablePerfil.SuspendLayout(); this.SuspendLayout();
33            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 23F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(32, 24); this.lblTitulo.Text = "Mi perfil laboral";
34            this.lblSubtitulo.AutoSize = true; this.lblSubtitulo.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblSubtitulo.Location = \
new System.Drawing.Point(36, 78); this.lblSubtitulo.Text = "Informacion registrada por Recursos Humanos.";
35            this.tablePerfil.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Left | System.Windows.Forms.AnchorStyles.Right))); this.tablePerfil.BackColor = System.Drawing.Color.FromArgb(24, \
25, 34); this.tablePerfil.ColumnCount = 2; this.tablePerfil.ColumnStyles.Add(new \
System.Windows.Forms.ColumnStyle(System.Windows.Forms.SizeType.Percent, 32F)); this.tablePerfil.ColumnStyles.Add(new \
System.Windows.Forms.ColumnStyle(System.Windows.Forms.SizeType.Percent, 68F)); this.tablePerfil.Controls.Add(this.lblNombre, 0, 0); \
this.tablePerfil.Controls.Add(this.lblNombreValor, 1, 0); this.tablePerfil.Controls.Add(this.lblCodigo, 0, 1); \
this.tablePerfil.Controls.Add(this.lblCodigoValor, 1, 1); this.tablePerfil.Controls.Add(this.lblCedula, 0, 2); \
this.tablePerfil.Controls.Add(this.lblCedulaValor, 1, 2); this.tablePerfil.Controls.Add(this.lblCargo, 0, 3); \
this.tablePerfil.Controls.Add(this.lblCargoValor, 1, 3); this.tablePerfil.Controls.Add(this.lblDepartamento, 0, 4); \
this.tablePerfil.Controls.Add(this.lblDepartamentoValor, 1, 4); this.tablePerfil.Controls.Add(this.lblSalario, 0, 5); \
this.tablePerfil.Controls.Add(this.lblSalarioValor, 1, 5); this.tablePerfil.Controls.Add(this.lblEstado, 0, 6); \
this.tablePerfil.Controls.Add(this.lblEstadoValor, 1, 6); this.tablePerfil.Controls.Add(this.lblIngreso, 0, 7); \
this.tablePerfil.Controls.Add(this.lblIngresoValor, 1, 7); this.tablePerfil.Location = new System.Drawing.Point(38, 126); this.tablePerfil.Name = \
"tablePerfil"; this.tablePerfil.Padding = new System.Windows.Forms.Padding(24); this.tablePerfil.RowCount = 8; for (int i = 0; i < 8; i++) \
this.tablePerfil.RowStyles.Add(new System.Windows.Forms.RowStyle(System.Windows.Forms.SizeType.Absolute, 58F)); this.tablePerfil.Size = new \
System.Drawing.Size(944, 514);
36        ConfigureCaption(this.lblNombre, "Nombre completo"); ConfigureValue(this.lblNombreValor); ConfigureCaption(this.lblCodigo, "Codigo de \

```


UI/FrmMiPerfil.Designer.cs (continuacion)

```
    empleado"); ConfigureValue(this.lblCodigoValor); ConfigureCaption(this.lblCedula, "Cedula"); ConfigureValue(this.lblCedulaValor); \
    ConfigureCaption(this.lblCargo, "Cargo"); ConfigureValue(this.lblCargoValor); ConfigureCaption(this.lblDepartamento, "Departamento"); \
    ConfigureValue(this.lblDepartamentoValor); ConfigureCaption(this.lblSalario, "Salario base"); ConfigureValue(this.lblSalarioValor); \
    ConfigureCaption(this.lblEstado, "Estado laboral"); ConfigureValue(this.lblEstadoValor); ConfigureCaption(this.lblIngreso, "Fecha de ingreso"); \
    ConfigureValue(this.lblIngresoValor);
37     this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F); this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font; this.BackColor = \
    System.Drawing.Color.FromArgb(13, 14, 20); this.ClientSize = new System.Drawing.Size(1020, 742); this.Controls.Add(this.tablePerfil); \
    this.Controls.Add(this.lblSubtitulo); this.Controls.Add(this.lblTitulo); this.Font = new System.Drawing.Font("Segoe UI", 9F); this.ForeColor = \
    System.Drawing.Color.FromArgb(229, 228, 245); this.Name = "FrmMiPerfil"; this.Text = "Mi perfil"; this.Load += new \
    System.EventHandler(this.FrmMiPerfil_Load); this.tablePerfil.ResumeLayout(false); this.ResumeLayout(false); this.PerformLayout();
38     }
39
40     private static void ConfigureCaption(System.Windows.Forms.Label label, string text) { label.Dock = System.Windows.Forms.DockStyle.Fill; \
    label.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); label.Text = text; label.TextAlign = System.Drawing.ContentAlignment.MiddleLeft; }
41     private static void ConfigureValue(System.Windows.Forms.Label label) { label.AutoEllipsis = true; label.Dock = \
    System.Windows.Forms.DockStyle.Fill; label.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold); label.Text = "-"; \
    label.TextAlign = System.Drawing.ContentAlignment.MiddleLeft; }
42 }
43 }
```

UI/FrmMisAsistencias.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Services;
8
9 namespace SistemaGestionNomina.UI
10 {
11     public partial class FrmMisAsistencias : Form
12     {
13         private readonly EmployeePortalService portalService = new EmployeePortalService();
14
15         public FrmMisAsistencias()
16         {
17             InitializeComponent();
18             ControlStyleHelper.ApplyModernForm(this);
19         }
20
21         private void FrmMisAsistencias_Load(object sender, EventArgs e)
22         {
23             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
24             dtpDesde.Value = new DateTime(DateTime.Today.Year, DateTime.Today.Month, 1);
25             dtpHasta.Value = DateTime.Today;
26             cmbEstado.SelectedIndex = 0;
27             LoadAttendance();
28         }
29
30         private void btnFiltrar_Click(object sender, EventArgs e) { LoadAttendance(); }
31
32         private void btnLimpiar_Click(object sender, EventArgs e)
33         {
34             dtpDesde.Checked = false;
35             dtpHasta.Checked = false;
36             cmbEstado.SelectedIndex = 0;
37             LoadAttendance();
38         }
39
40         private void LoadAttendance()
41         {
42             try
43             {
44                 DateTime? start = dtpDesde.Checked ? dtpDesde.Value.Date : (DateTime?)null;
45                 DateTime? end = dtpHasta.Checked ? dtpHasta.Value.Date : (DateTime?)null;
46                 List<Asistencia> items = portalService.GetMyAttendance(start, end,
47                     Convert.ToString(cmbEstado.SelectedItem));
48
49                 dgvAsistencias.Rows.Clear();
50                 foreach (Asistencia item in items)
51                 {
52                     dgvAsistencias.Rows.Add(item.Fecha.ToString("dd/MM/yyyy"),
53                         item.HoraEntrada.HasValue ? item.HoraEntrada.Value.ToString(@"hh\:mm") : "-",
54                         item.HoraSalida.HasValue ? item.HoraSalida.Value.ToString(@"hh\:mm") : "-",
55                         item.HorasTrabajadas.ToString("0.00"), item.Estado);
56                 }
57
58                 EmployeeAttendanceSummaryViewModel summary = EmployeePortalService.SummarizeAttendance(items);
59                 lblPuntualesValor.Text = summary.Puntuales + "\nPuntuales";
60                 lblTardanzasValor.Text = summary.Tardanzas + "\nTardanzas";
61             }
62             catch { }
63         }
64     }
65 }
```

UI/FrmMisAsistencias.cs (continuacion)

```
61         lblFaltasValor.Text = summary.Faltas + "\nFaltas";
62         lblPermisosValor.Text = summary.Permisos + "\nPermisos";
63         lblHorasExtraValor.Text = summary.HorasExtra.ToString("0.00") + " h\nHoras extra";
64     }
65     catch (Exception ex)
66     {
67         MessageBox.Show(ex.Message, "Mis asistencias", MessageBoxButtons.OK, MessageBoxIcon.Warning);
68     }
69 }
70 }
71 }
```

UI/FrmMisAsistencias.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmMisAsistencias
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelFiltros;
8         private System.Windows.Forms.DateTimePicker dtpDesde;
9         private System.Windows.Forms.DateTimePicker dtpHasta;
10        private System.Windows.Forms.ComboBox cmbEstado;
11        private System.Windows.Forms.Button btnFiltrar;
12        private System.Windows.Forms.Button btnLimpiar;
13        private System.Windows.Forms.FlowLayoutPanel panelResumen;
14        private System.Windows.Forms.Label lblPuntualesValor;
15        private System.Windows.Forms.Label lblTardanzasValor;
16        private System.Windows.Forms.Label lblFaltasValor;
17        private System.Windows.Forms.Label lblPermisosValor;
18        private System.Windows.Forms.Label lblHorasExtraValor;
19        private System.Windows.Forms.DataGridView dgvAsistencias;
20        private System.Windows.Forms.DataGridViewTextBoxColumn colFecha;
21        private System.Windows.Forms.DataGridViewTextBoxColumn colEntrada;
22        private System.Windows.Forms.DataGridViewTextBoxColumn colSalida;
23        private System.Windows.Forms.DataGridViewTextBoxColumn colHoras;
24        private System.Windows.Forms.DataGridViewTextBoxColumn colEstado;
25
26        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
27
28        private void InitializeComponent()
29        {
30            this.lblTitulo = new System.Windows.Forms.Label(); this.panelFiltros = new System.Windows.Forms.Panel(); this.dtpDesde = new \
System.Windows.Forms.DateTimePicker(); this.dtpHasta = new System.Windows.Forms.DateTimePicker(); this.cmbEstado = new \
System.Windows.Forms.ComboBox(); this.btnFiltrar = new System.Windows.Forms.Button(); this.btnLimpiar = new System.Windows.Forms.Button(); \
this.panelResumen = new System.Windows.Forms.FlowLayoutPanel(); this.lblPuntualesValor = new System.Windows.Forms.Label(); this.lblTardanzasValor = \
new System.Windows.Forms.Label(); this.lblFaltasValor = new System.Windows.Forms.Label(); this.lblPermisosValor = new System.Windows.Forms.Label(); \
this.lblHorasExtraValor = new System.Windows.Forms.Label(); this.dgvAsistencias = new System.Windows.Forms.DataGridView(); this.colFecha = new \
System.Windows.Forms.DataGridViewTextBoxColumn(); this.colEntrada = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.colSalida = new \
System.Windows.Forms.DataGridViewTextBoxColumn(); this.colHoras = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.colEstado = new \
System.Windows.Forms.DataGridViewTextBoxColumn();
31            this.panelFiltros.SuspendLayout(); this.panelResumen.SuspendLayout(); \
((System.ComponentModel.ISupportInitialize)(this.dgvAsistencias)).BeginInit(); this.SuspendLayout();
32            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 23F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Mis asistencias";
33            this.panelFiltros.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Left | System.Windows.Forms.AnchorStyles.Right))); this.panelFiltros.BackColor = System.Drawing.Color.FromArgb(24, \
25, 34); this.panelFiltros.Controls.Add(this.dtpDesde); this.panelFiltros.Controls.Add(this.cmbEstado); \
this.panelFiltros.Controls.Add(this.btnFiltrar); this.panelFiltros.Controls.Add(this.btnLimpiar); \
this.panelFiltros.Location = new System.Drawing.Point(34, 86); this.panelFiltros.Size = new System.Drawing.Size(952, 74);
34            this.dtpDesde.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpDesde.Location = new System.Drawing.Point(18, 22); \
this.dtpDesde.Name = "dtpDesde"; this.dtpDesde.ShowCheckBox = true; this.dtpDesde.Size = new System.Drawing.Size(190, 27);
35            this.dtpHasta.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpHasta.Location = new System.Drawing.Point(224, 22); \
this.dtpHasta.Name = "dtpHasta"; this.dtpHasta.ShowCheckBox = true; this.dtpHasta.Size = new System.Drawing.Size(190, 27);
36            this.cmbEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbEstado.Items.AddRange(new object[] { "Todos", \
"Puntual", "Tardanza", "Falta", "Permiso" }); this.cmbEstado.Location = new System.Drawing.Point(430, 21); this.cmbEstado.Name = "cmbEstado"; \
this.cmbEstado.Size = new System.Drawing.Size(176, 28);
37            this.btnFiltrar.Location = new System.Drawing.Point(630, 17); this.btnFiltrar.Name = "btnFiltrar"; this.btnFiltrar.Size = new \
System.Drawing.Size(142, 38); this.btnFiltrar.Text = "Aplicar filtros"; this.btnFiltrar.Click += new System.EventHandler(this.btnFiltrar_Click);
38            this.btnLimpiar.Location = new System.Drawing.Point(790, 17); this.btnLimpiar.Name = "btnLimpiar"; this.btnLimpiar.Size = new \
System.Drawing.Size(142, 38); this.btnLimpiar.Text = "Limpiar"; this.btnLimpiar.Click += new System.EventHandler(this.btnLimpiar_Click);
39            this.panelResumen.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Left | System.Windows.Forms.AnchorStyles.Right))); this.panelResumen.Controls.Add(this.lblPuntualesValor); \

```

UI/FrmMisAsistencias.Designer.cs (continuacion)

```

        this.panelResumen.Controls.Add(this.lblTardanzasValor); this.panelResumen.Controls.Add(this.lblFaltasValor); \
        this.panelResumen.Controls.Add(this.lblPermisosValor); this.panelResumen.Controls.Add(this.lblHorasExtraValor); this.panelResumen.FlowDirection = \
        System.Windows.Forms.FlowDirection.LeftToRight; this.panelResumen.Location = new System.Drawing.Point(34, 178); this.panelResumen.Name = \
        "panelResumen"; this.panelResumen.Size = new System.Drawing.Size(952, 80); this.panelResumen.WrapContents = false;
40         ConfigureSummary(this.lblPuntualesValor, "Puntuales", System.Drawing.Color.FromArgb(52, 211, 153)); ConfigureSummary(this.lblTardanzasValor, \
        "Tardanzas", System.Drawing.Color.FromArgb(251, 191, 36)); ConfigureSummary(this.lblFaltasValor, "Faltas", System.Drawing.Color.FromArgb(248, 113, \
        113)); ConfigureSummary(this.lblPermisosValor, "Permisos", System.Drawing.Color.FromArgb(96, 165, 250)); ConfigureSummary(this.lblHorasExtraValor, \
        "Horas extra", System.Drawing.Color.FromArgb(167, 139, 250));
41         this.dgvAsistencias.AllowUserToAddRows = false; this.dgvAsistencias.AllowUserToDeleteRows = false; this.dgvAsistencias.Anchor = \
        ((System.Windows.Forms.AnchorStyles)((((System.Windows.Forms.AnchorStyles.Top | System.Windows.Forms.AnchorStyles.Bottom) | \
        System.Windows.Forms.AnchorStyles.Left) | System.Windows.Forms.AnchorStyles.Right))); this.dgvAsistencias.AutoSizeColumnsMode = \
        System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill; this.dgvAsistencias.BackgroundColor = System.Drawing.Color.FromArgb(18, 19, 27); \
        this.dgvAsistencias.BorderStyle = System.Windows.Forms.BorderStyle.None; this.dgvAsistencias.ColumnHeadersHeight = 40; \
        this.dgvAsistencias.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] { this.colFecha, this.colEntrada, this.colSalida, this.colHoras, \
        this.colEstado }); this.dgvAsistencias.Location = new System.Drawing.Point(34, 276); this.dgvAsistencias.Name = "dgvAsistencias"; \
        this.dgvAsistencias.ReadOnly = true; this.dgvAsistencias.RowHeadersVisible = false; this.dgvAsistencias.RowTemplate.Height = 36; \
        this.dgvAsistencias.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect; this.dgvAsistencias.Size = new \
        System.Drawing.Size(952, 430);
42         this.colFecha.HeaderText = "Fecha"; this.colEntrada.HeaderText = "Entrada"; this.colSalida.HeaderText = "Salida"; this.colHoras.HeaderText = \
        "Horas trabajadas"; this.colEstado.HeaderText = "Estado";
43         this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F); this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font; this.BackColor = \
        System.Drawing.Color.FromArgb(13, 14, 20); this.ClientSize = new System.Drawing.Size(1020, 742); this.Controls.Add(this.dgvAsistencias); \
        this.Controls.Add(this.panelResumen); this.Controls.Add(this.panelFiltros); this.Controls.Add(this.lblTitulo); this.Font = new \
        System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.Name = "FrmMisAsistencias"; this.Text = "Mis \
        asistencias"; this.Load += new System.EventHandler(this.FrmMisAsistencias_Load); this.panelFiltros.ResumeLayout(false); \
        this.panelResumen.ResumeLayout(false); ((System.ComponentModel.ISupportInitialize)(this.dgvAsistencias)).EndInit(); this.ResumeLayout(false); \
        this.PerformLayout();
44     }
45
46     private static void ConfigureSummary(System.Windows.Forms.Label label, string title, System.Drawing.Color color) { label.BackColor = \
        System.Drawing.Color.FromArgb(24, 25, 34); label.Font = new System.Drawing.Font("Segoe UI", 13F, System.Drawing.FontStyle.Bold); label.ForeColor = \
        color; label.Margin = new System.Windows.Forms.Padding(0, 0, 12, 0); label.Size = new System.Drawing.Size(178, 76); label.Text = "0\n" + title; \
        label.TextAlign = System.Drawing.ContentAlignment.MiddleCenter; }
47 }
48 }

```

UI\FrmMisComprobantes.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Diagnostics;
5 using System.Drawing.Printing;
6 using System.Windows.Forms;
7 using SistemaGestionNomina.Helpers;
8 using SistemaGestionNomina.Models;
9 using SistemaGestionNomina.Services;
10
11 namespace SistemaGestionNomina.UI
12 {
13     public partial class FrmMisComprobantes : Form
14     {
15         private readonly EmployeePortalService portalService = new EmployeePortalService();
16         private readonly PrintDocument printDocument = new PrintDocument();
17         private List<Comprobante> currentItems = new List<Comprobante>();
18         private Comprobante selectedPayslip;
19
20         public FrmMisComprobantes()
21         {
22             InitializeComponent();
23             ControlStyleHelper.ApplyModernForm(this);
24             printDocument.PrintPage += PrintDocument_PrintPage;
25         }
26
27         private void FrmMisComprobantes_Load(object sender, EventArgs e)
28         {
29             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime) LoadPayslips();
30         }
31
32         private void txtBuscar_TextChanged(object sender, EventArgs e)
33         {
34             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime) LoadPayslips();
35         }
36
37         private void dgvComprobantes_SelectionChanged(object sender, EventArgs e)
38         {
39             if (dgvComprobantes.CurrentRow == null || dgvComprobantes.CurrentRow.Cells["colId"].Value == null) return;
40             try
41             {
42                 selectedPayslip = portalService.GetMyPayslipById(
43                     Convert.ToInt32(dgvComprobantes.CurrentRow.Cells["colId"].Value));
44                 RenderPreview();
45             }
46             catch (Exception ex)
47             {
48                 selectedPayslip = null;
49                 lblVistaPrevia.Text = ex.Message;
50             }
51         }
52
53         private void LoadPayslips()
54         {
55             try
56             {
57                 currentItems = portalService.GetMyPayslips(txtBuscar.Text);
58                 dgvComprobantes.Rows.Clear();
59                 foreach (Comprobante item in currentItems)
60                 {
```

UI/FrmMisComprobantes.cs (continuación)

```
61         dgvComprobantes.Rows.Add(item.IdComprobante, item.NumeroComprobante,
62             item.PeriodoNombre, item.FechaGeneracion.ToString("dd/MM/yyyy"),
63             item.TotalIngresos.ToString("0.00"), item.TotalDeducciones.ToString("0.00"),
64             item.NetoPagar.ToString("0.00"));
65     }
66
67     if (currentItems.Count == 0)
68     {
69         selectedPayslip = null;
70         lblVistaPrevia.Text = "No hay comprobantes disponibles.";
71     }
72 }
73 catch (Exception ex)
74 {
75     MessageBox.Show(ex.Message, "Mis comprobantes", MessageBoxButtons.OK, MessageBoxIcon.Warning);
76 }
77 }
78
79 private void RenderPreview()
80 {
81     if (selectedPayslip == null) return;
82     lblVistaPrevia.Text = "COMPROBANTE DE PAGO\n\n" +
83         "Numero: " + selectedPayslip.NumeroComprobante + "\n" +
84         "Periodo: " + selectedPayslip.PeriodoNombre + "\n" +
85         "Fecha: " + selectedPayslip.FechaGeneracion.ToString("dd/MM/yyyy") + "\n\n" +
86         "Ingresos: B/. " + selectedPayslip.TotalIngresos.ToString("#,##0.00") + "\n" +
87         "Deducciones: B/. " + selectedPayslip.TotalDeducciones.ToString("#,##0.00") + "\n\n" +
88         "NETO A PAGAR: B/. " + selectedPayslip.NetoPagar.ToString("#,##0.00");
89 }
90
91 private bool EnsureSelected()
92 {
93     if (selectedPayslip != null) return true;
94     MessageBox.Show("Selecione un comprobante.", "Mis comprobantes",
95         MessageBoxButtons.OK, MessageBoxIcon.Information);
96     return false;
97 }
98
99 private void btnDescargar_Click(object sender, EventArgs e)
100 {
101     if (!EnsureSelected()) return;
102     try
103     {
104         string path = portalService.DownloadMyPayslipPdf(selectedPayslip.IdComprobante);
105         if (string.IsNullOrEmpty(path)) return;
106         selectedPayslip.RutaPdf = path;
107         UiFactory.ShowExported(path);
108     }
109     catch (Exception ex)
110     {
111         MessageBox.Show(ex.Message, "Descargar PDF", MessageBoxButtons.OK, MessageBoxIcon.Warning);
112     }
113 }
114
115 private void btnAbrirPdf_Click(object sender, EventArgs e)
116 {
117     if (!EnsureSelected()) return;
118     try
119     {
120         string path = portalService.GetExistingMyPayslipPdf(selectedPayslip.IdComprobante);
```

UI/FrmMisComprobantes.cs (continuacion)

```
121         Process.Start(new ProcessStartInfo(path) { UseShellExecute = true });
122     }
123     catch (Exception ex)
124     {
125         MessageBox.Show(ex.Message, "Abrir PDF", MessageBoxButtons.OK, MessageBoxIcon.Warning);
126     }
127 }
128
129 private void btnImprimir_Click(object sender, EventArgs e)
130 {
131     if (!EnsureSelected()) return;
132     try
133     {
134         selectedPayslip = portalService.GetMyPayslipForPrint(selectedPayslip.IdComprobante);
135         using (PrintPreviewDialog preview = new PrintPreviewDialog())
136         {
137             preview.Document = printDocument;
138             preview.Width = 900;
139             preview.Height = 700;
140             preview.StartPosition = FormStartPosition.CenterParent;
141             preview.ShowDialog(this);
142         }
143     }
144     catch (Exception ex)
145     {
146         MessageBox.Show(ex.Message, "Imprimir comprobante", MessageBoxButtons.OK, MessageBoxIcon.Warning);
147     }
148 }
149
150 private void PrintDocument_PrintPage(object sender, PrintPageEventArgs e)
151 {
152     if (selectedPayslip == null)
153     {
154         e.HasMorePages = false;
155         return;
156     }
157
158     ComprobantePrintRenderer.Draw(e.Graphics, e.MarginBounds, selectedPayslip);
159     e.HasMorePages = false;
160 }
161 }
162 }
```


UI/FrmMisComprobantes.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmMisComprobantes
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.TextBox txtBuscar;
8         private System.Windows.Forms.DataGridView dgvComprobantes;
9         private System.Windows.Forms.DataGridViewTextBoxColumn colId;
10        private System.Windows.Forms.DataGridViewTextBoxColumn colNumero;
11        private System.Windows.Forms.DataGridViewTextBoxColumn colPeriodo;
12        private System.Windows.Forms.DataGridViewTextBoxColumn colFecha;
13        private System.Windows.Forms.DataGridViewTextBoxColumn colIngresos;
14        private System.Windows.Forms.DataGridViewTextBoxColumn colDeducciones;
15        private System.Windows.Forms.DataGridViewTextBoxColumn colNeto;
16        private System.Windows.Forms.Panel panelVistaPrevia;
17        private System.Windows.Forms.Label lblVistaPrevia;
18        private System.Windows.Forms.Button btnDescargar;
19        private System.Windows.Forms.Button btnAbrirPdf;
20        private System.Windows.Forms.Button btnImprimir;
21
22        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
23
24        private void InitializeComponent()
25        {
26            this.lblTitulo = new System.Windows.Forms.Label(); this.txtBuscar = new System.Windows.Forms.TextBox(); this.dgvComprobantes = new \
System.Windows.Forms.DataGridView(); this.colId = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.colNumero = new \
System.Windows.Forms.DataGridViewTextBoxColumn(); this.colPeriodo = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.colFecha = new \
System.Windows.Forms.DataGridViewTextBoxColumn(); this.colIngresos = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.colDeducciones = new \
System.Windows.Forms.DataGridViewTextBoxColumn(); this.colNeto = new System.Windows.Forms.DataGridViewTextBoxColumn(); this.panelVistaPrevia = new \
System.Windows.Forms.Panel(); this.lblVistaPrevia = new System.Windows.Forms.Label(); this.btnDescargar = new System.Windows.Forms.Button(); \
this.btnAbrirPdf = new System.Windows.Forms.Button(); this.btnImprimir = new System.Windows.Forms.Button();
27            ((System.ComponentModel.ISupportInitialize)(this.dgvComprobantes)).BeginInit(); this.panelVistaPrevia.SuspendLayout(); this.SuspendLayout();
28            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 23F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Mis comprobantes";
29            this.txtBuscar.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right))); this.txtBuscar.Location = new System.Drawing.Point(666, 36); this.txtBuscar.Name = "txtBuscar"; \
this.txtBuscar.Size = new System.Drawing.Size(320, 27); this.txtBuscar.TextChanged += new System.EventHandler(this.txtBuscar_TextChanged);
30            this.dgvComprobantes.AllowUserToAddRows = false; this.dgvComprobantes.AllowUserToDeleteRows = false; this.dgvComprobantes.Anchor = \
((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | System.Windows.Forms.AnchorStyles.Left) | \
System.Windows.Forms.AnchorStyles.Right)); this.dgvComprobantes.AutoSizeColumnsMode = System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill; \
this.dgvComprobantes.BackgroundColor = System.Drawing.Color.FromArgb(18, 19, 27); this.dgvComprobantes.BorderStyle = \
System.Windows.Forms.BorderStyle.None; this.dgvComprobantes.ColumnHeadersHeight = 40; this.dgvComprobantes.Columns.AddRange(new \
System.Windows.Forms.DataGridColumn[] { this.colId, this.colNumero, this.colPeriodo, this.colFecha, this.colIngresos, this.colDeducciones, \
this.colNeto }); this.dgvComprobantes.Location = new System.Drawing.Point(34, 92); this.dgvComprobantes.MultiSelect = false; \
this.dgvComprobantes.Name = "dgvComprobantes"; this.dgvComprobantes.ReadOnly = true; this.dgvComprobantes.RowHeadersVisible = false; \
this.dgvComprobantes.RowTemplate.Height = 34; this.dgvComprobantes.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect; \
this.dgvComprobantes.Size = new System.Drawing.Size(952, 262); this.dgvComprobantes.SelectionChanged += new \
System.EventHandler(this.dgvComprobantes_SelectionChanged);
31            this.colId.HeaderText = "Id"; this.colId.Name = "colId"; this.colId.ReadOnly = true; this.colId.Visible = false; this.colNumero.HeaderText = \
"Numero"; this.colPeriodo.HeaderText = "Periodo"; this.colFecha.HeaderText = "Fecha"; this.colIngresos.HeaderText = "Ingresos"; \
this.colDeducciones.HeaderText = "Deducciones"; this.colNeto.HeaderText = "Neto";
32            this.panelVistaPrevia.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Bottom) | System.Windows.Forms.AnchorStyles.Left) | System.Windows.Forms.AnchorStyles.Right)); \
this.panelVistaPrevia.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); this.panelVistaPrevia.Controls.Add(this.lblVistaPrevia); \
this.panelVistaPrevia.Location = new System.Drawing.Point(34, 374); this.panelVistaPrevia.Name = "panelVistaPrevia"; this.panelVistaPrevia.Size = new \
System.Drawing.Size(694, 332);
33            this.lblVistaPrevia.Dock = System.Windows.Forms.DockStyle.Fill; this.lblVistaPrevia.Font = new System.Drawing.Font("Consolas", 11F); \
this.lblVistaPrevia.Location = new System.Drawing.Point(0, 0); this.lblVistaPrevia.Name = "lblVistaPrevia"; this.lblVistaPrevia.Padding = new \
System.Windows.Forms.Padding(28); this.lblVistaPrevia.Size = new System.Drawing.Size(694, 332); this.lblVistaPrevia.Text = "Seleccione un comprobante \

```

UI/FrmMisComprobantes.Designer.cs (continuacion)

```
34     para ver el detalle.";
        this.btnDescargar.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right))); this.btnDescargar.Location = new System.Drawing.Point(754, 390); this.btnDescargar.Name = "btnDescargar"; \
this.btnDescargar.Size = new System.Drawing.Size(232, 44); this.btnDescargar.Text = "Descargar PDF"; this.btnDescargar.Click += new \
System.EventHandler(this.btnDescargar_Click);
35     this.btnAbrirPdf.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right))); this.btnAbrirPdf.Location = new System.Drawing.Point(754, 452); this.btnAbrirPdf.Name = "btnAbrirPdf"; \
this.btnAbrirPdf.Size = new System.Drawing.Size(232, 44); this.btnAbrirPdf.Text = "Abrir PDF existente"; this.btnAbrirPdf.Click += new \
System.EventHandler(this.btnAbrirPdf_Click);
36     this.btnImprimir.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right))); this.btnImprimir.Location = new System.Drawing.Point(754, 514); this.btnImprimir.Name = "btnImprimir"; \
this.btnImprimir.Size = new System.Drawing.Size(232, 44); this.btnImprimir.Text = "Vista previa e imprimir"; this.btnImprimir.Click += new \
System.EventHandler(this.btnImprimir_Click);
37     this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F); this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font; this.BackColor = \
System.Drawing.Color.FromArgb(13, 14, 20); this.ClientSize = new System.Drawing.Size(1020, 742); this.Controls.Add(this.btnImprimir); \
this.Controls.Add(this.btnAbrirPdf); this.Controls.Add(this.btnDescargar); this.Controls.Add(this.panelVistaPrevia); \
this.Controls.Add(this.dgvComprobantes); this.Controls.Add(this.txtBuscar); this.Controls.Add(this.lblTitulo); this.Font = new \
System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.Name = "FrmMisComprobantes"; this.Text = \
"Mis comprobantes"; this.Load += new System.EventHandler(this.FrmMisComprobantes_Load); this.panelVistaPrevia.ResumeLayout(false); \
((System.ComponentModel.ISupportInitialize)(this.dgvComprobantes)).EndInit(); this.ResumeLayout(false); this.PerformLayout();
38     }
39 }
40 }
```

UI/FrmNomina.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Services;
8 using SistemaGestionNomina.Security;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmNomina : Form
13     {
14         private readonly NominaService nominaService = new NominaService();
15         private readonly EmpleadoService empleadoService = new EmpleadoService();
16         private readonly ExcelExportService excelExportService = new ExcelExportService();
17         private readonly PdfExportService pdfExportService = new PdfExportService();
18         private readonly AuthorizationService authorizationService = new AuthorizationService();
19         private readonly PayrollLifecycleService payrollLifecycleService = new PayrollLifecycleService();
20         private Nomina currentNomina;
21         private int? currentNominaId;
22
23         public FrmNomina()
24         {
25             InitializeComponent();
26             ControlStyleHelper.ApplyModernForm(this);
27         }
28
29         private void FrmNomina_Load(object sender, EventArgs e)
30         {
31             if (LicenseManager.UsageMode == LicenseUsageMode.DesignTime) return;
32             btnCalcular.Visible = authorizationService.HasPermission(Permissions.PayrollCalculate);
33             btnConfirmarPago.Visible = authorizationService.HasPermission(Permissions.PayrollConfirm);
34             btnExportarExcel.Visible = authorizationService.HasPermission(Permissions.PayrollExport);
35             btnExportarPdf.Visible = authorizationService.HasPermission(Permissions.PayrollExport);
36             btnPagar.Visible = authorizationService.HasPermission(Permissions.PayrollPay);
37             btnAnular.Visible = authorizationService.HasPermission(Permissions.PayrollAnnul);
38             btnRecalcular.Visible = authorizationService.HasPermission(Permissions.PayrollRecalculate);
39             btnVerHistorial.Visible = authorizationService.HasPermission(Permissions.PayrollHistoryView);
40             LoadDepartments();
41             ActualizarEstadoBotones();
42         }
43
44         private void LoadDepartments()
45         {
46             List<Departamento> departments = new List<Departamento>();
47             departments.Add(new Departamento { IdDepartamento = 0, Nombre = "Todos los departamentos" });
48             departments.AddRange(empleadoService.GetDepartments());
49             cmbDepartamento.DataSource = departments;
50             cmbDepartamento.DisplayMember = "Nombre";
51             cmbDepartamento.ValueMember = "IdDepartamento";
52             cmbPeriodo.Text = "Nómina " + DateTime.Today.ToString("MMMM yyyy");
53         }
54
55         private void btnCalcular_Click(object sender, EventArgs e)
56         {
57             if (!ValidationHelper.ValidateDateRange(dtpFechaInicio.Value.Date, dtpFechaFin.Value.Date))
58             {
59                 return;
60             }
61         }
```

UI/FrmNomina.cs (continuacion)

```
61
62     try
63     {
64         int? dep = null;
65         if (cmbDepartamento.SelectedValue is int && (int)cmbDepartamento.SelectedValue > 0)
66         {
67             dep = (int)cmbDepartamento.SelectedValue;
68         }
69         currentNomina = nominaService.CalcularNomina(dtpFechaInicio.Value.Date, dtpFechaFin.Value.Date, dep);
70         if (!string.IsNullOrEmpty(cmbPeriodo.Text))
71         {
72             currentNomina.PeriodoNombre = cmbPeriodo.Text.Trim();
73         }
74         LoadNominaGrid();
75     }
76     catch (Exception ex)
77     {
78         MessageBox.Show(ex.Message, "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Warning);
79     }
80 }
81
82 private void LoadNominaGrid()
83 {
84     dgvNominaDetalle.Rows.Clear();
85     foreach (NominaDetalle d in currentNomina.Detalles)
86     {
87         dgvNominaDetalle.Rows.Add(d.CodigoEmpleado, d.EmpleadoNombre, "B/. " + d.SueldoBase.ToString("0.00"),
88             "B/. " + d.Bonos.ToString("0.00"), d.HorasExtra.ToString("0.00"),
89             "B/. " + d.TotalDeducciones.ToString("0.00"), "B/. " + d.Netopagar.ToString("0.00"));
90     }
91
92     lblTotalIngresos.Text = "Ingresos: B/. " + currentNomina.TotalIngresos.ToString("0.00");
93     lblTotalDeducciones.Text = "Deducciones: B/. " + currentNomina.TotalDeducciones.ToString("0.00");
94     lblTotalNeto.Text = "B/. " + currentNomina.TotalNeto.ToString("0.00");
95     lblProcesados.Text = currentNomina.Detalles.Count + " empleados procesados";
96     lblEstadoNomina.Text = "Estado: " + (currentNomina.Estado ?? "Calculada");
97     ActualizarEstadoBotones();
98 }
99
100 private void btnConfirmarPago_Click(object sender, EventArgs e)
101 {
102     if (!EnsureCalculated()) return;
103     if (MessageBox.Show("¿Desea confirmar el pago y generar comprobantes?", "Confirmar pago",
104         MessageBoxButtons.YesNo, MessageBoxIcon.Question) != DialogResult.Yes)
105     {
106         return;
107     }
108
109     try
110     {
111         int id = nominaService.ConfirmarPago(currentNomina, dtpFechaInicio.Value.Date, dtpFechaFin.Value.Date);
112         currentNominaId = id;
113         currentNomina.Estado = PayrollStates.Confirmed;
114         MessageBox.Show("Nómina confirmada correctamente. Id: " + id + "\nSe generaron los comprobantes.\nEl período ha sido cerrado.",
115             "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Information);
116         ActualizarEstadoBotones();
117     }
118     catch (Exception ex)
119     {
120         MessageBox.Show(ex.Message, "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Warning);
121     }
122 }
```

UI/FrmNomina.cs (continuacion)

```
121     }
122 }
123
124 private void btnPagar_Click(object sender, EventArgs e)
125 {
126     if (!currentNominaId.HasValue)
127     {
128         MessageBox.Show("Primero debe confirmar la nómina.", "Pago", MessageBoxButtons.OK, MessageBoxIcon.Information);
129         return;
130     }
131
132     if (MessageBox.Show("¿Registrar el pago de esta nómina? Esta acción no se puede deshacer.",
133         "Registrar pago", MessageBoxButtons.YesNo, MessageBoxIcon.Question) != DialogResult.Yes)
134     {
135         return;
136     }
137
138     try
139     {
140         nominaService.PagarNomina(currentNominaId.Value);
141         currentNomina.Estado = PayrollStates.Paid;
142         MessageBox.Show("Pago registrado correctamente.", "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Information);
143         ActualizarEstadoBotones();
144     }
145     catch (Exception ex)
146     {
147         MessageBox.Show(ex.Message, "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Warning);
148     }
149 }
150
151 private void btnAnular_Click(object sender, EventArgs e)
152 {
153     if (!currentNominaId.HasValue)
154     {
155         MessageBox.Show("Primero debe confirmar la nómina.", "Anular", MessageBoxButtons.OK, MessageBoxIcon.Information);
156         return;
157     }
158
159     using (FrmAnularNomina dialogo = new FrmAnularNomina(currentNomina))
160     {
161         if (dialogo.ShowDialog() != DialogResult.OK) return;
162
163         try
164         {
165             nominaService.AnularNomina(currentNominaId.Value, dialogo.Motivo);
166             currentNomina.Estado = PayrollStates.Anulled;
167             MessageBox.Show("Nómina anulada correctamente.\nEl período ha sido reabierto para corrección.",
168                 "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Information);
169             ActualizarEstadoBotones();
170         }
171         catch (Exception ex)
172         {
173             MessageBox.Show(ex.Message, "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Warning);
174         }
175     }
176 }
177
178 private void btnRecalcular_Click(object sender, EventArgs e)
179 {
180     if (!currentNominaId.HasValue)
```

UI/FrmNomina.cs (continuacion)

```
181         {
182             MessageBox.Show("Primero debe anular la nómina.", "Recalcular", MessageBoxButtons.OK, MessageBoxIcon.Information);
183             return;
184         }
185
186         try
187         {
188             int? dep = null;
189             if (cmbDepartamento.SelectedValue is int && (int)cmbDepartamento.SelectedValue > 0)
190                 dep = (int)cmbDepartamento.SelectedValue;
191
192             Nomina nuevaNomina = nominaService.CalcularNomina(dtpFechaInicio.Value.Date, dtpFechaFin.Value.Date, dep);
193             if (!string.IsNullOrEmpty(cmbPeriodo.Text))
194                 nuevaNomina.PeriodoNombre = cmbPeriodo.Text.Trim();
195
196             int idNueva = nominaService.RecalcularNomina(currentNominaId.Value,
197                 dtpFechaInicio.Value.Date, dtpFechaFin.Value.Date, nuevaNomina);
198             currentNomina = nuevaNomina;
199             currentNomina.Estado = PayrollStates.Confirmed;
200             currentNominaId = idNueva;
201             LoadNominaGrid();
202             MessageBox.Show("Nómina recalculada y confirmada correctamente. Id: " + idNueva,
203                 "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Information);
204             ActualizarEstadoBotones();
205         }
206         catch (Exception ex)
207         {
208             MessageBox.Show(ex.Message, "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Warning);
209         }
210     }
211
212     private void btnVerHistorial_Click(object sender, EventArgs e)
213     {
214         if (!currentNominaId.HasValue)
215         {
216             MessageBox.Show("Primero debe confirmar la nómina.", "Historial", MessageBoxButtons.OK, MessageBoxIcon.Information);
217             return;
218         }
219
220         List<NominaVersion> versiones = nominaService.ObtenerHistorialNomina(currentNominaId.Value);
221         if (versiones.Count == 0)
222         {
223             MessageBox.Show("No hay versiones registradas para esta nómina.", "Historial",
224                 MessageBoxButtons.OK, MessageBoxIcon.Information);
225             return;
226         }
227
228         using (FrmHistorialNomina dialogo = new FrmHistorialNomina(versiones))
229         {
230             dialogo.ShowDialog();
231         }
232     }
233
234     private void ActualizarEstadoBotones()
235     {
236         string estado = currentNomina?.Estado ?? string.Empty;
237         bool annulled = string.Equals(estado, PayrollStates.Annulled, StringComparison.OrdinalIgnoreCase);
238         btnCalcular.Enabled = authorizationService.HasPermission(Permissions.PayrollCalculate) &&
239             (!currentNominaId.HasValue || annulled);
240         btnConfirmarPago.Enabled = authorizationService.HasPermission(Permissions.PayrollConfirm) &&
```

UI/FrmNomina.cs (continuacion)

```
241         PayrollStateMachine.CanTransition(estado, PayrollStates.Confirmed);
242         btnPagar.Enabled = authorizationService.HasPermission(Permissions.PayrollPay) &&
243         PayrollStateMachine.CanTransition(estado, PayrollStates.Paid);
244         btnAnular.Enabled = authorizationService.HasPermission(Permissions.PayrollAnnul) &&
245         PayrollStateMachine.CanTransition(estado, PayrollStates.Annullled);
246         btnRecalcular.Enabled = authorizationService.HasPermission(Permissions.PayrollRecalculate) && annulled;
247     }
248
249     private void btnExportarExcel_Click(object sender, EventArgs e)
250     {
251         if (EnsureCalculated()) UiFactory.ShowExported(excelExportService.ExportarNomina(currentNomina));
252     }
253
254     private void btnExportarPdf_Click(object sender, EventArgs e)
255     {
256         if (EnsureCalculated()) UiFactory.ShowExported(pdfExportService.ExportarNomina(currentNomina));
257     }
258
259     private bool EnsureCalculated()
260     {
261         if (currentNomina != null && currentNomina.Detalles.Count > 0) return true;
262         MessageBox.Show("Primero calcule la nómina.", "Nómina", MessageBoxButtons.OK, MessageBoxIcon.Information);
263         return false;
264     }
265 }
266 }
```

UI\FrmNomina.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmNomina
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel panelParametros;
8         private System.Windows.Forms.ComboBox cmbPeriodo;
9         private System.Windows.Forms.Label lblPeriodo;
10        private System.Windows.Forms.Label lblFechaInicio;
11        private System.Windows.Forms.Label lblFechaFin;
12        private System.Windows.Forms.Label lblDepartamento;
13        private System.Windows.Forms.Label lblResumen;
14        private System.Windows.Forms.DateTimePicker dtpFechaInicio;
15        private System.Windows.Forms.DateTimePicker dtpFechaFin;
16        private System.Windows.Forms.ComboBox cmbDepartamento;
17        private System.Windows.Forms.Button btnCalcular;
18        private System.Windows.Forms.Panel cardTotales;
19        private System.Windows.Forms.Label lblTotalNeto;
20        private System.Windows.Forms.Label lblProcesados;
21        private System.Windows.Forms.Label lblTotalIngresos;
22        private System.Windows.Forms.Label lblTotalDeducciones;
23        private System.Windows.Forms.Button btnConfirmarPago;
24        private System.Windows.Forms.Button btnExportarExcel;
25        private System.Windows.Forms.Button btnExportarPdf;
26        private System.Windows.Forms.Button btnPagar;
27        private System.Windows.Forms.Button btnAnular;
28        private System.Windows.Forms.Button btnRecalcular;
29        private System.Windows.Forms.Button btnVerHistorial;
30        private System.Windows.Forms.Label lblEstadoNomina;
31        private System.Windows.Forms.DataGridView dgvNominaDetalle;
32
33        protected override void Dispose(bool disposing)
34        {
35            if (disposing)
36            {
37                if (components != null) components.Dispose();
38                if (btnPagar != null) btnPagar.Dispose();
39                if (btnAnular != null) btnAnular.Dispose();
40                if (btnRecalcular != null) btnRecalcular.Dispose();
41                if (btnVerHistorial != null) btnVerHistorial.Dispose();
42                if (lblEstadoNomina != null) lblEstadoNomina.Dispose();
43            }
44            base.Dispose(disposing);
45        }
46
47        private void InitializeComponent()
48        {
49            this.lblTitulo = new System.Windows.Forms.Label();
50            this.panelParametros = new System.Windows.Forms.Panel();
51            this.cmbPeriodo = new System.Windows.Forms.ComboBox();
52            this.lblPeriodo = new System.Windows.Forms.Label();
53            this.lblFechaInicio = new System.Windows.Forms.Label();
54            this.lblFechaFin = new System.Windows.Forms.Label();
55            this.lblDepartamento = new System.Windows.Forms.Label();
56            this.lblResumen = new System.Windows.Forms.Label();
57            this.dtpFechaInicio = new System.Windows.Forms.DateTimePicker();
58            this.dtpFechaFin = new System.Windows.Forms.DateTimePicker();
59            this.cmbDepartamento = new System.Windows.Forms.ComboBox();
60            this.btnCalcular = new System.Windows.Forms.Button();
```


UI/FrmNomina.Designer.cs (continuacion)

```
61         this.cardTotales = new System.Windows.Forms.Panel();
62         this.lblTotalNeto = new System.Windows.Forms.Label();
63         this.lblProcesados = new System.Windows.Forms.Label();
64         this.lblTotalIngresos = new System.Windows.Forms.Label();
65         this.lblTotalDeducciones = new System.Windows.Forms.Label();
66         this.btnConfirmarPago = new System.Windows.Forms.Button();
67         this.btnExportarExcel = new System.Windows.Forms.Button();
68         this.btnExportarPdf = new System.Windows.Forms.Button();
69         this.btnPagar = new System.Windows.Forms.Button();
70         this.btnAnular = new System.Windows.Forms.Button();
71         this.btnRecalcular = new System.Windows.Forms.Button();
72         this.btnVerHistorial = new System.Windows.Forms.Button();
73         this.lblEstadoNomina = new System.Windows.Forms.Label();
74         this.dgvNominaDetalle = new System.Windows.Forms.DataGridView();
75         this.panelParametros.SuspendLayout();
76         this.cardTotales.SuspendLayout();
77         ((System.ComponentModel.ISupportInitialize)(this.dgvNominaDetalle)).BeginInit();
78         this.SuspendLayout();
79         //
80         // lblTitulo
81         //
82         this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
83         this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
84         this.lblTitulo.Location = new System.Drawing.Point(34, 24);
85         this.lblTitulo.Name = "lblTitulo";
86         this.lblTitulo.Size = new System.Drawing.Size(500, 42);
87         this.lblTitulo.TabIndex = 0;
88         this.lblTitulo.Text = "Cálculo de nómina";
89         //
90         // panelParametros
91         //
92         this.panelParametros.BackColor = System.Drawing.Color.Black;
93         this.panelParametros.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
94         this.panelParametros.Controls.Add(this.lblPeriodo);
95         this.panelParametros.Controls.Add(this.lblFechaInicio);
96         this.panelParametros.Controls.Add(this.lblFechaFin);
97         this.panelParametros.Controls.Add(this.lblDepartamento);
98         this.panelParametros.Controls.Add(this.cmbPeriodo);
99         this.panelParametros.Controls.Add(this.dtpFechaInicio);
100        this.panelParametros.Controls.Add(this.dtpFechaFin);
101        this.panelParametros.Controls.Add(this.cmbDepartamento);
102        this.panelParametros.Controls.Add(this.btnCalcular);
103        this.panelParametros.Location = new System.Drawing.Point(36, 96);
104        this.panelParametros.Name = "panelParametros";
105        this.panelParametros.Size = new System.Drawing.Size(330, 250);
106        this.panelParametros.TabIndex = 1;
107        //
108        // cmbPeriodo
109        //
110        this.cmbPeriodo.BackColor = System.Drawing.Color.FromArgb(24, 25, 34);
111        this.cmbPeriodo.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
112        this.cmbPeriodo.Items.AddRange(new object[] { "Nómina mensual", "Primera quincena", "Segunda quincena" });
113        this.cmbPeriodo.Location = new System.Drawing.Point(22, 50);
114        this.cmbPeriodo.Name = "cmbPeriodo";
115        this.cmbPeriodo.Size = new System.Drawing.Size(280, 23);
116        this.cmbPeriodo.TabIndex = 0;
117        //
118        // dtpFechaInicio
119        //
120        this.dtpFechaInicio.Format = System.Windows.Forms.DateTimePickerFormat.Short;
```

UI/FrmNomina.Designer.cs (continuacion)

```
121         this.dtpFechaInicio.Location = new System.Drawing.Point(22, 104);
122         this.dtpFechaInicio.Name = "dtpFechaInicio";
123         this.dtpFechaInicio.Size = new System.Drawing.Size(130, 23);
124         this.dtpFechaInicio.TabIndex = 1;
125         //
126         // dtpFechaFin
127         //
128         this.dtpFechaFin.Format = System.Windows.Forms.DateTimePickerFormat.Short;
129         this.dtpFechaFin.Location = new System.Drawing.Point(172, 104);
130         this.dtpFechaFin.Name = "dtpFechaFin";
131         this.dtpFechaFin.Size = new System.Drawing.Size(130, 23);
132         this.dtpFechaFin.TabIndex = 2;
133         //
134         // cmbDepartamento
135         //
136         this.cmbDepartamento.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList;
137         this.cmbDepartamento.Location = new System.Drawing.Point(22, 160);
138         this.cmbDepartamento.Name = "cmbDepartamento";
139         this.cmbDepartamento.Size = new System.Drawing.Size(280, 23);
140         this.cmbDepartamento.TabIndex = 3;
141         //
142         // btnCalcular
143         //
144         this.btnCalcular.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
145         this.btnCalcular.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
146         this.btnCalcular.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold);
147         this.btnCalcular.Location = new System.Drawing.Point(22, 198);
148         this.btnCalcular.Name = "btnCalcular";
149         this.btnCalcular.Size = new System.Drawing.Size(280, 38);
150         this.btnCalcular.TabIndex = 4;
151         this.btnCalcular.Text = "Calcular nómina";
152         this.btnCalcular.UseVisualStyleBackColor = false;
153         this.btnCalcular.Click += new System.EventHandler(this.btnCalcular_Click);
154         this.lblPeriodo.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
155         this.lblPeriodo.Location = new System.Drawing.Point(22, 28);
156         this.lblPeriodo.Name = "lblPeriodo";
157         this.lblPeriodo.Size = new System.Drawing.Size(130, 18);
158         this.lblPeriodo.TabIndex = 10;
159         this.lblPeriodo.Text = "Periodo";
160         this.lblFechaInicio.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
161         this.lblFechaInicio.Location = new System.Drawing.Point(22, 82);
162         this.lblFechaInicio.Name = "lblFechaInicio";
163         this.lblFechaInicio.Size = new System.Drawing.Size(130, 18);
164         this.lblFechaInicio.TabIndex = 11;
165         this.lblFechaInicio.Text = "Fecha inicio";
166         this.lblFechaFin.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
167         this.lblFechaFin.Location = new System.Drawing.Point(172, 82);
168         this.lblFechaFin.Name = "lblFechaFin";
169         this.lblFechaFin.Size = new System.Drawing.Size(130, 18);
170         this.lblFechaFin.TabIndex = 12;
171         this.lblFechaFin.Text = "Fecha fin";
172         this.lblDepartamento.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
173         this.lblDepartamento.Location = new System.Drawing.Point(22, 138);
174         this.lblDepartamento.Name = "lblDepartamento";
175         this.lblDepartamento.Size = new System.Drawing.Size(130, 18);
176         this.lblDepartamento.TabIndex = 13;
177         this.lblDepartamento.Text = "Departamento";
178         //
179         // cardTotales
180         //
```

UI/FrmNomina.Designer.cs (continuacion)

```
181         this.cardTotales.BackColor = System.Drawing.Color.FromArgb(17, 18, 26);
182         this.cardTotales.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
183         this.cardTotales.Controls.Add(this.lblResumen);
184         this.cardTotales.Controls.Add(this.lblTotalNeto);
185         this.cardTotales.Controls.Add(this.lblProcesados);
186         this.cardTotales.Controls.Add(this.lblTotalIngresos);
187         this.cardTotales.Controls.Add(this.lblTotalDeducciones);
188         this.cardTotales.Controls.Add(this.btnConfirmarPago);
189         this.cardTotales.Controls.Add(this.btnPagar);
190         this.cardTotales.Controls.Add(this.btnAnular);
191         this.cardTotales.Controls.Add(this.btnRecalcular);
192         this.cardTotales.Controls.Add(this.btnVerHistorial);
193         this.cardTotales.Controls.Add(this.btnExportarExcel);
194         this.cardTotales.Controls.Add(this.btnExportarPdf);
195         this.cardTotales.Controls.Add(this.lblEstadoNomina);
196         this.cardTotales.Location = new System.Drawing.Point(392, 96);
197         this.cardTotales.Name = "cardTotales";
198         this.cardTotales.Size = new System.Drawing.Size(592, 250);
199         this.cardTotales.TabIndex = 2;
200         //
201         // lblTotalNeto
202         //
203         this.lblTotalNeto.Font = new System.Drawing.Font("Segoe UI", 24F, System.Drawing.FontStyle.Bold);
204         this.lblTotalNeto.ForeColor = System.Drawing.Color.FromArgb(139, 92, 246);
205         this.lblTotalNeto.Location = new System.Drawing.Point(24, 54);
206         this.lblTotalNeto.Name = "lblTotalNeto";
207         this.lblTotalNeto.Size = new System.Drawing.Size(280, 48);
208         this.lblTotalNeto.TabIndex = 0;
209         this.lblTotalNeto.Text = "B/. 0.00";
210         //
211         // lblProcesados
212         //
213         this.lblProcesados.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
214         this.lblProcesados.Location = new System.Drawing.Point(28, 104);
215         this.lblProcesados.Name = "lblProcesados";
216         this.lblProcesados.Size = new System.Drawing.Size(250, 24);
217         this.lblProcesados.TabIndex = 1;
218         this.lblProcesados.Text = "0 empleados procesados";
219         //
220         // lblTotalIngresos
221         //
222         this.lblTotalIngresos.ForeColor = System.Drawing.Color.FromArgb(34, 197, 94);
223         this.lblTotalIngresos.Location = new System.Drawing.Point(28, 146);
224         this.lblTotalIngresos.Name = "lblTotalIngresos";
225         this.lblTotalIngresos.Size = new System.Drawing.Size(230, 26);
226         this.lblTotalIngresos.TabIndex = 2;
227         this.lblTotalIngresos.Text = "Ingresos: B/. 0.00";
228         //
229         // lblTotalDeducciones
230         //
231         this.lblTotalDeducciones.ForeColor = System.Drawing.Color.FromArgb(239, 68, 68);
232         this.lblTotalDeducciones.Location = new System.Drawing.Point(28, 180);
233         this.lblTotalDeducciones.Name = "lblTotalDeducciones";
234         this.lblTotalDeducciones.Size = new System.Drawing.Size(250, 26);
235         this.lblTotalDeducciones.TabIndex = 3;
236         this.lblTotalDeducciones.Text = "Deducciones: B/. 0.00";
237         this.lblResumen.Font = new System.Drawing.Font("Segoe UI", 12F, System.Drawing.FontStyle.Bold);
238         this.lblResumen.ForeColor = System.Drawing.Color.FromArgb(170, 170, 185);
239         this.lblResumen.Location = new System.Drawing.Point(24, 20);
240         this.lblResumen.Name = "lblResumen";
```

UI/FrmNomina.Designer.cs (continuacion)

```
241         this.lblResumen.Size = new System.Drawing.Size(260, 26);
242         this.lblResumen.TabIndex = 7;
243         this.lblResumen.Text = "Neto a pagar";
244         //
245         // btnConfirmarPago
246         //
247         this.btnConfirmarPago.BackColor = System.Drawing.Color.FromArgb(167, 139, 250);
248         this.btnConfirmarPago.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
249         this.btnConfirmarPago.Location = new System.Drawing.Point(390, 34);
250         this.btnConfirmarPago.Name = "btnConfirmarPago";
251         this.btnConfirmarPago.Size = new System.Drawing.Size(150, 36);
252         this.btnConfirmarPago.TabIndex = 4;
253         this.btnConfirmarPago.Text = "Confirmar pago";
254         this.btnConfirmarPago.UseVisualStyleBackColor = false;
255         this.btnConfirmarPago.Click += new System.EventHandler(this.btnConfirmarPago_Click);
256         //
257         // btnExportarExcel
258         //
259         this.btnExportarExcel.BackColor = System.Drawing.Color.Black;
260         this.btnExportarExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
261         this.btnExportarExcel.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
262         this.btnExportarExcel.Location = new System.Drawing.Point(390, 96);
263         this.btnExportarExcel.Name = "btnExportarExcel";
264         this.btnExportarExcel.Size = new System.Drawing.Size(80, 32);
265         this.btnExportarExcel.TabIndex = 5;
266         this.btnExportarExcel.Text = "Excel";
267         this.btnExportarExcel.UseVisualStyleBackColor = false;
268         this.btnExportarExcel.Click += new System.EventHandler(this.btnExportarExcel_Click);
269         //
270         // btnExportarPdf
271         //
272         this.btnExportarPdf.BackColor = System.Drawing.Color.Black;
273         this.btnExportarPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
274         this.btnExportarPdf.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
275         this.btnExportarPdf.Location = new System.Drawing.Point(480, 96);
276         this.btnExportarPdf.Name = "btnExportarPdf";
277         this.btnExportarPdf.Size = new System.Drawing.Size(60, 32);
278         this.btnExportarPdf.TabIndex = 6;
279         this.btnExportarPdf.Text = "PDF";
280         this.btnExportarPdf.UseVisualStyleBackColor = false;
281         this.btnExportarPdf.Click += new System.EventHandler(this.btnExportarPdf_Click);
282         //
283         // btnPagar
284         //
285         this.btnPagar.BackColor = System.Drawing.Color.FromArgb(34, 197, 94);
286         this.btnPagar.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
287         this.btnPagar.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
288         this.btnPagar.ForeColor = System.Drawing.Color.White;
289         this.btnPagar.Location = new System.Drawing.Point(390, 140);
290         this.btnPagar.Name = "btnPagar";
291         this.btnPagar.Size = new System.Drawing.Size(80, 32);
292         this.btnPagar.TabIndex = 7;
293         this.btnPagar.Text = "Pagar";
294         this.btnPagar.UseVisualStyleBackColor = false;
295         this.btnPagar.Click += new System.EventHandler(this.btnPagar_Click);
296         //
297         // btnAnular
298         //
299         this.btnAnular.BackColor = System.Drawing.Color.FromArgb(239, 68, 68);
300         this.btnAnular.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
```

UI/FrmNomina.Designer.cs (continuacion)

```
301         this.btnAnular.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
302         this.btnAnular.ForeColor = System.Drawing.Color.White;
303         this.btnAnular.Location = new System.Drawing.Point(480, 140);
304         this.btnAnular.Name = "btnAnular";
305         this.btnAnular.Size = new System.Drawing.Size(80, 32);
306         this.btnAnular.TabIndex = 8;
307         this.btnAnular.Text = "Anular";
308         this.btnAnular.UseVisualStyleBackColor = false;
309         this.btnAnular.Click += new System.EventHandler(this.btnAnular_Click);
310         //
311         // btnRecalcular
312         //
313         this.btnRecalcular.BackColor = System.Drawing.Color.FromArgb(250, 204, 21);
314         this.btnRecalcular.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
315         this.btnRecalcular.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
316         this.btnRecalcular.ForeColor = System.Drawing.Color.Black;
317         this.btnRecalcular.Location = new System.Drawing.Point(390, 180);
318         this.btnRecalcular.Name = "btnRecalcular";
319         this.btnRecalcular.Size = new System.Drawing.Size(100, 32);
320         this.btnRecalcular.TabIndex = 9;
321         this.btnRecalcular.Text = "Recalcular";
322         this.btnRecalcular.UseVisualStyleBackColor = false;
323         this.btnRecalcular.Click += new System.EventHandler(this.btnRecalcular_Click);
324         //
325         // btnVerHistorial
326         //
327         this.btnVerHistorial.BackColor = System.Drawing.Color.Black;
328         this.btnVerHistorial.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
329         this.btnVerHistorial.ForeColor = System.Drawing.Color.FromArgb(167, 139, 250);
330         this.btnVerHistorial.Location = new System.Drawing.Point(500, 180);
331         this.btnVerHistorial.Name = "btnVerHistorial";
332         this.btnVerHistorial.Size = new System.Drawing.Size(80, 32);
333         this.btnVerHistorial.TabIndex = 10;
334         this.btnVerHistorial.Text = "Historial";
335         this.btnVerHistorial.UseVisualStyleBackColor = false;
336         this.btnVerHistorial.Click += new System.EventHandler(this.btnVerHistorial_Click);
337         //
338         // lblEstadoNomina
339         //
340         this.lblEstadoNomina.Font = new System.Drawing.Font("Segoe UI", 9F, System.Drawing.FontStyle.Bold);
341         this.lblEstadoNomina.ForeColor = System.Drawing.Color.FromArgb(167, 139, 250);
342         this.lblEstadoNomina.Location = new System.Drawing.Point(24, 214);
343         this.lblEstadoNomina.Name = "lblEstadoNomina";
344         this.lblEstadoNomina.Size = new System.Drawing.Size(260, 20);
345         this.lblEstadoNomina.TabIndex = 11;
346         this.lblEstadoNomina.Text = "Estado: -";
347         //
348         // dgvNominaDetalle
349         //
350         this.dgvNominaDetalle.AllowUserToAddRows = false;
351         this.dgvNominaDetalle.AllowUserToDeleteRows = false;
352         this.dgvNominaDetalle.BackColor = System.Drawing.Color.Black;
353         this.dgvNominaDetalle.ColumnHeadersHeightSizeMode = System.Windows.Forms.DataGridViewColumnHeadersHeightSizeMode.AutoSize;
354         this.dgvNominaDetalle.Location = new System.Drawing.Point(36, 382);
355         this.dgvNominaDetalle.Name = "dgvNominaDetalle";
356         this.dgvNominaDetalle.ReadOnly = true;
357         this.dgvNominaDetalle.RowHeadersVisible = false;
358         this.dgvNominaDetalle.Size = new System.Drawing.Size(948, 318);
359         this.dgvNominaDetalle.TabIndex = 3;
360         this.dgvNominaDetalle.Columns.Add("Codigo", "Código");
```

UI/FrmNomina.Designer.cs (continuacion)

```
361         this.dgvNominaDetalle.Columns.Add("Empleado", "Empleado");
362         this.dgvNominaDetalle.Columns.Add("SueldoBase", "Sueldo base");
363         this.dgvNominaDetalle.Columns.Add("Bonos", "Bonos");
364         this.dgvNominaDetalle.Columns.Add("HorasExtra", "Horas extra");
365         this.dgvNominaDetalle.Columns.Add("Deducciones", "Deducciones");
366         this.dgvNominaDetalle.Columns.Add("Neto", "Neto a pagar");
367         //
368         // FrmNomina
369         //
370         this.AutoScaleDimensions = new System.Drawing.SizeF(7F, 15F);
371         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
372         this.BackColor = System.Drawing.Color.FromArgb(13, 14, 20);
373         this.ClientSize = new System.Drawing.Size(1020, 742);
374         this.Controls.Add(this.dgvNominaDetalle);
375         this.Controls.Add(this.cardTotales);
376         this.Controls.Add(this.panelParametros);
377         this.Controls.Add(this.lblTitulo);
378         this.Font = new System.Drawing.Font("Segoe UI", 9F);
379         this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245);
380         this.Name = "FrmNomina";
381         this.Text = "Nómina";
382         this.Load += new System.EventHandler(this.FrmNomina_Load);
383         this.panelParametros.ResumeLayout(false);
384         this.cardTotales.ResumeLayout(false);
385         ((System.ComponentModel.ISupportInitialize)(this.dgvNominaDetalle)).EndInit();
386         this.ResumeLayout(false);
387     }
388 }
389 }
```

UI/FrmPortalTrabajador.cs

```
1 using System;
2 using System.ComponentModel;
3 using System.Windows.Forms;
4 using SistemaGestionNomina.Helpers;
5 using SistemaGestionNomina.Models;
6 using SistemaGestionNomina.Services;
7
8 namespace SistemaGestionNomina.UI
9 {
10     public partial class FrmPortalTrabajador : Form
11     {
12         private readonly EmployeePortalService portalService = new EmployeePortalService();
13         private readonly Action openProfile;
14         private readonly Action openAttendance;
15         private readonly Action openPayslips;
16         private readonly Action openPassword;
17         private readonly Action openAbsences;
18
19         public FrmPortalTrabajador() : this(null, null, null, null, null) { }
20
21         public FrmPortalTrabajador(Action profile, Action attendance, Action payslips, Action password,
22             Action absences)
23         {
24             openProfile = profile;
25             openAttendance = attendance;
26             openPayslips = payslips;
27             openPassword = password;
28             openAbsences = absences;
29             InitializeComponent();
30             ControlStyleHelper.ApplyModernForm(this);
31         }
32
33         private void FrmPortalTrabajador_Load(object sender, EventArgs e)
34         {
35             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
36             try
37             {
38                 EmployeePortalDashboardViewModel model = portalService.GetDashboard();
39                 lblBienvenida.Text = "Hola, " + model.Profile.NombreCompleto;
40                 lblCargoValor.Text = model.Profile.Cargo;
41                 lblDepartamentoValor.Text = model.Profile.Departamento;
42                 lblCodigoValor.Text = model.Profile.Codigo;
43                 lblNetoValor.Text = model.LatestPayslip == null
44                     ? "Sin comprobantes"
45                     : "B/. " + model.LatestPayslip.NetoPagar.ToString("#,##0.00");
46                 lblPeriodoValor.Text = model.LatestPayslip == null ? "No disponible" : model.LatestPayslip.PeriodoNombre;
47                 lblAsistenciaValor.Text = model.AttendanceSummary.Puntuales + " puntuales / " +
48                     model.AttendanceSummary.Tardanzas + " tardanzas";
49
50                 dgvRecientes.Rows.Clear();
51                 foreach (Asistencia item in model.RecentAttendance)
52                 {
53                     dgvRecientes.Rows.Add(item.Fecha.ToString("dd/MM/yyyy"), item.Estado,
54                         item.HoraEntrada.HasValue ? item.HoraEntrada.Value.ToString(@"hh\:mm") : "-",
55                         item.HorasTrabajadas.ToString("0.00"));
56                 }
57             }
58             catch (Exception ex)
59             {
60                 MessageBox.Show(ex.Message, "Portal del trabajador", MessageBoxButtons.OK, MessageBoxIcon.Warning);
61             }
62         }
63     }
64 }
```

UI/FrmPortalTrabajador.cs (continuación)

```
61         }
62     }
63
64     private void btnMiPerfil_Click(object sender, EventArgs e) { if (openProfile != null) openProfile(); }
65     private void btnMisAsistencias_Click(object sender, EventArgs e) { if (openAttendance != null) openAttendance(); }
66     private void btnMisComprobantes_Click(object sender, EventArgs e) { if (openPayslips != null) openPayslips(); }
67     private void btnCambiarPassword_Click(object sender, EventArgs e) { if (openPassword != null) openPassword(); }
68     private void btnMisSolicitudes_Click(object sender, EventArgs e) { if (openAbsences != null) openAbsences(); }
69 }
70 }
```


UI\FrmPortalTrabajador.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmPortalTrabajador
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblBienvenida;
7         private System.Windows.Forms.Label lblSubtitulo;
8         private System.Windows.Forms.Panel panelPerfil;
9         private System.Windows.Forms.Label lblPerfilTitulo;
10        private System.Windows.Forms.Label lblCargoValor;
11        private System.Windows.Forms.Label lblDepartamentoValor;
12        private System.Windows.Forms.Label lblCodigoValor;
13        private System.Windows.Forms.Panel panelPago;
14        private System.Windows.Forms.Label lblPagoTitulo;
15        private System.Windows.Forms.Label lblNetoValor;
16        private System.Windows.Forms.Label lblPeriodoValor;
17        private System.Windows.Forms.Panel panelAsistencia;
18        private System.Windows.Forms.Label lblAsistenciaTitulo;
19        private System.Windows.Forms.Label lblAsistenciaValor;
20        private System.Windows.Forms.DataGridView dgvRecientes;
21        private System.Windows.Forms.DataGridViewTextBoxColumn colFecha;
22        private System.Windows.Forms.DataGridViewTextBoxColumn colEstado;
23        private System.Windows.Forms.DataGridViewTextBoxColumn colEntrada;
24        private System.Windows.Forms.DataGridViewTextBoxColumn colHoras;
25        private System.Windows.Forms.Button btnMiPerfil;
26        private System.Windows.Forms.Button btnMisAsistencias;
27        private System.Windows.Forms.Button btnMisComprobantes;
28        private System.Windows.Forms.Button btnCambiarPassword;
29        private System.Windows.Forms.Button btnMisSolicitudes;
30
31        protected override void Dispose(bool disposing)
32        {
33            if (disposing && components != null) components.Dispose();
34            base.Dispose(disposing);
35        }
36
37        private void InitializeComponent()
38        {
39            this.lblBienvenida = new System.Windows.Forms.Label();
40            this.lblSubtitulo = new System.Windows.Forms.Label();
41            this.panelPerfil = new System.Windows.Forms.Panel();
42            this.lblPerfilTitulo = new System.Windows.Forms.Label();
43            this.lblCargoValor = new System.Windows.Forms.Label();
44            this.lblDepartamentoValor = new System.Windows.Forms.Label();
45            this.lblCodigoValor = new System.Windows.Forms.Label();
46            this.panelPago = new System.Windows.Forms.Panel();
47            this.lblPagoTitulo = new System.Windows.Forms.Label();
48            this.lblNetoValor = new System.Windows.Forms.Label();
49            this.lblPeriodoValor = new System.Windows.Forms.Label();
50            this.panelAsistencia = new System.Windows.Forms.Panel();
51            this.lblAsistenciaTitulo = new System.Windows.Forms.Label();
52            this.lblAsistenciaValor = new System.Windows.Forms.Label();
53            this.dgvRecientes = new System.Windows.Forms.DataGridView();
54            this.colFecha = new System.Windows.Forms.DataGridViewTextBoxColumn();
55            this.colEstado = new System.Windows.Forms.DataGridViewTextBoxColumn();
56            this.colEntrada = new System.Windows.Forms.DataGridViewTextBoxColumn();
57            this.colHoras = new System.Windows.Forms.DataGridViewTextBoxColumn();
58            this.btnMiPerfil = new System.Windows.Forms.Button();
59            this.btnMisAsistencias = new System.Windows.Forms.Button();
60            this.btnMisComprobantes = new System.Windows.Forms.Button();
```

UI/FrmPortalTrabajador.Designer.cs (continuacion)

```

61         this.btnCambiarPassword = new System.Windows.Forms.Button();
62         this.btnMisSolicitudes = new System.Windows.Forms.Button();
63         this.panelPerfil.SuspendLayout();
64         this.panelPago.SuspendLayout();
65         this.panelAsistencia.SuspendLayout();
66         ((System.ComponentModel.ISupportInitialize)(this.dgvRecientes)).BeginInit();
67         this.SuspendLayout();
68         this.lblBienvenida.AutoSize = true; this.lblBienvenida.Font = new System.Drawing.Font("Segoe UI", 23F, System.Drawing.FontStyle.Bold); \
this.lblBienvenida.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.lblBienvenida.Location = new System.Drawing.Point(30, 22); \
this.lblBienvenida.Name = "lblBienvenida"; this.lblBienvenida.Size = new System.Drawing.Size(355, 52); this.lblBienvenida.Text = "Portal del \
trabajador";
69         this.lblSubtitulo.AutoSize = true; this.lblSubtitulo.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblSubtitulo.Location = \
new System.Drawing.Point(34, 76); this.lblSubtitulo.Name = "lblSubtitulo"; this.lblSubtitulo.Size = new System.Drawing.Size(352, 20); \
this.lblSubtitulo.Text = "Consulta tu informacion laboral y comprobantes.";
70         this.panelPerfil.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); this.panelPerfil.Controls.Add(this.lblPerfilTitulo); \
this.panelPerfil.Controls.Add(this.lblCargoValor); this.panelPerfil.Controls.Add(this.lblDepartamentoValor); \
this.panelPerfil.Controls.Add(this.lblCodigoValor); this.panelPerfil.Location = new System.Drawing.Point(34, 116); this.panelPerfil.Name = \
"panelPerfil"; this.panelPerfil.Size = new System.Drawing.Size(292, 154);
71         this.lblPerfilTitulo.AutoSize = true; this.lblPerfilTitulo.Font = new System.Drawing.Font("Segoe UI", 11F, System.Drawing.FontStyle.Bold); \
this.lblPerfilTitulo.Location = new System.Drawing.Point(18, 17); this.lblPerfilTitulo.Text = "Resumen laboral";
72         this.lblCargoValor.AutoEllipsis = true; this.lblCargoValor.Font = new System.Drawing.Font("Segoe UI", 10F, System.Drawing.FontStyle.Bold); \
this.lblCargoValor.Location = new System.Drawing.Point(18, 56); this.lblCargoValor.Size = new System.Drawing.Size(252, 25); this.lblCargoValor.Text = \
"Cargo";
73         this.lblDepartamentoValor.AutoEllipsis = true; this.lblDepartamentoValor.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); \
this.lblDepartamentoValor.Location = new System.Drawing.Point(18, 86); this.lblDepartamentoValor.Size = new System.Drawing.Size(252, 24); \
this.lblDepartamentoValor.Text = "Departamento";
74         this.lblCodigoValor.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblCodigoValor.Location = new System.Drawing.Point(18, \
116); this.lblCodigoValor.Size = new System.Drawing.Size(252, 24); this.lblCodigoValor.Text = "Codigo";
75         this.panelPago.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); this.panelPago.Controls.Add(this.lblPagoTitulo); \
this.panelPago.Controls.Add(this.lblNetoValor); this.panelPago.Controls.Add(this.lblPeriodoValor); this.panelPago.Location = new \
System.Drawing.Point(346, 116); this.panelPago.Name = "panelPago"; this.panelPago.Size = new System.Drawing.Size(292, 154);
76         this.lblPagoTitulo.AutoSize = true; this.lblPagoTitulo.Font = new System.Drawing.Font("Segoe UI", 11F, System.Drawing.FontStyle.Bold); \
this.lblPagoTitulo.Location = new System.Drawing.Point(18, 17); this.lblPagoTitulo.Text = "Ultimo comprobante";
77         this.lblNetoValor.AutoEllipsis = true; this.lblNetoValor.Font = new System.Drawing.Font("Segoe UI", 19F, System.Drawing.FontStyle.Bold); \
this.lblNetoValor.ForeColor = System.Drawing.Color.FromArgb(52, 211, 153); this.lblNetoValor.Location = new System.Drawing.Point(18, 52); \
this.lblNetoValor.Size = new System.Drawing.Size(252, 44); this.lblNetoValor.Text = "Sin comprobantes";
78         this.lblPeriodoValor.ForeColor = System.Drawing.Color.FromArgb(161, 161, 175); this.lblPeriodoValor.Location = new System.Drawing.Point(18, \
111); this.lblPeriodoValor.Size = new System.Drawing.Size(252, 24); this.lblPeriodoValor.Text = "Periodo";
79         this.panelAsistencia.Anchor = ((System.Windows.Forms.AnchorStyles)((System.Windows.Forms.AnchorStyles.Top | \
System.Windows.Forms.AnchorStyles.Right))); this.panelAsistencia.BackColor = System.Drawing.Color.FromArgb(24, 25, 34); \
this.panelAsistencia.Controls.Add(this.lblAsistenciaTitulo); this.panelAsistencia.Controls.Add(this.lblAsistenciaValor); \
this.panelAsistencia.Location = new System.Drawing.Point(658, 116); this.panelAsistencia.Name = "panelAsistencia"; this.panelAsistencia.Size = new \
System.Drawing.Size(328, 154);
80         this.lblAsistenciaTitulo.AutoSize = true; this.lblAsistenciaTitulo.Font = new System.Drawing.Font("Segoe UI", 11F, \
System.Drawing.FontStyle.Bold); this.lblAsistenciaTitulo.Location = new System.Drawing.Point(18, 17); this.lblAsistenciaTitulo.Text = "Asistencia del \
mes";
81         this.lblAsistenciaValor.Font = new System.Drawing.Font("Segoe UI", 13F, System.Drawing.FontStyle.Bold); this.lblAsistenciaValor.ForeColor = \
System.Drawing.Color.FromArgb(167, 139, 250); this.lblAsistenciaValor.Location = new System.Drawing.Point(18, 59); this.lblAsistenciaValor.Size = new \
System.Drawing.Size(286, 62); this.lblAsistenciaValor.Text = "0 puntuales / 0 tardanzas";
82         this.btnMiPerfil.Location = new System.Drawing.Point(34, 292); this.btnMiPerfil.Name = "btnMiPerfil"; this.btnMiPerfil.Size = new \
System.Drawing.Size(176, 42); this.btnMiPerfil.Text = "Mi perfil"; this.btnMiPerfil.Click += new System.EventHandler(this.btnMiPerfil_Click);
83         this.btnMisAsistencias.Location = new System.Drawing.Point(228, 292); this.btnMisAsistencias.Name = "btnMisAsistencias"; \
this.btnMisAsistencias.Size = new System.Drawing.Size(176, 42); this.btnMisAsistencias.Text = "Mis asistencias"; this.btnMisAsistencias.Click += new \
System.EventHandler(this.btnMisAsistencias_Click);
84         this.btnMisComprobantes.Location = new System.Drawing.Point(422, 292); this.btnMisComprobantes.Name = "btnMisComprobantes"; \
this.btnMisComprobantes.Size = new System.Drawing.Size(176, 42); this.btnMisComprobantes.Text = "Mis comprobantes"; this.btnMisComprobantes.Click += \
new System.EventHandler(this.btnMisComprobantes_Click);
85         this.btnMisSolicitudes.Location = new System.Drawing.Point(616, 292); this.btnMisSolicitudes.Name = "btnMisSolicitudes"; \
this.btnMisSolicitudes.Size = new System.Drawing.Size(176, 42); this.btnMisSolicitudes.Text = "Mis solicitudes"; this.btnMisSolicitudes.Click += new \
System.EventHandler(this.btnMisSolicitudes_Click);

```

UI/FrmPortalTrabajador.Designer.cs (continuacion)

```
86         this.btnCambiarPassword.Location = new System.Drawing.Point(810, 292); this.btnCambiarPassword.Name = "btnCambiarPassword"; \
this.btnCambiarPassword.Size = new System.Drawing.Size(176, 42); this.btnCambiarPassword.Text = "Cambiar contraseña"; this.btnCambiarPassword.Click \
+= new System.EventHandler(this.btnCambiarPassword_Click);
87         this.dgvRecientes.AllowUserToAddRows = false; this.dgvRecientes.AllowUserToDeleteRows = false; this.dgvRecientes.Anchor = \
((System.Windows.Forms.AnchorStyles)((((System.Windows.Forms.AnchorStyles.Top | System.Windows.Forms.AnchorStyles.Bottom) | \
System.Windows.Forms.AnchorStyles.Left) | System.Windows.Forms.AnchorStyles.Right))); this.dgvRecientes.AutoSizeColumnsMode = \
System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill; this.dgvRecientes.BackgroundColor = System.Drawing.Color.FromArgb(18, 19, 27); \
this.dgvRecientes.BorderStyle = System.Windows.Forms.BorderStyle.None; this.dgvRecientes.ColumnHeadersHeight = 40; \
this.dgvRecientes.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] { this.colFecha, this.colEstado, this.colEntrada, this.colHoras }); \
this.dgvRecientes.Location = new System.Drawing.Point(34, 362); this.dgvRecientes.Name = "dgvRecientes"; this.dgvRecientes.ReadOnly = true; \
this.dgvRecientes.RowHeadersVisible = false; this.dgvRecientes.RowTemplate.Height = 36; this.dgvRecientes.SelectionMode = \
System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect; this.dgvRecientes.Size = new System.Drawing.Size(952, 340);
88         this.colFecha.HeaderText = "Fecha"; this.colFecha.Name = "colFecha"; this.colFecha.ReadOnly = true;
89         this.colEstado.HeaderText = "Estado"; this.colEstado.Name = "colEstado"; this.colEstado.ReadOnly = true;
90         this.colEntrada.HeaderText = "Entrada"; this.colEntrada.Name = "colEntrada"; this.colEntrada.ReadOnly = true;
91         this.colHoras.HeaderText = "Horas"; this.colHoras.Name = "colHoras"; this.colHoras.ReadOnly = true;
92         this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F); this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font; this.BackColor = \
System.Drawing.Color.FromArgb(13, 14, 20); this.ClientSize = new System.Drawing.Size(1020, 742); this.Controls.Add(this.dgvRecientes); \
this.Controls.Add(this.btnCambiarPassword); this.Controls.Add(this.btnMisSolicitudes); this.Controls.Add(this.btnMisComprobantes); \
this.Controls.Add(this.btnMisAsistencias); this.Controls.Add(this.btnMiPerfil); this.Controls.Add(this.panelAsistencia); \
this.Controls.Add(this.panelPago); this.Controls.Add(this.panelPerfil); this.Controls.Add(this.lblSubtitulo); this.Controls.Add(this.lblBienvenida); \
this.Font = new System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.Name = \
"FrmPortalTrabajador"; this.Text = "Portal del trabajador"; this.Load += new System.EventHandler(this.FrmPortalTrabajador_Load);
93         this.panelPerfil.ResumeLayout(false); this.panelPerfil.PerformLayout(); this.panelPago.ResumeLayout(false); this.panelPago.PerformLayout(); \
this.panelAsistencia.ResumeLayout(false); this.panelAsistencia.PerformLayout(); \
((System.ComponentModel.ISupportInitialize)(this.dgvRecientes)).EndInit(); this.ResumeLayout(false); this.PerformLayout();
94     }
95 }
96 }
```

UI/FrmReportes.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Services;
8 using SistemaGestionNomina.Security;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmReportes : Form
13     {
14         private readonly ReporteService reporteService = new ReporteService();
15         private readonly EmpleadoService empleadoService = new EmpleadoService();
16         private readonly ExcelExportService excelExportService = new ExcelExportService();
17         private readonly PdfExportService pdfExportService = new PdfExportService();
18         private readonly AuthorizationService authorizationService = new AuthorizationService();
19
20         public FrmReportes()
21         {
22             InitializeComponent();
23             ControlStyleHelper.ApplyModernForm(this);
24         }
25
26         private void FrmReportes_Load(object sender, EventArgs e)
27         {
28             if (LicenseManager.UsageMode != LicenseUsageMode.Designtime)
29             {
30                 bool personal = authorizationService.HasPermission(Permissions.ReportsPersonal);
31                 bool financial = authorizationService.HasPermission(Permissions.ReportsFinancial);
32                 cardEmpleados.Visible = personal;
33                 cardNomina.Visible = financial;
34                 cardPagos.Visible = financial;
35                 cardDeducciones.Visible = financial;
36                 btnExportarExcel.Visible = personal && financial;
37                 btnExportarPdf.Visible = personal && financial;
38                 LoadReports();
39             }
40         }
41
42         private void btnReporteNomina_Click(object sender, EventArgs e)
43         {
44             authorizationService.DemandPermission(Permissions.ReportsFinancial);
45             GenerateReport("Reporte general de nómina", "General", "PDF");
46         }
47
48         private void btnReporteEmpleados_Click(object sender, EventArgs e)
49         {
50             authorizationService.DemandPermission(Permissions.ReportsPersonal);
51             GenerateReport("Empleados activos", "Personal", "PDF");
52         }
53
54         private void btnReportePagos_Click(object sender, EventArgs e)
55         {
56             authorizationService.DemandPermission(Permissions.ReportsFinancial);
57             GenerateReport("Historial de pagos", "Histórico", "PDF");
58         }
59
60         private void btnReporteDeducciones_Click(object sender, EventArgs e)
```

UI/FrmReportes.cs (continuacion)

```
61     {
62         authorizationService.DemandPermission(Permissions.ReportsFinacial);
63         GenerateReport("Resumen de deducciones", "Deducciones", "PDF");
64     }
65
66     private void btnExportarExcel_Click(object sender, EventArgs e)
67     {
68         GenerateReport("Reportes administrativos", "General", "Excel");
69     }
70
71     private void btnExportarPdf_Click(object sender, EventArgs e)
72     {
73         GenerateReport("Reportes administrativos", "General", "PDF");
74     }
75
76     private void GenerateReport(string name, string type, string format)
77     {
78         try
79         {
80             string path;
81             if (name == "Empleados activos" && format == "Excel")
82             {
83                 List<Empleado> empleados = empleadoService.GetActive(null);
84                 if (empleados.Count == 0)
85                 {
86                     MessageBox.Show("No hay empleados activos para generar el reporte.", "Reportes", MessageBoxButtons.OK, MessageBoxIcon.Information);
87                     return;
88                 }
89
90                 path = excelExportService.ExportarEmpleados(empleados);
91             }
92             else if (name == "Empleados activos")
93             {
94                 List<Empleado> empleados = empleadoService.GetActive(null);
95                 if (empleados.Count == 0)
96                 {
97                     MessageBox.Show("No hay empleados activos para generar el reporte.", "Reportes", MessageBoxButtons.OK, MessageBoxIcon.Information);
98                     return;
99                 }
100
101                 path = pdfExportService.ExportarEmpleados(empleados);
102             }
103             else if (format == "Excel")
104             {
105                 List<ReporteGenerado> reportes = reporteService.GetAll();
106                 if (name == "Reportes administrativos" && reportes.Count == 0)
107                 {
108                     MessageBox.Show("No hay reportes registrados para exportar.", "Reportes", MessageBoxButtons.OK, MessageBoxIcon.Information);
109                     return;
110                 }
111
112                 path = excelExportService.ExportarReportes(reportes);
113             }
114             else
115             {
116                 List<ReporteGenerado> reportes = reporteService.GetAll();
117                 if (name == "Reportes administrativos" && reportes.Count == 0)
118                 {
119                     MessageBox.Show("No hay reportes registrados para exportar.", "Reportes", MessageBoxButtons.OK, MessageBoxIcon.Information);
120                     return;
121                 }
122             }
123         }
124     }
125 }
```

UI/FrmReportes.cs (continuacion)

```
121         }
122
123         path = pdfExportService.ExportarReportes(reportes);
124     }
125
126     if (string.IsNullOrEmpty(path)) return;
127
128     reporteService.Register(name, type + " / " + format, path);
129     UiFactory.ShowExported(path);
130     LoadReports();
131 }
132 catch (Exception ex)
133 {
134     MessageBox.Show(ex.Message, "Reportes", MessageBoxButtons.OK, MessageBoxIcon.Warning);
135 }
136 }
137
138 private void LoadReports()
139 {
140     List<ReporteGenerado> reports = reporteService.GetAll();
141     dgvReportesRecientes.Rows.Clear();
142     foreach (ReporteGenerado r in reports)
143     {
144         dgvReportesRecientes.Rows.Add(r.NombreReporte, r.Tipo, r.GeneradoPor,
145             r.FechaGeneracion.ToString("dd/MM/yyyy HH:mm"), r.RutaArchivo);
146     }
147 }
148 }
149 }
```

UI/FrmReportes.Designer.cs

```
1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmReportes
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo;
7         private System.Windows.Forms.Panel cardNomina;
8         private System.Windows.Forms.Panel cardEmpleados;
9         private System.Windows.Forms.Panel cardPagos;
10        private System.Windows.Forms.Panel cardDeducciones;
11        private System.Windows.Forms.Label lblCardNomina;
12        private System.Windows.Forms.Label lblCardEmpleados;
13        private System.Windows.Forms.Label lblCardPagos;
14        private System.Windows.Forms.Label lblCardDeducciones;
15        private System.Windows.Forms.Button btnReporteNomina;
16        private System.Windows.Forms.Button btnReporteEmpleados;
17        private System.Windows.Forms.Button btnReportePagos;
18        private System.Windows.Forms.Button btnReporteDeducciones;
19        private System.Windows.Forms.Button btnExportarExcel;
20        private System.Windows.Forms.Button btnExportarPdf;
21        private System.Windows.Forms.DataGridView dgvReportesRecientes;
22
23        protected override void Dispose(bool disposing)
24        {
25            if (disposing && (components != null)) components.Dispose();
26            base.Dispose(disposing);
27        }
28
29        private void InitializeComponent()
30        {
31            this.lblTitulo = new System.Windows.Forms.Label();
32            this.cardNomina = new System.Windows.Forms.Panel();
33            this.lblCardNomina = new System.Windows.Forms.Label();
34            this.btnReporteNomina = new System.Windows.Forms.Button();
35            this.cardEmpleados = new System.Windows.Forms.Panel();
36            this.lblCardEmpleados = new System.Windows.Forms.Label();
37            this.btnReporteEmpleados = new System.Windows.Forms.Button();
38            this.cardPagos = new System.Windows.Forms.Panel();
39            this.lblCardPagos = new System.Windows.Forms.Label();
40            this.btnReportePagos = new System.Windows.Forms.Button();
41            this.cardDeducciones = new System.Windows.Forms.Panel();
42            this.lblCardDeducciones = new System.Windows.Forms.Label();
43            this.btnReporteDeducciones = new System.Windows.Forms.Button();
44            this.btnExportarExcel = new System.Windows.Forms.Button();
45            this.btnExportarPdf = new System.Windows.Forms.Button();
46            this.dgvReportesRecientes = new System.Windows.Forms.DataGridView();
47            this.dataGridViewTextBoxColumn1 = new System.Windows.Forms.DataGridViewTextBoxColumn();
48            this.dataGridViewTextBoxColumn2 = new System.Windows.Forms.DataGridViewTextBoxColumn();
49            this.dataGridViewTextBoxColumn3 = new System.Windows.Forms.DataGridViewTextBoxColumn();
50            this.dataGridViewTextBoxColumn4 = new System.Windows.Forms.DataGridViewTextBoxColumn();
51            this.dataGridViewTextBoxColumn5 = new System.Windows.Forms.DataGridViewTextBoxColumn();
52            this.cardNomina.SuspendLayout();
53            this.cardEmpleados.SuspendLayout();
54            this.cardPagos.SuspendLayout();
55            this.cardDeducciones.SuspendLayout();
56            ((System.ComponentModel.ISupportInitialize)(this.dgvReportesRecientes)).BeginInit();
57            this.SuspendLayout();
58            //
59            // lblTitulo
60            //
```

UI/FrmReportes.Designer.cs (continuacion)

```
61         this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 22F, System.Drawing.FontStyle.Bold);
62         this.lblTitulo.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
63         this.lblTitulo.Location = new System.Drawing.Point(34, 24);
64         this.lblTitulo.Name = "lblTitulo";
65         this.lblTitulo.Size = new System.Drawing.Size(520, 42);
66         this.lblTitulo.TabIndex = 0;
67         this.lblTitulo.Text = "Centro de reportes";
68         //
69         // cardNomina
70         //
71         this.cardNomina.BackColor = System.Drawing.Color.Black;
72         this.cardNomina.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
73         this.cardNomina.Controls.Add(this.lblCardNomina);
74         this.cardNomina.Controls.Add(this.btnReporteNomina);
75         this.cardNomina.Location = new System.Drawing.Point(36, 100);
76         this.cardNomina.Name = "cardNomina";
77         this.cardNomina.Size = new System.Drawing.Size(220, 150);
78         this.cardNomina.TabIndex = 1;
79         //
80         // lblCardNomina
81         //
82         this.lblCardNomina.Font = new System.Drawing.Font("Segoe UI", 12F, System.Drawing.FontStyle.Bold);
83         this.lblCardNomina.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
84         this.lblCardNomina.Location = new System.Drawing.Point(24, 22);
85         this.lblCardNomina.Name = "lblCardNomina";
86         this.lblCardNomina.Size = new System.Drawing.Size(170, 58);
87         this.lblCardNomina.TabIndex = 1;
88         this.lblCardNomina.Text = "Reporte general\r\nde nómina";
89         //
90         // btnReporteNomina
91         //
92         this.btnReporteNomina.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
93         this.btnReporteNomina.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
94         this.btnReporteNomina.Location = new System.Drawing.Point(28, 92);
95         this.btnReporteNomina.Name = "btnReporteNomina";
96         this.btnReporteNomina.Size = new System.Drawing.Size(160, 34);
97         this.btnReporteNomina.TabIndex = 0;
98         this.btnReporteNomina.Text = "Generar PDF";
99         this.btnReporteNomina.UseVisualStyleBackColor = false;
100        this.btnReporteNomina.Click += new System.EventHandler(this.btnReporteNomina_Click);
101        //
102        // cardEmpleados
103        //
104        this.cardEmpleados.BackColor = System.Drawing.Color.Black;
105        this.cardEmpleados.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
106        this.cardEmpleados.Controls.Add(this.lblCardEmpleados);
107        this.cardEmpleados.Controls.Add(this.btnReporteEmpleados);
108        this.cardEmpleados.Location = new System.Drawing.Point(278, 100);
109        this.cardEmpleados.Name = "cardEmpleados";
110        this.cardEmpleados.Size = new System.Drawing.Size(220, 150);
111        this.cardEmpleados.TabIndex = 2;
112        //
113        // lblCardEmpleados
114        //
115        this.lblCardEmpleados.Font = new System.Drawing.Font("Segoe UI", 12F, System.Drawing.FontStyle.Bold);
116        this.lblCardEmpleados.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
117        this.lblCardEmpleados.Location = new System.Drawing.Point(24, 22);
118        this.lblCardEmpleados.Name = "lblCardEmpleados";
119        this.lblCardEmpleados.Size = new System.Drawing.Size(170, 58);
120        this.lblCardEmpleados.TabIndex = 1;
```


UI/FrmReportes.Designer.cs (continuacion)

```
121         this.lblCardEmpleados.Text = "Empleados\r\nactivos";
122         //
123         // btnReporteEmpleados
124         //
125         this.btnReporteEmpleados.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167))))), ((int)(((byte)(139))))), ((int)(((byte)(250)))));
126         this.btnReporteEmpleados.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
127         this.btnReporteEmpleados.Location = new System.Drawing.Point(28, 92);
128         this.btnReporteEmpleados.Name = "btnReporteEmpleados";
129         this.btnReporteEmpleados.Size = new System.Drawing.Size(160, 34);
130         this.btnReporteEmpleados.TabIndex = 0;
131         this.btnReporteEmpleados.Text = "Generar PDF";
132         this.btnReporteEmpleados.UseVisualStyleBackColor = false;
133         this.btnReporteEmpleados.Click += new System.EventHandler(this.btnReporteEmpleados_Click);
134         //
135         // cardPagos
136         //
137         this.cardPagos.BackColor = System.Drawing.Color.Black;
138         this.cardPagos.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
139         this.cardPagos.Controls.Add(this.lblCardPagos);
140         this.cardPagos.Controls.Add(this.btnReportePagos);
141         this.cardPagos.Location = new System.Drawing.Point(520, 100);
142         this.cardPagos.Name = "cardPagos";
143         this.cardPagos.Size = new System.Drawing.Size(220, 150);
144         this.cardPagos.TabIndex = 3;
145         //
146         // lblCardPagos
147         //
148         this.lblCardPagos.Font = new System.Drawing.Font("Segoe UI", 12F, System.Drawing.FontStyle.Bold);
149         this.lblCardPagos.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229))))), ((int)(((byte)(228))))), ((int)(((byte)(245)))));
150         this.lblCardPagos.Location = new System.Drawing.Point(24, 22);
151         this.lblCardPagos.Name = "lblCardPagos";
152         this.lblCardPagos.Size = new System.Drawing.Size(170, 67);
153         this.lblCardPagos.TabIndex = 1;
154         this.lblCardPagos.Text = "Historial\r\nnde pagos";
155         //
156         // btnReportePagos
157         //
158         this.btnReportePagos.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167))))), ((int)(((byte)(139))))), ((int)(((byte)(250)))));
159         this.btnReportePagos.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
160         this.btnReportePagos.Location = new System.Drawing.Point(28, 92);
161         this.btnReportePagos.Name = "btnReportePagos";
162         this.btnReportePagos.Size = new System.Drawing.Size(160, 34);
163         this.btnReportePagos.TabIndex = 0;
164         this.btnReportePagos.Text = "Generar PDF";
165         this.btnReportePagos.UseVisualStyleBackColor = false;
166         this.btnReportePagos.Click += new System.EventHandler(this.btnReportePagos_Click);
167         //
168         // cardDeducciones
169         //
170         this.cardDeducciones.BackColor = System.Drawing.Color.Black;
171         this.cardDeducciones.BorderStyle = System.Windows.Forms.BorderStyle.FixedSingle;
172         this.cardDeducciones.Controls.Add(this.lblCardDeducciones);
173         this.cardDeducciones.Controls.Add(this.btnReporteDeducciones);
174         this.cardDeducciones.Location = new System.Drawing.Point(762, 100);
175         this.cardDeducciones.Name = "cardDeducciones";
176         this.cardDeducciones.Size = new System.Drawing.Size(220, 150);
177         this.cardDeducciones.TabIndex = 4;
178         //
179         // lblCardDeducciones
180         //
```

UI/FrmReportes.Designer.cs (continuacion)

```
181         this.lblCardDeducciones.Font = new System.Drawing.Font("Segoe UI", 12F, System.Drawing.FontStyle.Bold);
182         this.lblCardDeducciones.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
183         this.lblCardDeducciones.Location = new System.Drawing.Point(24, 22);
184         this.lblCardDeducciones.Name = "lblCardDeducciones";
185         this.lblCardDeducciones.Size = new System.Drawing.Size(170, 58);
186         this.lblCardDeducciones.TabIndex = 1;
187         this.lblCardDeducciones.Text = "Resumen de\r\rndeducciones";
188         //
189         // btnReporteDeducciones
190         //
191         this.btnReporteDeducciones.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(167)))), ((int)(((byte)(139)))), ((int)(((byte)(250))))));
192         this.btnReporteDeducciones.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
193         this.btnReporteDeducciones.Location = new System.Drawing.Point(28, 92);
194         this.btnReporteDeducciones.Name = "btnReporteDeducciones";
195         this.btnReporteDeducciones.Size = new System.Drawing.Size(160, 34);
196         this.btnReporteDeducciones.TabIndex = 0;
197         this.btnReporteDeducciones.Text = "Generar PDF";
198         this.btnReporteDeducciones.UseVisualStyleBackColor = false;
199         this.btnReporteDeducciones.Click += new System.EventHandler(this.btnReporteDeducciones_Click);
200         //
201         // btnExportarExcel
202         //
203         this.btnExportarExcel.BackColor = System.Drawing.Color.Black;
204         this.btnExportarExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
205         this.btnExportarExcel.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
206         this.btnExportarExcel.Location = new System.Drawing.Point(776, 284);
207         this.btnExportarExcel.Name = "btnExportarExcel";
208         this.btnExportarExcel.Size = new System.Drawing.Size(95, 34);
209         this.btnExportarExcel.TabIndex = 5;
210         this.btnExportarExcel.Text = "Excel";
211         this.btnExportarExcel.UseVisualStyleBackColor = false;
212         this.btnExportarExcel.Click += new System.EventHandler(this.btnExportarExcel_Click);
213         //
214         // btnExportarPdf
215         //
216         this.btnExportarPdf.BackColor = System.Drawing.Color.Black;
217         this.btnExportarPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat;
218         this.btnExportarPdf.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
219         this.btnExportarPdf.Location = new System.Drawing.Point(886, 284);
220         this.btnExportarPdf.Name = "btnExportarPdf";
221         this.btnExportarPdf.Size = new System.Drawing.Size(95, 34);
222         this.btnExportarPdf.TabIndex = 6;
223         this.btnExportarPdf.Text = "PDF";
224         this.btnExportarPdf.UseVisualStyleBackColor = false;
225         this.btnExportarPdf.Click += new System.EventHandler(this.btnExportarPdf_Click);
226         //
227         // dgvReportesRecientes
228         //
229         this.dgvReportesRecientes.AllowUserToAddRows = false;
230         this.dgvReportesRecientes.AllowUserToDeleteRows = false;
231         this.dgvReportesRecientes.BackgroundColor = System.Drawing.Color.Black;
232         this.dgvReportesRecientes.ColumnHeadersHeightSizeMode = System.Windows.Forms.DataGridViewColumnHeadersHeightSizeMode.AutoSize;
233         this.dgvReportesRecientes.Columns.AddRange(new System.Windows.Forms.DataGridViewColumn[] {
234             this.dataGridViewTextBoxColumn1,
235             this.dataGridViewTextBoxColumn2,
236             this.dataGridViewTextBoxColumn3,
237             this.dataGridViewTextBoxColumn4,
238             this.dataGridViewTextBoxColumn5});
239         this.dgvReportesRecientes.Location = new System.Drawing.Point(36, 334);
240         this.dgvReportesRecientes.Name = "dgvReportesRecientes";
```

UI/FrmReportes.Designer.cs (continuacion)

```
241         this.dgvReportesRecientes.ReadOnly = true;
242         this.dgvReportesRecientes.RowHeadersVisible = false;
243         this.dgvReportesRecientes.RowHeadersWidth = 51;
244         this.dgvReportesRecientes.Size = new System.Drawing.Size(946, 330);
245         this.dgvReportesRecientes.TabIndex = 7;
246         //
247         // dataGridViewTextBoxColumn1
248         //
249         this.dataGridViewTextBoxColumn1.HeaderText = "Nombre del reporte";
250         this.dataGridViewTextBoxColumn1.MinimumWidth = 6;
251         this.dataGridViewTextBoxColumn1.Name = "dataGridViewTextBoxColumn1";
252         this.dataGridViewTextBoxColumn1.ReadOnly = true;
253         this.dataGridViewTextBoxColumn1.Width = 125;
254         //
255         // dataGridViewTextBoxColumn2
256         //
257         this.dataGridViewTextBoxColumn2.HeaderText = "Tipo";
258         this.dataGridViewTextBoxColumn2.MinimumWidth = 6;
259         this.dataGridViewTextBoxColumn2.Name = "dataGridViewTextBoxColumn2";
260         this.dataGridViewTextBoxColumn2.ReadOnly = true;
261         this.dataGridViewTextBoxColumn2.Width = 125;
262         //
263         // dataGridViewTextBoxColumn3
264         //
265         this.dataGridViewTextBoxColumn3.HeaderText = "Generado por";
266         this.dataGridViewTextBoxColumn3.MinimumWidth = 6;
267         this.dataGridViewTextBoxColumn3.Name = "dataGridViewTextBoxColumn3";
268         this.dataGridViewTextBoxColumn3.ReadOnly = true;
269         this.dataGridViewTextBoxColumn3.Width = 125;
270         //
271         // dataGridViewTextBoxColumn4
272         //
273         this.dataGridViewTextBoxColumn4.HeaderText = "Fecha";
274         this.dataGridViewTextBoxColumn4.MinimumWidth = 6;
275         this.dataGridViewTextBoxColumn4.Name = "dataGridViewTextBoxColumn4";
276         this.dataGridViewTextBoxColumn4.ReadOnly = true;
277         this.dataGridViewTextBoxColumn4.Width = 125;
278         //
279         // dataGridViewTextBoxColumn5
280         //
281         this.dataGridViewTextBoxColumn5.HeaderText = "Ruta";
282         this.dataGridViewTextBoxColumn5.MinimumWidth = 6;
283         this.dataGridViewTextBoxColumn5.Name = "dataGridViewTextBoxColumn5";
284         this.dataGridViewTextBoxColumn5.ReadOnly = true;
285         this.dataGridViewTextBoxColumn5.Width = 125;
286         //
287         // FrmReportes
288         //
289         this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 20F);
290         this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
291         this.BackColor = System.Drawing.Color.FromArgb(((int)(((byte)(13)))), ((int)(((byte)(14)))), ((int)(((byte)(20))))));
292         this.ClientSize = new System.Drawing.Size(1020, 742);
293         this.Controls.Add(this.dgvReportesRecientes);
294         this.Controls.Add(this.btnExportarPdf);
295         this.Controls.Add(this.btnExportarExcel);
296         this.Controls.Add(this.cardDeducciones);
297         this.Controls.Add(this.cardPagos);
298         this.Controls.Add(this.cardEmpleados);
299         this.Controls.Add(this.cardNomina);
300         this.Controls.Add(this.lblTitulo);
```

UI/FrmReportes.Designer.cs (continuacion)

```
301         this.Font = new System.Drawing.Font("Segoe UI", 9F);
302         this.ForeColor = System.Drawing.Color.FromArgb(((int)(((byte)(229)))), ((int)(((byte)(228)))), ((int)(((byte)(245))))));
303         this.Name = "FrmReportes";
304         this.Text = "Reportes";
305         this.Load += new System.EventHandler(this.FrmReportes_Load);
306         this.cardNomina.ResumeLayout(false);
307         this.cardEmpleados.ResumeLayout(false);
308         this.cardPagos.ResumeLayout(false);
309         this.cardDeducciones.ResumeLayout(false);
310         ((System.ComponentModel.ISupportInitialize)(this.dgvReportesRecientes)).EndInit();
311         this.ResumeLayout(false);
312
313     }
314
315     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn1;
316     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn2;
317     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn3;
318     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn4;
319     private System.Windows.Forms.DataGridViewTextBoxColumn dataGridViewTextBoxColumn5;
320 }
321 }
```

UI\FrmSolicitudAusenciaDetalle.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmSolicitudAusenciaDetalle : Form
13     {
14         private readonly SolicitudAusencia existingRequest;
15         private readonly AbsenceRequestService absenceService = new AbsenceRequestService();
16         private readonly EmpleadoService employeeService = new EmpleadoService();
17         public bool RequestCreated { get; private set; }
18
19         public FrmSolicitudAusenciaDetalle() : this(null) { }
20
21         public FrmSolicitudAusenciaDetalle(SolicitudAusencia item)
22         {
23             existingRequest = item;
24             InitializeComponent();
25             ControlStyleHelper.ApplyModernForm(this);
26         }
27
28         private void FrmSolicitudAusenciaDetalle_Load(object sender, EventArgs e)
29         {
30             if (LicenseManager.UsageMode == LicenseUsageMode.Designtime) return;
31             cmbTipo.DataSource = new List<string>(AbsenceTypes.All);
32             if (existingRequest != null)
33             {
34                 Text = "Detalle de solicitud";
35                 lblTitulo.Text = "Detalle de la solicitud";
36                 cmbEmpleado.Items.Add(existingRequest.CodigoEmpleado + " - " + existingRequest.EmpleadoNombre);
37                 cmbEmpleado.SelectedIndex = 0;
38                 cmbTipo.SelectedItem = existingRequest.Tipo;
39                 dtpInicio.Value = existingRequest.FechaInicio;
40                 dtpFin.Value = existingRequest.FechaFin;
41                 txtMotivo.Text = existingRequest.Motivo;
42                 txtResolucion.Text = existingRequest.ObservacionResolucion;
43                 lblEstado.Text = "Estado: " + existingRequest.Estado;
44                 SetReadOnly();
45                 return;
46             }
47
48             lblEstado.Text = "Estado inicial: Pendiente";
49             if (string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase))
50             {
51                 cmbEmpleado.Items.Add("Empleado asociado a la sesión");
52                 cmbEmpleado.SelectedIndex = 0;
53                 cmbEmpleado.Enabled = false;
54             }
55             else
56             {
57                 List<Empleado> employees = employeeService.GetActive(null);
58                 cmbEmpleado.DataSource = employees;
59                 cmbEmpleado.DisplayMember = "NombreCompleto";
60                 cmbEmpleado.ValueMember = "IdEmpleado";
```

UI/FrmSolicitudAusenciaDetalle.cs (continuacion)

```
61     }
62 }
63
64 private void btnGuardar_Click(object sender, EventArgs e)
65 {
66     try
67     {
68         string type = Convert.ToString(cmbTipo.SelectedItem);
69         if (string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase))
70         {
71             absenceService.CreateOwn(new OwnAbsenceRequest
72             {
73                 Type = type,
74                 StartDate = dtpInicio.Value.Date,
75                 EndDate = dtpFin.Value.Date,
76                 Reason = txtMotivo.Text
77             });
78         }
79         else
80         {
81             if (!(cmbEmpleado.SelectedValue is int))
82                 throw new ArgumentException("Seleccione un empleado.");
83             absenceService.CreateForEmployee(new SolicitudAusencia
84             {
85                 IdEmpleado = (int)cmbEmpleado.SelectedValue,
86                 Tipo = type,
87                 FechaInicio = dtpInicio.Value.Date,
88                 FechaFin = dtpFin.Value.Date,
89                 Motivo = txtMotivo.Text
90             });
91         }
92         RequestCreated = true;
93         MessageBox.Show("Solicitud registrada en estado Pendiente.", "Ausencias",
94             MessageBoxButtons.OK, MessageBoxIcon.Information);
95         DialogResult = DialogResult.OK;
96         Close();
97     }
98     catch (Exception ex)
99     {
100         MessageBox.Show(ex.Message, "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Warning);
101     }
102 }
103
104 private void btnCerrar_Click(object sender, EventArgs e) { Close(); }
105
106 private void SetReadOnly()
107 {
108     cmbEmpleado.Enabled = false;
109     cmbTipo.Enabled = false;
110     dtpInicio.Enabled = false;
111     dtpFin.Enabled = false;
112     txtMotivo.ReadOnly = true;
113     txtResolucion.ReadOnly = true;
114     btnGuardar.Visible = false;
115 }
116 }
117 }
```

UI/FrmSolicitudAusenciaDetalle.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmSolicitudAusenciaDetalle
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo, lblEmpleado, lblTipo, lblInicio, lblFin, lblMotivo, lblResolucion, lblEstado;
7         private System.Windows.Forms.ComboBox cmbEmpleado, cmbTipo;
8         private System.Windows.Forms.DateTimePicker dtpInicio, dtpFin;
9         private System.Windows.Forms.TextBox txtMotivo, txtResolucion;
10        private System.Windows.Forms.Button btnGuardar, btnCerrar;
11        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
12        private void InitializeComponent()
13        {
14            this.lblTitulo = new System.Windows.Forms.Label(); this.lblEmpleado = new System.Windows.Forms.Label(); this.lblTipo = new \
System.Windows.Forms.Label(); this.lblInicio = new System.Windows.Forms.Label(); this.lblFin = new System.Windows.Forms.Label(); this.lblMotivo = new \
System.Windows.Forms.Label(); this.lblResolucion = new System.Windows.Forms.Label(); this.lblEstado = new System.Windows.Forms.Label();
15            this.cmbEmpleado = new System.Windows.Forms.ComboBox(); this.cmbTipo = new System.Windows.Forms.ComboBox(); this.dtpInicio = new \
System.Windows.Forms.DateTimePicker(); this.dtpFin = new System.Windows.Forms.DateTimePicker(); this.txtMotivo = new System.Windows.Forms.TextBox(); \
System.Windows.Forms.Button(); this.SuspendLayout();
16            this.BackColor = System.Drawing.Color.FromArgb(18, 18, 27); this.ClientSize = new System.Drawing.Size(650, 510); this.Font = new \
System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.FormBorderStyle = \
System.Windows.Forms.FixedDialog; this.MaximizeBox = false; this.MinimizeBox = false; this.Name = "FrmSolicitudAusenciaDetalle"; \
this.StartPosition = System.Windows.Forms.FormStartPosition.CenterParent; this.Text = "Nueva solicitud de ausencia"; this.Load += new \
System.EventHandler(this.FrmSolicitudAusenciaDetalle_Load);
17            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 20F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Nueva solicitud de ausencia";
18            this.lblEstado.AutoSize = true; this.lblEstado.ForeColor = System.Drawing.Color.FromArgb(167, 139, 250); this.lblEstado.Location = new \
System.Drawing.Point(32, 66); this.lblEstado.Text = "Estado inicial: Pendiente";
19            this.lblEmpleado.AutoSize = true; this.lblEmpleado.Location = new System.Drawing.Point(32, 104); this.lblEmpleado.Text = "Empleado"; \
this.cmbEmpleado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbEmpleado.Location = new System.Drawing.Point(35, 126); \
this.cmbEmpleado.Size = new System.Drawing.Size(278, 23);
20            this.lblTipo.AutoSize = true; this.lblTipo.Location = new System.Drawing.Point(329, 104); this.lblTipo.Text = "Tipo"; \
this.cmbTipo.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbTipo.Location = new System.Drawing.Point(332, 126); \
this.cmbTipo.Size = new System.Drawing.Size(278, 23);
21            this.lblInicio.AutoSize = true; this.lblInicio.Location = new System.Drawing.Point(32, 170); this.lblInicio.Text = "Fecha inicial"; \
this.dtpInicio.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpInicio.Location = new System.Drawing.Point(35, 192); \
this.dtpInicio.Size = new System.Drawing.Size(180, 23);
22            this.lblFin.AutoSize = true; this.lblFin.Location = new System.Drawing.Point(230, 170); this.lblFin.Text = "Fecha final"; this.dtpFin.Format \
= System.Windows.Forms.DateTimePickerFormat.Short; this.dtpFin.Location = new System.Drawing.Point(233, 192); this.dtpFin.Size = new \
System.Drawing.Size(180, 23);
23            this.lblMotivo.AutoSize = true; this.lblMotivo.Location = new System.Drawing.Point(32, 236); this.lblMotivo.Text = "Motivo"; \
this.txtMotivo.Location = new System.Drawing.Point(35, 258); this.txtMotivo.MaxLength = 500; this.txtMotivo.Multiline = true; this.txtMotivo.Size = \
new System.Drawing.Size(575, 76);
24            this.lblResolucion.AutoSize = true; this.lblResolucion.Location = new System.Drawing.Point(32, 350); this.lblResolucion.Text = "Observación \
de resolución"; this.txtResolucion.Location = new System.Drawing.Point(35, 372); this.txtResolucion.Multiline = true; this.txtResolucion.ReadOnly = \
true; this.txtResolucion.Size = new System.Drawing.Size(575, 58);
25            this.btnGuardar.BackColor = System.Drawing.Color.FromArgb(124, 58, 237); this.btnGuardar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnGuardar.Location = new System.Drawing.Point(414, 452); this.btnGuardar.Size = new System.Drawing.Size(94, 34); this.btnGuardar.Text = \
"Registrar"; this.btnGuardar.UseVisualStyleBackColor = false; this.btnGuardar.Click += new System.EventHandler(this.btnGuardar_Click);
26            this.btnCerrar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnCerrar.Location = new System.Drawing.Point(516, 452); \
this.btnCerrar.Size = new System.Drawing.Size(94, 34); this.btnCerrar.Text = "Cerrar"; this.btnCerrar.Click += new \
System.EventHandler(this.btnCerrar_Click);
27            this.Controls.Add(this.lblTitulo); this.Controls.Add(this.lblEstado); this.Controls.Add(this.lblEmpleado); \
this.Controls.Add(this.cmbEmpleado); this.Controls.Add(this.lblTipo); this.Controls.Add(this.cmbTipo); this.Controls.Add(this.lblInicio); \
this.Controls.Add(this.dtpInicio); this.Controls.Add(this.lblFin); this.Controls.Add(this.dtpFin); this.Controls.Add(this.lblMotivo); \
this.Controls.Add(this.txtMotivo); this.Controls.Add(this.lblResolucion); this.Controls.Add(this.txtResolucion); this.Controls.Add(this.btnGuardar); \
this.Controls.Add(this.btnCerrar); this.ResumeLayout(false); this.PerformLayout();
28        }
29    }

```

30 }

UI/FrmSolicitudesAusencia.cs

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Windows.Forms;
5 using SistemaGestionNomina.Helpers;
6 using SistemaGestionNomina.Models;
7 using SistemaGestionNomina.Security;
8 using SistemaGestionNomina.Services;
9
10 namespace SistemaGestionNomina.UI
11 {
12     public partial class FrmSolicitudesAusencia : Form
13     {
14         private readonly AbsenceRequestService service = new AbsenceRequestService();
15         private readonly EmpleadoService employeeService = new EmpleadoService();
16         private readonly ExcelExportService excelExportService = new ExcelExportService();
17         private readonly PdfExportService pdfExportService = new PdfExportService();
18         private readonly AuthorizationService authorizationService = new AuthorizationService();
19         private List<SolicitudAusencia> currentRequests = new List<SolicitudAusencia>();
20
21         public FrmSolicitudesAusencia()
22         {
23             InitializeComponent();
24             ControlStyleHelper.ApplyModernForm(this);
25         }
26
27         private void FrmSolicitudesAusencia_Load(object sender, EventArgs e)
28         {
29             if (LicenseManager.UsageMode == LicenseUsageMode.DesignTime) return;
30             cmbTipo.Items.Add("Todos");
31             cmbTipo.Items.AddRange(AbsenceTypes.All);
32             cmbTipo.SelectedIndex = 0;
33             cmbEstado.Items.Add("Todos");
34             cmbEstado.Items.AddRange(AbsenceStates.All);
35             cmbEstado.SelectedIndex = 0;
36
37             bool worker = string.Equals(SessionContext.Role, Roles.Trabajador, StringComparison.OrdinalIgnoreCase);
38             if (worker)
39             {
40                 cmbDepartamento.Items.Add("Mi información");
41                 cmbDepartamento.SelectedIndex = 0;
42                 cmbDepartamento.Enabled = false;
43                 btnNuevo.Visible = authorizationService.HasPermission(Permissions.OwnAbsencesCreate);
44                 btnGestionar.Visible = false;
45                 btnCancelar.Visible = authorizationService.HasPermission(Permissions.OwnAbsencesCancel);
46                 btnExcel.Visible = false;
47                 btnPdf.Visible = false;
48                 lblTitulo.Text = "Mis solicitudes de ausencia";
49             }
50             else
51             {
52                 List<Departamento> departments = new List<Departamento>();
53                 departments.Add(new Departamento { IdDepartamento = 0, Nombre = "Todos los departamentos" });
54                 departments.AddRange(employeeService.GetDepartments());
55                 cmbDepartamento.DataSource = departments;
56                 cmbDepartamento.DisplayMember = "Nombre";
57                 cmbDepartamento.ValueMember = "IdDepartamento";
58                 btnNuevo.Visible = authorizationService.HasPermission(Permissions.AbsencesCreate);
59                 btnGestionar.Visible = authorizationService.HasPermission(Permissions.AbsencesApprove) ||
60                     authorizationService.HasPermission(Permissions.AbsencesReject);
61             }
62         }
63     }
64 }
```

UI/FrmSolicitudesAusencia.cs (continuacion)

```
61         btnCancelar.Visible = authorizationService.HasPermission(Permissions.AbsencesCancel);
62         btnExcel.Visible = authorizationService.HasPermission(Permissions.AbsencesExport);
63         btnPdf.Visible = btnExcel.Visible;
64     }
65     LoadRequests();
66 }
67
68 private void btnBuscar_Click(object sender, EventArgs e) { LoadRequests(); }
69
70 private void LoadRequests()
71 {
72     try
73     {
74         int? departmentId = null;
75         if (cmbDepartamento.SelectedValue is int && (int)cmbDepartamento.SelectedValue > 0)
76             departmentId = (int)cmbDepartamento.SelectedValue;
77         currentRequests = service.Search(new AbsenceQuery
78         {
79             DepartmentId = departmentId,
80             Type = cmbTipo.SelectedIndex > 0 ? Convert.ToString(cmbTipo.SelectedItem) : string.Empty,
81             Status = cmbEstado.SelectedIndex > 0 ? Convert.ToString(cmbEstado.SelectedItem) : string.Empty,
82             From = dtpDesde.Checked ? (DateTime?)dtpDesde.Value.Date : null,
83             To = dtpHasta.Checked ? (DateTime?)dtpHasta.Value.Date : null,
84             Search = txtBuscar.Text.Trim()
85         });
86         dgvSolicitudes.Rows.Clear();
87         for (int i = 0; i < currentRequests.Count; i++)
88         {
89             SolicitudAusencia item = currentRequests[i];
90             dgvSolicitudes.Rows.Add(item.IdSolicitud, item.CodigoEmpleado, item.EmpleadoNombre,
91                 item.DepartamentoNombre, item.Tipo, item.FechaInicio.ToString("dd/MM/yyyy"),
92                 item.FechaFin.ToString("dd/MM/yyyy"), item.Estado, item.UsuarioSolicitante,
93                 item.FechaCreacion.ToString("dd/MM/yyyy HH:mm"));
94         }
95         lblResultado.Text = currentRequests.Count + " solicitudes";
96     }
97     catch (Exception ex)
98     {
99         MessageBox.Show(ex.Message, "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Warning);
100     }
101 }
102
103 private void btnNuevo_Click(object sender, EventArgs e)
104 {
105     using (FrmSolicitudAusenciaDetalle form = new FrmSolicitudAusenciaDetalle())
106     {
107         if (form.ShowDialog() == DialogResult.OK) LoadRequests();
108     }
109 }
110
111 private void btnVer_Click(object sender, EventArgs e)
112 {
113     int id = SelectedRequestId();
114     if (id == 0) return;
115     try
116     {
117         using (FrmSolicitudAusenciaDetalle form = new FrmSolicitudAusenciaDetalle(service.GetById(id)))
118             form.ShowDialog();
119     }
120     catch (Exception ex) { MessageBox.Show(ex.Message, "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Warning); }
```

UI/FrmSolicitudesAusencia.cs (continuacion)

```
121     }
122
123     private void btnGestionar_Click(object sender, EventArgs e) { OpenManagement(false); }
124     private void btnCancelar_Click(object sender, EventArgs e) { OpenManagement(true); }
125
126     private void OpenManagement(bool cancelOnly)
127     {
128         int id = SelectedRequestId();
129         if (id == 0) return;
130         using (FrmGestionAusencias form = new FrmGestionAusencias(id, cancelOnly))
131         {
132             if (form.ShowDialog() == DialogResult.OK) LoadRequests();
133         }
134     }
135
136     private int SelectedRequestId()
137     {
138         if (dgvSolicitudes.CurrentRow == null || dgvSolicitudes.CurrentRow.Cells["IdSolicitud"].Value == null)
139         {
140             MessageBox.Show("Seleccione una solicitud.", "Ausencias", MessageBoxButtons.OK, MessageBoxIcon.Information);
141             return 0;
142         }
143         return Convert.ToInt32(dgvSolicitudes.CurrentRow.Cells["IdSolicitud"].Value);
144     }
145
146     private void btnExcel_Click(object sender, EventArgs e)
147     {
148         if (currentRequests.Count > 0) UiFactory.ShowExported(excelExportService.ExportarAusencias(currentRequests));
149     }
150
151     private void btnPdf_Click(object sender, EventArgs e)
152     {
153         if (currentRequests.Count > 0) UiFactory.ShowExported(pdfExportService.ExportarAusencias(currentRequests));
154     }
155 }
156 }
```

UI/FrmSolicitudesAusencia.Designer.cs

```

1 namespace SistemaGestionNomina.UI
2 {
3     partial class FrmSolicitudesAusencia
4     {
5         private System.ComponentModel.IContainer components = null;
6         private System.Windows.Forms.Label lblTitulo, lblResultado;
7         private System.Windows.Forms.TextBox txtBuscar;
8         private System.Windows.Forms.ComboBox cmbTipo, cmbEstado, cmbDepartamento;
9         private System.Windows.Forms.DateTimePicker dtpDesde, dtpHasta;
10        private System.Windows.Forms.Button btnBuscar, btnNuevo, btnVer, btnGestionar, btnCancelar, btnExcel, btnPdf;
11        private System.Windows.Forms.DataGridView dgvSolicitudes;
12        protected override void Dispose(bool disposing) { if (disposing && components != null) components.Dispose(); base.Dispose(disposing); }
13        private void InitializeComponent()
14        {
15            this.lblTitulo = new System.Windows.Forms.Label(); this.lblResultado = new System.Windows.Forms.Label(); this.txtBuscar = new \
System.Windows.Forms.TextBox(); this.cmbTipo = new System.Windows.Forms.ComboBox(); this.cmbEstado = new System.Windows.Forms.ComboBox(); \
this.cmbDepartamento = new System.Windows.Forms.ComboBox(); this.dtpDesde = new System.Windows.Forms.DateTimePicker(); this.dtpHasta = new \
System.Windows.Forms.DateTimePicker(); this.btnBuscar = new System.Windows.Forms.Button(); this.btnNuevo = new System.Windows.Forms.Button(); \
this.btnVer = new System.Windows.Forms.Button(); this.btnGestionar = new System.Windows.Forms.Button(); this.btnCancelar = new \
System.Windows.Forms.Button(); this.btnExcel = new System.Windows.Forms.Button(); this.btnPdf = new System.Windows.Forms.Button(); \
this.dgvSolicitudes = new System.Windows.Forms.DataGridView(); ((System.ComponentModel.ISupportInitialize)(this.dgvSolicitudes)).BeginInit(); \
this.SuspendLayout();
16            this.BackColor = System.Drawing.Color.FromArgb(12, 12, 18); this.ClientSize = new System.Drawing.Size(1120, 720); this.Font = new \
System.Drawing.Font("Segoe UI", 9F); this.ForeColor = System.Drawing.Color.FromArgb(229, 228, 245); this.Name = "FrmSolicitudesAusencia"; this.Text = \
"Solicitudes de ausencia"; this.Load += new System.EventHandler(this.FrmSolicitudesAusencia_Load);
17            this.lblTitulo.AutoSize = true; this.lblTitulo.Font = new System.Drawing.Font("Segoe UI", 24F, System.Drawing.FontStyle.Bold); \
this.lblTitulo.Location = new System.Drawing.Point(28, 20); this.lblTitulo.Text = "Solicitudes de ausencia";
18            this.txtBuscar.Location = new System.Drawing.Point(34, 82); this.txtBuscar.Size = new System.Drawing.Size(160, 23);
19            this.cmbDepartamento.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbDepartamento.Location = new \
System.Drawing.Point(204, 82); this.cmbDepartamento.Size = new System.Drawing.Size(170, 23);
20            this.cmbTipo.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbTipo.Location = new System.Drawing.Point(384, 82); \
this.cmbTipo.Size = new System.Drawing.Size(155, 23);
21            this.cmbEstado.DropDownStyle = System.Windows.Forms.ComboBoxStyle.DropDownList; this.cmbEstado.Location = new System.Drawing.Point(549, 82); \
this.cmbEstado.Size = new System.Drawing.Size(112, 23);
22            this.dtpDesde.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpDesde.Location = new System.Drawing.Point(671, 82); \
this.dtpDesde.ShowCheckBox = true; this.dtpDesde.Checked = false; this.dtpDesde.Size = new System.Drawing.Size(126, 23);
23            this.dtpHasta.Format = System.Windows.Forms.DateTimePickerFormat.Short; this.dtpHasta.Location = new System.Drawing.Point(807, 82); \
this.dtpHasta.ShowCheckBox = true; this.dtpHasta.Checked = false; this.dtpHasta.Size = new System.Drawing.Size(126, 23);
24            this.btnBuscar.BackColor = System.Drawing.Color.FromArgb(124, 58, 237); this.btnBuscar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnBuscar.Location = new System.Drawing.Point(943, 78); this.btnBuscar.Size = new System.Drawing.Size(80, 31); this.btnBuscar.Text = "Buscar"; \
this.btnBuscar.UseVisualStyleBackColor = false; this.btnBuscar.Click += new System.EventHandler(this.btnBuscar_Click);
25            this.btnNuevo.BackColor = System.Drawing.Color.FromArgb(124, 58, 237); this.btnNuevo.FlatStyle = System.Windows.Forms.FlatStyle.Flat; \
this.btnNuevo.Location = new System.Drawing.Point(34, 126); this.btnNuevo.Size = new System.Drawing.Size(100, 32); this.btnNuevo.Text = "Nueva"; \
this.btnNuevo.UseVisualStyleBackColor = false; this.btnNuevo.Click += new System.EventHandler(this.btnNuevo_Click);
26            this.btnVer.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnVer.Location = new System.Drawing.Point(142, 126); this.btnVer.Size = \
new System.Drawing.Size(100, 32); this.btnVer.Text = "Ver detalle"; this.btnVer.Click += new System.EventHandler(this.btnVer_Click);
27            this.btnGestionar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnGestionar.Location = new System.Drawing.Point(250, 126); \
this.btnGestionar.Size = new System.Drawing.Size(100, 32); this.btnGestionar.Text = "Gestionar"; this.btnGestionar.Click += new \
System.EventHandler(this.btnGestionar_Click);
28            this.btnCancelar.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnCancelar.Location = new System.Drawing.Point(358, 126); \
this.btnCancelar.Size = new System.Drawing.Size(100, 32); this.btnCancelar.Text = "Cancelar"; this.btnCancelar.Click += new \
System.EventHandler(this.btnCancelar_Click);
29            this.btnExcel.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnExcel.Location = new System.Drawing.Point(884, 126); \
this.btnExcel.Size = new System.Drawing.Size(66, 32); this.btnExcel.Text = "Excel"; this.btnExcel.Click += new System.EventHandler(this.btnExcel_Click);
30            this.btnPdf.FlatStyle = System.Windows.Forms.FlatStyle.Flat; this.btnPdf.Location = new System.Drawing.Point(958, 126); this.btnPdf.Size = \
new System.Drawing.Size(66, 32); this.btnPdf.Text = "PDF"; this.btnPdf.Click += new System.EventHandler(this.btnPdf_Click);
31            this.dgvSolicitudes.AllowUserToAddRows = false; this.dgvSolicitudes.AllowUserToDeleteRows = false; this.dgvSolicitudes.AutoSizeColumnsMode = \
System.Windows.Forms.DataGridViewAutoSizeColumnsMode.Fill; this.dgvSolicitudes.BackgroundColor = System.Drawing.Color.FromArgb(18, 18, 27); \
this.dgvSolicitudes.Location = new System.Drawing.Point(34, 176); this.dgvSolicitudes.MultiSelect = false; this.dgvSolicitudes.ReadOnly = true; \
this.dgvSolicitudes.RowHeadersVisible = false; this.dgvSolicitudes.SelectionMode = System.Windows.Forms.DataGridViewSelectionMode.FullRowSelect; \

```

UI/FrmSolicitudesAusencia.Designer.cs (continuacion)

```
32         this.dgvSolicitudes.Size = new System.Drawing.Size(990, 478);
            this.dgvSolicitudes.Columns.Add("IdSolicitud", "Id"); this.dgvSolicitudes.Columns["IdSolicitud"].Visible = false; \
this.dgvSolicitudes.Columns.Add("Codigo", "Código"); this.dgvSolicitudes.Columns.Add("Empleado", "Empleado"); \
this.dgvSolicitudes.Columns.Add("Departamento", "Departamento"); this.dgvSolicitudes.Columns.Add("Tipo", "Tipo"); \
this.dgvSolicitudes.Columns.Add("Desde", "Desde"); this.dgvSolicitudes.Columns.Add("Hasta", "Hasta"); this.dgvSolicitudes.Columns.Add("Estado", \
"Estado"); this.dgvSolicitudes.Columns.Add("Solicitante", "Solicitante"); this.dgvSolicitudes.Columns.Add("Creacion", "Creada");
33         this.lblResultado.AutoSize = true; this.lblResultado.Location = new System.Drawing.Point(34, 674); this.lblResultado.Text = "0 solicitudes";
34         this.Controls.Add(this.lblTitulo); this.Controls.Add(this.txtBuscar); this.Controls.Add(this.cmbDepartamento); \
this.Controls.Add(this.cmbTipo); this.Controls.Add(this.cmbEstado); this.Controls.Add(this.dtpDesde); this.Controls.Add(this.dtpHasta); \
this.Controls.Add(this.btnBuscar); this.Controls.Add(this.btnNuevo); this.Controls.Add(this.btnVer); this.Controls.Add(this.btnGestionar); \
this.Controls.Add(this.btnCancelar); this.Controls.Add(this.btnExcel); this.Controls.Add(this.btnPdf); this.Controls.Add(this.dgvSolicitudes); \
this.Controls.Add(this.lblResultado); ((System.ComponentModel.ISupportInitialize)(this.dgvSolicitudes)).EndInit(); this.ResumeLayout(false); \
this.PerformLayout();
35     }
36 }
37 }
```

UI/UiFactory.cs

```
1 using System.Drawing;
2 using System.Windows.Forms;
3 using SistemaGestionNomina.Helpers;
4 using SistemaGestionNomina.UI.Controls;
5
6 namespace SistemaGestionNomina.UI
7 {
8     public static class UiFactory
9     {
10         public static Label Title(string text, int x, int y, int width)
11         {
12             return ThemeHelper.Label(text, x, y, width, 40, 20F, FontStyle.Bold, ThemeHelper.Text);
13         }
14
15         public static Label Subtitle(string text, int x, int y, int width)
16         {
17             return ThemeHelper.Label(text, x, y, width, 28, 10.5F, FontStyle.Regular, ThemeHelper.TextMuted);
18         }
19
20         public static RoundedPanel Card(int x, int y, int width, int height)
21         {
22             RoundedPanel panel = new RoundedPanel();
23             panel.Location = new Point(x, y);
24             panel.Size = new Size(width, height);
25             return panel;
26         }
27
28         public static RoundedButton PrimaryButton(string text, int x, int y, int width)
29         {
30             RoundedButton button = new RoundedButton();
31             button.Text = text;
32             button.Location = new Point(x, y);
33             button.Width = width;
34             button.BackColor = ThemeHelper.Accent;
35             button.ForeColor = Color.FromArgb(48, 25, 92);
36             button.BorderColor = ThemeHelper.Accent;
37             return button;
38         }
39
40         public static RoundedButton SecondaryButton(string text, int x, int y, int width)
41         {
42             RoundedButton button = PrimaryButton(text, x, y, width);
43             button.BackColor = Color.Black;
44             button.ForeColor = ThemeHelper.Text;
45             button.BorderColor = ThemeHelper.Border;
46             return button;
47         }
48
49         public static DateTimePicker DatePicker(string name, int x, int y, int width)
50         {
51             DateTimePicker picker = new DateTimePicker();
52             picker.Name = name;
53             picker.Location = new Point(x, y);
54             picker.Size = new Size(width, 30);
55             picker.Format = DateTimePickerFormat.Short;
56             picker.CalendarForeColor = ThemeHelper.Text;
57             picker.CalendarMonthBackground = ThemeHelper.CardAlt;
58             return picker;
59         }
60     }
```

UI/UiFactory.cs (continuacion)

```
61     public static DataGridView Grid(int x, int y, int width, int height)
62     {
63         DataGridView grid = new DataGridView();
64         grid.Location = new Point(x, y);
65         grid.Size = new Size(width, height);
66         ThemeHelper.ApplyGrid(grid);
67         return grid;
68     }
69
70     public static void ShowExported(string path)
71     {
72         if (string.IsNullOrEmpty(path))
73         {
74             return;
75         }
76
77         MessageBox.Show("Archivo generado correctamente:\n" + path, "Exportación",
78             MessageBoxButtons.OK, MessageBoxIcon.Information);
79     }
80 }
81 }
```